

Probabilistic Methods for Phylogenetic Inference, Adaptive Sampling, and Combinatorial
Limits

by

Zachary McNulty

A dissertation submitted in partial satisfaction of the

requirements for the degree of

Doctor of Philosophy

in

Applied Mathematics

in the

Graduate Division

of the

University of California, Berkeley

Committee in charge:

Professor Steven N. Evans, Chair

Professor Shirshendu Ganguly

Professor Fraydoun Rezakhanlou

Summer 2026

The dissertation of Zachary McNulty, titled Probabilistic Methods for Phylogenetic Inference, Adaptive Sampling, and Combinatorial Limits, is approved:

Chair	_____	Date	_____
	_____	Date	_____
	_____	Date	_____

University of California, Berkeley

Probabilistic Methods for Phylogenetic Inference, Adaptive Sampling, and Combinatorial
Limits

Copyright 2026
by
Zachary McNulty

Abstract

Probabilistic Methods for Phylogenetic Inference, Adaptive Sampling, and Combinatorial Limits

by

Zachary McNulty

Doctor of Philosophy in Applied Mathematics

University of California, Berkeley

Professor Steven N. Evans, Chair

This thesis develops probabilistic methods for studying how discrete stochastic systems can be sampled, inferred, and approximated. Across problems in phylogenetics, adaptive sampling, combinatorial growth, we exploit the structure of the underlying stochastic processes to obtain finite-sample guarantees, characterize optimal procedures, and identify informative asymptotic limits.

In Chapter 1, we study bipartition coverage probabilities under the multispecies coalescent model. These bipartitions are used by some species-tree inference procedures to restrict the relevant search space, and thus our bounds provide meaningful estimates of the sample complexity of these restricted algorithms. Practically, our bounds remain below biologically realistic numbers of loci across a substantially broader range of parameter settings, expanding their usefulness for empirical datasets. Theoretically, our analysis sharpens our understanding of coalescence under the multispecies coalescent model and develops new asymptotics for these bounds and absorption times under Kingman's coalescent in the natural short branch regime.

In Chapter 2, in joint work with Daniel Raban we study a family of sequential sampling rules that adaptively bias sampling probabilities in order to achieve faster convergence of the empirical distribution to a desired target law, while keeping the resulting samples as unpredictable as possible. Such trade-offs arise naturally in compliance-testing settings, including health inspections and anti-doping programs. The resulting scheme arises naturally among a class of Markovian samplers sharing a certain translation-invariance, converges at the fastest possible $O(n^{-1})$ rate, and is the unique solution to a natural entropy-regularized optimization problem. We characterize the influence of a natural bias parameter on the trade-off between convergence and unpredictability, as measured by an adversarial prediction accuracy.

In Chapter 3, we study phylogenetic inference under the PCH model of [14], a stochastic model for the evolution of linguistic characters which explicitly incorporates polymorphism. The foundation of this model is a simple birth-death process which describes how states evolve down the underlying phylogenetic tree. Building off the work of [28], we analyze various probabilistic properties of this process, translating them into measures of quartet support and candidate bipartitions relevant to phylogenetic inference. These results motivate ASTRAL-based inference procedures and provide preliminary sample-complexity bounds for recovering the underlying language phylogeny from independently evolving characters.

Lastly, in Chapter 4, we connect the modern theory of hypergraphons to the classical theory of exchangeable arrays and Doob-Martin boundary theory. We show a broad range of combinatorial growth processes can be encoded in such arrays, and that the graphon notion of left-convergence via embedding densities easily generalizes. Moreover, we show the generated topology is equivalent to the Doob-Martin topology, generalizing some of the work of [32, 33]. Lastly, we characterize the limiting objects in terms of a natural ergodicity condition.

Contents

Contents	i
List of Figures	iii
1 Improved Bipartition Cover Bound	1
1.1 Introduction	1
1.2 Topology-Free Improvements to the Original Bound	5
1.3 The Balanced-Tree Bound	9
1.4 Asymptotic Growth and Improvement Factors	16
1.5 Numerical Results and Simulations	20
1.6 Conclusion	26
1.A Supporting Results for Coalescence and Stochastic Orders	27
1.B Technical Results for Extremal Tree Topologies	34
1.C Coalescent-Time Asymptotics	36
2 Self-Balancing Sampling	42
2.1 Introduction	42
2.2 Model and Structural Properties	46
2.3 Fast Convergence of the Empirical Distribution	51
2.4 Optimization Characterizations	58
2.5 Expected Cover Time	61
2.6 Weak-Bias Diffusive Limits and the Balance–Predictability Trade-Off	64
2.7 Concluding Remarks	71
2.A Empirical Results	73
2.B Optimal Control	79
2.C Technical Proofs	82
3 Inference under the PCH Model	90
3.1 Introduction	90
3.2 Basic Results	96
3.3 A Restricted Search-Space	104
3.4 Alternative ASTRAL Procedure	106

3.5	Sample-Complexity Bounds	108
3.6	Discussion and Concluding Remarks	110
3.A	The Anomaly-Free Zone	112
3.B	Immigration-Death Process Proofs	114
4	Exchangeable Array Processes	118
4.1	Introduction	118
4.2	Exchangeable Array Processes	120
4.3	Doob-Martin Boundary of Exchangeable Array Processes	124
4.4	Embedding Density of the Representation Functions	129
4.5	Conclusions and Future Directions	134
4.A	Candidate Cut Metrics for Exchangeable Array Processes	136
4.B	Background on Doob-Martin Boundary Theory	141
	Bibliography	146

List of Figures

1.1	Two different gene trees generated from the same species tree under the multispecies coalescent model. The black lines outline the underlying species tree topology, and the colored lines indicate the gene lineages	3
1.2	The two main extremal tree topologies: caterpillar and balanced trees	4
1.3	Example of the descendant counts $\{\alpha_i\}$ associated to the nontrivial bipartitions/edges of a phylogenetic tree. Note there is some subjectivity in deciding which edge adjacent to the root to take: depending on our choice our descendant counts would either be $\{\alpha_i\} = \{4, 2, 2, 2\}$ or $\{\alpha_i\} = \{3, 2, 2, 2\}$. To remove ambiguity, we always choose the edge with the least descendants.	6
1.4	Example of the inductive step in Lemma 1.2.1	7
1.5	Coalescent tree structure showing edge e with two subtrees of sizes m and $k - m$. The subtrees are connected to e via a branch with minimal length $T_{\min}$	10
1.6	Bipartition cover bounds as functions of the two key parameters k and $T_{\min}$. (top) The original bound [80]. (bottom) The balanced bound (Theorem 1.3.6).	21
1.7	Improvement ratios. (top) Improvement of the caterpillar bound (Corollary 1.2.3) over the original bound (Theorem 1.1.1). (bottom) Improvement of the balanced bound (Theorem 1.3.6) over the original bound.	22
1.8	Empirical overestimation ratios for caterpillar trees. (top) The original bound (Theorem 1.1.1). (bottom) The balanced bound (Theorem 1.3.6).	23
1.9	Empirical overestimation ratios for balanced trees. (top) The original bound (Theorem 1.1.1). (bottom) The balanced bound (Theorem 1.3.6).	23
1.10	Log-scaled overestimation ratios for Yule trees. Rows vary $T_{\min} \in \{0.2, 0.5, 1\}$ and columns vary $k \in \{8, 16\}$; nested boxes show progressively wider quantile intervals around the median (25–75%, 12.5–87.5%, 6.25–93.75%, and 3.125–96.875%).	25
1.11	Side-by-side illustration of the initial tree T (left) and the tree T' after balancing (right). The pruning path $\{e_i^p\}$ in red and the attachment path $\{e_i^a\}$ in blue are highlighted	34
2.1	Schematic of the self-balancing update starting from a uniform distribution: choose an outcome, downweight it by α , then renormalize.	44

2.2	Log-log plots of Nd_N as a function of the scaling factor c for (a) uniform $\beta = [0.5, 0.5, 0.5, 0.5]$ and (b) non-uniform $\beta \propto [1.1, 0.8, 0.9, 1.2]$. In both cases $p(0)$ is the uniform distribution. Dotted lines show the expected $\Theta(c^{-1})$ and $\Theta(c^{-1/2})$ scalings in the $p(0) \neq p^*$ and $p(0) = p^*$ regimes respectively.	74
2.3	Predictability plots for different choices of β . (left) Relationship between prediction accuracy and the scaling c of β . (right) The ratio of the gap $\Delta(c) := \text{accuracy} - p_{max}^*$ from the uniform baseline and the first order $\sqrt{c}$ expected scaling.	75
2.4	Log-log plots of the trade-off between adversarial accuracy and the convergence rate $(Nd_N)^{-1}$. (a) uniform $\beta = [0.5, 0.5, 0.5, 0.5]$ and (b) non-uniform $\beta \propto [1.1, 0.8, 0.9, 1.2]$	76
2.5	Citizen's assembly simulation results. The effect of the rescaled bias parameter $\beta' = c\beta$ on the probability that the quota constraints are met and on the worst-case individual inclusion probability.	78
3.1	The two rooted quartet topologies with split $uv xy$	100

Acknowledgments

I would like to express my gratitude to the many individuals who have supported me throughout my Ph.D. journey.

First and foremost, I would like to thank my advisor, Professor Steven Evans, for his patience, guidance, and support throughout these many challenging years. His kind words, gentle encouragement, and insightful feedback have been invaluable in shaping my research and guiding me through this tumultuous process. I am grateful for the opportunity to work with him and for his mentorship. Thanks to my committee members, Professors Shirshendu Ganguly and Fraydoun Rezakhanlou, for their insightful feedback and encouragement, for starting me off at the beginning of my Ph.D. journey, and now for seeing it to a close. Thanks to my collaborator, Daniel Raban, for his help on part of this thesis, and for his friendship along the way.

I would like to thank my partner Julia Kim for her unwavering love and encouragement. She has been a constant source of strength, believing in me on the many days I did not believe in myself, and for that I am eternally grateful. I am not sure how I would have completed this journey without her by my side. To my cat, Zack Junior, for being cute, and spending many afternoons in his cozy cat bed by my side.

Last, but not least, I would like to thank my family and friends for their unwavering support and encouragement. To the many members of the math department soccer (go Possums!) and frisbee team, and our regular group of spire slayers and bird collectors, for keeping me sane throughout this process, and for helping me find a sense of belonging in the department. To Eric Jankowski, Andrew Scharf, Carlos Esparza, Connor Halleck-Dube, Tom Schang, Lucy Horowitz, Spencer Taylor, Ely Sandine, and the other members of my cohort: Rose Lopez, John Nolan, Erik Herrera, and everyone else. To Zack Stier and our many afternoons of basketball. I wish you all the best of luck on the next step of your journey.

Chapter 1

An Improved Bipartition Cover Bound for the Multispecies Coalescent Model

1.1 Introduction

Consensus methods have grown in popularity in phylogenetics as large genomic datasets have revealed the limits of single-gene tree inference. Evolutionary processes such as incomplete lineage sorting, gene duplication and loss, and horizontal gene transfer can cause individual gene trees to differ from the true species tree. Consensus and summary-based methods aim to reconcile this discordance by aggregating information across many loci, providing statistically consistent estimates of species relationships under realistic models of gene tree variation.

However, even for a relatively small number of species, the space of possible species tree topologies is intractably large. Because of this, it is often necessary to reduce the search space somehow. One common strategy uses the *bipartitions* of the underlying phylogenetic tree $\mathcal{T}^*$. Each edge e of $\mathcal{T}^*$ induces a bipartition of its leaf set $\mathcal{L}$: deleting e produces two connected components, whose leaf sets form the two parts of the bipartition. By Buneman’s splits-equivalence theorem (see, e.g., [72, Theorem 3.1.4]), these bipartitions uniquely characterize $\mathcal{T}^*$. It is therefore natural to restrict the search to species trees whose bipartitions belong to a candidate set obtained from the input gene trees, possibly extended using additional heuristics. This strategy is employed by several modern phylogenetic methods, including FastRFS [81] and the widely used ASTRAL family: ASTRAL [59], ASTRAL-II [58], ASTRAL-III [87], and wASTRAL [86].

In the case the true species tree lies in this restricted space, we say the gene trees form a *bipartition cover* of the species tree. When this occurs, ASTRAL comes with some strong finite-sample guarantees [74]. When it does not occur, ASTRAL provides no guarantees on how the estimated tree will relate to the true species tree. As a result, it is important to

understand the likelihood of such a cover when assessing whether these finite-sample guarantees apply. Moreover, since an experimenter must decide how many loci to collect before the underlying species tree topology is known, they cannot calibrate the required sample size to that topology. A topology-free upper bound provides a conservative benchmark for the sample size required to obtain a bipartition cover with a prescribed probability. Under the *multispecies coalescent model (MSC)*, the paper by Uricchio, Warnow, and Rosenberg [80] develops such a bound as a function only of two key parameters: the number of species k and the minimum branch length $T_{\min}$ of the tree.

In this chapter, we identify a few areas where this existing bound is lossy and develop some topology-free improvements. Our main contributions come from careful analysis of various “worst cases” of the MSC model: species tree topologies that somehow make coalescence difficult and certain bipartitions challenging to recover. Beyond this, since all these bounds are complicated functions of the underlying parameters, we develop some asymptotics to describe the growth rates of these bounds as functions of these parameters in a few natural regimes.

Throughout, suppose we have a species tree $\mathcal{T}^*$ with k species and minimum branch length $T_{\min}$. For $q \in (0, 1)$, define

$$N_{\text{cov}}(q; \mathcal{T}^*) := \inf\{n \in \mathbb{N} : \mathbb{P}(G_1, \dots, G_n \text{ form a bipartition cover of } \mathcal{T}^*) \geq q\},$$

where the G_i are independent gene trees generated from $\mathcal{T}^*$ under the MSC model, as described below. Namely, $N_{\text{cov}}(q; \mathcal{T}^*)$ represents the minimal number of independent gene trees required to recover the underlying species tree $\mathcal{T}^*$ with the prescribed probability q . In this chapter, our goal is to upper bound this quantity.

1.1.1 The Multispecies Coalescent Model

We start with a brief discussion of the *multispecies coalescent (MSC) model*, the probabilistic model underlying all the theoretical guarantees of ASTRAL and many other phylogenetic inference algorithms. The MSC model describes how gene tree topologies are generated from a given species tree [67, 85]. It is in a sense the simplest model that can explain the phenomenon of *incomplete lineage sorting (ILS)*, the observation that the most common gene tree topology can differ from the species tree topology. This phenomenon makes species tree inference difficult: the intuitive “majority vote” estimator is not consistent.

The MSC model is essentially just a constrained version of Kingman’s coalescent: going backwards in time along each branch of the species tree, we assume the surviving gene lineages coalesce according to Kingman’s coalescent. Namely, each pair of lineages coalesce at rate one. Figure 1.1 gives an example of the MSC model and illustrates how gene tree topologies can differ from the species tree. More concretely, let

$$g_{ij}(T) := \sum_{k=j}^i \frac{\exp(-\binom{k}{2}T)(2k-1)(-1)^{k-j}j_{(k-1)}i_{[k]}}{j!(k-j)!i_{(k)}} \quad (1.1.1)$$

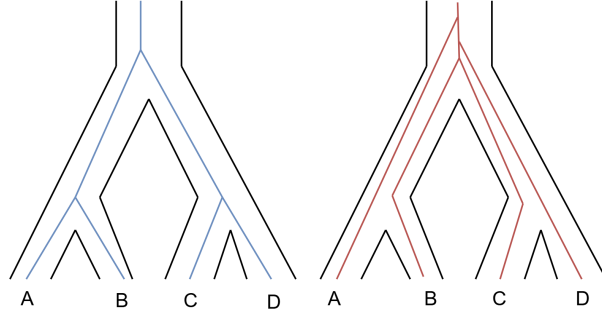


Figure 1.1: Two different gene trees generated from the same species tree under the multi-species coalescent model. The black lines outline the underlying species tree topology, and the colored lines indicate the gene lineages

be the probability that i lineages coalesce into exactly j lineages in time T under Kingman's coalescent. Here $a_{(k)} = a(a+1) \cdot \dots \cdot (a+k-1)$ and $a_{[k]} = a(a-1) \cdot \dots \cdot (a-k+1)$ denote the rising and falling factorial respectively, and the time T is measured in coalescent units. The function $g_{ij}(T)$ is the basis of the original bound in Uricchio, Warnow, and Rosenberg [80] and all the new bounds we develop. In the setting of the MSC model, T represents the length of a given branch of the species tree, and $g_{ij}(T)$ describes the distribution of the number of lineages that remain uncoalesced at the end of that branch if we start with i lineages.

1.1.2 The Original Bound and Main Contributions

Under the MSC model, [80] establishes the following bound.

Theorem 1.1.1 (Original Bipartition Cover upper bound).

$$M_o(k, T_{\min}) := \frac{\log\left(\frac{1-q}{k-3}\right)}{\log(1 - g_{k-2,1}(T_{\min}))} \quad (1.1.2)$$

satisfies $N_{\text{cov}}(q; \mathcal{T}^*) \leq \lceil M_o(k, T_{\min}) \rceil$.

In this work, we aim to develop a topology-free improvement of this upper bound. Our work relies on careful analysis of various “worst-case” topologies when it comes to phylogenetic inference. For this purpose, we define two tree types: a *caterpillar tree* and a *balanced tree*.

A *balanced tree* is any rooted tree so that for every vertex the number of leaves in its two subtrees differs by at most one. A *caterpillar tree* is a tree where every non-leaf vertex has one child who is a leaf. Figure 1.2 provides a simple illustration.

These two tree topologies represent opposite extremes for coalescent behavior: one maximally balanced and the other maximally unbalanced. Caterpillar trees present a combinatorial bottleneck for phylogenetic inference, as many of their bipartitions involve very large subsets of taxa. Balanced trees, however, present a more subtle and severe difficulty: they

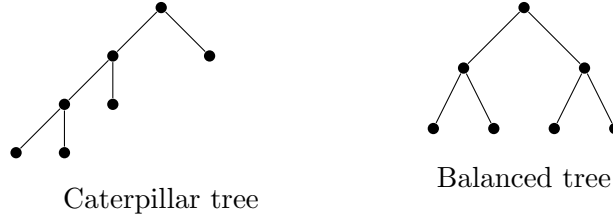


Figure 1.2: The two main extremal tree topologies: caterpillar and balanced trees

induce a *coalescent bottleneck*, in which lineages are so evenly dispersed that coalescence is systematically delayed.

This distinction is made precise in Lemma 1.2.1, which identifies caterpillar trees as extremal for increasing functions of descendant counts, and in Lemma 1.3.5, which shows that balanced trees are stochastically worst-case for the number of surviving lineages under the multispecies coalescent. Together, these lemmas underpin our main result, which appears as Theorem 1.3.6 below.

Theorem. For $\ell \geq 2$, let $w_\ell(T)$ denote the probability that the lineages descending from a balanced ℓ -taxon tree, with all branch lengths equal to T , coalesce to a single lineage. Then

$$N_{\text{cov}}(q; \mathcal{T}^*) \leq \inf \left\{ n \in \mathbb{N} : \sum_{\ell=2}^{k-2} (1 - w_\ell(T_{\min}))^n \leq 1 - q \right\}.$$

In particular,

$$N_{\text{cov}}(q; \mathcal{T}^*) \leq \left\lceil \frac{\log\left(\frac{k-3}{1-q}\right)}{-\log(1 - w_{k-2}(T_{\min}))} \right\rceil.$$

Due to the recursive structure of a balanced tree, these probabilities w_ℓ are easy to compute. Empirically, we observe this improves over Theorem 1.1.1 by several orders of magnitude in some regimes, especially when the minimum branch length $T_{\min}$ is small.

Analytically, we show in this short-branch regime this new bound improves upon the original by a factor exponential in the inverse minimum branch length; Lemma 1.4.3 makes this precise.

For clarity of exposition, we postpone several short technical results and their proofs to the appendix.

1.2 Topology-Free Improvements to the Original Bound

1.2.1 Sources of Loss in the Original Bound

To guide our work, we outline the proof of the original bound, and identify where it is lossy.

Proof Outline of Theorem 1.1.1. Every tree contains the *trivial bipartitions* $\{x\} | (\mathcal{L} - \{x\})$, so we may restrict attention to nontrivial bipartitions. There are $k - 3$ *nontrivial bipartitions* in a species tree: one for each internal edge¹.

1. By union bound we can just bound the probability a specific species tree bipartition occurs in the gene trees.
2. By independence of gene trees we can reduce to the case of a single gene tree.
3. Suppose φ_e is the bipartition associated to edge e in a species tree. One way φ_e can occur in the gene tree is if all lineages below e coalesce by the time they exit e . If there are α_e such lineages, we can lower bound this probability by the probability α_e lineages coalesce down to one along just the single edge e .
4. The worst-case for the previous step is if edge e is as short as possible (length $T_{\min}$) and the number of lineages that have to coalesce is as large as possible ($\alpha_e = k - 2$ since φ_e is nontrivial).

If we let E_n be the event all bipartitions occur in n gene trees and $E_{i,n}$ be the event bipartition φ_i occurs in at least one of the n gene trees, then combining the above gives

$$\begin{aligned}
 \mathbb{P}(E_n) &\geq 1 - \sum_{i=1}^{k-3} (1 - \mathbb{P}(E_{i,n})) \\
 &\geq 1 - (k-3) \cdot \max_i (1 - \mathbb{P}(E_{i,n})) \\
 &= 1 - (k-3) (1 - \min_i \mathbb{P}(E_{i,1}))^n \\
 &\geq 1 - (k-3) (1 - g_{k-2,1}(T_{\min}))^n
 \end{aligned}$$

from which we can invert to find the bound on n . □

¹There are $k - 2$ internal edges, but the two edges adjacent to the root induce the same, possibly trivial, bipartition.

1.2.2 Descendant Counts and Caterpillar Extremality

For an edge e , let α_e denote the number of leaves below e , namely leaves in the connected component of $\mathcal{T}^* \setminus \{e\}$ not containing the root. When an edge e_i is indexed, we define $\alpha_i := \alpha_{e_i}$. We call the set

$$\{\alpha_i\}_{i=1}^{k-3} := \{\alpha_e : e \text{ generates a nontrivial bipartition}\}$$

the *descendant counts* of $\mathcal{T}^*$. Since both edges adjacent to the root generate the same bipartition, there is some ambiguity in how we define the descendant counts. To resolve this, we always choose the edge adjacent to the root with fewer descendants, although this choice is inconsequential. An example is given in Figure 1.3.

One step at which the proof of Theorem 1.1.1 is lossy is that we essentially bound

$$\mathbb{P}(E_{i,1}) \geq g_{\alpha_i,1}(T_{e_i}) \geq g_{k-2,1}(T_{\min}),$$

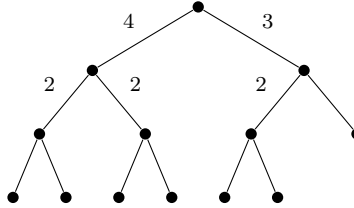


Figure 1.3: Example of the descendant counts $\{\alpha_i\}$ associated to the nontrivial bipartitions/edges of a phylogenetic tree. Note there is some subjectivity in deciding which edge adjacent to the root to take: depending on our choice our descendant counts would either be $\{\alpha_i\} = \{4, 2, 2, 2\}$ or $\{\alpha_i\} = \{3, 2, 2, 2\}$. To remove ambiguity, we always choose the edge with the least descendants.

For most edges e , the number of descendants α_e is far smaller than the total number of species k . By Lemma 1.A.4, we know $\ell \mapsto g_{\ell,1}(T_{\min})$ is a decreasing function of ℓ : the more lineages we start with, the harder it is to coalesce down to one. As a result, this second approximation $g_{\alpha_i,1} \geq g_{k-2,1}$ can be quite lossy. Nonetheless, to derive a topology-free bound we must assume these α_i are in a sense as large as possible. The following makes this precise.

Lemma 1.2.1 (Caterpillars Maximize Increasing Sums). *Suppose $\{\alpha_i\}_{i=1}^{k-3}$ are the descendant counts of our rooted binary tree T . Moreover, suppose $f : \mathbb{N} \rightarrow \mathbb{R}$ is nondecreasing. Then $\sum_i f(\alpha_i)$ is maximized when T is the caterpillar tree, in which case $\{\alpha_i\} = \{2, 3, \dots, k-2\}$.*

The result is true regardless of our convention for choosing the edge adjacent to the root. Moreover, it is clearly also true if we use *all* descendant counts $\{\alpha_i\}_{i=1}^{2k-2}$ rather than just those corresponding to nontrivial bipartitions: all but one of the additional α_i are equal to one, and it is clear from the proof below that we can handle the exceptional value as well.

Proof of Lemma 1.2.1. Since f is nondecreasing, it suffices to show we can reindex $\{\alpha_i\}_{i=1}^{k-3}$ so $\alpha_i \leq i + 1$. Heuristically, we are just pairing up each edge e in our tree T with an edge e' in our caterpillar tree so that e has at most as many descendants as e' . We induct on k .

If $k = 4$, the two unique rooted trees, shown in Figure 1.2, both have $\alpha_1 = 2$ for their only nontrivial bipartition, so the statement is trivially true.

Otherwise, suppose we have a tree $\mathcal{T}$ with $k > 4$ leaves. Suppose the two subtrees of the root contain m, r leaves respectively. There are two cases:

- (a) *Either $m \leq 2$ or $r \leq 2$.* If $\mathcal{T}'$ is the larger subtree of the root, by induction we can reindex its descendant counts $\{\alpha_i\}_{i=1}^{k-4}$ so $\alpha_i \leq i + 1$. The edge e connecting $\mathcal{T}'$ to the root trivially has $\alpha_e \leq k - 2$, as all nontrivial bipartitions in $\mathcal{T}$ do, we can pair this remaining edge e with the top nontrivial edge of our caterpillar tree.
- (b) *Both $m, r \geq 3$.* Let $\mathcal{T}', \mathcal{T}''$ be the two subtrees of the root with m, r leaves respectively, and let $\{\kappa_i\}_{i=1}^{r-3}$ and $\{\beta_j\}_{j=1}^{m-3}$ be their descendant counts. If $r < 4$ (respectively, $m < 4$), the corresponding subtree has no nontrivial bipartitions, and we take the family $\{\kappa_i\}$ (respectively, $\{\beta_j\}$) to be empty. By induction, these can be ordered so $\kappa_i \leq i + 1$ and $\beta_j \leq j + 1$. Clearly then defining

$$\alpha_i = \begin{cases} \kappa_i, & 1 \leq i \leq r - 3, \\ \beta_{i-r+3}, & r - 3 < i \leq k - 6. \end{cases}$$

Namely, we list the edges of $\mathcal{T}'$ first, and then the edges of $\mathcal{T}''$. This accounts for all but three of the edges of $\mathcal{T}$: one of the edges e adjacent to the root of $\mathcal{T}$, one edge e' in $\mathcal{T}'$, and one edge e'' in $\mathcal{T}''$.

The descendant counts $\alpha_{e'}, \alpha_{e''}$ of e', e'' are at most $r - 1, m - 1$ respectively. Moreover,

$$r - 1 = k - m - 1 \leq k - 4, \quad m - 1 = k - r - 1 \leq k - 4.$$

Thus we can let $\alpha_{k-5}, \alpha_{k-4}$ correspond to edges e', e'' respectively. Trivially e has descendant count at most $k - 2$, as all nontrivial bipartitions do. Thus we can let α_{k-3} correspond to edge e .

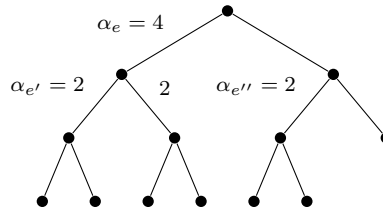


Figure 1.4: Example of the inductive step in Lemma 1.2.1

□

1.2.3 The Caterpillar Bipartition Cover Bound

Lemma 1.2.1 immediately implies the following main bound.

Theorem 1.2.2. *Suppose $h : \mathbb{N} \rightarrow [0, 1]$ is decreasing and $\mathbb{P}(E_{i,1}) \geq h(\alpha_i)$. Then*

$$n_h := \inf \left\{ n \in \mathbb{N} : \sum_{\ell=2}^{k-2} [1 - h(\ell)]^n \leq 1 - q \right\} \quad (1.2.1)$$

satisfies $N_{\text{cov}}(q; \mathcal{T}^) \leq n_h$.*

Proof. By the union bound and independence of the gene trees,

$$1 - \mathbb{P}(E_n) \leq \sum_{i=1}^{k-3} (1 - \mathbb{P}(E_{i,n})) = \sum_{i=1}^{k-3} [1 - \mathbb{P}(E_{i,1})]^n \leq \sum_{i=1}^{k-3} [1 - h(\alpha_i)]^n.$$

Since h is decreasing and bounded in $[0, 1]$, the function $f(x) = [1 - h(x)]^n$ is increasing. Applying Lemma 1.2.1 to this f gives

$$\mathbb{P}(E_n) \geq 1 - \sum_{\ell=2}^{k-2} [1 - h(\ell)]^n.$$

Thus, for every n in the set defining n_h , we have $\mathbb{P}(E_n) \geq q$, and hence $N_{\text{cov}}(q; \mathcal{T}^*) \leq n_h$. $\square$

If h is independent of the species tree topology, then so is equation (1.2.1). Letting $h(x) = g_{x,1}(T_{\min}) \in [0, 1]$, which is decreasing by Corollary 1.A.5, then Theorem 1.2.2 and the original bound $\mathbb{P}(E_{i,1}) \geq g_{\alpha_i,1}(T_{\min})$ of Uricchio, Warnow, and Rosenberg [80] give a new topology-free bound.

Corollary 1.2.3 (Caterpillar Bipartition Cover Bound).

$$M_c(k, T_{\min}) := \inf \left\{ n \in \mathbb{N} : \sum_{\ell=2}^{k-2} [1 - g_{\ell,1}(T_{\min})]^n \leq 1 - q \right\} \quad (1.2.2)$$

satisfies $N_{\text{cov}}(q; \mathcal{T}^) \leq M_c(k, T_{\min})$.*

As with the original bound (1.1.2), equation (1.2.2) depends on $\mathcal{T}^*$ only through k and $T_{\min}$. Heuristically, this replaces the worst-case $g_{k-2,1}(T_{\min})$ with the corresponding order- n power mean of $\{g_{\ell,1}(T_{\min})\}_{\ell=2}^{k-2}$.

1.3 The Balanced-Tree Bound

1.3.1 Coalescence Below an Edge

Theorem 1.2.2 suggests another avenue for improvement is to obtain a tighter bound $\mathbb{P}(E_{i,1}) \geq h(\alpha_i)$ than the original $h(x) = g_{x,1}(T_{\min})$. For small values of $T_{\min}$ the bound in Corollary 1.2.3 is dominated by the terms $g_{\alpha_i,1}$ for which α_i are quite large. These are the bipartitions corresponding to edges e close to the root of the tree which have many descendants. In Uricchio, Warnow, and Rosenberg [80], the authors essentially assume *none* of the descendants coalesce before they reach edge e . When e is far from the leaves, descendants have a long distance to travel through the tree and potentially coalesce, making this estimate very conservative.

Hence to improve this bound we should try to account for coalescent events that occur below the edge e . Namely, we ought to somehow replace $g_{\alpha_e,1}(T_{\min})$ by $\mathbb{E}g_{X_e,1}(T_{\min})$ where X_e is the (random) number of remaining lineages entering edge e . Clearly $\mathbb{P}(E_{i,1}) \geq \mathbb{E}g_{X_{e_i},1}(T_{\min})$ by the same logic that Theorem 1.1.1 employs. In general, X_e depends on the topology of $\mathcal{T}^*$ below the edge e . To develop a topology-free bound, we must somehow identify the worst-case topology in this setting.

Recall a random variable X is said to *first-order stochastically dominate* Y if $\mathbb{P}(X > t) \geq \mathbb{P}(Y > t)$ for all t , or equivalently if $\mathbb{E}u(X) \geq \mathbb{E}u(Y)$ for all nondecreasing functions u . We denote this by $X \geq_{st} Y$. Similarly, X is said to *second-order stochastically dominate* Y if $\mathbb{E}u(X) \geq \mathbb{E}u(Y)$ for all nondecreasing concave functions, and we denote this by $X \geq_{sst} Y$. First-order stochastic dominance naturally implies second-order.

Since $x \mapsto g_{x,1}(T)$ is decreasing, convex (Corollary 1.A.5, Lemma 1.A.7), we see

Lemma 1.3.1. *If $U_{e_i} \geq_{sst} X_{e_i}$ then $\mathbb{P}(E_{i,1}) \geq \mathbb{E}g_{U_{e_i},1}(T_{\min})$.*

Remark 1.3.2. If U_e is a function of α_e alone and is (second order) stochastically increasing in this parameter, applying Theorem 1.2.2 with $h(\ell) = \mathbb{E}g_{U_e,1}(T_{\min})$ yields a new topology-free bound.

To apply this observation, we begin by analyzing the coalescent events occurring in the two edges directly below e .

Let Z_i^T be the random variable with distribution $\mathbb{P}(Z_i^T = j) = g_{ij}(T)$. Namely, if we start i lineages then Z_i^T represents the (random) number of lineages that remain uncoalesced after running Kingman's coalescent for time T . When we drop the T superscript, it is implicitly assumed that $T = T_{\min}$. For a positive integer-valued random variable A , let Z_A^T denote the independent mixture

$$\mathbb{P}(Z_A^T = i) = \sum_{\ell \geq 1} \mathbb{P}(A = \ell) \mathbb{P}(Z_\ell^T = i).$$

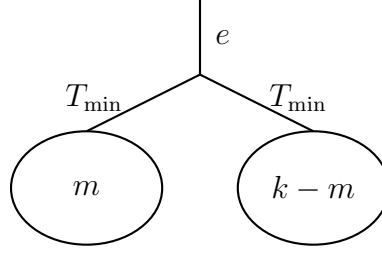


Figure 1.5: Coalescent tree structure showing edge e with two subtrees of sizes m and $k - m$. The subtrees are connected to e via a branch with minimal length $T_{\min}$.

When we write expressions such as $Z_A^T + Z_B^T$ including multiple such mixtures, we implicitly mean we are mixing two independent families $\{Z_\ell\}_{\ell \geq 1}, \{Z'_\ell\}_{\ell \geq 1}$ so

$$\mathbb{P}(Z_A^T + Z_B^T = i) = \sum_{m=1}^{i-1} \mathbb{P}(Z_A^T = m) \mathbb{P}(Z_B^T = i - m).$$

Suppose below the edge e our tree splits into two subtrees of size $m, k - m$. Clearly then

$$X_e \leq_{st} Z_m^{T_1} + Z_{k-m}^{T_2},$$

where T_1, T_2 are the lengths of the branches connecting the two subtrees to e . Moreover

$$Z_m^{T_1} + Z_{k-m}^{T_2} \leq_{st} Z_m^{T_{\min}} + Z_{k-m}^{T_{\min}},$$

by Lemma 1.A.4. Figure 1.5 illustrates the setup. Next, we show more balanced splits $(m, k - m)$ tend to have less coalescence.

1.3.2 Deterministic Local Balancing

Lemma 1.3.3 (Deterministic Balancing Lemma). *Let k and T be fixed. Then*

$$Z_i^T + Z_{k-i}^T \leq_{st} Z_j^T + Z_{k-j}^T, \quad \forall 1 \leq i \leq j \leq \frac{k}{2}.$$

In particular, the stochastic maximizer is $i = \lfloor k/2 \rfloor$.

Heuristically, $i = \lfloor k/2 \rfloor$, where the two subtrees in Figure 1.5 are as evenly balanced as possible, is the maximizer because it minimizes the total number of pairs $\binom{m}{2} + \binom{k-m}{2}$ available in the two subpopulations, hampering coalescence. Lemma 1.3.3 implies

$$X_e \leq_{st} Z_{\lfloor \alpha_e/2 \rfloor} + Z_{\lceil \alpha_e/2 \rceil} \tag{1.3.1}$$

Proof of Lemma 1.3.3. The statement is vacuously true for $k = 2, 3$, so for induction assume it is true for all $k' < k$. Let $S_{k,m}^T := Z_m^T + Z_{k-m}^T$ denote the total number of lineages entering an edge e with subtrees of size $m, k - m$, the setup in Figure 1.5. Let $\tau_{k,m}$ be the time of the first coalescent event in either of our two subpopulations of size $m, k - m$, allowing the possibility $\tau_{k,m} > T$.

The basic idea is that if we start with k lineages, we can reduce to $k - 1$ lineages by just waiting for the first coalescence event to occur in either subpopulation. There is of course the chance no coalescent event occurs, so we must handle this separately. The proof then relies on a few main ideas. Firstly, for more balanced splits we tend to wait longer until the first coalescent event.

Claim 1. For fixed k , $\tau_{k,m}$ is stochastically increasing in m for $1 \leq m \leq k/2$.

Next, conditional on a specific first coalescent time, the more balanced our split is, the more lineages tend to remain uncoalesced by the end of the branch.

Claim 2. For fixed k, x, t , $\mathbb{P}(S_{k,m}^T \geq x \mid \tau_{k,m} = t)$ is an increasing function of m for $1 \leq m \leq k/2$.

Lastly, the more time we wait until the first coalescent event, the less time the remaining lineages have to coalesce and the more lineages we tend to have by the end of the branch.

Claim 3. For fixed k, x, m , $\mathbb{P}(S_{k,m}^T \geq x \mid \tau_{k,m} = t)$ is an increasing function of t .

To prove Claim 1, let $h_{k,m} := \binom{m}{2} + \binom{k-m}{2}$. Then $\tau_{k,m} \sim \text{Exp}(h_{k,m})$. Since

$$\mathbb{P}(\tau_{k,m} > t) = \exp(-th_{k,m})$$

and $h_{k,m}$ is decreasing in m for $1 \leq m \leq k/2$, these tails are increasing in m . Hence $\tau_{k,m}$ is stochastically increasing in m as desired.

To prove Claim 2, first suppose $t > T$. No coalescent event occurs before time T , so

$$\mathbb{P}(S_{k,m}^T \geq x \mid \tau_{k,m} = t) = \mathbf{1}\{x \leq k\},$$

which is independent of m , so the claim is trivial. For $t \leq T$, by the strong memoryless property of the exponential distribution underlying Kingman's coalescent, we know

$$\mathbb{P}(S_{k,m}^T \geq x \mid \tau_{k,m} = t) = p_{k,m} \cdot \mathbb{P}(S_{k-1,m-1}^{T-t} \geq x) + (1 - p_{k,m}) \cdot \mathbb{P}(S_{k-1,m}^{T-t} \geq x), \quad (1.3.2)$$

where

$$p_{k,m} = \frac{\binom{m}{2}}{\binom{m}{2} + \binom{k-m}{2}}$$

is the probability the first coalescence occurs in the population of size m . Namely, after the first coalescent event occurs in either subpopulation, what remains is a Kingman's coalescent with one less lineage. For $1 \leq m \leq k/2 - 1$, by induction

$$\mathbb{P}(S_{k,m}^T \geq x \mid \tau_{k,m} = t)$$

$$\begin{aligned}
 &= p_{k,m} \cdot \mathbb{P}(S_{k-1,m-1}^{T-t} \geq x) + (1 - p_{k,m}) \cdot \mathbb{P}(S_{k-1,m}^{T-t} \geq x) \\
 &\leq \mathbb{P}(S_{k-1,m}^{T-t} \geq x) \\
 &\leq p_{k,m+1} \cdot \mathbb{P}(S_{k-1,m}^{T-t} \geq x) + (1 - p_{k,m+1}) \cdot \mathbb{P}(S_{k-1,m+1}^{T-t} \geq x) \\
 &= \mathbb{P}(S_{k,m+1}^T \geq x \mid \tau_{k,m+1} = t).
 \end{aligned}$$

Thus $\mathbb{P}(S_{k,m}^T \geq x \mid \tau_{k,m} = t)$ is increasing in m for each fixed t .

To prove Claim 3, by equation (1.3.2) it suffices to show for fixed k, m that $S_{k,m}^T$ is stochastically decreasing in T . But this follows from a simple coupling argument. For $s \leq T$, couple $S_{k,m}^T$ to $S_{k,m}^s$ by using the same exponential clocks for the coalescent events. At time s the number of lineages remaining agree, and $S_{k,m}^T$ can only decrease from here if more coalescents occur between times s and T .

With the claims proved, we are ready to finally finish our proof of Lemma 1.3.3. Note that our above claims are exactly the conditions of the Mixture Lemma (Lemma 1.A.11), which shows first-order stochastic dominance is preserved by increasing mixtures, where $\{S_{k,m}^T\}_{m=1}^{\lfloor k/2 \rfloor}$ are playing the role of our V_i and $\{\tau_{k,m}\}_{m=1}^{\lfloor k/2 \rfloor}$ are playing the role of our Θ_i . Hence the result just follows from applying this lemma. $\square$

1.3.3 Recursive Balancing and the Balanced Worst Case

By iterating the ideas from the previous section, it is natural to believe the topology minimizing coalescence is the one where *all* splits are even, namely when the topology below edge e is the balanced tree. This is true in the following sense.

Given a binary tree $\mathcal{T}$ (with branch lengths) let $X_{\mathcal{T}}$ denote the number of lineages that have not coalesced by the time they reach the root. We first prove an extension of our deterministic balancing lemma, Lemma 1.3.3, which allows for random subpopulations.

To do so, we require a slightly stronger stochastic order than $\leq_{st}$. We say a random variable X is less than Y with respect to the *likelihood ratio ordering* (LRO), and write $X \leq_{lr} Y$, if the likelihood ratio $f_Y(t)/f_X(t)$ is nondecreasing in t , where f_X and f_Y are the respective mass functions. Equivalently,

$$f_X(s)f_Y(t) \geq f_X(t)f_Y(s), \quad s \leq t.$$

The LRO implies first-order stochastic dominance.

Lemma 1.3.4 (Balancing Lemma). *If A, A', B, B' are mutually independent, positive integer-valued random variables so that $A \leq_{lr} A'$ and $B \leq_{lr} B'$, then*

$$Z_{A+B} + Z_{A'+B'} \leq_{st} Z_{A+B'} + Z_{A'+B}.$$

Note when A, B, A', B' are all deterministic, this reduces to Lemma 1.3.3.

Proof of Balancing Lemma 1.3.4. First observe that for a fixed sum s , the deterministic balancing Lemma 1.3.3 implies the random variables

$$Y_d := Z_{\frac{s+d}{2}} + Z_{\frac{s-d}{2}}$$

are decreasing with respect to $\leq_{st}$ for $0 \leq d < s$. By the Mixture Lemma (Lemma 1.A.11), we see $Y_D \geq_{st} Y_{D'}$ whenever $0 \leq D \leq_{st} D' < s$. Namely, if we fix the total number of lineages s , then if the imbalance D between our two subpopulations tends to be less than imbalance D' , we tend to have less coalescence.

From here, the idea is simple: we condition on the total number of lineages across the two populations, and attempt to control the conditional differences. For convenience, define

$$\Delta_A := A' - A, \quad \Delta_B := B' - B, \quad S_A := A' + A, \quad S_B := B' + B.$$

Now, note in both $Z_{A+B} + Z_{A'+B'}$ and $Z_{A+B'} + Z_{A'+B}$ the total number of lineages $S = S_A + S_B$ is the same. Hence, by conditioning on $S_A, S_B, |\Delta_A|$, and $|\Delta_B|$, it suffices to show $D' \geq_{st} D$, where D, D' are the conditional differences

$$D' = |\Delta_A + \Delta_B| \mid \{|\Delta_A|, |\Delta_B|, S_A, S_B\}, \quad D = |\Delta_A - \Delta_B| \mid \{|\Delta_A|, |\Delta_B|, S_A, S_B\}.$$

After this conditioning, $\{D, D'\} = \{|\Delta_A| + |\Delta_B|, ||\Delta_A| - |\Delta_B||\}$. If $|\Delta_A| = 0$ or $|\Delta_B| = 0$, then $D = D'$, and the conclusion is immediate. Hence suppose both absolute differences are positive. Let

$$q = \mathbb{P}(D' = |\Delta_A| + |\Delta_B|) = \mathbb{P}(\Delta_A \Delta_B \geq 0 \mid |\Delta_A|, |\Delta_B|, S_A, S_B)$$

be the probability D' takes on the larger of these two values. Then it suffices to show $q \geq 1/2$. Since Δ_A, Δ_B are conditionally independent given S_A, S_B

$$q = p_A p_B + (1 - p_A)(1 - p_B) \geq \frac{1}{2} \iff \left(p_A - \frac{1}{2}\right) \left(p_B - \frac{1}{2}\right) \geq 0,$$

where $p_A = \mathbb{P}(\Delta_A \geq 0 \mid |\Delta_A|, S_A)$ and likewise for p_B . We will show that $p_A, p_B \geq 1/2$, completing the proof. Note conditional on $\{|\Delta_A| = m, S_A = s\}$ then $p_A \geq 1/2$ occurs precisely when

$$\mathbb{P}\left(A' = \frac{s+m}{2}, A = \frac{s-m}{2}\right) \geq \mathbb{P}\left(A' = \frac{s-m}{2}, A = \frac{s+m}{2}\right).$$

As A, A' are independent, this follows from $A \leq_{lr} A'$. Similarly, $p_B \geq 1/2$, so $D' \geq_{st} D$. The result follows then by taking expectations with respect to $\Delta_A, \Delta_B, S_A, S_B$. $\square$

With the random balancing lemma in hand, we can now iterate local balancing through the tree. This shows that the balanced topology minimizes coalescence among all trees with the same number of leaves. The following result makes this precise.

Lemma 1.3.5 (Balanced Tree Is Worst Case). *If $\mathcal{B}_k$ is the balanced tree with k leaves and all branch lengths equal to $T_{\min}$ then $X_{\mathcal{B}_k} \geq_{st} X_{\mathcal{T}}$ for any other tree $\mathcal{T}$ with k leaves and minimum branch length $T_{\min}$.*

Proof of Lemma 1.3.5. First, replacing every branch length in $\mathcal{T}$ by $T_{\min}$ can only increase the number of lineages reaching the root. It therefore suffices to assume that all branch lengths are $T_{\min}$.

Throughout the proof, write $X_r := X_{\mathcal{B}_r}$ and $U_r := Z_{X_r}$, with all occurrences of these random variables understood to be independent copies when they arise from disjoint subtrees. We first establish the balanced-subtree split comparison

$$U_m + U_{k-m} \leq_{st} U_{\lfloor k/2 \rfloor} + U_{\lceil k/2 \rceil} \stackrel{d}{=} X_k, \quad 1 \leq m \leq \lfloor k/2 \rfloor. \quad (1.3.3)$$

The idea of the proof is simple: both X_m and X_{k-m} consist of two subtrees of (approximately) the same size. By taking one subtree from each of X_m, X_{k-m} and swapping them using the balancing lemma, we create two balanced subpopulations. From there, we can inductively balance these two subpopulations separately to get the result.

More formally, we prove equation (1.3.3) by strong induction on k . The cases $k \leq 3$ are immediate. For $k = 4$, the only nontrivial comparison is

$$U_1 + U_3 = Z_1 + Z_{X_3} \leq_{st} Z_1 + Z_3 \leq_{st} Z_2 + Z_2 = U_2 + U_2,$$

where the first inequality follows from Corollary 1.A.10 because $X_3 \leq 3$ almost surely, and the second from the deterministic balancing lemma, Lemma 1.3.3.

Now suppose equation (1.3.3) holds for every $k' < k$, and fix $1 \leq m < \lfloor k/2 \rfloor$. Set

$$a = \lfloor m/2 \rfloor, \quad a' = \lceil m/2 \rceil, \quad b = \lfloor (k-m)/2 \rfloor, \quad b' = \lceil (k-m)/2 \rceil.$$

Then $a \leq a' \leq b \leq b'$, and the recursive construction of the balanced trees gives

$$X_m \stackrel{d}{=} U_a + U_{a'}, \quad X_{k-m} \stackrel{d}{=} U_b + U_{b'}.$$

By Lemma 1.A.14,

$$U_a \leq_{lr} U_{a'} \leq_{lr} U_b \leq_{lr} U_{b'}.$$

Suppose first that k is even. Since $a + b' = a' + b = k/2$, apply the Balancing Lemma with $(A, A', B, B') = (U_a, U_b, U_{a'}, U_{b'})$ to obtain

$$U_m + U_{k-m} \stackrel{d}{=} Z_{U_a + U_{a'}} + Z_{U_b + U_{b'}} \leq_{st} Z_{U_a + U_{b'}} + Z_{U_{a'} + U_b}.$$

Applying the inductive hypothesis with total $k/2$ shows both $U_a + U_{b'} \leq_{st} X_{k/2}$ and $U_{a'} + U_b \leq_{st} X_{k/2}$. Hence Corollary 1.A.10 yields

$$U_m + U_{k-m} \leq_{st} Z_{X_{k/2}} + Z_{X_{k/2}} = U_{k/2} + U_{k/2},$$

which proves equation (1.3.3) for even k .

If k is odd, then $a + b = (k - 1)/2$ and $a' + b' = (k + 1)/2$. Applying the Balancing Lemma with $(A, A', B, B') = (U_a, U_{b'}, U_{a'}, U_b)$ gives

$$U_m + U_{k-m} \stackrel{d}{=} Z_{U_a+U_{a'}} + Z_{U_b+U_{b'}} \leq_{st} Z_{U_a+U_b} + Z_{U_{a'}+U_{b'}}.$$

The inductive comparisons with totals $(k - 1)/2$ and $(k + 1)/2$ bound the two inner sums by $X_{(k-1)/2}$ and $X_{(k+1)/2}$, respectively. Therefore

$$U_m + U_{k-m} \leq_{st} Z_{X_{(k-1)/2}} + Z_{X_{(k+1)/2}} = U_{\lfloor k/2 \rfloor} + U_{\lceil k/2 \rceil}.$$

This completes the proof of the balanced-subtree split comparison.

We now prove the stated worst-case result, again by induction on k . The case $k = 1$ is immediate. Suppose the result holds for every smaller tree, and let L and R be the two root subtrees of $\mathcal{T}$, with m and $k - m$ leaves, respectively, where $m \leq k - m$. The inductive hypothesis gives $X_L \leq_{st} X_m$ and $X_R \leq_{st} X_{k-m}$. Consequently, Corollary 1.A.10 and equation (1.3.3) give

$$X_{\mathcal{T}} = Z_{X_L} + Z_{X_R} \leq_{st} U_m + U_{k-m} \leq_{st} U_{\lfloor k/2 \rfloor} + U_{\lceil k/2 \rceil} = X_{\mathcal{B}_k}.$$

□

1.3.4 The Balanced Bipartition Cover Bound

Lemma 1.3.5 suggests a natural way of upper-bounding the number of lineages X_e entering a given edge e : $X_e \leq_{st} X_{\mathcal{B}_{\alpha_e}}$. From this, we get our final bound.

Theorem 1.3.6 (Balanced Bipartition Cover Bound). *Let*

$$M_b(k, T_{\min}) := \inf \left\{ n \in \mathbb{N} : \sum_{\ell=2}^{k-2} [1 - w_{\ell}(T_{\min})]^n \leq 1 - q \right\} \quad (1.3.4)$$

where $w_{\ell} := \mathbb{P}(Z_{X_{\mathcal{B}_{\ell}}} = 1)$. Then $N_{\text{cov}}(q; \mathcal{T}^*) \leq M_b(k, T_{\min})$. In particular,

$$N_{\text{cov}}(q; \mathcal{T}^*) \leq \left\lceil \frac{\log\left(\frac{k-3}{1-q}\right)}{-\log(1 - w_{k-2}(T_{\min}))} \right\rceil.$$

Proof of Theorem 1.3.6. Lemma 1.3.5 implies $X_e \leq_{st} X_{\mathcal{B}_{\alpha_e}}$ so by the remark below Lemma 1.3.1 it suffices to show that $U_{\ell} := X_{\mathcal{B}_{\ell}}$ is stochastically increasing in ℓ . This follows from induction as

$$U_{\ell} \stackrel{d}{=} Z_{U_{\lfloor \ell/2 \rfloor}} + Z'_{U_{\lceil \ell/2 \rceil}}.$$

When ℓ increases by one, exactly one of $\lfloor \ell/2 \rfloor$ and $\lceil \ell/2 \rceil$ increases by one. Hence the inductive hypothesis and Corollary 1.A.10 imply $U_{\ell+1} \geq_{st} U_\ell$. It follows that

$$h(\ell) := \mathbb{E} g_{U_\ell, 1}(T_{\min}) = w_\ell(T_{\min})$$

is decreasing. Applying Theorem 1.2.2 with this choice of h gives the first bound. Moreover, $w_\ell(T_{\min}) \geq w_{k-2}(T_{\min})$ for $2 \leq \ell \leq k-2$, so

$$\sum_{\ell=2}^{k-2} [1 - w_\ell(T_{\min})]^n \leq (k-3) [1 - w_{k-2}(T_{\min})]^n.$$

Solving the resulting inequality for n gives the stated explicit bound. $\square$

As the subtrees of a balanced tree are balanced, we can calculate $w_\ell(T_{\min})$ easily in a recursive fashion. Letting $W_\ell := Z_{X_{B_\ell}}$, we see $W_1 \equiv 1$ and $W_\ell =^d Z_{W_{\lfloor \ell/2 \rfloor} + W_{\lceil \ell/2 \rceil}}$, so

$$\mathbb{P}(W_\ell = j) = \sum_{a \geq 1} \sum_{b \geq 1} \mathbb{P}(W_{\lfloor \ell/2 \rfloor} = a) \mathbb{P}(W_{\lceil \ell/2 \rceil} = b) g_{a+b, j}(T_{\min}), \quad \ell \geq 2. \quad (1.3.5)$$

1.4 Asymptotic Growth and Improvement Factors

Since the bounds are complicated functions of $T_{\min}$ and k , we develop estimates of their scaling with respect to these two parameters.

Here we confirm our intuition that our new bounds drastically improve performance in the small T regime. For fixed T , we show that all the bounds are $\Theta_T(\log(k))$, and that this is essentially the best one can do while relying on the union bound. We start with a result that specifies the asymptotics of the function $g_{k,1}(T)$ underlying all our bounds.

1.4.1 Coalescence-Probability Asymptotics

Lemma 1.4.1 (Asymptotics of $g_{k,1}(T)$). *The function $T \mapsto g_{k,1}(T)$ has asymptotics*

$$\begin{cases} 1 - g_{k,1}(T) \sim \frac{3(k-1)}{k+1} \exp(-T), & T \uparrow \infty, \\ g_{k,1}(T) \sim \frac{k!}{2^{k-1}} T_k^{k-1}, & k^2 T_k = o(1). \end{cases} \quad (1.4.1)$$

Moreover, for fixed T , we have $g_{k,1}(T) \downarrow \mathbb{P}(S_\infty \leq T) =: s(T)$ where $S_\infty := \sum_{i=2}^\infty \tau_i$ for $\tau_i \sim \text{Exp}\left(\binom{i}{2}\right)$ independent. Furthermore, the gap $\Delta_k(T) := g_{k,1}(T) - s(T)$ satisfies

$$\Delta_k(T) = \frac{2f_{S_\infty}(T)}{k} + o\left(\frac{1}{k}\right) \sim \frac{2f_{S_\infty}(T)}{k}.$$

The proof can be found in Appendix 1.C.

1.4.2 Asymptotic Growth of the Bounds

Using this, we develop the following asymptotics for our two bounds. Recall that $M_o(k, T)$ denotes the original bound of Uricchio, Warnow, and Rosenberg [80] and $M_b(k, T)$ our balanced bound (Theorem 1.3.6). Let

$$u(T) := \frac{2 - \exp(-T/2)}{1 - \exp(-T/2)} \geq \mathbb{E}X_{\mathcal{B}_k}^T$$

be our upper bound on the number of lineages reaching the root of a balanced tree, which we prove in Corollary 1.A.9 below.

Jensen's inequality then gives the convenient upper bound

$$M_b(k, T) \leq \frac{\log\left(\frac{k-3}{1-q}\right)}{-\log\left(1 - g_{\mathbb{E}X_{\mathcal{B}_{k-2}}, 1}(T)\right)}.$$

Thus the balanced-tree analysis replaces $k-2$ by $\mathbb{E}X_{\mathcal{B}_{k-2}}$, which may be substantially smaller.

Lemma 1.4.2 (Bound Asymptotics). *Let $\kappa_{q,k} := \log((k-3)/(1-q))$. Then*

$$M_o(k, T) \sim \begin{cases} \frac{\kappa_{q,k}}{T}, & T \uparrow \infty, \\ \frac{2^{k-3}\kappa_{q,k}}{(k-2)!T^{k-3}}, & T = o(k^{-2}), \\ \frac{\kappa_{q,k}}{-\log(1-s(T))}, & T \text{ fixed}, k \rightarrow \infty \end{cases}$$

$$M_b(k, T) \lesssim \frac{\kappa_{q,k}}{-\log(1 - g_{u(T), 1}(T))}, \quad T \text{ fixed}, k \rightarrow \infty$$

Proof. Let $s(T) := \mathbb{P}(S_\infty \leq T)$ be our limiting CDF, $\Delta_k(T) := g_{k,1}(T) - s(T)$ our gap, and $c(T) := 2f_{S_\infty}(T)$ our gap's asymptotic constant.

We start with M_o . Note Lemma 1.4.1 implies

$$\log(1 - g_{k,1}(T)) \sim \begin{cases} -T, & T \uparrow \infty, \\ -\frac{k!}{2^{k-1}}T^{k-1}, & T = o(k^{-2}). \end{cases} \quad (1.4.2)$$

From this

$$M_o(k, T) \sim \begin{cases} \frac{\kappa_{q,k}}{T}, & T \uparrow \infty, \\ \frac{2^{k-3}\kappa_{q,k}}{(k-2)!T^{k-3}}, & T = o(k^{-2}). \end{cases}$$

In particular, we recover the small T bound found in Appendix B of [74], although here we exactly quantify the asymptotic constant.

For the fixed T and asymptotic k bound, we Taylor expand $\log(x)$ about $1-s$ to observe

$$\log(1 - g_{k,1}(T)) = \log(1 - s) - \sum_{n=1}^{\infty} \frac{\Delta^n}{n(1-s)^n} = \log(1 - s) - \frac{\Delta}{1-s} - O(\Delta^2).$$

Hence this and Lemma 1.4.1 imply as $k \rightarrow \infty$

$$M_o(k, T) = \frac{\kappa_{q,k}}{\left(\frac{c(T)}{1-s} + o(1)\right) \cdot \frac{1}{k} - \log(1-s)} \sim \frac{\kappa_{q,k}}{-\log(1-s)}.$$

For $M_b(k, T)$, note as $g_{k,1}(T)$ is decreasing in k we have

$$\sum_{\ell=2}^{k-2} (1 - w_\ell(T))^n \geq (k-3)(1 - g_{2,1}(T))^n = (k-3) \exp(-nT).$$

Hence $M_b(k, T) \geq \kappa_{q,k} T^{-1}$. As $k \mapsto g_{k,1}(T)$ is decreasing and convex by Lemma 1.A.7,

$$\sum_{\ell=2}^{k-2} (1 - \mathbb{E}g_{U_\ell,1}(T))^n \leq (k-3)(1 - g_{\mathbb{E}U_{k-2},1}(T))^n.$$

Applying Corollary 1.A.9,

$$M_b(k, T) \leq \frac{\kappa_{q,k}}{-\log(1 - g_{u(T),1}(T))}.$$

Combining these two previous results,

$$\frac{\kappa_{q,k}}{T} \leq M_b(k, T) \leq \frac{\kappa_{q,k}}{-\log(1 - g_{u(T),1}(T))}.$$

Since $M_b(k, T) \leq M_o(k, T)$, the lower bound shows us $M_b(k, T) \sim \kappa_{q,k} T^{-1}$ as $T \rightarrow \infty$. Since $u(T) \sim 2/T$ as $T \downarrow 0$, the preceding Taylor asymptotics, which require $k^2 T = o(1)$, do not describe the relevant joint regime. Lemma 1.4.3 analyzes this regime directly.

□

1.4.3 The Improvement Factor

Lemma 1.4.3 (Improvement Factor). *For fixed T , the improvement factor satisfies*

$$\liminf_{k \rightarrow \infty} \frac{M_o(k, T)}{M_b(k, T)} \geq \beta_T := \frac{\log(1 - g_{u(T),1}(T))}{\log(1 - s(T))}.$$

Moreover, as $T \downarrow 0$,

$$\log \beta_T = \Theta(T^{-1}).$$

Proof. The fixed- T estimates in Lemma 1.4.2 give

$$M_o(k, T) \sim \frac{\kappa_{q,k}}{-\log(1 - s(T))}, \quad M_b(k, T) \lesssim \frac{\kappa_{q,k}}{-\log(1 - g_{u(T),1}(T))},$$

which imply the stated lower bound on the limiting improvement factor.

Since $-\log(1-x) \sim x$ as $x \downarrow 0$ and both $g_{u(T),1}(T), s(T) \downarrow 0$, we have

$$\log \beta_T = \log g_{u(T),1}(T) - \log s(T) + o(1).$$

We first control $\log s(T)$. Let $K(x) := \log \mathbb{E} \exp(-xS_\infty)$. By independence, for $x > 0$,

$$K(x) = -\log \left(\prod_{n=2}^{\infty} \left[1 + \frac{x}{\lambda_n} \right] \right).$$

Since $(n-1)^2 \leq 2\lambda_n \leq n^2$, the infinite-product expansion of $\sinh$ gives

$$0 \leq K(x) + \log \left(\frac{\sinh(\pi\sqrt{2x})}{\pi\sqrt{2x}} \right) \leq \log(1+2x).$$

Thus $K(x) \sim -\pi\sqrt{2x}$ as $x \rightarrow \infty$. Applying Corollary 2 of [13] with $A = 1$, $P(x) = s(x)$, $B = -\pi^2/2$, and $\beta = -1$ yields

$$\log s(T) \sim -\frac{\pi^2}{2}T^{-1} \quad \text{as } T \downarrow 0.$$

Next, $\log g_{u(T),1}(T) \leq 0$, which already gives $\log \beta_T = O(T^{-1})$. For the lower bound, let $\tau_i \sim \text{Exp}(\binom{i}{2})$ be independent, so

$$g_{k,1}(T) = \mathbb{P} \left(\sum_{i=2}^k \tau_i \leq T \right).$$

Using $1 - e^{-x} \geq x/(1+x)$,

$$g_{k,1}(T) \geq \mathbb{P} \left(\bigcap_{i=2}^k \left\{ \tau_i \leq \frac{T}{k-1} \right\} \right) \geq \prod_{i=2}^k \frac{\frac{T}{k-1} \binom{i}{2}}{1 + \frac{T}{k-1} \binom{i}{2}}.$$

If $x_{i,T} := \frac{T}{k-1} \binom{i}{2}$, then

$$T \log g_{k,1}(T) \geq Tk \cdot \frac{1}{k} \sum_{i=2}^k \log \frac{x_{i,T}}{1 + x_{i,T}}.$$

Since $u(T) \sim 2/T$, a Riemann-sum approximation gives

$$T \log g_{u(T),1}(T) \geq 2 \int_0^1 \log \frac{y^2}{1+y^2} dy + o(1) = -(\pi + 2 \log 2) + o(1).$$

Combining this with the asymptotics for $\log s(T)$ gives

$$\log \beta_T \geq T^{-1} \left(-\pi - 2 \log 2 + \frac{\pi^2}{2} + o(1) \right) = \Omega(T^{-1}).$$

Together with the upper bound, this proves $\log \beta_T = \Theta(T^{-1})$. $\square$

Thus, in this regime, the asymptotic lower bound on the improvement factor grows exponentially in T^{-1} .

1.4.4 Limits of the Union-Bound Framework

Repeating the above analysis shows that *any* bound built using the union bound and the coalescent function $g_{k,1}(T)$ cannot be asymptotically better than $\log(k)/T$. This is true, *even if the descendant counts α_i are known*, as we can still upper bound $g_{\alpha_i,1}(T) \leq g_{2,1}(T)$. Hence the asymptotics in k are fundamentally limited to $\Theta(\log(k))$ by our union bound argument, and we can only hope to improve upon the constant down to the limit T^{-1} . The next section examines the corresponding finite-parameter improvements numerically.

To see why the $\Theta(\log(k))$ growth rate is natural for the k asymptotics, note that $g_{k,1}(T)$ is roughly constant and equal to $s(T) > 0$ for all large k . Hence in large caterpillar trees, where most of the edges have large descendant counts, most of their bipartitions have $g_{k,1}(T)$ saturating in this way.

In this setting, we are essentially saying each bipartition corresponds to a $\text{Geometric}(s(T))$ random variable which specifies how many gene trees we need for that specific bipartition to show up. Each such variable has tail $\mathbb{P}(\text{Geo}(s) > m) = (1 - s)^m$. Solving for $p(m) = k^{-1}$ we see $m = \log(k)/[-\log(1 - s)]$, so the maximum of k independent such geometric random variables ought to be roughly of this order.

1.5 Numerical Results and Simulations

In this section we compare the performance of our new bounds both to the original bound of Uricchio, Warnow, and Rosenberg [80] and to empirical results coming from simulations of the MSC model under various species tree topologies. Code for reproducing the below figures can be found on GitHub [54].

1.5.1 Numerical Growth of the Bounds

First, we numerically explore how the original and balanced bounds grow as functions of the two key parameters: the minimum branch length $T_{\min}$ and the species count k . We evaluated the bounds at coverage probability $q = 0.99$, for $T_{\min} \in \{0.1, 0.2, 0.5, 1, 2\}$ and $k \in \{5, 10, 15, 20, 25, 30\}$. Figure 1.6 highlights our results. Both bounds increase with k and decrease with $T_{\min}$, as expected. However, the balanced bound is dramatically lower in essentially all regimes.

While it varies species to species, the maximal number of independent genes is typically on the order of 10^3 to 10^5 [50, 64]. One drawback of the original bipartition cover bound (1.1.2) that Figure 1.6 highlights is that it dips above this threshold even for relatively few species k , especially when the minimum branch length is short. This can limit its practicability because scientists might not have access to enough genes to achieve their desired coverage probability. One attractive feature of our new bound is that it remains below these biologically reasonable thresholds for a much wider parameter range, greatly improving its utility.

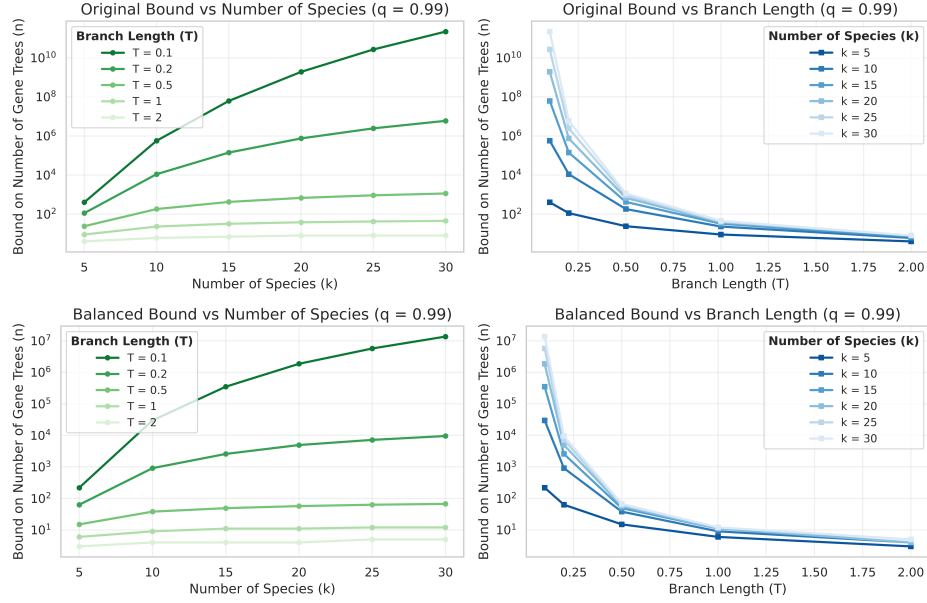


Figure 1.6: Bipartition cover bounds as functions of the two key parameters k and $T_{\min}$. (top) The original bound [80]. (bottom) The balanced bound (Theorem 1.3.6).

1.5.2 Numerical Improvement Ratios

To quantify how much these new bounds improve on the bound of [80], we plot the ratios $I := M_{\text{old}}/M_{\text{new}}$ again as functions of the two nontrivial parameters $T_{\min}$, k . Values of $I > 1$ indicate an improvement over the original bound, and values $I < 1$ a degradation. Figure 1.7 shows these improvement ratios over the same parameter range as the previous subsection.

As Figure 1.7 highlights, most of the improvement comes from the more sophisticated balanced bound (Theorem 1.3.6), which improves on the original bound by several orders of magnitude in the challenging high k and low $T_{\min}$ regimes.

On the other hand, the caterpillar bound only improves by a small constant factor. This is due to the fact

$$s(T) := \lim_{\ell \rightarrow \infty} g_{\ell,1}(T) > 0,$$

so most terms in the sum $\sum_{\ell=2}^{k-2} (1 - g_{\ell,1}(T))^n$ quickly saturate near $s(T)$ and we gain little by replacing the maximum $g_{k-2,1}(T)$ by $g_{\ell,1}(T)$.

1.5.3 Simulation-Based Overestimation

In this subsection, we aim to quantify the level to which our bounds overestimate the number of gene trees required to obtain a bipartition cover. If n_b is one of our upper bounds, define the *overestimation ratio* to be the quantity $n_b/N_{\text{cov}}(q; \mathcal{T}^*)$. To estimate $N_{\text{cov}}(q; \mathcal{T}^*)$, we sample gene trees independently from the MSC model until we get a bipartition cover, and

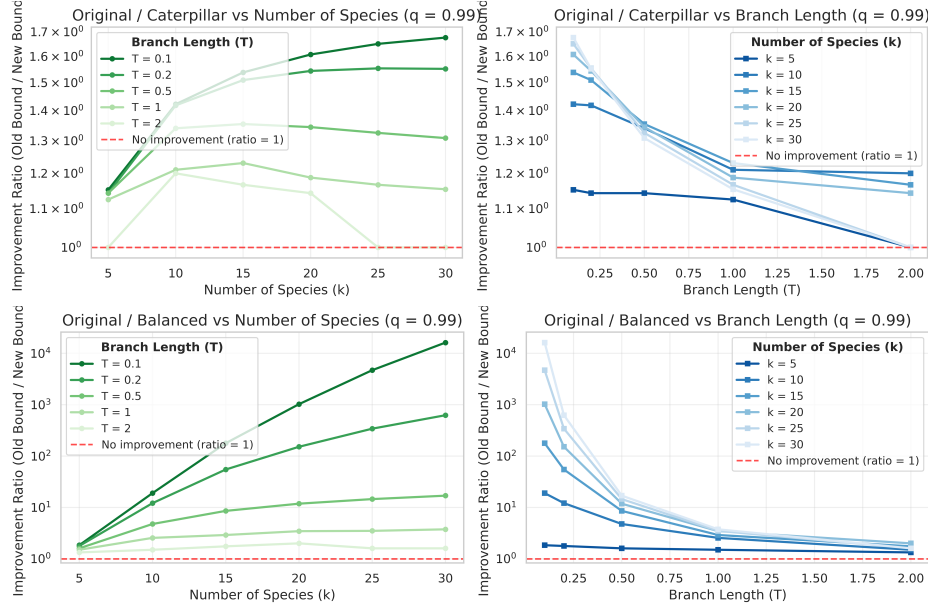


Figure 1.7: Improvement ratios. (top) Improvement of the caterpillar bound (Corollary 1.2.3) over the original bound (Theorem 1.1.1). (bottom) Improvement of the balanced bound (Theorem 1.3.6) over the original bound.

repeat this for N independent trials to get an estimate of the desired quantile. We used $N = 10^4$ and $q = 0.9$. These were chosen so a significant number of trials (roughly 1000) would generate data above the true 90th quantile. Each trial was capped at 10^5 gene trees, and we required fewer than 1% of trials in every parameter setting to reach this cap.

We explore a few different species tree topologies: the caterpillar tree with equal branch lengths, the balanced tree with equal branch lengths, and trees generated randomly from the Yule model. For the first two topologies, we used $T_{\min} \in \{0.1, 0.2, 0.5, 1\}$ and $k \in \{10, 15, 20, 25, 30\}$. We generated 100 Yule trees with birth rate one for each pair $T_{\min} \in \{0.2, 0.5, 1\}$ and $k \in \{8, 16\}$, and rescaled all branch lengths so that the shortest internal branch had length $T_{\min}$. We then again ran N trials on each tree.

Because caterpillar and balanced trees realize the two distinct extremal mechanisms used in our topology-free analysis, they help measure how close the resulting bound is to optimal in that sense. We expect the bound to be tighter for these structured cases than for the more typical Yule trees. Our results for caterpillar and balanced trees are displayed in Figures 1.8 and 1.9, respectively.

The overestimation ratio is significantly higher for balanced trees as compared to caterpillars. Since our bound is topology-free, this aligns with the empirical observation that caterpillar trees are difficult to recover under the MSC.

Examining its dependence on the model parameters, we observe that the overestimation ratio of our new bound varies less with the species parameter k as compared to the original

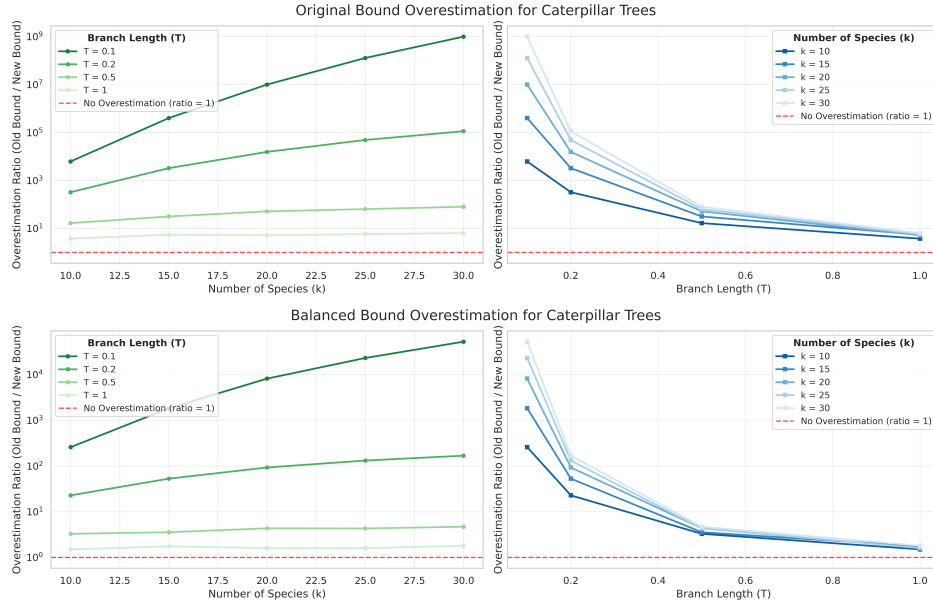


Figure 1.8: Empirical overestimation ratios for caterpillar trees. (top) The original bound (Theorem 1.1.1). (bottom) The balanced bound (Theorem 1.3.6).

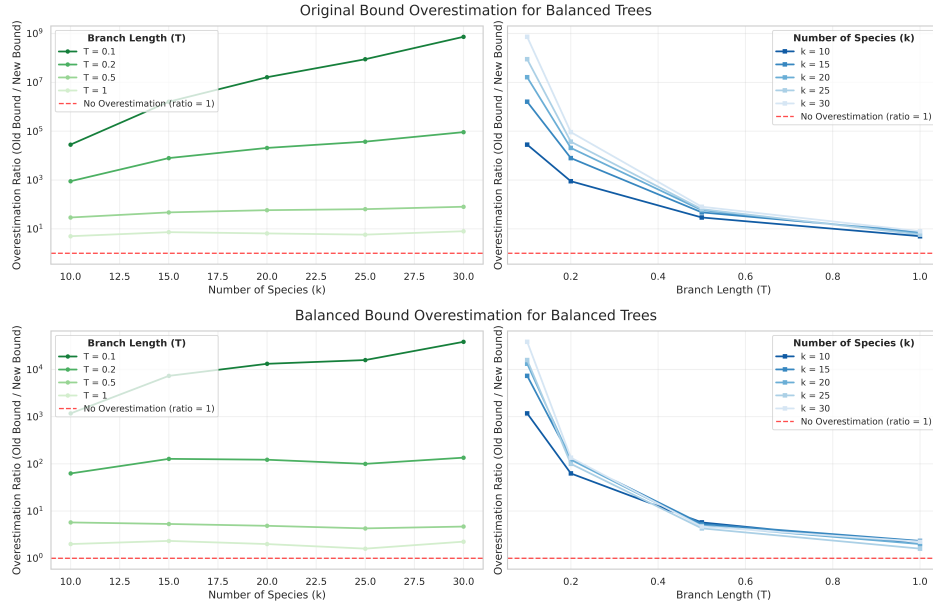


Figure 1.9: Empirical overestimation ratios for balanced trees. (top) The original bound (Theorem 1.1.1). (bottom) The balanced bound (Theorem 1.3.6).

bound, indicating that the balanced bound should scale more favorably to larger values of k compared to the original bound. On the other hand, performance appears to deteriorate significantly as $T_{\min}$ decreases, suggesting less favorable scaling in the small $T_{\min}$ regime.

Since most trees differ substantially from these worst-case configurations, the performance on Yule trees provides a more realistic measure of the typical behavior of the bound. To assess this, we sample a range of species tree topologies from the Yule model and compute the corresponding overestimation ratios. Figure 1.10 shows the resulting distribution of these ratios across various choices of k and $T_{\min}$.

As expected, performance in this setting is noticeably worse than in the structured worst-case scenarios, but it remains substantially better than that of the original bound. Increasing the species parameter k appears to have a much larger impact than in the previous settings, suggesting worse scaling. Overall, the new bound is still fairly far from tight, suggesting incorporating even partial topological information might be necessary.

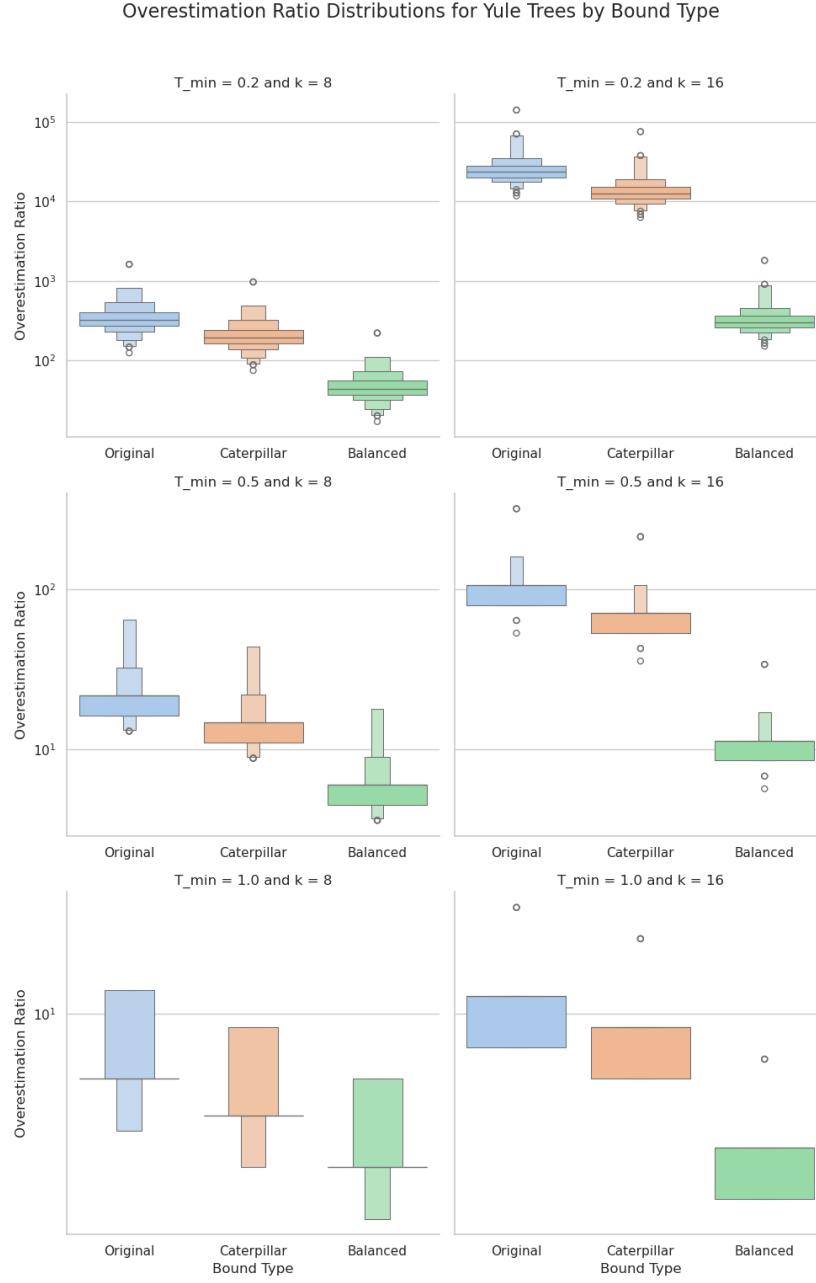


Figure 1.10: Log-scaled overestimation ratios for Yule trees. Rows vary $T_{\min} \in \{0.2, 0.5, 1\}$ and columns vary $k \in \{8, 16\}$; nested boxes show progressively wider quantile intervals around the median (25–75%, 12.5–87.5%, 6.25–93.75%, and 3.125–96.875%).

1.6 Conclusion

We improved the bipartition cover bound of Uricchio, Warnow, and Rosenberg [80] by identifying two topology-free extremal features of species trees. Caterpillar trees maximize the descendant-count profiles that enter the union-bound argument, whereas balanced trees stochastically maximize the number of lineages that remain uncoalesced for a fixed number of descendants. Combining these observations yields a recursively computable balanced bound that can improve on the original bound by several orders of magnitude across the parameter regimes considered in our simulations.

In analyzing coalescence under the MSC model, our key technical tools are the balancing lemmas, Lemmas 1.3.3 and 1.3.4. These formalize the intuition that, when a fixed collection of lineages is distributed more evenly across descendant populations, more lineages tend to remain uncoalesced. While short internal branches and anomaly-zone behavior are familiar obstacles to species-tree inference, our analysis identifies a distinct topology-dependent mechanism: balanced splits systematically delay coalescence, allowing ancestral lineages to persist longer and thereby increasing the potential for incomplete lineage sorting. Moreover, these balancing lemmas may be interesting in their own right, providing new insight into coalescence under the MSC model.

For fixed branch length, the new bound grows as $\Theta_T(\log k)$, which is the best possible order within the union-bound framework used here. The bound need not be globally tight, however, because its two extremal ingredients are attained by different tree topologies. As the branch length tends to zero, our asymptotic lower bound on the improvement factor grows as $\exp(\Theta(T^{-1}))$. Our simulations likewise show these bounds are still fairly conservative, particularly in the challenging short branch length regime. Incorporating partial information about the species tree topology therefore provides a natural direction for obtaining sharper finite-sample guarantees.

1.A Supporting Results for Coalescence and Stochastic Orders

1.A.1 Log-Concavity and Ultra Log-Concavity

A sequence $\{x_k\}$ is *log-concave* if $x_k^2 \geq x_{k-1}x_{k+1}$. We say a random variable X is log-concave if the sequence $x_k = \mathbb{P}(X = k)$ is log-concave.

In order to apply Theorem 1.A.12, we will need to determine whether various variables of interest are log-concave. For this purpose, we introduce a few basic closure properties of log-concave random variables. It will turn out to be useful to work with a strengthening of log-concavity. We say a sequence $\{x_k\}$ is *ultra log-concave (ULC)* if the sequence $\{k! \cdot x_k\}$ is log-concave, or equivalently $kx_k^2 \geq (k+1)x_{k-1}x_{k+1}$. Clearly ultra log-concavity implies log-concavity. The following result, and much more, can be found in Chapter 4 of [70].

Theorem 1.A.1 (Prékopa). *If X, Y are independent, (ultra) log-concave, integer-valued random variables, then $X + Y$ is (ultra) log-concave. Equivalently, the class of (ultra) log-concave distributions is closed under convolution.*

The following shows us that Kingman's coalescent preserves log-concavity in our setting of interest.

Theorem 1.A.2 (Preservation of Ultra Log-Concavity). *Suppose P_t is the transition semi-group of a pure-death process with death rates λ_i so that (a) λ_i is convex (b) $\eta_i := \lambda_i/i$ is concave and (c) at least one of these is strictly convex/concave. Moreover, suppose that π is ultra log-concave with either $\text{supp}(\pi) = \{2, \dots, M\}$ or $\text{supp}(\pi) = \{1, \dots, M\}$ for some $M \in \mathbb{N} \cup \{\infty\}$. Then $\pi(t) := \pi P_t$ is ultra log-concave for all $t \geq 0$. In particular, this is true for Kingman's coalescent with $\lambda_i = \binom{i}{2}$.*

Proof of Theorem 1.A.2. Recall Kolmogorov's forward equations imply

$$\frac{d}{dt}\pi_i(t) = \lambda_{i+1}\pi_{i+1}(t) - \lambda_i\pi_i(t).$$

First, we handle the interior. For $i \in \text{Int}(\text{supp}(\pi))$ we can define the ratios

$$v_i := \lambda_i \cdot \frac{\pi_i}{\pi_{i-1}}, \quad r_i := \frac{i\pi_i^2}{(i+1)\pi_{i-1}\pi_{i+1}} = \frac{v_i}{v_{i+1}} \cdot \frac{\eta_{i+1}}{\eta_i}.$$

From here, using the forward equations we can see

$$\begin{aligned} \frac{\dot{r}_i}{r_i} &= \frac{d}{dt} \log(r_i) \\ &= 2 \frac{\dot{\pi}_i}{\pi_i} - \frac{\dot{\pi}_{i-1}}{\pi_{i-1}} - \frac{\dot{\pi}_{i+1}}{\pi_{i+1}} \end{aligned}$$

$$\begin{aligned}
 &= (-2\lambda_i + \lambda_{i-1} + \lambda_{i+1}) + 2\lambda_{i+1} \frac{\pi_{i+1}}{\pi_i} - \lambda_i \frac{\pi_i}{\pi_{i-1}} - \lambda_{i+2} \frac{\pi_{i+2}}{\pi_{i+1}} \\
 &= (\lambda_{i-1} - 2\lambda_i + \lambda_{i+1}) + 2v_{i+1} - v_i - v_{i+2}.
 \end{aligned}$$

By assumption, $r_i(0) \geq 1$ for all i . Since $r_i(t)$ is continuous in t , the only way for $r_i(t) < 1$ to occur is if $r_i(s) = 1$ for some $0 \leq s < t$. We will show at this time that $\dot{r}_i(s) > 0$, and hence we can never drop below one. Suppose $r_i(s) = 1$ and $r_k(s) \geq 1$ for all other k . From these, we see respectively

$$v_i = \frac{\eta_i}{\eta_{i+1}} \cdot v_{i+1}, \quad v_{i+2} \leq \frac{\eta_{i+2}}{\eta_{i+1}} \cdot v_{i+1}.$$

Using these two bounds, we can see

$$\frac{\dot{r}_i}{r_i} \geq (\lambda_{i-1} - 2\lambda_i + \lambda_{i+1}) - [\eta_i - 2\eta_{i+1} + \eta_{i+2}] \cdot \frac{v_{i+1}}{\eta_{i+1}}.$$

Since λ_i is convex, the first term is nonnegative. Since η_i is concave, the second term is nonnegative. Moreover, by assumption one of these must be strictly positive, and thus $\dot{r}_i/r_i > 0$ as desired.

Now, we handle the boundaries. In the case $M < \infty$, we trivially have

$$M\pi_M^2(t) \geq (M+1)\pi_{M-1}(t)\pi_{M+1}(t) = 0$$

as $\pi_{M+1}(t) = 0$ for all t – in a pure-death process on $\mathbb{N}$ the support can never increase. For the lower boundary, first consider the case $\text{supp}(\pi) = \{1, \dots, M\}$. Then again we trivially have $\pi_1(t) \geq 2\pi_0(t)\pi_2(t) = 0$ for all t as we can never drop below one in a pure-death process on $\mathbb{N}$. In the case $\text{supp}(\pi) = \{2, \dots, M\}$ note that $2\pi_2(0) > 3\pi_1(0)\pi_3(0) = 0$. By continuity, we must have $2\pi_2(t) > 3\pi_1(t)\pi_3(t)$ for small enough $t > 0$. For such a t , we see that $\pi(t)$ is ultra log-concave. Moreover, since $\pi_1(t) > 0$ for any positive t we see $\text{supp}(\pi(t)) = \{1, \dots, M\}$ so we can now apply the previous case to show ultra log-concavity is preserved for all further times. $\square$

1.A.2 Coalescent Transition Probabilities and Lineage Counts

We will start by proving some basic properties about the function $g_{i,j}(T)$ and the associated random variables Z_i^T with distribution $\mathbb{P}(Z_i^T = j) = g_{i,j}(T)$. Recall that Z_i^T gives the distribution of the number of lineages remaining after starting with i lineages and running Kingman's coalescent for T seconds.

Moreover, recall $X_{\mathcal{B}_k}$ denotes the number of lineages reaching the root of a balanced tree with k leaves and branch lengths all equal to T . For convenience, we denote $X_k := X_{\mathcal{B}_k}$. Using the above, we will show that all the variables of interest in this chapter are log-concave.

Lemma 1.A.3. *The random variables X_m, Z_{X_m} are ultra log-concave, and hence log-concave.*

Proof. We will prove this by induction on m . Trivially $X_1 = Z_1 \equiv 1$ are ultra log-concave. Now suppose for induction that X_i, Z_{X_i} are log-concave for $i < m$. Note

$$X_m = Z_{X_{\lfloor m/2 \rfloor}} + Z_{X_{\lceil m/2 \rceil}}.$$

By Prékopa's Theorem 1.A.1, the sum of independent ultra log-concave distributions is ultra log-concave, so X_m is ultra log-concave. Since $\text{supp}(X_m) = \{2, \dots, m\}$, by Theorem 1.A.2, Kingman's coalescent preserves ultra log-concavity, so Z_{X_m} is ultra log-concave. This completes the induction. $\square$

Now we show our variables Z_i^T have some simple monotonicity properties.

Lemma 1.A.4. *Let $i \leq j$ and $t \leq T$. Then $Z_i^T \leq_{st} Z_j^T$ and $Z_i^T \leq_{st} Z_i^t$.*

Heuristically this just states the obvious: the more lineages we start with at the beginning of a branch and the less time we have to coalesce, the more lineages we tend to end with.

Proof of Lemma 1.A.4. Again the fact the function is decreasing in T follows immediately via a coupling argument. To see that it is increasing in i , we just condition on the first coalescence event. Assume $z \leq i$, otherwise both $\mathbb{P}(Z_i^T \geq z) = \mathbb{P}(Z_{i-1} \geq z) = 0$. Note if $\tau \sim \exp\left(\binom{i}{2}\right)$ is the time of the first coalescent event

$$\mathbb{P}(Z_i^T \geq z) = \int_0^\infty \mathbb{P}(Z_{i-1}^{T-t} \geq z) \tau(dt) \geq \int_0^\infty \mathbb{P}(Z_{i-1}^T \geq z) \tau(dt) = \mathbb{P}(Z_{i-1}^T \geq z)$$

where we let $\mathbb{P}(Z_{i-1}^T \geq z) = 1$ if $T < 0$. This gives $Z_i^T \geq Z_{i-1}^T$ stochastically as desired. $\square$

Note that $g_{i1}(T) = 1 - \mathbb{P}(Z_i^T \geq 2)$, from which we get the following immediate corollary.

Corollary 1.A.5. *The function $g_{i1}(T)$ is decreasing in i and increasing in T .*

Thus $g_{i,j}(T)$ has some natural monotonicity properties. The following Lemmas show that most of the variability in $g_{i,j}(T)$ occurs when both T and i are small.

Lemma 1.A.6. *For fixed n, k , the function $\mathbb{P}(Z_n^T = k) = g_{n,k}(T)$ is log-concave in T .*

Proof of Lemma 1.A.6. We prove this by induction on n . For $n = 1$ we see $g_{n,k}(T) = \mathbf{1}\{k = 1\}$ which is trivially log-concave. For the inductive step, recall that by conditioning on the first coalescent event and using the memoryless property of Kingman's coalescent, we observed

$$g_{n,k}(T) = \int_0^T g_{n-1,k}(T-t) \binom{n}{2} \exp\left(-\binom{n}{2}t\right) dt.$$

Namely, we are just convolving the log-concave function $g_{n-1,k}$ with the log concave function $\exp(-\binom{n}{2}t)$. By Prékopa's theorem 1.A.1, convolutions preserve log-concavity. Thus $g_{n,k}$ is log-concave. $\square$

Lemma 1.A.7. *For fixed T , the function $i \mapsto g_{i,1}(T)$ is convex.*

Proof of Lemma 1.A.7. It is a classical result of Karlin that the transition matrix $P_T := [g_{i,j}(T)]_{i,j \in \mathbb{N}}$ is totally positive (of any order) for any fixed T . See for example [42]. Theorem 2 of the paper [26], again a result of Karlin, then shows such matrices preserve convexity. Since $e_1 = (1, 0, 0, \dots)$ is convex and $(g_{i,1}(T)) = P_T e_1$ this implies the sequence $\{g_{i,1}(T)\}_{i \in \mathbb{N}}$ is convex. $\square$

Next, we study the expected number of lineages that survive under Kingman's coalescent when the number of starting lineages is large. Since the initial coalescent rate $\binom{i}{2}$ grows so fast in the number of lineages, eventually adding more lineages has barely any influence on the number remaining at time T . This is essentially the well-known observation that Kingman's coalescent can come down from infinity in any positive time.

Lemma 1.A.8 (Upper Bound for Expected Number of Lineages). *For any $i \in \mathbb{N}$ and $T > 0$*

$$\mathbb{E}Z_i^T \leq \frac{1}{1 - (1 - 1/i) \exp(-T/2)} \leq \frac{1}{1 - \exp(-T/2)}.$$

In particular, for any fixed $T > 0$ we see $\mathbb{E}Z_i^T = \Theta_T(1)$.

Note that in the small T regime, where $\exp(-T/2) \approx 1 - T/2$, this bound is roughly $2/T$, recovering the well-known asymptotics of Kingman's coalescent as it comes down from infinity. Hence in a sense this bound is roughly tight in that regime.

Proof of Lemma 1.A.8. Kolmogorov's backwards equations and Jensen's inequality imply

$$\left. \frac{d}{dt} \mathbb{E}Z_i^t \right|_{t=T} = \mathbb{E} \left(- \binom{Z_i^T}{2} \right) \leq - \binom{\mathbb{E}Z_i^T}{2}.$$

Hence letting $z(t) := \mathbb{E}Z_i^t$ we see $z(t)$ satisfies

$$\dot{z}(t) \leq -\frac{z(z-1)}{2}, \quad z(0) = i.$$

Let $\dot{y} = -y(y-1)/2$ with $y(0) = i$. Applying separation of variables

$$y(t) = \frac{1}{1 - (1 - 1/i) \exp(-t/2)}.$$

Since $\dot{z} \leq \dot{y}$ and $z(0) = y(0)$, applying the comparison principle yields

$$z(t) \leq y(t) \leq \frac{1}{1 - \exp(-t/2)},$$

as we desired. $\square$

Corollary 1.A.9. *Suppose that $k \in (2^{m-1}, 2^m]$. Then*

$$\mathbb{E}X_{\mathcal{B}_k}^T \leq \frac{2 - \exp(-T/2)}{1 - \exp(-T/2) + \exp(-mT/2)/2^m} \leq \frac{2 - \exp(-T/2)}{1 - \exp(-T/2)}$$

Proof of Corollary 1.A.9. Fix T and let

$$u(n) := \frac{1}{1 - (1 - 1/n)\alpha}, \quad \alpha := \exp(-T/2)$$

be from our upper bound in Lemma 1.A.8. Then u is increasing and concave in n . Let $Y_m := X_{\mathcal{B}_{2^m}}$ and $a_m := \mathbb{E}Y_m$. Clearly by a coupling argument $\mathbb{E}X_{\mathcal{B}_k} \leq a_m$. Moreover, by concavity

$$a_{m+1} = \mathbb{E}(Z_{Y_m} + Z_{Y_m}) \leq 2\mathbb{E}u(Y_m) \leq 2u(a_m)$$

In particular, this implies

$$\frac{1}{a_{m+1}} \geq \frac{1 - \alpha}{2} + \frac{\alpha}{2} \cdot \frac{1}{a_m}.$$

Iterating this, we see

$$a_m \leq \frac{2 - \alpha}{1 - \alpha + (\alpha/2)^m} \leq \frac{2 - \alpha}{1 - \alpha}$$

□

1.A.3 First-Order Stochastic-Dominance Tools

The following results collect general properties of first-order stochastic dominance as they apply to the random variables Z_i^T . The book [73] contains a broader treatment of stochastic orders. Below we typically suppress the T notation unless it is explicitly necessary. The following generalizes Lemma 1.A.4 by allowing the number of initial lineages to be random.

Corollary 1.A.10 (Stochastic Dominance of Compositions). *If $X \geq_{st} Y$ then*

$$Z_X \geq_{st} Z_Y.$$

Moreover, if $X' \geq_{st} Y'$ and $X \perp\!\!\!\perp Y, X' \perp\!\!\!\perp Y'$, or more generally if $(X, Y) \geq_{st} (X', Y')$ coordinate-wise then

$$Z_X + Z_Y \geq_{st} Z_{X'} + Z_{Y'}.$$

In our context, the first statement above just says that if we tend to have more lineages entering a given edge of our tree, then we tend to also have more lineages exiting it. The second statement says that if an edge e tends to have more lineages uncoalesced in both its subtrees, then the edge e tends to have more lineages entering it.

Proof of Corollary 1.A.10. For the first statement, note that

$$\mathbb{P}(Z_X \geq t) = \mathbb{E}_{i \sim X} \mathbb{P}(Z_i \geq t) \geq \mathbb{E}_{i \sim Y} \mathbb{P}(Z_i \geq t) = \mathbb{P}(Z_Y \geq t),$$

as by Lemma 1.A.4 the function $i \mapsto \mathbb{P}(Z_i \geq t)$ is increasing. Hence $Z_X \geq Z_Y$ stochastically. The second claim follows immediately by conditioning on Z_X

$$\begin{aligned} \mathbb{P}(Z_X + Z_Y \geq t) &= \mathbb{E}_{z \sim Z_X} \mathbb{P}(Z_Y \geq t - z) \\ &\geq \mathbb{E}_{z \sim Z_X} \mathbb{P}(Z_{Y'} \geq t - z) \\ &\geq \mathbb{E}_{z \sim Z_{X'}} \mathbb{P}(Z_{Y'} \geq t - z) \\ &= \mathbb{P}(Z_{X'} + Z_{Y'} \geq t) \end{aligned}$$

so that $Z_X + Z_Y \geq_{st} Z_{X'} + Z_{Y'}$. The joint statement follows similarly just from the fact $\varphi(a, b) := \mathbb{P}(Z_a + Z_b \geq t)$ is trivially increasing coordinate-wise. $\square$

The following tells us that increasing mixtures of increasing variables are again increasing.

Lemma 1.A.11 (Mixture Lemma). *Suppose that we have collections of random variables $\{V_i\}_{i \in \mathbb{N}}, \{\Theta_i\}_{i \in \mathbb{N}}$ so that for each $i \leq j$ we have:*

- a) $\Theta_i \leq_{st} \Theta_j$,
- b) $V_i \mid \{\Theta_i = \theta\} \leq_{st} V_j \mid \{\Theta_j = \theta\}$ for every fixed θ ,
- c) $V_i \mid \{\Theta_i = \theta\} \leq_{st} V_i \mid \{\Theta_i = \theta'\}$ for every $\theta \leq \theta'$ and i .

Then $V_i \leq_{st} V_j$ for $i \leq j$.

Proof of the Mixture Lemma (Lemma 1.A.11). For $i < j$ note by conditioning on Θ_i we get

$$\begin{aligned} \mathbb{P}(V_i \geq x) &= \int_{\mathbb{R}} \mathbb{P}(V_i \geq x \mid \Theta_i = \theta) \Theta_i(d\theta) \\ &\leq \int_{\mathbb{R}} \mathbb{P}(V_j \geq x \mid \Theta_j = \theta) \Theta_i(d\theta) \\ &\leq \int_{\mathbb{R}} \mathbb{P}(V_j \geq x \mid \Theta_j = \theta) \Theta_j(d\theta) \\ &= \mathbb{P}(V_j \geq x) \end{aligned}$$

by first using property (b) and then properties (a),(c) together. For the latter, we are using the fact that (c) implies $\mathbb{P}(V_j \geq x \mid \Theta_j = \theta)$ is an increasing function in θ , and since $\Theta_i \leq_{st} \Theta_j$ this ordering is preserved when you take the expectation with respect to any increasing function. $\square$

1.A.4 Likelihood-Ratio-Order Tools

The likelihood ratio ordering defined above has some convenient closure properties. The following appear as Theorems 1.C.9 and 1.C.17 of [73], respectively.

Theorem 1.A.12 (LRO preserved by Convolutions). *Suppose we have two collections of independent, log-concave random variables, $\{X_i\}_{i=1}^n$ and $\{Y_i\}_{i=1}^n$. If $X_i \leq_{lr} Y_i$ for all i , then $\sum_{i=1}^n X_i \leq_{lr} \sum_{i=1}^n Y_i$.*

Theorem 1.A.13 (LRO preserved by Mixtures). *Suppose we have a collection $\{V_i\}_{i \in \mathbb{N}}$ of random variables so that $V_i \leq_{lr} V_j$ for $i \leq j$. Moreover, suppose that we have random variables M, N independent of $\{V_i\}$ with $M \leq_{lr} N$. Then $V_M \leq_{lr} V_N$.*

Using these results gives the following monotonicity lemma for our random variables of interest.

Lemma 1.A.14. *For $i \leq j$ we have $Z_i \leq_{lr} Z_j$, $X_i \leq_{lr} X_j$ and $Z_{X_i} \leq_{lr} Z_{X_j}$.*

Proof. First, we show $\{Z_i\}_{i \in \mathbb{N}}$ is increasing in the LRO, namely

$$\mathbb{P}(Z_i = m) \cdot \mathbb{P}(Z_j = n) \geq \mathbb{P}(Z_i = n) \cdot \mathbb{P}(Z_j = m), \quad \forall i \leq j, \forall m \leq n.$$

But this is a consequence of the total positivity of the transition kernels for birth-death processes (see for example [42]).

We will prove the other statements by induction. Trivially $X_1 \equiv 1 \leq_{lr} 2 \equiv X_2$. Thus suppose for induction that $X_1 \leq_{lr} \dots \leq_{lr} X_i$. Then Theorem 1.A.13 implies $Z_{X_1} \leq_{lr} \dots \leq_{lr} Z_{X_i}$. Using this, Theorem 1.A.12 implies

$$X_i = Z_{X_{\lfloor i/2 \rfloor}} + Z_{X_{\lceil i/2 \rceil}} \leq_{lr} Z_{X_{\lfloor (i+1)/2 \rfloor}} + Z_{X_{\lceil (i+1)/2 \rceil}} = X_{i+1}$$

since Z_{X_m} are log-concave by Lemma 1.A.3. Hence the result follows from induction. $\square$

1.B Technical Results for Extremal Tree Topologies

1.B.1 Balanced Trees Minimize Convex Descendant-Count Sums

We can provide a partial complement to Lemma 1.2.1 analyzing the other extreme. More concretely, we will show that the more balanced the tree is, the smaller these sums tend to be. We will not use this result in this chapter, but it adds to the idea that the balanced tree and the caterpillar tree are in a sense the two extreme points in our space of trees.

Lemma 1.B.1. *Suppose $\{\alpha_i\}_{i=1}^{2k-2}$ are the descendant counts of our rooted binary tree T corresponding to all the edges of our tree (not just ones corresponding to nontrivial bipartitions). Moreover, suppose $f : \mathbb{N} \rightarrow \mathbb{R}$ is nondecreasing and convex. Then $\sum_i f(\alpha_i)$ is minimized when T is the balanced tree.*

Proof of Lemma 1.B.1. We will describe a natural greedy balancing operation on the tree and prove it decreases $\sum f(\alpha_i)$. Recall that a *cherry* in a binary tree refers to a subtree consisting of two leaves.

Say that a given vertex $v \in V(T)$ is *unbalanced* if the number of leaves in its two subtrees differs by more than one. In order to make the tree more balanced, we will construct a new tree T' by pruning one of the cherries of the larger subtree and attaching it to one of the leaves of the smaller subtree.

To locate which subtree to prune, start at v and traverse the edge leading to the larger subtree. Iteratively descend the tree in this fashion until you reach a leaf ℓ . At this point, necessarily ℓ and its sibling form a cherry. If this were not the case, then the sibling of ℓ forms a larger subtree, and we would have traversed into that subtree rather than towards ℓ_p . This will be the cherry we prune.

To locate the leaf at which to reattach, start at v and traverse the edge leading to the smaller subtree. Iterate this procedure until you reach a leaf ℓ_a . Attach the cherry at this leaf ℓ to form a new tree T' with the same number of leaves overall as T . An illustration of this algorithm is provided in Figure 1.11.

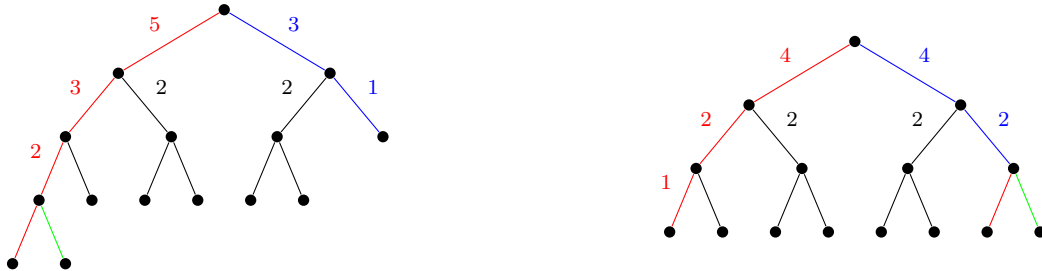


Figure 1.11: Side-by-side illustration of the initial tree T (left) and the tree T' after balancing (right). The pruning path $\{e_i^p\}$ in red and the attachment path $\{e_i^a\}$ in blue are highlighted

Suppose that $\{\alpha_i\}_{i=1}^{2k-2}, \{\alpha'_i\}_{i=1}^{2k-2}$ are the descendant counts of T, T' respectively. We claim that $\sum f(\alpha'_i) \leq \sum f(\alpha_i)$, namely that this balancing operation can only decrease the overall sum. To see why, let

$$v = u_0 \xrightarrow{e_1^p} \dots \xrightarrow{e_n^p} u_n = \ell_p, \quad v = w_0 \xrightarrow{e_1^a} \dots \xrightarrow{e_m^a} w_m = \ell_a$$

be the paths we traverse the tree in our pruning (from $v \rightarrow \ell_p$) and grafting (from $v \rightarrow \ell_a$) operations respectively. Note that $\alpha_e = \alpha'_e$ for all edges $e \notin \{e_i^p\} \cup \{e_i^a\}$ not along either of these paths. Moreover, every edge along the pruning path loses exactly one descendant leaf and every edge along the attachment path gains exactly one leaf. The edges in T, T' are exactly the same, except for the relocation of the cherry, and hence we pair the descendant counts α_e, α'_e

$$\sum_i (f(\alpha_i) - f(\alpha'_i)) = \sum_{i=1}^{n-1} [f(\alpha_{e_i^p}) - f(\alpha_{e_i^p} - 1)] + \sum_{i=1}^{m-1} [f(\alpha_{e_i^a}) - f(\alpha_{e_i^a} + 1)].$$

Moreover, by construction $m \leq n$ and $\alpha_{e_i^p} > \alpha_{e_i^a}$. This is because we always choose the larger subtree for pruning and the smaller for attaching. Thus $\alpha_{e_1^p} > \alpha_{e_1^a}$ and hence by induction

$$\alpha_{e_i^p} \geq \left\lceil \frac{\alpha_{e_{i-1}^p}}{2} \right\rceil > \left\lfloor \frac{\alpha_{e_{i-1}^a}}{2} \right\rfloor \geq \alpha_{e_i^a}$$

so $\alpha_{e_i^p} > \alpha_{e_i^a}$ is true for all i . Using this, we can see

$$= \sum_{i=1}^{m-1} [f(\alpha_{e_i^p}) - f(\alpha_{e_i^p} - 1)] + [f(\alpha_{e_i^a}) - f(\alpha_{e_i^a} + 1)] + \sum_{i=m}^{n-1} [f(\alpha_{e_i^p}) - f(\alpha_{e_i^p} - 1)].$$

Each term in the first summand is nonnegative as f is convex, and each term in the second summand is nonnegative as f is increasing. Hence $\sum (f(\alpha_i) - f(\alpha'_i)) \geq 0$ and the result follows.

Lastly, we argue that any tree can be converted into a balanced tree by iterating this procedure. If the subtrees of a given unbalanced vertex have size L, R respectively where $L - R \geq 2$ then clearly after this procedure they have sizes $L - 1, R + 1$ respectively with imbalance $(L - 1) - (R + 1) = (L - R) - 2$. Hence the imbalance decreases by two, and if we iterate this procedure enough times we can get the imbalance to be $(L - R) \bmod(2)$, namely the vertex is balanced. Moreover, the descendant counts of any vertices not contained in one of the subtrees of v are left unchanged. Hence we can start at the root and iteratively apply the procedure until the root is balanced. Then we can work our way down the tree. $\square$

This type of iterative subtree balancing is reminiscent of measures of tree balance in phylogenetics, where metrics such as the Colless or Sackin indices quantify how balanced a tree is relative to the maximally balanced or maximally imbalanced (caterpillar) configurations.

Tree balance statistics have long been used in phylogenetics to characterize the shapes of evolutionary trees, compare them to null models, and infer evolutionary processes such as speciation or extinction biases [45].

Classical tree-balancing procedures in computer science—such as AVL trees [1] and weight-balanced trees [61]—use local rotations to enforce constraints on subtree depths or sizes. Our algorithm shares the same conceptual goal of redistributing leaves to reduce imbalance, but it is purely combinatorial and does not maintain any search property.

1.C Coalescent-Time Asymptotics

1.C.1 Hypoexponential Representation and Derivative Bounds

In this appendix, we prove Lemma 1.4.1 by developing asymptotics for the time to coalescence in Kingman’s coalescent. More properties of this coalescent time are discussed in [60] and [15]. The paper [44] further generalizes some of these ideas to Λ -coalescents.

We will work in slightly more generality than this chapter requires. Namely, suppose that we have positive **distinct** rates $\{\lambda_i\}_{i=1}^\infty$. Define the random variables $\{S_k\}_{k=1}^\infty$ by $S_k = \sum_{i=1}^k X_i$ for $X_i \sim \text{Exp}(\lambda_i)$ independent (namely X_i has density $\tau_i(x) := \lambda_i \exp(-\lambda_i x)$). Moreover, define $F_k(T) := \mathbb{P}(S_k \leq T)$ to be their CDFs and $f_k(T)$ their densities. If we define

$$C_{i,k} := \prod_{\substack{m=1 \\ m \neq i}}^k \frac{\lambda_m}{\lambda_m - \lambda_i}$$

then we have the following well-known explicit formulas for the CDFs and densities

$$F_k(T) = 1 - \sum_{i=1}^k C_{i,k} \exp(-\lambda_i T), \quad f_k(T) = \sum_{i=1}^k C_{i,k} \lambda_i \exp(-\lambda_i T). \quad (1.C.1)$$

See [84], for example. Moreover, we have the natural recursive relation

$$F_k(T) = (F_{k-1} * \tau_k)(T) := \int_0^T F_{k-1}(x) \lambda_k \exp(-\lambda_k(T-x)) dx. \quad (1.C.2)$$

Since S_k are monotonically increasing, we see $S_k \uparrow S_\infty := \sum_{i=1}^\infty X_i$ almost surely, where the limit may be infinite. To quantify the rate of convergence, define the partial sums $s_k = \sum_{i=1}^k \lambda_i$, the remainder $R_k := \sum_{i=k+1}^\infty X_i$, and the remainder’s expectation $r_k := \mathbb{E}R_k = \sum_{i=k+1}^\infty \lambda_i^{-1}$. We begin with explicit formulas for the derivatives of the CDF in terms of these rates $\{\lambda_i\}$.

Lemma 1.C.1. *If we have positive, distinct rates $\{\lambda_i\}_{i=1}^\infty$:*

(i) At zero, the one-sided derivatives $F_k^{(m)}(0) = 0$ for $0 \leq m \leq k-1$. The nonzero derivatives take the form

$$F_k^{(k+r)}(0) = (-1)^r \left(\prod_{i=1}^k \lambda_i \right) (h_r(\lambda_1, \dots, \lambda_k)),$$

where h_r are the complete homogeneous symmetric polynomials

$$h_r(\lambda_1, \dots, \lambda_k) := \sum_{1 \leq i_1 \leq i_2 \leq \dots \leq i_r \leq k} \prod_{j=1}^r \lambda_{i_j}.$$

In particular, for $r = 0, 1$ we have

$$F_k^{(k)}(0) = \prod_{i=1}^k \lambda_i, \quad F_k^{(k+1)}(0) = - \left(\sum_{i=1}^k \lambda_i \right) \left(\prod_{i=1}^k \lambda_i \right).$$

(ii) The derivatives are uniformly bounded. Namely, for fixed m

$$\|F_k^{(m)}\|_\infty \leq M_m = \max_{i=1, \dots, m} |F_i^{(m)}(0)|.$$

Proof. (i) It is easy to see that for $m \geq 1$

$$F_k^{(m)}(0) = (-1)^{m-1} \sum_{i=1}^k C_{i,k} \lambda_i^m$$

which vanishes by Lemma 2 of [84] when $m \leq k-1$. To calculate the first nonzero derivatives, we will rely on the recurrence in Equation 1.C.2. Note that given two smooth functions $g, h : [0, \infty) \rightarrow \mathbb{R}$ and their convolution

$$g * h(T) := \int_0^T g(T-t)h(t) dt = \int_0^T g(t)h(T-t) dt.$$

Hence, by Leibniz's integral rule we have

$$(g * h)'(T) = (g' * h)(T) + g(0)h(T) = (g * h')(T) + g(T)h(0). \quad (1.C.3)$$

By iteratively applying this to $F_k = F_{k-1} * \tau_k$, we see

$$F_k^{(m)}(T) = (F_{k-1}^{(m)} * \tau_k)(T) + \sum_{j=0}^{m-k} F_{k-1}^{(k-1+j)}(0) \tau_k^{(m-k-j)}(T), \quad (1.C.4)$$

since by (ii) the first $k-2$ derivatives of F_{k-1} vanish. In particular, if $k > m$ we see $F_k^{(m)} = F_{k-1}^{(m)} * \tau_k$. Using this, we can prove $F_k^{(k+m)}(0)$ has the desired form by induction.

Note $F_1(T) = 1 - \exp(-\lambda_1 T)$ so $F_1^{(m+1)}(0) = (-1)^m \lambda_1 \cdot \lambda_1^m$ as desired. Moreover, $\tau_k^{(r)}(0) = (-1)^r \lambda_k^{r+1}$. Thus by induction and equation (1.C.4)

$$F_k^{(k+m)}(0) = \sum_{j=0}^m F_{k-1}^{(k-1+j)}(0) \tau_k^{(m-j)}(0) = (-1)^m \left(\prod_{i=1}^k \lambda_i \right) \sum_{j=0}^m \lambda_k^{m-j} a_{k-1,j}$$

where $a_{k,m} := h_m(\lambda_1, \dots, \lambda_k)$ are our symmetric polynomials. Hence it suffices to show these polynomials satisfy the recurrence

$$a_{k,m} = \sum_{j=0}^m \lambda_k^{m-j} a_{k-1,j}.$$

But this is almost immediate, just by grouping terms according to the number of i_ℓ that equal k

$$a_{k,m} = \sum_{1 \leq i_1 \leq i_2 \leq \dots \leq i_m \leq k} \prod_{\ell=1}^m \lambda_{i_\ell} = \sum_{j=0}^m \lambda_k^{m-j} \sum_{1 \leq i_1 \leq i_2 \leq \dots \leq i_j \leq k-1} \prod_{\ell=1}^j \lambda_{i_\ell} = \sum_{j=0}^m \lambda_k^{m-j} a_{k-1,j},$$

where the inner sum and product for $j = 0$ are understood to equal one. Hence

$$F_k^{(k+m)}(0) = (-1)^m a_{k,m} \prod_{i=1}^k \lambda_i$$

takes the desired form.

(ii) Since we saw in the previous section that for $k > m$ we have $F_k^{(m)} = F_{k-1}^{(m)} * \tau_k$, by iteratively applying Young's convolution inequality we get

$$\|F_k^{(m)}\|_\infty \leq \|F_{k-1}^{(m)}\|_\infty \cdot \|\tau_k\|_1 = \|F_{k-1}^{(m)}\|_\infty \implies \|F_k^{(m)}\|_\infty \leq \|F_m^{(m)}\|_\infty, \quad \forall k \geq m.$$

Hence we can see $\|F_k^{(m)}\|_\infty \leq \max_{i=1, \dots, m} \|F_i^{(m)}\|_\infty =: M_m < \infty$.

To quantify M_m , note that using the expression for the derivatives of τ_k , we can see equation (1.C.4) can be re-expressed as

$$F_k^{(m)}(T) = F_{k-1}^{(m)} * \tau_k(T) + F_k^{(m)}(0) \exp(-\lambda_k T).$$

Then Young's convolution inequality implies

$$|F_k^{(m)}(T)| \leq \|F_{k-1}^{(m)}\|_\infty \cdot (1 - \exp(-\lambda_k T)) + |F_k^{(m)}(0)| \exp(-\lambda_k T) \leq \max\{\|F_{k-1}^{(m)}\|_\infty, |F_k^{(m)}(0)|\}.$$

for any $T \geq 0$. Taking the supremum over T yields

$$\|F_k^{(m)}\|_\infty \leq \max\{\|F_{k-1}^{(m)}\|_\infty, |F_k^{(m)}(0)|\}.$$

By induction we see $M_m = \max_{i=1, \dots, m} |F_i^{(m)}(0)|$ as desired. $\square$

1.C.2 Small- and Large-Time CDF Asymptotics

Now that we have these bounds, the main results follow. We in fact show something slightly stronger: that the relative error in the Taylor series approximation is negligible in the $T = o((s_k/k)^{-1})$ regime no matter where we truncate it.

Theorem 1.C.2. *If $m = \arg \min_{i=1,\dots,k} \lambda_i$, then the CDFs have asymptotics*

$$\begin{cases} 1 - F_k(T) \sim C_{m,k} \exp(-\lambda_m T), & T \uparrow \infty \text{ and } k \text{ fixed,} \\ F_k(T) \sim \frac{\prod_{i=1}^k \lambda_i}{k!} T^k, & T = o((s_k/k)^{-1}), \end{cases}$$

where $T = o((s_k/k)^{-1})$ denotes any sequence of pairs (T_n, k_n) so that $\lim_{n \rightarrow \infty} s_{k_n} T_n / k_n = 0$. More generally, in the same $T = o((s_k/k)^{-1})$ regime and for any finite $m \geq 0$

$$\frac{F_k(T) - \sum_{r=0}^m a_{r,k}}{a_{m,k}} \rightarrow 0, \quad a_{r,k} := (-1)^r \frac{h_r(\lambda_1, \dots, \lambda_k) \prod_{i=1}^k \lambda_i}{(k+r)!} T^{k+r}.$$

Proof. For k fixed, the exponential in equation 1.C.1 with the smallest rate decays the slowest, so $1 - F_k(T) \sim C_{m,k} \exp(-\lambda_m T)$ as $T \rightarrow \infty$. For the other asymptotics, first fix k and let $h_{r,k} := h_r(\lambda_1, \dots, \lambda_k)$ be our symmetric polynomials and $p_k = \prod_{i=1}^k \lambda_i$. Since $F_k(T)$ is the sum of a finite number of exponentials, its Taylor series at zero converges to $F_k(T)$ for all T . Moreover, (i) of Lemma 1.C.1 above tells us $F_k^{(k+r)}(0) = (-1)^r p_k h_{r,k}$. Thus, let

$$a_{r,k} = (-1)^r \frac{p_k h_{r,k}}{(k+r)!} T^{k+r}$$

denote the $(r+1)^{th}$ nonzero term in the Taylor series expansion about zero. Suppose that we approximate $F_k(T)$ by the first $(m+1)$ nonzero terms of its Taylor series about zero. We can measure the approximation error relative to the last term $a_{m,k}$ to see

$$\frac{F_k(T) - \sum_{r=0}^m a_{r,k}}{a_{m,k}} = \sum_{r=1}^{\infty} (-1)^r \frac{h_{m+r}/h_m T^r}{(k+m+r)_r}$$

where $(x)_r = x(x-1) \cdots (x-r+1)$ denotes the falling factorial. We will now bound the right hand side by a geometric series.

Note that trivially $h_{r+1} \leq h_r s_k$ as the former includes all monomials of degree $(r+1)$ in $\{\lambda_i\}_{i=1}^k$ with coefficient one, and the latter has all the monomials with coefficient at least one. Hence $h_{m+r}/h_m = \prod_{i=0}^{r-1} h_{m+i+1}/h_{m+i} \leq s_k^r$. Using this and the simple bound $(k+m+r)_r \geq k^r$ we get

$$\left| \sum_{r=1}^{\infty} (-1)^r \frac{h_{m+r}/h_m T^r}{(k+m+r)_r} \right| \leq \sum_{r=1}^{\infty} \left(\frac{s_k T}{k} \right)^r = \frac{s_k T/k}{1 - s_k T/k}$$

for $|s_k T/k| < 1$. When $T = o((s_k/k)^{-1})$, this tends to zero. Thus in this regime

$$\frac{F_k(T) - \sum_{r=0}^m a_{r,k}}{a_{m,k}} \rightarrow 0,$$

as desired. In particular

$$F_k(T) = (1 + o(1)) \frac{\prod_{i=1}^k \lambda_i}{k!} T^k.$$

□

1.C.3 Gap Asymptotics

The following statement gives us a sense of how fast our coalescent probabilities $g_{k,1}(T)$ will saturate as $k \rightarrow \infty$.

Theorem 1.C.3 (Gap Asymptotics). *If $\sum_{i=1}^{\infty} \lambda_i^{-1} < \infty$, then $F_k \rightarrow F_{\infty}$ uniformly. Quantitatively, the gap $\Delta_k(T) = F_k(T) - F_{\infty}(T)$ has asymptotics*

$$\Delta_k = (f_{\infty} + o(1))r_k.$$

Proof. Dini's theorem already guarantees that $F_k \rightarrow F_{\infty}$ uniformly. Here, we will get quantitative control on the rate of convergence. Note that (ii) of Lemma 1.C.1 implies the functions f_k are uniformly bounded by M_1 and uniformly M_2 -Lipschitz. Thus they form a uniformly equicontinuous family which is uniformly bounded. Since $S_k \Rightarrow S_{\infty}$, as a consequence of [9] converse to Scheffe's lemma, we see $f_k \rightarrow f_{\infty}$ uniformly on $[0, \infty)$. Now, note that by conditioning on the tail R_k , as $S_k \perp\!\!\!\perp R_k$

$$\mathbb{P}(S_{\infty} \leq T) = \int_0^{\infty} \mathbb{P}(S_k \leq T - r) \mathbb{P}(R_k \in dr) = \mathbb{E}F_k(T - R_k).$$

Hence by Taylor expanding the CDF F_k we get

$$\Delta = \mathbb{E}_{R_k}[F_k(T) - F_k(T - R_k)] = f_k(T) \cdot \mathbb{E}R_k + \frac{1}{2} \mathbb{E}(f'_k(\xi(R_k))R_k^2),$$

where $\xi(r) \in (T - r, T)$ for every r . Note that $\text{Var}(R_k) = \sum_{i=k+1}^{\infty} \lambda_i^{-2} \leq r_k^2$, implying $\mathbb{E}R_k^2 \leq 2r_k^2$. Since $\|f'_k\|_{\infty} \leq M_2$ is uniformly bounded by (iii) and as $f_k(T) \rightarrow f_{\infty}(T)$ uniformly as $k \rightarrow \infty$

$$\Delta_k(T) = f_{\infty}(T)r_k + (f_k(T) - f_{\infty}(T))r_k + O(r_k^2) = f_{\infty}(T)r_k + o(r_k).$$

More specifically, since $M_2 = \max\{|F_1''(0)|, |F_2''(0)|\} = \lambda_1 \max(\lambda_1, \lambda_2)$

$$\|\Delta_k(T) - f_{\infty}r_k\|_{\infty} \leq \|f_k - f_{\infty}\|_{\infty}r_k + \lambda_1 \max(\lambda_1, \lambda_2)r_k^2 = o(r_k)$$

so the rate of convergence is uniform in T .

□

Lemma 1.4.1 then just follows from the special case of Kingman's coalescent, where we have rates $\tilde{\lambda}_i = \binom{i+1}{2}$. Note the indexing is slightly different here than in the main paper: it is shifted down by one, but otherwise the same. Here we have

$$r_k = \sum_{i=k+1}^{\infty} \frac{1}{\binom{i+1}{2}} = \frac{2}{k+1}, \quad s_k = \sum_{i=1}^k \binom{i+1}{2} = \frac{k(k+1)(k+2)}{6} = \Theta(k^3).$$

Moreover, a simple induction shows that $C_{1,k} = 3k/(k+2)$. Hence we get asymptotics

$$F_k(T) \sim \begin{cases} 1 - \frac{3k}{k+2} \exp(-T), & T \uparrow \infty \text{ and } k \text{ fixed,} \\ \frac{(k+1)!}{2^k} T^k, & T = o(k^{-2}). \end{cases} \quad \Delta_k(T) = (f_{\infty}(T) + o(1)) \cdot \frac{2}{k+1}$$

Chapter 2

Self-Balancing Sampling

2.1 Introduction

Consider the problem of sequentially sampling X_i from d outcomes $[d] := \{1, 2, \dots, d\}$ in a way that allows the empirical distributions

$$\mu(n) := \frac{1}{n} \sum_{i=1}^n \delta_{X_i}$$

to converge to the uniform distribution. Two naive ways of doing this are:

- *Uniformly at random*: Sample $X_i \stackrel{\text{iid}}{\sim} \text{Uniform}([d])$.
- *Deterministically/Greedily*: Sample $X_i = i \pmod{d} + 1$.

For these two sampling schemes, there's a trade-off between the *speed* of convergence and the *predictability* of the samples. Namely, the former converges at rate $O(n^{-1/2})$ in the total variation distance, while the latter converges at the much faster rate $O(n^{-1})$. On the other hand, the former process is as unpredictable as possible, maximizing the entropy at each sampling step, while the latter is perfectly predictable.

This tradeoff becomes readily apparent in the problem of scheduling audits or inspections, such as testing for doping in professional sports. A sports regulatory body is tasked with routinely testing teams for the use of performance enhancing drugs, but any deterministic schedule has the potential to be taken advantage of, since the players can temporarily cease doping in anticipation of the tests. On the other hand, sampling teams IID to receive doping tests can lead to wasteful behavior, such as spending resources to test a team multiple times in relatively short succession, while letting other teams remain temporarily untested; in effect, temporary imbalances in the empirical distribution of audited teams reflect inefficiencies in the order of teams chosen. Moreover, empirical evidence suggests *biasing* inspections towards individuals with certain risk factors can improve outcomes [49]. Similar tradeoffs arise in the scheduling of restaurant health inspections, where inspection agencies often seek

to prioritize higher-risk establishments or those with poor inspection histories while still preserving enough unpredictability to prevent strategic compliance behavior [12].

Another closely related problem is how to sample from a population while keeping the realized sample balanced across several overlapping strata. For example, suppose we are constructing a citizen’s assembly of 100 members out of a volunteer pool of 1000, and we would like our assembly to be demographically representative across multiple overlapping characteristics, such as gender, age, and education level. We may attempt to balance these characteristics using a deterministic or quota-constrained selection algorithm for choosing a committee, but, as noted in [29], some algorithms can lead to somewhat undemocratic outcomes in which certain members of the volunteer pool have a minuscule chance of being chosen for the committee. On the other hand, a simple random sample can result in a committee which is not properly representative for each characteristic.

Lastly, in randomized controlled trials there is inherent tension between ensuring covariates known to influence treatment effects are balanced across groups, while maintaining enough randomness to provide robustness against unknown prognostic factors. [35] explore this trade-off between covariate balance and robustness, but their sampling strategy is inherently non-sequential. On the other hand, the classical work of [66] provides a sequential analogue, biasing each assignment toward the treatment group that would reduce observed covariate imbalance. While effective as a balancing heuristic, the resulting procedure is not derived from optimizing any explicit objective, making the balance/robustness tradeoff relatively opaque.

In this chapter, we explore this balance–unpredictability trade-off for a natural family of sequential sampling rules. Heuristically, we show that by adaptively biasing our sampling process, we can limit predictability while maintaining fast convergence rates. For this purpose, consider the following sequential sampling rule, informally described as “downweight the outcome’s probability and renormalize”: Fix some $\alpha \in (0, 1)$, and start with some initial distribution $p(0) \in \Delta_+^d$, where

$$\Delta_+^d := \left\{ x \in \mathbb{R}^d : x_i > 0, \sum x_i = 1 \right\}$$

is the d -dimensional unit simplex with full support. At step n , recursively do the following:

Step 1: Sample an outcome $X_n \sim p(n-1)$.

Step 2: Define $p(n)$ by downweighting $p_{X_n}(n-1)$ by a factor of α and renormalizing the probability vector, i.e.

$$p_i(n) := \begin{cases} \frac{\alpha p_i(n-1)}{1 - (1-\alpha) \cdot p_{X_n}(n-1)} & \text{if } i = X_n, \\ \frac{p_i(n-1)}{1 - (1-\alpha) \cdot p_{X_n}(n-1)} & \text{if } i \neq X_n. \end{cases}$$

See Figure 2.1 for a visual schematic of the update rule.

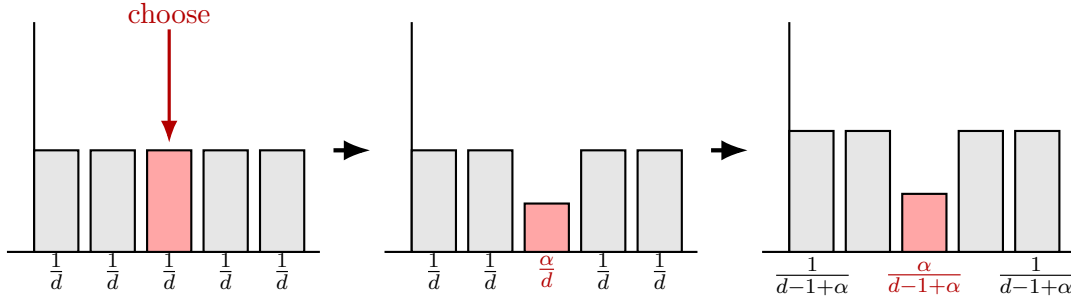


Figure 2.1: Schematic of the self-balancing update starting from a uniform distribution: choose an outcome, downweight it by α , then renormalize.

This sequential sampling rule is still random, but intuitively it generates empirical distributions which are closer to uniform because it de-emphasizes outcomes which are currently over-represented in the empirical distribution. More precisely, if we denote the *counts process*

$$N_i = N_i(n) := \#\{1 \leq j \leq n : X_j = i\},$$

then this rule generates samples according to

$$p_i = p_i(n) := \frac{p_i(0) \cdot \alpha^{N_i(n)}}{\sum_{j=1}^d p_j(0) \cdot \alpha^{N_j(n)}}.$$

Hence this sampler interpolates the two naive sampling methods: when $\alpha = 1$ (and $p(0)$ is uniform), we recover the uniform sampler, and when $\alpha \downarrow 0$, we recover greedy least-count sampling with uniformly random tie-breaking; if ties are broken deterministically, we recover deterministic sampling.

As we will later see, the same scheme works for non-uniform target distributions by replacing a uniform α by a suitable α_i for each i ; more precisely, for a target distribution p^* , one can set $\alpha_i = e^{-r/p_i^*}$ for every i , where the choice of $r > 0$ determines the trade-off between balance and predictability. Perhaps most unexpectedly, this simple sampling rule admits two complementary characterizations: as the essentially unique invariant potential-based sampler (Proposition 2.2.2) and as the Gibbs optimizer of an entropy-regularized count-imbalance objective (Theorem 2.4.3).

Related Work. Our sampling rule is related to several bodies of work on negatively dependent and history-dependent sampling. Negatively reinforced and de-preferential urn models assign lower selection probabilities to colors that are currently overrepresented, typically by applying a fixed decreasing weight function to the vector of urn proportions [7, 43]. These works establish laws of large numbers and central limit theorems under general replacement mechanisms. Our model differs in that observations directly increment the

selection counts and the sampling probabilities depend exponentially on the raw counts, $p_i(n) \propto p_i(0)e^{-(\log \alpha_i)N_i(n)}$, rather than sample proportions. Equivalently, the weight applied to the empirical proportion $\mu_i(n)$ is $e^{-n(\log \alpha_i)\mu_i(n)}$, so for fixed α_i , the strength of the repulsion on the empirical measure scale grows with time. This produces a positive-recurrent centered-count process and empirical convergence on the n^{-1} scale, rather than ordinary central-limit fluctuations.

A second closely related line of work concerns self-interacting and self-repellent random walks. Recent self-repellent random-walk constructions modify a base Markov kernel by downweighting states having large empirical occupation proportions and establish almost-sure convergence and central limit theorems with reduced asymptotic variance [21]. Exponentially interacting reinforced walks on complete graphs have also been studied, particularly for systems of several attracting or repelling walkers [69]. In contrast, our sampler has no graph constraint or dependence on the current location, and its exponential weights are applied to cumulative counts rather than to occupation proportions with a fixed interaction strength.

The trade-off between balance and unpredictability has a long history in restricted randomization for sequential experiments. Beginning with Efron’s biased-coin design and its subsequent extensions, these procedures increase the probability of assigning the next subject to an underrepresented treatment while retaining sufficient randomness to limit selection bias [3, 4, 23, 52, 75]. This literature commonly evaluates a design through both its resulting treatment imbalance and its *selection bias*, the probability with which an observer can predict the next assignment. The paper [38] provides a generalized framework for extending these classical ideas to an arbitrary number of treatment groups and target allocation proportions. Within this framework, our sampler uses assignment probabilities that depend exponentially on the weighted cumulative counts. The resulting strength of correction increases with the observed count imbalance, rather than remaining at a fixed bias level. We further derive this exponential rule from an explicit entropy-regularized count-imbalance objective and analyze its convergence, centered-count dynamics, and weak-bias diffusion behavior. In this sense, the self-balancing sampler may be viewed as a multidimensional family of biased-coin designs for which the balance–unpredictability trade-off is both explicitly parameterized and amenable to a unified stochastic-process analysis.

Outline. Section 2.2.2 shows that such samplers arise naturally by enforcing an invariance property on the sampling probabilities $p(n)$, namely that they depend only on *differences* in the counts N_i rather than their absolute magnitudes. We show that after properly centering the counts process, it descends to an irreducible and positive recurrent Markov chain on the centered counts, a process which exhibits strong mean-reversion properties.

Then, in Section 2.3 we show our first main result: that for any choice of α , the self-balancing sampler converges in total variation at rate $O(1/n)$ in expectation, matching the fastest possible rate up to constants, while almost surely it converges at rate $O(\log n/n)$ of its target law almost surely, which is optimal up to a logarithmic factor. Hence, the

self-balancing property of this sampler drastically improves upon the convergence rate of IID sampling, and matches the convergence rate of the naive greedy sampler. Moreover, we quantify the influence the bias parameter α has on the leading order constant and show there are two regimes: the leading constant is order $(\log(1/\alpha))^{-1/2}$ when $p(0) = p^*$, where p^* is the asymptotic sampling distribution, and is order $(\log(1/\alpha))^{-1}$ otherwise. A key technical ingredient is a quantitative specialization of the drift criterion of [63], which provides the explicit parameter dependence needed in our convergence bounds.

In Section 2.4, we show that the self-balancing sampler admits a natural interpretation through an entropy-regularized optimization problem, balancing minimization of the mean-squared imbalance in the counts with maximizing entropy. This formalizes the balance/convergence versus robustness/unpredictability trade off in terms of an explicit objective, a limitation of the classical procedure [66]. Appendix Section 2.B later shows that this sampler further fits within the broader framework of entropy-regularized optimal control.

In an effort to further quantify the trade-off between convergence and unpredictability, we develop diffusion limits in Section 2.6 and an adversarial notion of predictability in Section 2.6.2. The latter defines predictability as the accuracy an adversary trying to predict the next sampled index would achieve if they behaved optimally. We are particularly interested in a *weak-biasing regime*, in which $\alpha \uparrow 1$, which is the natural regime under which diffusion limits arise. Using these tools, we provide some non-rigorous analysis into how the behavior of the sampler at stationarity changes as we vary the bias parameter α . In this weak-biasing regime, the diffusion heuristics approximate the centered counts by an Ornstein-Uhlenbeck process with stationary fluctuations of order $(\log(1/\alpha))^{-1/2}$, matching the upper-bounds derived in Section 2.3.

Complementing this, the adversarial analysis in Section 2.6.2 translates these fluctuations into prediction accuracy. In the $d = 2$ case with uniform initialization and uniform target law p^* , through explicit analysis of the stationary law of the centered counts process, we show an adversary can predict the next sample with accuracy $\frac{1}{2} + \Theta(\sqrt{\log(1/\alpha)})$ in the weak-bias regime $\alpha \uparrow 1$. More generally, we conjecture this is true when several outcomes maximize p^* , and we provide non-rigorous analysis suggesting an adversary can exploit fluctuations among the tied outcomes in order to achieve excess prediction accuracy above $\max_i p_i^*$ again of order $\sqrt{\log(1/\alpha)}$, as in the $d = 2$ case.

Empirical demonstrations and applications to the citizen's assembly model of our results are provided in Appendix 2.A.

2.2 Model and Structural Properties

2.2.1 Definitions

A *sequential sampling rule* on $[d]$ consists of a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_n)_{n \geq 0}, \mathbb{P})$, an $\mathcal{F}_n$ -adapted sequence $(\rho_n)_{n \geq 0}$ of random distributions on $[d]$, and random variables $(X_n)_{n \geq 1}$

on $[d]$ so that

$$\mathbb{P}(X_{n+1} = i \mid \mathcal{F}_n) = \rho_n(i), \quad \forall i \in [d].$$

Most commonly, we have $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$. Such a sequential sampling rule selects its distribution ρ_n based solely on the previously observed outcomes, and then conditional on ρ_n generates the next outcome X_{n+1} by sampling from ρ_n .

We say such a rule is *Markovian* if it selects its next sample $X_{n+1} \in [d]$ based solely on the current selection counts $\{N_i(n)\}_{i=1}^d$ where

$$N_i(n) := |\{1 \leq j \leq n : X_j = i\}|.$$

Formally, such a sampling rule is given by a measurable function $\pi : \mathbb{N}^d \rightarrow \Delta^d$ where Δ^d is the unit simplex in $\mathbb{R}^d$, and samples are generated via

$$\mathbb{P}(X_{n+1} = i \mid \{X_j\}_{j \leq n}) = \pi_i(N(n)).$$

The self-balancing sampler is then a Markovian sampling rule with sampling function $\pi_i = p_i$. Note we can express this sampling rule as a softmax

$$p_i = p_i(n) = \frac{p_i(0) \cdot e^{-\beta_i N_i(n)}}{\sum_{j=1}^d p_j(0) \cdot e^{-\beta_j N_j(n)}}, \quad (2.2.1)$$

where $\beta_i = -\log \alpha_i$. As we will see below, these β_i are in some sense a more natural parameterization. Heuristically, they can be interpreted as follows: for $i, j \in [d]$, if $\beta_i^{-1}/\beta_j^{-1} = \frac{p}{q}$ is rational, then p samples of i are weighted the same as q samples of j . Thus β_i^{-1} measures the relative weight assigned to sampling $i \in [d]$: higher weighted indices tend to be selected more often.

Throughout the chapter, unless stated otherwise, we assume that the ratios β_i/β_j are rational for every $i, j \in [d]$. Equivalently, there exist $r > 0$ and $v \in \mathbb{N}_{>0}^d$ such that $\beta_i = r/v_i$ for every i . This *rational-ratio* assumption ensures that the projection-centered counts evolve on a countable state space.

We also use $\beta \downarrow 0$ to denote radial convergence along a fixed bias vector. More precisely, this means that $\beta^{(m)} = r_m \beta$ for a fixed $\beta \in (0, \infty)^d$ and a sequence $r_m > 0$ satisfying $r_m \rightarrow 0$. Thus the ratios among the coordinates of β remain fixed while their common scale tends to zero. In this chapter, we are interested in how the empirical sampling distribution

$$\mu(n) := \frac{1}{n} \sum_{i=1}^n \delta_{X_i}$$

evolves over time. Under the self-balancing sampler, we show the limiting distribution is

$$p_i^* := \frac{\beta_i^{-1}}{\sum_{j=1}^d \beta_j^{-1}}.$$

Namely, asymptotically we just select each i proportional to its relative weight β_i^{-1} .

Since the counts $N_i(n)$ drift off to infinity over time, we will require some form of centering to properly study this process. There are two natural centering schemes. The first, which we call the *mean-centered counts*, just centers each count by its expected limiting proportion

$$Y_i(n) := N_i(n) - np_i^*.$$

The second, which we call the *projection-centered counts*, reweights the counts according to their relative weights β_i^{-1} , fixes the coordinate d as a baseline, and compares the counts against this baseline

$$\tilde{Y}_i(n) := \beta_i N_i(n) - \beta_d N_d(n) = \beta_i \beta_d \cdot [\beta_d^{-1} N_i(n) - \beta_i^{-1} N_d(n)].$$

We can express the sample probabilities p_i directly in terms of these centered counts via

$$p_i = \frac{p_i(0) \cdot e^{-\beta_i Y_i(n)}}{\sum_{j=1}^d p_j(0) \cdot e^{-\beta_j Y_j(n)}} = \frac{p_i(0) \cdot e^{-\tilde{Y}_i(n)}}{\sum_{j=1}^d p_j(0) \cdot e^{-\tilde{Y}_j(n)}}.$$

2.2.2 Potential-based sampling

In this section, we show the self-balancing sampler arises naturally by enforcing a certain invariance property on our class of Markovian sampling rules, and that under this relation the counts process $(N(n))_{n \in \mathbb{N}}$ descends to our projection-centered counts $(\tilde{Y}(n))_{n \in \mathbb{N}}$. This projected chain is a Markov chain in its own right, and under mild assumptions is irreducible, positive recurrent, and admits a unique stationary distribution.

For a Markovian sampling rule, one natural choice for the sampling function π is to define a potential $\varphi_i : \mathbb{N} \rightarrow (0, \infty)$ for each coordinate and to select a sample proportional to its potential

$$\pi_i^\varphi(x) := \frac{\varphi_i(x_i)}{\sum_{j=1}^d \varphi_j(x_j)}, \quad \forall x \in \mathbb{N}^d.$$

We say a Markovian sampling rule π is *translation invariant* if

$$\pi(x + \mathbf{1}) = \pi(x), \quad \forall x \in \mathbb{N}^d.$$

where $\mathbf{1} = (1, \dots, 1)$. Translation-invariance expresses the opinion that our next sample X_{n+1} should depend only on the differences in the sampling counts $\{N_i(n)\}_{i=1}^d$ rather than their absolute sizes. More generally, we say π is *v-invariant* for some $v \in \mathbb{N}^d$ if

$$\pi(x + v) = \pi(x), \quad \forall x \in \mathbb{N}^d.$$

Here v -invariance expresses the heuristic that sampling i a total of v_i times ought to be equivalent to sampling j a total of v_j times. Thus (v_i) plays an analogous role to the relative weights (β_i^{-1}) .

Note we can view v -invariance as inducing an equivalence relation $\sim_v$ on $\mathbb{N}^d$

$$x \sim_v y \iff x = y + kv, \quad k \in \mathbb{Z}.$$

Under this relation and a slight technical assumption, our counts process $(N(n))_{n \in \mathbb{N}}$ descends to a Markov chain on $\mathbb{Z}^{d-1}$.

Proposition 2.2.1 (Projection-Centered Chain). *If π is v -invariant and $\gcd(v_1, \dots, v_d) = 1$ then $(v_d N_i(n) - v_i N_d(n))_{i=1}^{d-1}$ is a Markov chain on $\mathbb{Z}^{d-1}$.*

Proof. Let $x, y \in \mathbb{N}^d$. Define

$$T_v(x) := (v_d x_i - v_i x_d)_{i=1}^{d-1}.$$

First we show that $x \sim_v y$ if and only if $T_v(x) = T_v(y)$. If $y = x + kv$ then

$$T_v(y) = v_d y_i - v_i y_d = (v_d x_i - v_i x_d) + k(v_d v_i - v_i v_d) = v_d x_i - v_i x_d = T_v(x),$$

as desired. Conversely, if $T_v(x) = T_v(y)$ then re-arranging shows

$$\frac{x_i - y_i}{v_i} = \frac{x_d - y_d}{v_d}, \quad \forall i \in [d-1].$$

Since the right hand side is independent of i , we get $x = y + kv$ for some constant $k \in \mathbb{R}$. Since $x, y \in \mathbb{N}^d$ and $\gcd(v_1, \dots, v_d) = 1$, we must in fact have $k \in \mathbb{Z}$. Thus $x \sim_v y$.

Now we show that the counts process descends to a Markov chain under T_v . If $T_v(x) = T_v(y)$, then $y = x + kv$ for some $k \in \mathbb{Z}$, and hence v -invariance gives $\pi(y) = \pi(x)$. Therefore, the conditional law of $T_v(N(n+1))$ given $N(n) = x$ depends on x only through $T_v(x)$. Thus $(T_v(N(n)))_{n \geq 0}$ is a Markov chain on $T_v(\mathbb{N}^d)$. $\square$

Under the assumptions of the above proposition, the relation $\sim_v$ is equivalent to

$$x \sim_v y \iff v_d x_i - v_i x_d = v_d y_i - v_i y_d.$$

Alternatively, we can express the above equivalence as

$$\frac{x_i}{v_i} - \frac{1}{d} \sum_{j=1}^d \frac{x_j}{v_j} = \frac{y_i}{v_i} - \frac{1}{d} \sum_{j=1}^d \frac{y_j}{v_j}, \quad \forall i \in [d], \quad (2.2.2)$$

which shows the projection-centered counts $\tilde{Y}(n)$ center the reweighted counts $\beta_i N_i(n)$ by their mean, rather than the long-term average np^* as $Y(n)$ does. Now, we show the self-balancing sampler is essentially the only v -invariant sampling rule generated by a potential.

Proposition 2.2.2. *If π^φ is v -invariant, there exists $b, c_{i,k} > 0$ so*

$$\varphi_i(mv_i + k) = c_{i,k} b^m, \quad \forall m \in \mathbb{N}, k \in \{0, 1, \dots, v_i - 1\}.$$

Moreover, if φ_i is either log-convex or log-concave, then $\varphi_i(n) = c_i \alpha_i^n$ where

$$\log(\alpha_i) = \log(b) v_i^{-1}.$$

Proof. Denote $S(x) := \sum_j \varphi_j(x_j)$. Then v -invariance implies

$$\frac{\varphi_i(x_i)}{S(x)} = \frac{\varphi_i(x_i + v_i)}{S(x + v)}, \quad \frac{\varphi_j(x_j)}{S(x)} = \frac{\varphi_j(x_j + v_j)}{S(x + v)}.$$

Dividing these equalities and rearranging yields

$$\frac{\varphi_i(x_i + v_i)}{\varphi_i(x_i)} = \frac{\varphi_j(x_j + v_j)}{\varphi_j(x_j)}.$$

As the left side is independent of x_j and the right side is independent of x_i , these ratios must equal some constant $b > 0$ independent of x_i, x_j . Hence for every $i \in [d]$ and $k \in \{0, 1, \dots, v_i - 1\}$ we must have

$$\varphi_i(mv_i + k) = \varphi_i(k) \cdot b^m, \quad \forall m \in \mathbb{N}.$$

Log-concavity implies for $k \in \{0, 1, \dots, v_i - 2\}$

$$\frac{\varphi_i(0)}{\varphi_i(1)} \leq \frac{\varphi_i(k)}{\varphi_i(k+1)} \leq \frac{\varphi_i(v_i)}{\varphi_i(v_i+1)} = \frac{\varphi_i(0)b}{\varphi_i(1)b}.$$

If instead φ_i is log-convex, the direction of the inequalities all swap. Either way, we see these ratios are constant, and thus $\varphi_i(m) = c_i \alpha_i^m$ where $\alpha_i = b^{1/v_i}$. Thus we have

$$\log(\alpha_i) = \log(b) \cdot v_i^{-1},$$

giving the desired result. $\square$

In the case of translation-invariance, the log-convexity/concavity condition is unnecessary and we recover the self-balancing sampler in the case of uniform $\beta_i \equiv \beta$. Note that the sampling function for the self-balancing sampler is v -invariant only in the case the ratios $\beta_i^{-1}/\beta_j^{-1}$ are all rational, enforcing a relatively minor constraint on β .

When we have potential $\varphi_i(n) = c_i \alpha_i^n$ so that π^φ is v -invariant, it is easy to see π^φ is invariant with respect to

$$v' := v/\gcd(v_1, \dots, v_d),$$

as well. Hence in this setting of interest in this chapter, we may always assume $\gcd(v) = 1$ so by Proposition 2.2.1 our projection-centered counts are Markovian.

As the self-balancing sampler has strong mean-reversion properties, it is natural to expect the projection-centered counts are positive recurrent, and hence have some stationary distribution. This is indeed the case.

Lemma 2.2.3. *For $\beta_i = rv_i^{-1}$, where $v \in \mathbb{N}_{>0}^d$ and $r > 0$, the projected chain*

$$(v_d N_i(n) - v_i N_d(n))_{i=1}^{d-1}$$

with $p_i(0) > 0$ and potential $\varphi_i(n) = p_i(0)e^{-\beta_i n}$ as in (2.2.1) is irreducible and positive recurrent, and thus has a unique stationary distribution η on

$$\mathcal{X}_v := \left\{ (v_d n_i - v_i n_d)_{i=1}^{d-1} : n \in \mathbb{N}^d \right\} \subseteq \mathbb{Z}^{d-1}.$$

The proof of this technical recurrence result is deferred to Appendix 2.C.1.

In general, determining the stationary distribution η is challenging, as the chain is non-reversible for $d > 2$. However, in the special case $d = 2$, we can calculate η explicitly, which we will do in Lemma 2.6.3 below. Unsurprisingly, at stationarity the sampler generates samples according to the target p^* .

Lemma 2.2.4. *Suppose π is v -invariant and the projection-centered chain*

$$(v_d N_i(n) - v_i N_d(n))_{i=1}^{d-1}$$

has stationary distribution η on $\mathbb{Z}^{d-1}$. If we sample $y \sim \eta$ and $X \mid y \sim \pi(y)$, then

$$\mathbb{P}(X = i) = \frac{v_i}{\sum_j v_j} = p_i^*.$$

Proof. Suppose $\tilde{Y}^v \sim \eta$. Let $p_i := \mathbb{P}(X = i)$ and define $\Delta_n \tilde{Y}^v := \tilde{Y}^v(n+1) - \tilde{Y}^v(n)$. If $\tilde{Y}^v(n) \sim \eta$, then $\tilde{Y}^v(n+1) \sim \eta$ as well. Hence $\mathbb{E} \Delta_n \tilde{Y}^v = 0 \in \mathbb{Z}^{d-1}$. As discussed above, we can view $\tilde{Y}^v(n)$ as the image of $N(n)$ under the map

$$x \mapsto (v_d x_i - v_i x_d)_{i=1}^{d-1}.$$

Hence

$$(\Delta_n \tilde{Y}^v)_i = \begin{cases} v_d & w.p. & p_i, \\ -v_i & w.p. & p_d, \\ 0 & w.p. & 1 - p_i - p_d. \end{cases}$$

Hence the i^{th} coordinate of this expectation is

$$(\mathbb{E} \Delta_n \tilde{Y}^v)_i = v_d p_i - v_i p_d = 0.$$

Thus $p_i = \frac{v_i}{v_d} p_d$. Since $\sum_{i=1}^d p_i = 1$, we see

$$\sum_{i=1}^d \frac{v_i}{v_d} = \frac{1}{p_d} \rightarrow p_d = \frac{v_d}{\sum_{j=1}^d v_j},$$

which yields the desired result. □

2.3 Fast Convergence of the Empirical Distribution

In this section, we show for any choice of $\alpha_i \in (0, 1)$, the self-balancing sampler converges to the desired limiting distribution at a rate faster than IID sampling, and in fact converges as fast as possible. Below, for functions $f, g : \mathbb{N} \rightarrow [0, \infty)$ we say $f = O(g)$ if $\limsup_{n \rightarrow \infty} f/g < \infty$. Further, we say $f = \Omega(g)$ if $g = O(f)$ and $f = \Theta(g)$ if $f = O(g)$ and $g = O(f)$.

Any empirical distribution sampled on $d \geq 2$ values (each with nonzero probability) converges at least as slow as $1/n$, because for any distribution $q \in \Delta^d$,

$$\|\mu_n - q\|_{\text{TV}} = \frac{1}{2} \sum_{i=1}^d \left| \frac{N_i}{n} - q_i \right| \geq \frac{1}{2n} \sum_{i=1}^d \min_{z \in \mathbb{Z}} |z - nq_i|.$$

If q_i is irrational for some i , then $\{nq_i \pmod{1}\}_{n=1}^\infty$ is dense in $[0, 1]$, so there are infinitely many n such that $\min_{z \in \mathbb{Z}} |z - nq_i| \geq 1/3$, giving an overall lower bound of $\|\mu_n - q\|_{\text{TV}} \geq \frac{1}{6n}$ for infinitely many n . On the other hand, if all q_i are rational with least common denominator m , then for any n not divisible by m , $\min_{z \in \mathbb{Z}} |z - nq_i| \geq 1/m$ for some i . Thus, for infinitely many n , we get $\|\mu_n - q\|_{\text{TV}} \geq \frac{1}{2mn}$. Thus there exists a deterministic subsequence (n_k) depending only on q such that, for every sampling scheme,

$$\|\mu_{n_k} - q\|_{\text{TV}} = \Omega(n_k^{-1}).$$

IID sampling converges at the rate $\Theta(1/\sqrt{n})$ in expectation, as for samples from a distribution q ,

$$\mathbb{E}[\|\mu_n - q\|_{\text{TV}}] = \frac{1}{2\sqrt{n}} \sum_{i=1}^d \mathbb{E} \left[\left| \frac{N_i - nq_i}{\sqrt{n}} \right| \right],$$

where each expectation term is $\Theta(1)$ by the central limit theorem. However, for our self-balancing sampler, the empirical distribution converges much faster than IID sampling. Theorem 2.3.2 shows that in expectation, the convergence actually matches the fastest possible rate of $O(1/n)$, regardless of the choice of $\alpha_i \in (0, 1)$.

Such a rate commonly appears in such biased sampling procedures. By specializing the classical Foster-Lyapunov drift condition to the sequential sampling setting of interest, [38] provides a sufficient drift condition for this to occur. However, by developing a more quantitative version (Lemma 2.3.1) of the classical Foster-Lyapunov drift conditions, we are able to get more explicit control on the leading order constants. More details on such drift conditions can be found in [56], for example.

We will use the following quantitative drift lemma, whose technical proof is deferred to Appendix 2.C.2.

Lemma 2.3.1. *Let $U_n \geq 0$ be $\mathcal{F}_n$ -adapted for $n \geq 0$. Suppose there exists $a, b, c > 0$ with $a \leq c$ so*

$$(i) \quad \mathbb{E}[U_{n+1} - U_n \mid \mathcal{F}_n] \leq -a \text{ on } \{U_n \geq b\},$$

$$(ii) \quad |U_{n+1} - U_n| \leq c \text{ a.s.}$$

Then

$$\sup_{n \geq 0} \mathbb{E}[U_n] \leq \mathbb{E}[U_0] + b + c + \frac{c^2}{2a}.$$

Theorem 2.3.2 (Average-case convergence rate). *For the self-balancing sampling rule, the empirical distribution $\mu(n)$ converges at a rate of $1/n$:*

$$\mathbb{E}\|\mu(n) - p^*\|_{\text{TV}} = O(1/n).$$

More granularly, for $\max_i \beta_i \leq 1$,

$$\mathbb{E}\|\mu(n) - p^*\|_{\text{TV}} \lesssim \begin{cases} \frac{1}{n} \sum_i \frac{1}{\beta_i} & \text{if } p(0) \neq p^*, \\ \frac{1}{n} \sum_i \frac{1}{\sqrt{\beta_i}} & \text{if } p(0) = p^*. \end{cases}$$

where the constant depends only on $p(0)$, p^* , $\kappa := \frac{\max_i \beta_i}{\min_i \beta_i}$, and d .

As one might expect, increasing the bias β tends to decrease the gap between the target sampling law p^* and the empirical sampling distribution. To see why these two regimes arise, note when $p(0) = p^*$, the sampling law when $Y(0) = 0$ is already p^* . Hence it only has to correct the fluctuations about this stationary sampling law. When $p(0) \neq p^*$, the sampler must additionally correct this initial gap caused by this mismatch.

The following shows that, almost surely, the self-balancing sampler improves upon the $O(n^{-1/2})$ convergence rate of IID sampling and converges at rate $O(\log n/n)$ to its target p^* .

Theorem 2.3.3 (Almost-sure convergence rate). *For the self-balancing sampling rule,*

$$\limsup_{n \rightarrow \infty} \frac{n}{\log n} \|\mu(n) - p^*\|_{\text{TV}} \leq 2 \sum_{i=1}^d \frac{1}{\beta_i}$$

almost surely.

Hence, almost every realization converges at a rate of $O(\log n/n)$, which is only a logarithmic factor worse than the optimal $O(1/n)$ rate, and drastically improves over the rate of $O(1/\sqrt{n})$ for IID sampling.

The proofs of Theorems 2.3.2 and 2.3.3 are given together in the following subsection.

2.3.1 Proofs of the Convergence-Rate Theorems

Proof of Theorem 2.3.2. We treat the general case by essentially applying shifts to all of the counts to compensate for the “wrong” initial distribution. In particular, let

$$r_i := \log \frac{p_i(0)}{p_i^*}, \quad \bar{r} := \sum_j p_j^* r_j, \quad c_i := r_i - \bar{r}, \quad \xi_i := \beta_i Y_i - c_i.$$

Then

$$\|Y(n)\|_1 = \sum_i \left| \frac{c_i + \xi_i(n)}{\beta_i} \right| \leq \sum_i \frac{|c_i|}{\beta_i} + \sum_i \frac{|\xi_i(n)|}{\beta_i}.$$

When $p_i(0) = p_i^*$, the c_i terms disappear, so it suffices to show that $\mathbb{E}[|\xi_i(n)|] \lesssim \sqrt{\beta_i}$ when $p(0) = p^*$ and $\lesssim 1$ otherwise.

To handle all the $\xi_i(n)$ at once, consider $V_n := \sum_i \frac{\xi_i^2}{\beta_i}$; we will control $\mathbb{E}[V_n]$ via a drift bound. If $X_{n+1} = k$, then

$$\Delta \xi_i(n) := \xi_i(n+1) - \xi_i(n) = \beta_i(\mathbf{1}_{\{i=k\}} - p_i^*),$$

so we may compute (keeping in mind that $\sum_i p_i^* \xi_i(n) = 0$ for all n)

$$V_{n+1} - V_n = \sum_i \frac{\Delta \xi_i(n) (\xi_i(n+1) + \xi_i(n))}{\beta_i} = 2\xi_{X_{n+1}}(n) + \beta_{X_{n+1}}(1 - p_{X_{n+1}}^*).$$

and hence

$$\begin{aligned} \mathbb{E}[V_{n+1} - V_n \mid N(n)] &= \sum_i p_i(n) [2\xi_i(n) + \beta_i(1 - p_i^*)] \\ &\leq 2 \sum_i \xi_i(n) p_i(n) + \sum_i \beta_i p_i(n) \\ &\leq 2 \sum_i \xi_i(n) (p_i(n) - p_i^*) + \max_i \beta_i. \end{aligned}$$

To upper bound the first term in terms of (negative) V_n , recognize that it is an expectation with respect to a p_i^* , tilted in the direction of $-\xi$; in particular, $p_i(n) = \frac{p_i^* e^{-\xi_i(n)}}{\sum_j p_j^* e^{-\xi_j(n)}}$, and the term is $\mathbb{E}[\Xi]$, where $\Xi = \xi_i$ with probability $p_i(n)$. The more we tilt the distribution in the direction of $-\xi$, the lower the expectation of Ξ will be. To make this intuition precise, define the following interpolation for $0 \leq t \leq 1$

$$Z(t) := \sum_i p_i^* e^{-t\xi_i}, \quad q_i(t) := \frac{p_i^* e^{-t\xi_i}}{Z(t)}, \quad f(t) := \log Z(t).$$

Then $q(0) = p^*$, $q(1) = p(n)$, and if $\Xi(t) = \xi_i$ with probability $q_i(t)$ we have

$$f'(t) = - \sum_i \xi_i q_i(t) = -\mathbb{E}[\Xi(t)], \quad f''(t) = \text{Var}(\Xi(t)) \geq 0.$$

So we can calculate the desired expectation by tracking this interpolation

$$\sum_i \xi_i (p_i(n) - p_i^*) = -(f'(1) - f'(0)) = - \int_0^1 f''(t) dt \leq - \int_0^{\min\{1, 1/\|\xi(n)\|_\infty\}} f''(t) dt.$$

For $0 \leq t \leq \min\{1, 1/\|\xi(n)\|_\infty\}$, p_i^* and $q_i(t)$ are readily comparable

$$\frac{1}{e} \leq Z(t) \leq e,$$

which gives

$$\frac{p_i^*}{e^2} \leq q_i(t) \leq e^2 p_i^*.$$

For such t , by the variational characterization of variance

$$\begin{aligned} f''(t) &= \min_a \sum_i q_i(t) (\xi_i(n) - a)^2 \\ &\geq \frac{1}{e^2} \min_a \sum_i p_i^* (\xi_i(n) - a)^2 \\ &= \frac{1}{e^2} \text{Var}(\Xi(0)) \\ &= \frac{1}{e^2} \sum_i \xi_i^2 p_i^* \\ &= \frac{1}{e^2} S_\beta^{-1} V_n. \end{aligned}$$

for $S_\beta = \sum \beta_i^{-1}$. Putting it all together, we get a bound of

$$\mathbb{E}[V_{n+1} - V_n \mid N(n)] \leq -\frac{2}{e^2} \min\{1, 1/\|\xi(n)\|_\infty\} S_\beta^{-1} V_n + \max_i \beta_i.$$

We know that $\|\xi(n)\|_\infty \leq \sqrt{V_n \max_i \beta_i}$, so this is

$$\leq -\frac{2}{e^2} S_\beta^{-1} \min\left\{V_n, \sqrt{\frac{V_n}{\max_i \beta_i}}\right\} + \max_i \beta_i.$$

We have $S_\beta = \sum_j \beta_j^{-1} \leq \frac{d}{\min_i \beta_i} = d\kappa \frac{1}{\max_i \beta_i}$, so that for $\max_i \beta_i \leq 1$,

$$\leq -\frac{2}{e^2 d \kappa} (\max_i \beta_i) \min\{V_n, \sqrt{V_n}\} + \max_i \beta_i.$$

Since $\sqrt{V_n} \leq V_n$ for $V_n \geq 1$, we will actually establish our drift bound in terms of $U_n := \sqrt{1 + V_n}$. Notice that by the tangent-line inequality for the concave function $x \mapsto \sqrt{1 + x}$,

$$\begin{aligned} \mathbb{E}[U_{n+1} - U_n \mid N(n)] &\leq \frac{\mathbb{E}[V_{n+1} - V_n \mid N(n)]}{2\sqrt{1 + V_n}} \\ &\leq \frac{-\frac{2}{e^2 d \kappa} (\max_i \beta_i) \min\{V_n, \sqrt{V_n}\} + \max_i \beta_i}{2\sqrt{1 + V_n}}. \end{aligned}$$

For $U_n \geq \sqrt{1 + e^4 d^2 \kappa^2}$, we have $V_n \geq 1$ and $\frac{2}{e^2 d \kappa} \sqrt{V_n} \geq 2$, so we get

$$\leq -\frac{\max_i \beta_i}{2\sqrt{2} e^2 d \kappa}.$$

We also need a bound on the jump size for U_n : The map $x \mapsto \sqrt{1 + \|x\|_\beta^2}$ is 1-Lipschitz with respect to the norm $\|x\|_\beta := \sqrt{\sum_i x_i^2 / \beta_i}$, so again as $\sum_i \xi_i p_i^* = 0$ we get

$$\begin{aligned} |U_{n+1} - U_n| &\leq \sqrt{\sum_i \frac{(\xi_i(n+1) - \xi_i(n))^2}{\beta_i}} \\ &= \sqrt{\sum_i (\mathbf{1}_{\{X_{n+1}=i\}} - p_i^*) \Delta \xi_i(n)} \\ &\leq \sqrt{\max_i \beta_i}. \end{aligned}$$

Applying Lemma 2.3.1 with

$$a = \frac{\max_i \beta_i}{2\sqrt{2}e^2 d \kappa}, \quad b = \sqrt{1 + e^4 d^2 \kappa^2}, \quad c = \sqrt{\max_i \beta_i},$$

and using $\xi_i(0) = -c_i$, we obtain

$$\begin{aligned} \sup_n \mathbb{E}[\sqrt{V_n}] &\leq \sup_n \mathbb{E}[U_n] \\ &\leq \mathbb{E}[U_0] + \sqrt{1 + e^4 d^2 \kappa^2} + \sqrt{\max_i \beta_i} + \sqrt{2}e^2 d \kappa \\ &= \sqrt{1 + \sum_j \frac{c_j^2}{\beta_j}} + \sqrt{1 + e^4 d^2 \kappa^2} + \sqrt{\max_i \beta_i} + \sqrt{2}e^2 d \kappa \\ &\leq \sqrt{1 + \frac{\kappa}{\max_i \beta_i} \sum_j c_j^2} + \sqrt{1 + e^4 d^2 \kappa^2} + 1 + \sqrt{2}e^2 d \kappa \\ &\lesssim \begin{cases} 1, & \text{if } p(0) = p^*, \\ 1/\sqrt{\max_i \beta_i}, & \text{if } p(0) \neq p^*. \end{cases} \end{aligned}$$

where we have absorbed constants that depend only on $p(0), p^*, \kappa$, and d . Note, in particular, that the specific form of the constants in Lemma 2.3.1 makes the $\max_i \beta_i$ dependence cancel in the last term.

We finish by applying this bound to ξ_i

$$\mathbb{E}[|\xi_i(n)|] = \sqrt{\beta_i} \mathbb{E}\left[\frac{|\xi_i|}{\sqrt{\beta_i}}\right] \leq \sqrt{\beta_i} \mathbb{E}[\sqrt{V_n}] \lesssim \begin{cases} \sqrt{\beta_i} & \text{if } p(0) = p^* \\ 1 & \text{if } p(0) \neq p^* \end{cases}$$

for every i . Therefore,

$$\mathbb{E}\|Y(n)\|_1 \leq \sum_i \frac{|c_i| + \mathbb{E}|\xi_i(n)|}{\beta_i}$$

$$\lesssim \begin{cases} \sum_i \beta_i^{-1/2}, & \text{if } p(0) = p^*, \\ \sum_i \beta_i^{-1}, & \text{if } p(0) \neq p^*. \end{cases}$$

Since $\mathbb{E}\|\mu(n) - p^*\|_{\text{TV}} = (2n)^{-1}\mathbb{E}\|Y(n)\|_1$, this proves the claimed bounds. $\square$

Proof of Theorem 2.3.3. The idea is to show that the log odds ratios of the sampling probabilities never get too large; this will force $\beta_i N_i \approx \beta_j N_j$, which will make $N_i/n \propto 1/\beta_i$. In particular, let

$$R_{i,j}(n) := -\log \frac{p_i(n)}{p_j(n)} + \log \frac{p_i(0)}{p_j(0)} = \beta_i N_i - \beta_j N_j.$$

It is very unlikely to sample index i when $R_{i,j}$ already exceeds $2 \log n$, as the following one-step analysis shows:

$$\begin{aligned} \mathbb{P}(X_{n+1} = i, R_{i,j}(n) \geq 2 \log n) &= \mathbb{E}[\mathbb{P}(X_{n+1} = i, R_{i,j}(n) \geq 2 \log n \mid p(n))] \\ &= \mathbb{E}[p_i(n) \mathbf{1}_{\{R_{i,j}(n) \geq 2 \log n\}}] \\ &\leq \mathbb{E}\left[p_j(n) \frac{p_i(0)}{p_j(0)} \frac{1}{n^2} \mathbf{1}_{\{R_{i,j}(n) \geq 2 \log n\}}\right] \\ &\lesssim \frac{1}{n^2}. \end{aligned}$$

Note $|R_{i,j}(n+1) - R_{i,j}(n)| \leq B := \max_k \beta_k$. For $R_{i,j}(n)$ to exceed $2 \log n + B$ infinitely often, index i must be sampled infinitely often. As $2 \log n$ is increasing, we then see

$$\mathbb{P}(R_{i,j}(n) \geq 2 \log(n) + B \text{ i.o.}) \leq \mathbb{P}(X_{n+1} = i, R_{i,j}(n) \geq 2 \log(n) \text{ i.o.}) = 0,$$

by the Borel–Cantelli theorem. Applying this argument to $R_{j,i} = -R_{i,j}$, we get $|R_{i,j}(n)| \leq 2 \log n + B$ all but finitely often with probability one.

Now analyze the counts in terms of the $R_{i,1}$:

$$\frac{N_i}{n} = \frac{1}{\beta_i} \cdot \frac{\beta_1 N_1}{n} + \frac{R_{i,1}}{\beta_i n},$$

and summing over i gives

$$1 = \sum_{k=1}^d \frac{N_k}{n} = \frac{\beta_1 N_1}{n} \left(\sum_{k=1}^d \frac{1}{\beta_k} \right) + \frac{1}{n} \sum_{k=1}^d \frac{R_{k,1}}{\beta_k},$$

which we rearrange into

$$\frac{\beta_1 N_1}{n} = \frac{1}{\sum_{k=1}^d 1/\beta_k} \left(1 - \frac{1}{n} \sum_{k=1}^d \frac{R_{k,1}}{\beta_k} \right).$$

Plugging this back into the expression for N_i/n gives

$$\frac{N_i}{n} = \underbrace{\frac{1/\beta_i}{\sum_{k=1}^d 1/\beta_k}}_{p_i^*} \left(1 - \frac{1}{n} \sum_{k=1}^d \frac{R_{k,1}}{\beta_k} \right) + \frac{R_{i,1}}{\beta_i n}.$$

So the total variation distance is

$$\begin{aligned} \|\mu(n) - p^*\|_{\text{TV}} &= \frac{1}{2} \sum_{i=1}^d \left| \frac{N_i}{n} - p_i^* \right| \\ &= \frac{1}{2n} \sum_{i=1}^d \left| \frac{R_{i,1}}{\beta_i} - p_i^* \sum_{k=1}^d \frac{R_{k,1}}{\beta_k} \right| \\ &\leq \frac{1}{2n} \left(\sum_{i=1}^d \frac{|R_{i,1}|}{\beta_i} + \sum_{i=1}^d p_i^* \sum_{k=1}^d \frac{|R_{k,1}|}{\beta_k} \right) \\ &= \frac{1}{n} \sum_{i=1}^d \frac{|R_{i,1}|}{\beta_i}. \end{aligned}$$

Since $\limsup_{n \rightarrow \infty} |R_{i,1}(n)|/\log n \leq 2$ almost surely for every i , we conclude that

$$\limsup_{n \rightarrow \infty} \frac{n}{\log n} \|\mu(n) - p^*\|_{\text{TV}} \leq 2 \sum_{i=1}^d \frac{1}{\beta_i} \quad \text{almost surely.}$$

□

2.4 Optimization Characterizations

Here, we formalize the interpretation of the self-balancing sampler as optimizing the trade-off between the convergence rate of the empirical distribution and its unpredictability at each step. To this end, we show that this sampling rule solves a greedy optimization problem that balances a one-step reduction in count imbalance against preserving entropy in the next sample.

Let $H(q) := -\sum_{i=1}^d q_i \log q_i$ denote the Shannon entropy, let $H(q, p) := -\sum_{i=1}^d q_i \log p_i$ denote the cross-entropy, and let $\text{KL}(q||p) := H(q, p) - H(q) = \sum_{i=1}^d q_i \log \frac{q_i}{p_i}$.

Theorem 2.4.1. *Let $L(n) := \frac{1}{2} \sum_{i=1}^d (N_i(n) - \frac{n}{d})^2$ denote the quadratic loss for the empirical counts and $D(q) := \mathbb{E}[L(n+1) - L(n) \mid N(n), q]$ denote the one-step increase in L when sampling $X_{n+1} \sim q$. Then the unique solution to the optimization problem*

$$\arg \max_{q \in \Delta^d} (H(q) - \beta D(q))$$

is the self-balancing sampling method $p_i(n) \propto e^{-\beta N_i(n)}$, initialized at the uniform distribution.

Proof. Up to terms independent of q , the one-step loss is $D(q) = \sum_i q_i N_i(n)$. Thus the objective is equivalent to maximizing

$$H(q) - \beta \sum_i q_i N_i(n).$$

The Lagrange multiplier condition gives $q_i \propto e^{-\beta N_i(n)}$, and strict concavity gives uniqueness. $\square$

The following more general version allows the α_i (and hence the β_i) to be non-constant in i .

Theorem 2.4.2. *Let $L_\beta(n) := \frac{1}{2} \sum_{i=1}^d \beta_i (N_i(n) - np_i^*)^2$ denote the weighted quadratic loss for the empirical counts and $D_\beta(q) := \mathbb{E}[L_\beta(n+1) - L_\beta(n) \mid N(n), q]$ denote the one-step increase in L_β when sampling $X_{n+1} \sim q$. Then the unique solution to the optimization problem*

$$\arg \max_{q \in \Delta^d} (H(q) - H(q, p(0)) - D_\beta(q)) = \arg \min_{q \in \Delta^d} (\text{KL}(q \| p(0)) + D_\beta(q))$$

is the 1/2-offset self-balancing sampling method $p_i(n) \propto p_i(0) e^{-\beta_i (N_i(n) + 1/2)}$.

Proof. First, expand D to isolate the terms depending on q

$$\begin{aligned} D(q) &= \frac{1}{2} \sum_{i=1}^d \beta_i \mathbb{E}[N_i(n+1)^2 - 2(n+1)p_i^* N_i(n+1) + ((n+1)p_i^*)^2 \\ &\quad - N_i(n)^2 + 2np_i^* N_i(n) - (np_i^*)^2 \mid N(n), q] \\ &= \frac{1}{2} \sum_{i=1}^d \beta_i [q_i (N_i(n) + 1)^2 + (1 - q_i) N_i(n)^2 - N_i(n)^2 \\ &\quad - 2p_i^* ((n+1)(N_i(n) + q_i) - nN_i(n)) + (2n+1)(p_i^*)^2] \\ &= \frac{1}{2} \sum_{i=1}^d \beta_i [2q_i N_i(n) + q_i - 2(n+1)p_i^* q_i - 2p_i^* N_i(n) + (2n+1)(p_i^*)^2] \\ &= \sum_{i=1}^d \left(\beta_i N_i(n) q_i + \frac{1}{2} \beta_i q_i + \frac{2n+1}{2} \beta_i (p_i^*)^2 - \frac{N_i(n)}{\sum_{k=1}^d 1/\beta_k} \right) - \frac{n+1}{\sum_{k=1}^d 1/\beta_k}, \end{aligned}$$

so it is equivalent to maximize $H(q) + \sum_{i=1}^d q_i \log p_i(0) - \sum_{i=1}^d (\beta_i N_i(n) q_i + \frac{1}{2} \beta_i q_i)$. When all the β_i are equal and $p(0)$ is uniform, this just looks like maximizing $H(q) - \beta \langle N(n), q \rangle$.

Now solve the optimization problem using the Lagrange multiplier λ . The Lagrangian is

$$\mathcal{L}(q) := \sum_{i=1}^d \left(-q_i \log q_i + q_i \log p_i(0) - \beta_i N_i q_i - \frac{1}{2} \beta_i q_i + \lambda q_i \right) - \lambda.$$

Setting $\nabla_q \mathcal{L}(q) = 0$ gives

$$0 = -\log q_i - 1 + \log p_i(0) - \beta_i N_i - \frac{\beta_i}{2} + \lambda \quad \implies \quad q_i = p_i(0) e^{-\beta_i N_i - \beta_i/2 + \lambda - 1}.$$

Therefore, $q_i \propto p_i(0) e^{-\beta_i(N_i(n)+1/2)}$, as claimed. $\square$

The entropy-regularized optimization problem found in Theorem 2.4.2 appears naturally in the context of optimal control and reinforcement learning. In this setting, such regularization is used to quantify the classical exploration versus exploitation trade-off [34, 76, 83]. Moreover, as [79] observes, such exponential reweighting naturally appears as a result of this entropy penalty. We briefly summarize some of the main results in discrete-time in Appendix 2.B, and show that the self-balancing sampler and the above entropy-regularized optimization problem arise naturally in this framework when you penalize imbalance using mean-squared error.

The curious offset of $1/2$ in the counts appears to be an effect of greedy optimization over *discrete* time steps with a quadratic cost. Indeed, $\frac{\beta_i}{2}((x+1)^2 - x^2) = \beta_i x + \frac{\beta_i}{2}$. When all β_i are equal, this factor is absorbed into the normalization of the density. Theorem 2.4.3 gives an “instantaneous” optimization perspective, which does not include the $1/2$ offset.

Theorem 2.4.3. *Let $L_\beta(x) := \frac{1}{2} \sum_{i=1}^d \beta_i (x_i - np_i^*)^2$ denote the weighted quadratic loss as a function of counts x . Then the unique solution to the optimization problem*

$$\arg \min_{q \in \Delta^d} (\langle \nabla L_\beta(N(n)), q \rangle + \text{KL}(q \| p(0)))$$

is the self-balancing sampling method $p_i \propto p_i(0) e^{-\beta_i N_i(n)}$, initialized at the distribution $p_i(0)$.

Proof. Calculate the gradient

$$\frac{\partial L_\beta}{\partial x_i}(x) = \beta_i (x_i - np_i^*).$$

so we aim to optimize

$$\sum_{i=1}^d \beta_i (N_i(n) - np_i^*) q_i + \sum_{i=1}^d q_i \log q_i - q_i \log p_i(0).$$

Define the Lagrangian

$$\mathcal{L}(q) := \left(\sum_{i=1}^d \beta_i (N_i(n) - np_i^*) q_i + q_i \log q_i - q_i \log p_i(0) + \lambda q_i \right) - \lambda$$

with Lagrange multiplier λ . Setting $\nabla_q \mathcal{L}(q) = 0$ gives

$$0 = \beta_i (N_i(n) - np_i^*) + \log q_i + 1 - \log p_i(0) + \lambda,$$

$$= \beta_i N_i(n) - \frac{n}{\sum_{j=1}^d 1/\beta_j} + \log q_i + 1 - \log p_i(0) + \lambda,$$

which we solve to get

$$q_i \propto p_i(0) e^{-\beta_i N_i(n)}.$$

□

It is somewhat remarkable that the same sampling rule characterized by a global invariance rule (as in Proposition 2.2.2) can also be uniquely characterized by a myopic, one-step optimization. However, Theorem 2.4.3 can be viewed as an instance of the usual Gibbs variational principle [19]: a linear cost regularized by relative entropy has an exponentially weighted optimizer. The cost assigned to index i is $\beta_i(N_i(n) - np_i^*)$, and since $\beta_i p_i^*$ is constant in i , the target-centering term does not affect the optimizer after normalization. The resulting policy therefore depends only on the weighted counts $\beta_i N_i(n)$.

2.5 Expected Cover Time

Beyond asymptotic convergence of the empirical distribution, it is also important in applications that the sampler does not neglect any index for too long. A natural way to quantify this is via the cover time, namely the number of steps required before every outcome has been sampled at least once. In audit and inspection settings, this measures how long a team can go entirely untested; in sampling and representation problems, it measures how long a category can remain completely unrepresented in the realized sample.

For deterministic sampling, the number of steps to see all d outcomes is d , and for IID sampling, the expected number of steps to see all the outcomes, known as the “coupon collector problem,” is $\Theta(d \log d)$. The self-balancing sampler has behavior which interpolates between these two extremes, as we now show.

Theorem 2.5.1. *Let $C := \max_{1 \leq i \leq d} \min\{n \geq 1 : X_n = i\}$ be the number of steps until the self-balancing sampler sees every outcome at least once. Then, starting from the initial condition $p_i(0) = 1/d$,*

$$\mathbb{E}[C] = \sum_{i=1}^d \frac{\log(1 + \beta_i \log d)}{\beta_i} + O(d).$$

Proof. Writing $C = C_1 + \dots + C_d$, where $C_i := |\{1 \leq n \leq C : X_n = i\}|$ is the number of times we see outcome i before covering all the outcomes, we need only show

$$\mathbb{E}[C_i] = \frac{\log(1 + \beta_i \log d)}{\beta_i} + O(1).$$

The key insight is to embed the self-balancing sampler into a continuous time stochastic process with the sampling order determined by independent exponential clocks. For $1 \leq i \leq$

d and $j \geq 0$, let $W_{i,j} \sim \text{Exp}(\alpha_i^j)$ be independent, and let $Y_{i,j} := \sum_{k=0}^j W_{i,k}$. Then the order of the $Y_{i,j}$ and their associated i indices has the same distribution as the order of the X_n , e.g. $\mathbb{P}(Y_{1,0} < Y_{1,1} < Y_{6,0} < \dots) = \mathbb{P}(X_1 = 1, X_2 = 1, X_3 = 6, \dots)$. Letting

$$\tilde{C}_i(t) := \min\{j \geq 0 : Y_{i,j} > t\},$$

be the number of times index i is sampled by time t , we claim

$$\frac{\log(1 + \beta_i t)}{\beta_i} \leq \mathbb{E}[\tilde{C}_i(t)] \leq \frac{\log(1 + \beta_i t)}{\beta_i} + 1.$$

The lower bound is equivalent to $1 + \beta_i t \leq e^{\beta_i \mathbb{E}[\tilde{C}_i(t)]}$, which follows from

$$\begin{aligned} e^{\beta_i \mathbb{E}[\tilde{C}_i(t)]} &= 1 + \int_0^t \frac{d}{du} e^{\beta_i \mathbb{E}[\tilde{C}_i(u)]} \Big|_{u=s} ds \\ &= 1 + \beta_i \int_0^t \frac{d}{du} \mathbb{E}[\tilde{C}_i(u)] \Big|_{u=s} \cdot e^{\beta_i \mathbb{E}[\tilde{C}_i(s)]} ds \\ &= 1 + \beta_i \int_0^t \mathbb{E}[e^{-\beta_i \tilde{C}_i(s)}] e^{\beta_i \mathbb{E}[\tilde{C}_i(s)]} ds \\ &\geq 1 + \beta_i \int_0^t e^{-\beta_i \mathbb{E}[\tilde{C}_i(s)]} e^{\beta_i \mathbb{E}[\tilde{C}_i(s)]} ds \\ &= 1 + \beta_i t, \end{aligned}$$

where we used that the derivative of the expectation of a birth process is the expected birth rate. For the upper bound, we have

$$\begin{aligned} e^{\beta_i \mathbb{E}[\tilde{C}_i(t)]} &\leq \mathbb{E}[e^{\beta_i \tilde{C}_i(t)}] \\ &= 1 + \int_0^t \frac{d}{du} \mathbb{E}[e^{\beta_i \tilde{C}_i(u)}] \Big|_{u=s} ds. \end{aligned}$$

By Kolmogorov's forward equation, $\frac{d}{du} \mathbb{E}[e^{\beta_i \tilde{C}_i(u)}]$ is the expectation of the generator of the birth process $\tilde{C}_i(t)$ applied to the function $f(z) = e^{\beta_i z}$.

$$\begin{aligned} &= 1 + \int_0^t \mathbb{E}[e^{-\beta_i \tilde{C}_i(s)} (e^{\beta_i (\tilde{C}_i(s)+1)} - e^{\beta_i \tilde{C}_i(s)})] ds \\ &= 1 + \int_0^t e^{\beta_i} - 1 ds \\ &= 1 + (e^{\beta_i} - 1)t. \end{aligned}$$

This gives an upper bound of $\mathbb{E}[\tilde{C}_i(t)] \leq \frac{\log(1+(e^{\beta_i}-1)t)}{\beta_i} \leq \frac{\log(1+\beta_i t)}{\beta_i} + 1$.

Now, we incorporate the random time in the continuous process which corresponds to the cover time in the self-balancing sampler. Let $M_{-i} := \max\{W_{j,0} : j \neq i\}$ be the maximum waiting time for all $j \neq i$ to be sampled at least once, and note

$$C_i \stackrel{d}{=} \max\{1, \tilde{C}_i(M_{-i})\},$$

so that

$$\mathbb{E}[C_i] = \mathbb{E}[\tilde{C}_i(M_{-i})] + \underbrace{\mathbb{P}(W_{i,0} > W_{j,0} \forall j \neq i)}_{=1/d}.$$

Applying our previous inequalities gives

$$\left| \mathbb{E}[C_i] - \mathbb{E} \left[\frac{\log(1 + \beta_i M_{-i})}{\beta_i} \right] \right| \leq 1 + \frac{1}{d},$$

and all that remains now is to show that $\mathbb{E}[\tilde{C}_i(M_{-i})] \approx \mathbb{E}[\tilde{C}_i(\log d)]$, where $\mathbb{E}[M_{-i}] = \sum_{k=1}^{d-1} \frac{1}{k} \approx \log d$. We have

$$\begin{aligned} \left| \mathbb{E}[C_i] - \frac{\log(1 + \beta_i \log d)}{\beta_i} \right| &\leq \left| \mathbb{E}[C_i] - \mathbb{E} \left[\frac{\log(1 + \beta_i M_{-i})}{\beta_i} \right] \right| \\ &\quad + \left| \mathbb{E} \left[\frac{\log(1 + \beta_i M_{-i})}{\beta_i} \right] - \frac{\log(1 + \beta_i \log d)}{\beta_i} \right|. \end{aligned}$$

The latter term can be bounded using the fact that $g(x) := \frac{\log(1 + \beta_i x)}{\beta_i}$ is 1-Lipschitz, which follows from $|g'(x)| = \left| \frac{1}{1 + \beta_i x} \right| \leq 1$ for all $x \geq 0$.

$$\begin{aligned} &\leq 1 + \frac{1}{d} + \mathbb{E}[|M_{-i} - \log d|] \\ &\leq 1 + \frac{1}{d} + \sqrt{\mathbb{E}[(M_{-i} - \log d)^2]} \\ &= 1 + \frac{1}{d} + \sqrt{(\mathbb{E}[M_{-i} - \log d])^2 + \text{Var}(M_{-i})} \\ &= 1 + \frac{1}{d} + \sqrt{\left(\log d - \sum_{k=1}^{d-1} \frac{1}{k} \right)^2 + \sum_{k=1}^{d-1} \frac{1}{k^2}} \\ &\leq 1 + \frac{1}{d} + \sqrt{1 + \frac{\pi^2}{6}}. \end{aligned} \quad \square$$

For fixed β_i , the cover time is $\Theta(d \log \log d)$. Moreover, in the uniform case $\beta_i \equiv \beta$, the self-balancing sampler's behavior converges to the deterministic and IID behaviors under the respective limits $\beta \uparrow \infty$ and $\beta \downarrow 0$

$$\limsup_{\beta \uparrow \infty} \mathbb{E}[C] = \Theta(d), \quad \liminf_{\beta \downarrow 0} \mathbb{E}[C] = \Theta(d \log d).$$

Thus, for fixed bias, the self-balancing sampler has a cover time much closer to deterministic sampling than to IID sampling, but still larger than the deterministic cover time by a diverging $\log \log d$ factor.

2.6 Weak-Bias Diffusive Limits and the Balance–Predictability Trade-Off

In this section, we explore a few different diffusion approximations of our self-balancing sampler, highlighting its mean-reversion and other asymptotic properties. We are primarily interested in the *weak-biasing regime* where $\beta \downarrow 0$, which is the natural regime under which a diffusive limit can arise.

Our first result shows that in this weak-biasing regime each coordinate is in a sense just a time-changed Ornstein–Uhlenbeck (OU) process. The technical proofs of the two diffusion limits are deferred to Appendix 2.C.3.

Proposition 2.6.1. *Fix $i \in [d]$ and let $\tilde{Y}_i^\beta(n) := \beta_i N_i(n) - \beta_d N_d(n)$ be the projection-centered version of the self-balancing sampler with the given β . Let*

$$\tau_k^\beta := \inf \left\{ n > \tau_{k-1}^\beta : \tilde{Y}_i^\beta(n) \neq \tilde{Y}_i^\beta(\tau_{k-1}^\beta) \right\}, \quad \tau_0^\beta = 0.$$

For $h > 0$, define the continuous-time chain

$$Z_t^h := h^{-1/2} \left(\tilde{Y}_i^{h\beta}(\tau_{[t/h]}^{h\beta}) - c_\beta \right), \quad c_\beta := \log \left(\frac{p_i(0)}{p_d(0)} \right) - \log \left(\frac{p_i^*}{p_d^*} \right).$$

If $Z_0^h \rightarrow z_0$, then we have $Z_t^h \Rightarrow Z_t$ as $h \downarrow 0$ where Z_t is the diffusion

$$dZ_t = -\frac{\beta_i \beta_d}{\beta_i + \beta_d} Z_t dt + \sqrt{\beta_i \beta_d} \cdot dW_t, \quad Z_0 = z_0.$$

Here W_t denotes a standard 1-dimensional Brownian motion and $\Rightarrow$ denotes convergence in the Skorokhod space $D([0, T], \mathbb{R})$ for any $T < \infty$.

In the uniform case $\beta_i \equiv \beta$, the diffusion simplifies to

$$dZ_t = -\frac{\beta}{2} Z_t dt + \beta dW_t,$$

a simple OU process. The OU process is well-known for its strong mean-reversion properties, so in a sense it is unsurprising it appears as a limit of our discrete self-balancing sampler.

The correction c_β shifts the starting point of $\tilde{Y}(n)$ to the location where it has zero drift and where the sampling law $p(n)$ agrees with the target law p^* . Except in the case where $p(0) = p^*$ and thus $c_\beta = 0$, this zero-drift location is *not* the natural initial condition where all counts are zero. Consequently, Proposition 2.6.1 should not be interpreted as a diffusion limit from the all-zero count initialization. Rather, it describes the local fluctuations of the sampler near the zero-drift point, or heuristically the behavior after the process has already entered its equilibrium scale. In a sense, c_β represents a penalty we pay for starting our process away from stationarity. As we saw in Theorem 2.3.2, this penalty has a strong impact on leading-order constants of the convergence rate.

We develop an analogous diffusion-approximation for the mean-centered counts, this time without needing a time-change. Here, the interaction between the different coordinates becomes visible, and again the process exhibits mean-reversion.

Proposition 2.6.2. *Let $M^\beta(n) := (\beta_i[N_i(n) - np_i^*])_{i=1}^d$ be the (weighted) mean-centered version of the self-balancing sampler with the given β . For $h > 0$, define the continuous-time chain*

$$Z_t^h := h^{-1/2} (M^{h\beta}(\lfloor t/h \rfloor) - c_\beta), \quad (c_\beta)_i := \log \left(\frac{p_i(0)}{p_i^*} \right) + \text{KL}(p^* \| p(0)).$$

If $Z_0^h \rightarrow z_0$, then $Z_t^h \Rightarrow Z_t$ as $h \downarrow 0$ where Z_t is the diffusion

$$dZ_t = -\frac{1}{S_\beta} Z_t dt + \Sigma^{1/2} d\mathbf{W}_t, \quad Z_0 = z_0,$$

for $S_\beta = \sum_j \beta_j^{-1}$ and $\Sigma = S_\beta^{-1} \text{diag}(\beta) - S_\beta^{-2} \mathbf{1}\mathbf{1}^T$. Here $\mathbf{W}_t$ denotes a d -dimensional standard Brownian motion.

As a consequence of the centering, this diffusion is constrained to the hyperplane

$$\{z : \langle z, p^* \rangle = 0\}.$$

Again, the situation simplifies considerably in the uniform $\beta_i \equiv \beta$ case.

$$dZ_t = -\frac{\beta}{d} Z_t dt + \frac{\beta}{d} (dI_d - \mathbf{1}\mathbf{1}^T)^{1/2} dW_t.$$

The centering c_β in Proposition 2.6.2 satisfies

$$\langle c_\beta, p(0) \rangle = \text{KL}(p(0) \| p^*) + \text{KL}(p^* \| p(0)), \quad (2.6.1)$$

the symmetrized KL divergence. Hence again c_β is a measure of the typical discrepancy between the initial sampling law $p(0)$ and the target sampling law p^* .

2.6.1 Interpretation of Diffusive Limits

Using our diffusion limits, we can heuristically investigate how our two centered-counts processes behave in this weak-biasing regime. This section is non-rigorous and mainly serves to provide evidence that the upper bounds appearing in Theorem 2.3.2 may be tight in their dependence on the bias parameters. We start with the projection-centered case. Moreover, by studying the process's autocorrelation we aim to provide an alternative measure of its predictability in contrast to the adversarial notion discussed in Section 2.6.2.

Below, we will rely on some basic facts about OU processes, details of which can be found in Section 6.5 of [68]. Recall the one-dimensional OU process

$$dZ_t = -\theta Z_t dt + \sigma dW_t, \quad Z_0 = z_0,$$

has stationary distribution $N(0, \sigma^2/2\theta)$ and autocorrelation

$$\text{Corr}(Z_t, Z_{t+s}) = e^{-\theta s}.$$

For small β , where sampling is close to uniform, the waiting times $\tau_k^\beta - \tau_{k-1}^\beta \approx \text{Geo}(2/d)$, so we wait roughly time $d/2$ between samples. Hence using Proposition 2.6.1, we heuristically see that in the uniform β case the (time-changed) projection-centered counts are well-approximated by

$$N_i(n) - N_d(n) \approx \frac{Z_{2nh/d}}{\beta\sqrt{h}}.$$

which is an OU process with stationary distribution $N(0, (\beta h)^{-1})$. At stationarity

$$\text{Corr}(N_i(n+k) - N_d(n+k), N_i(n) - N_d(n)) \approx \exp\left(-\frac{\beta h k}{d}\right).$$

Since βh was just our rescaled bias, this approximation suggests that, in the small- β regime at stationarity, the process's autocorrelations decay exponentially at rate $O(\beta)$. Since $p^*(\beta) = p^*(c\beta)$ for any $c > 0$, scaling β has no effect on the limiting distribution.

For the mean-centered counts, the story is similar. Specifically, for an arbitrary $d \times d$ matrix Θ and positive-semidefinite covariance matrix Σ , the d -dimensional OU process

$$dZ_t = -\Theta Z_t dt + \Sigma^{1/2} dW_t,$$

has stationary distribution $N(0, \omega)$ for ω solving the Lyapunov equation

$$\Theta \omega + \omega \Theta^T = \Sigma.$$

Moreover, stationary covariance is

$$\text{Cov}(Z_{t+s}, Z_t) = e^{-\Theta s} \omega$$

In the context of Proposition 2.6.2, it is easy to see $\Theta = S_\beta^{-1} I_d$ and

$$\omega = \frac{1}{2}[D_\beta - S_\beta^{-1} 11^T] = \frac{1}{2}D_\beta[I_d - p^* 1^T],$$

for $D_\beta = \text{diag}(\beta_i)$. Thus, Proposition 2.6.2 motivates the stationary approximation

$$N(n) - np^* \approx D_\beta^{-1} \left[\frac{c_\beta}{h} + \frac{Z_{nh}}{\sqrt{h}} \right] \sim N \left(\frac{D_\beta^{-1} c_\beta}{h}, \frac{D_\beta^{-1} \omega D_\beta^{-1}}{h} \right).$$

Under this approximation, the stationary autocorrelation of any nonzero $a \in \mathbb{R}^d$ is

$$\text{Corr}(a^T(N(n+k) - (n+k)p^*), a^T(N(n) - np^*)) = e^{-kh/S_\beta},$$

Hence the diffusion heuristic again suggests an autocorrelation decay rate of order $O(\beta)$.

Note

$$\mathbb{E}\|\mu_n - p^*\|_{\text{TV}} = \frac{1}{2n} \sum_i \mathbb{E}|N_i(n) - np_i^*| = \frac{\mathbb{E}\|Y(n)\|_1}{2n}.$$

Using the above stationary distribution, our diffusion approximation suggests

$$\mathbb{E}\|\mu_n - p^*\|_{\text{TV}} \approx \frac{1}{2n} \left(\sum_{i=1}^d \mathbb{E} \left| N \left(\frac{(c_\beta)_i}{\beta_i h}, \frac{1 - p_i^*}{2\beta_i h} \right) \right| \right).$$

Hence, the mean of a folded-normal

$$\mathbb{E}|N(\mu, \sigma^2)| = \sigma \sqrt{\frac{2}{\pi}} \exp\left(-\frac{\mu^2}{2\sigma^2}\right) + \mu \left[1 - 2\Phi\left(-\frac{\mu}{\sigma}\right)\right],$$

suggests there are two regimes

$$\mathbb{E}\|\mu_n - p^*\|_{\text{TV}} \approx \begin{cases} O\left(\frac{d}{n\sqrt{\beta}h}\right) & c_\beta = 0, \\ O\left(\frac{1}{nh} \sum \frac{|(c_\beta)_i|}{\beta_i}\right) & c_\beta \neq 0. \end{cases}$$

Since $c_\beta = 0$ if and only if $p(0) = p^*$, this predicts the same dependence on the bias parameters as the upper bound in Theorem 2.3.2. It therefore suggests that this dependence may be tight, and we supplement the heuristic with empirical evidence in Appendix 2.A.

2.6.2 Adversarial Interpretation of Predictability

In order to develop a notion of how “predictable” such a sampler is, consider the case where an adversary attempts to guess the next sample given the current sample history. Then one way of quantifying the predictability of the sampler is to study how accurate such an adversary can be if they behave optimally. In this section, we formalize this idea.

Predictability criteria of this form have been studied extensively for restricted randomization designs, where selection bias is commonly measured by the probability that an observer can correctly guess the next treatment assignment [4, 23, 75]. In the special $d = 2$ case with uniform initialization and uniform bias, we prove below the adversary’s accuracy is of order $\sqrt{\beta}$ above the naive baseline in the weak-biasing $\beta \downarrow 0$ regime. We conjecture this to be true in general whenever several coordinates maximize p^* , and we provide some heuristic evidence of this conjecture using our diffusive limits.

Given the sample history, the adversary’s optimal strategy is to guess an index maximizing the next-step sampling probability. The resulting asymptotic accuracy is

$$A_\beta := \liminf_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^{n-1} \mathbb{E}_{y \sim \tilde{Y}^{(k)}} \left[\max_{i \in [d]} p_i(y) \right].$$

Here $\tilde{Y}(k)$ denotes our projection-centered counts. Lemma 2.2.3 shows these centered counts form an ergodic Markov chain with stationary distribution η . By the Markov-chain ergodic theorem [56, Theorem 17.0.1] and the bounded convergence theorem

$$A_\beta = \mathbb{E}_{y \sim \eta} \left[\max_{i \in [d]} p_i(y) \right],$$

for any initial state. In general, determining the stationary distribution η is difficult. However, in the special case of $d = 2$, then $\tilde{Y}$ becomes a simple birth-death process on $\mathbb{Z}$, giving

Lemma 2.6.3. *When $d = 2$ and uniform $\beta_1 = \beta_2 = \beta > 0$, the projection-centered counts $\tilde{Y}(n)$ form a reversible Markov chain on $\beta\mathbb{Z}$ with stationary distribution*

$$\eta(\beta z) \propto \frac{p_2(0) + p_1(0)e^{-\beta z}}{p_1(0) + p_2(0)} \left(\frac{p_1(0)}{p_2(0)} \right)^z e^{-\beta \binom{z}{2}}$$

Proof. In the $d = 2$ case then $\tilde{Y}_n$ is a simple birth-death chain with birth probability

$$\lambda_n = \frac{p_1(0)e^{-\beta n}}{p_2(0) + p_1(0)e^{-\beta n}}$$

and death otherwise. The detailed balance equations then imply

$$\frac{\pi(n)}{\pi(n-1)} = \frac{\lambda_{n-1}}{1 - \lambda_n}.$$

Hence

$$\pi(n) = \pi_0 \prod_{j=1}^n \frac{\lambda_{j-1}}{1 - \lambda_j} = \pi_0 \frac{p_2(0) + p_1(0)e^{-\beta n}}{p_1(0) + p_2(0)} \left(\frac{p_1(0)}{p_2(0)} \right)^n e^{-\beta \binom{n}{2}}$$

If $\beta > 0$ then this is normalizable as it has discrete Gaussian tails. □

In the case $p_1(0) = p_2(0) = \frac{1}{2}$, we get the symmetric and especially simple form

$$\eta(\beta z) \propto \cosh \left(\frac{\beta z}{2} \right) e^{-\frac{\beta z^2}{2}}.$$

When $d = 2$, we can quantify how this accuracy depends on β in the weak-biasing regime.

Lemma 2.6.4. *When $d = 2$, $p_1(0) = p_2(0) = 1/2$, and $\beta_1 = \beta_2 = \beta$, as $\beta \downarrow 0$ we have*

$$A_\beta = \frac{1}{2} + \Theta(\sqrt{\beta}).$$

Proof. Note in the $d = 2$ case with uniform β , we have

$$A_\beta = \liminf_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^{n-1} \mathbb{E}_{y \sim \tilde{Y}(k)} \left[\frac{1}{1 + e^{-\beta|y|}} \right].$$

Since $Y(n)$ is ergodic, the ergodic theorem and Lemma 2.6.3 imply

$$A_\beta = \mathbb{E}_{y \sim \eta} \left[\frac{1}{1 + e^{-\beta|y|}} \right].$$

While no simple closed form exists, the bound

$$\frac{1}{2} + \frac{x}{4} - \frac{x^3}{48} \leq (1 + e^{-x})^{-1} \leq \frac{1}{2} + \frac{x}{4}, \quad \forall x \geq 0,$$

implies for small β that the accuracy is of order

$$A_\beta = \frac{1}{2} + \frac{\beta}{4} \mathbb{E}_\eta |Y| + O(\beta^3 \mathbb{E}_\eta |Y|^3).$$

To control this, note the stationary distribution of Y is a mixture of two (discrete) Gaussians

$$\cosh\left(\frac{\beta y}{2}\right) e^{-\beta y^2/2} \propto \frac{1}{2} \left[e^{-\frac{\beta}{2}(y-1/2)^2} + e^{-\frac{\beta}{2}(y+1/2)^2} \right].$$

centered at $\pm \frac{1}{2}$ with scale $\beta^{-1/2}$. A standard Riemann-sum comparison shows that, for each fixed $r \geq 0$, a discrete Gaussian with fixed center and scale $\beta^{-1/2}$ satisfies

$$\mathbb{E}|Y|^r \sim \beta^{-r/2} \mathbb{E}|G|^r, \quad G \sim N(0, 1).$$

Since the stationary law is an equal mixture of two such discrete Gaussians, it follows that

$$\mathbb{E}_\eta |Y| \sim \sqrt{\frac{2}{\pi\beta}}, \quad \mathbb{E}_\eta |Y|^3 = O(\beta^{-3/2}).$$

Combining this with the above yields

$$A_\beta = \frac{1}{2} + c\sqrt{\beta} + O(\beta^{3/2}) = \frac{1}{2} + \Theta(\sqrt{\beta}).$$

as desired. □

Even when $d = 2$, if β is non-uniform we lose this simple birth-death structure. Instead, to investigate the general β case we rely on our diffusive limits. The following analysis is non-rigorous and aims only to provide heuristic evidence for why a square-root correction may persist more generally.

We saw in Section 2.6 that in the weak-biasing regime the weighted mean-centered counts are approximated at stationarity by

$$M^\beta(n) = D_\beta[N(n) - np^*] \approx c_\beta + \sqrt{h}Z, \quad Z \sim N(0, \omega),$$

where ω is the stationary covariance from Section 2.6. Since

$$p_i(M) = \frac{p_i(0)e^{-M_i}}{\sum_{j=1}^d p_j(0)e^{-M_j}}, \quad p(c_\beta) = p^*,$$

this suggests

$$\mathbb{E} \left[\max_i p_i(n) \right] \approx \mathbb{E} \left[\max_i p_i(c_\beta + \sqrt{h}Z) \right].$$

Let J_β denote the Jacobian of the softmax map at c_β . A direct calculation gives

$$J_\beta = -[\text{diag}(p^*) - p^*(p^*)^T],$$

and hence the first-order Taylor expansion yields

$$p(c_\beta + \sqrt{h}Z) = p^* + \sqrt{h}J_\beta Z + O(h).$$

so heuristically

$$\mathbb{E} \left[\max_i p_i(n) \right] \approx \mathbb{E} \left[\max_i \left(p_i^* + \sqrt{h}G_i \right) \right], \quad G = J_\beta Z \sim N(0, J_\beta \omega J_\beta^T).$$

If several coordinates tie for the maximum of p^* , then the max is not differentiable and the first-order Gaussian fluctuation should contribute. Writing $I^* = \{i : p_i^* = \max_j p_j^*\}$ for the set of maximizers, the corresponding first-order approximation is

$$\mathbb{E} \left[\max_i p_i(n) \right] = \max_i p_i^* + \sqrt{h} \mathbb{E} \left[\max_{i \in I^*} G_i \right] + O(h).$$

Using the fact $\langle Z, p^* \rangle = 0$, this heuristic further gives

$$G = -\text{diag}(p^*)Z \stackrel{d}{=} -\text{diag}(p^*)(W - \langle W, p^* \rangle 1)$$

for $W \sim N(0, D_\beta/2)$. Hence as the second term is constant on I^* and mean-zero

$$\mathbb{E} \left[\max_i p_i(n) \right] - \max_i p_i^* = \sqrt{\frac{h}{2\beta_{\min} S_\beta^2}} \cdot \mathbb{E} \max_{i \in I^*} B_i + O(h) \leq \sqrt{\frac{h \log |I^*|}{\beta_{\min} S_\beta^2}} + O(h).$$

for $B_i \stackrel{iid}{\sim} N(0, 1)$. The bound is motivated by classical asymptotics for the maximum of IID Gaussians (e.g. [82]), and is approximately tight when $|I^*|$ is large. Hence

$$A_\beta \approx \max_i p_i^* + \sqrt{\frac{h \log |I^*|}{\beta_{\min} S_\beta^2}} + O(h).$$

In the uniform β case, this becomes

$$A_\beta \approx \frac{1}{d} + \frac{\sqrt{\beta h \log d}}{d} + O(h)$$

Again, this approximation is most informative when d is large. Thus, the diffusion heuristic suggests that, in the weak-biasing regime, the adversary's accuracy is influenced primarily by the maximum of p^* and by the number of coordinates sharing this maximum value. It suggests that increasing either should increase the adversary's accuracy. Moreover, for fixed p^* , it predicts an excess accuracy above the baseline $\max_i p_i^*$ of order $\sqrt{\beta}$ as $\beta \downarrow 0$, consistent with the rigorous $d = 2$ result in Lemma 2.6.4.

In particular, the heuristic suggests that the adversary's accuracy should be larger when several coordinates share the minimal value of β_i , since the adversary can exploit the largest realized fluctuation among these tied coordinates.

2.7 Concluding Remarks

We introduced a family of self-balancing sequential samplers that adaptively downweight previously sampled outcomes. The resulting process interpolates between IID sampling, which is maximally random but poorly balanced, and deterministic sampling, which is optimally balanced but perfectly predictable. Despite retaining randomness at every step, the self-balancing sampler achieves empirical convergence at the deterministic scale $O(n^{-1})$ in expectation and $O(\log n/n)$ almost surely.

A central theme of the chapter is that this sampler is not merely a heuristic. It arises naturally from an invariance principle for potential-based Markovian samplers, is the unique solution to a one-step entropy-regularized imbalance minimization problem, and shows up informally in actual policy decisions. Together, these perspectives give complementary evidence that such exponential sampling rules arise naturally.

The weak-biasing analysis helps illustrate the convergence–predictability tradeoff. It suggests that, as the bias parameter decreases, the centered counts fluctuate on a larger scale and mix more slowly, preserving more unpredictability; as the bias parameter increases, the process becomes more tightly mean-reverting, improving balance but making the next sample easier to predict. The diffusion approximation provides a heuristic quantification of this tradeoff and suggests that tied maximizers of p^* produce first-order Gaussian corrections to predictability.

Several extensions remain open. One natural direction is to replace categorical balance with covariate balance, sampling units according to their contribution to a cumulative covariate imbalance. Namely, suppose each unit $i \in [d]$ has some covariate vector v_i assigned to it. If we have samples $X_1, \dots, X_n \in [d]$ define the cumulative (centered) covariate vector as

$$S_n := \sum_{i=1}^n v_{X_i} - n\bar{v}, \quad \bar{v} := \frac{1}{d} \sum_{i=1}^d v_i.$$

We can then generalize our self-balancing sampler so that index i is sampled with probability

$$p_i(n) \propto \exp \left(- \sum_j \beta_j v_{ij} (S_n)_j \right).$$

When $v_i = e_i$ are the standard basis vectors, this recovers our original self-balancing sampler. This provides a sequential analog to the Gram-Schmidt walk of [8], which in the context of [35] provides the key tool for balancing covariates across treatment groups in randomized controlled trials.

Another is to study risk-weighted versions of the sampler, where the target distribution depends on observed outcomes. Such extensions would bring the model closer to applications in inspections, audits, randomized experiments, and other settings where one seeks to balance coverage, responsiveness, and resistance to prediction. For instance, in both our health-inspections and anti-doping examples, follow-up inspections are scheduled more frequently for those performing poorly on the initial inspection.

Overall, we have shown that self-balancing exponential sampling rules provide a simple yet principled framework for designing sequential randomized procedures that balance coverage, unpredictability, and long-run empirical stability.

2.A Empirical Results

In this section, we provide some additional empirical evidence for some of the β scalings we developed heuristically in Section 2.6 above. Simulation code can be found in the GitHub repository `self_balancing_sampler`.

In the first few sections, we study the behavior of the self-balancing sampler under two different unit-norm choices of β when $d = 4$: the uniform β and the non-uniform $\beta \propto [1.1, 0.8, 0.9, 1.2]$, the latter of which also has a unique maximizer of $p^*(\beta)$. In the last section, we give an example of how the self-balancing sampler can be applied to the citizen's assembly problem, and analyze its behavior in this setting.

2.A.1 Convergence Rate

For $d_n := \mathbb{E}\|\mu_n - p^*\|_{\text{TV}}$, Theorem 2.3.2 implies that the following quantity is finite

$$C(\beta) := \limsup_{n \rightarrow \infty} n d_n < \infty.$$

The goal of this section is to determine how the bias parameter β influences the coefficient of the leading $O(n^{-1})$ term. Theorem 2.3.2 and our heuristic analysis in Section 2.6 suggests

$$C(c\beta) = \begin{cases} \Theta(c^{-1/2}) & p(0) = p^*, \\ \Theta(c^{-1}) & p(0) \neq p^*. \end{cases}$$

We provide empirical evidence these are the correct scalings. To do so, we generate estimates $\hat{d}_N$ by simulating M trajectories of the self-balancing sampler and taking the empirical average

$$\hat{d}_N = \frac{1}{M} \sum_{i=1}^M \|\mu_N^{(i)} - p^*\|_{\text{TV}},$$

for some large N . From here, we estimate

$$\hat{C}(\beta) = N \hat{d}_N.$$

To get a sense of how this constant varies with β , we fix some unit-norm β and study how $C(c\beta)$ varies as we vary the scaling $c > 0$. Since $p^*(\beta) = p^*(c\beta)$ for any $c > 0$, all these samplers have the same limiting distribution.

For our simulation, we chose $M = 1000$ and $N = 10000$. The latter was chosen large enough so $N\|\mu_N - p^*\|_{\text{TV}}$ approximately stabilizes across the range of c values tested. The results are displayed in Figure 2.2. Empirically, this supports the $\Theta(c^{-1})$ and $\Theta(c^{-1/2})$ scalings our analysis suggests, at least in the small β regime.

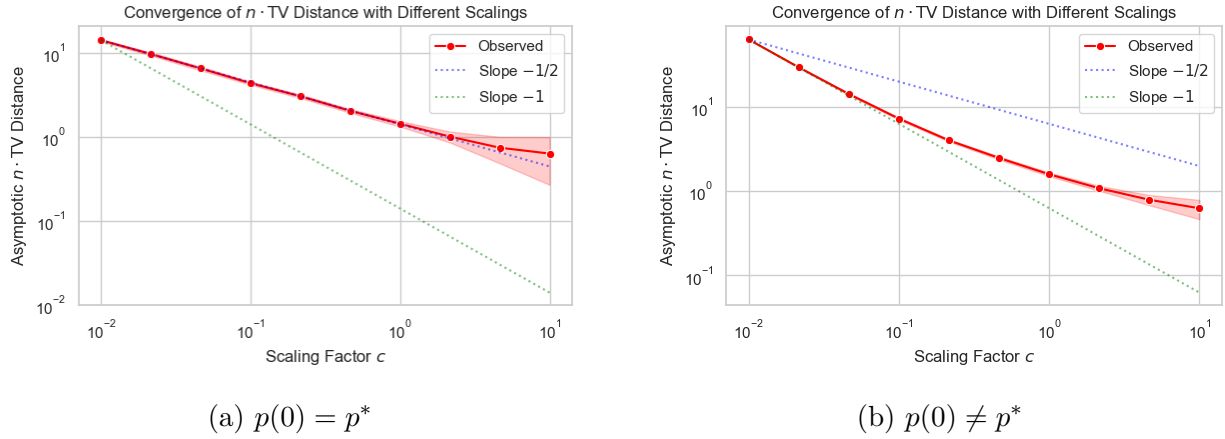


Figure 2.2: Log-log plots of Nd_N as a function of the scaling factor c for (a) uniform $\beta = [0.5, 0.5, 0.5, 0.5]$ and (b) non-uniform $\beta \propto [1.1, 0.8, 0.9, 1.2]$. In both cases $p(0)$ is the uniform distribution. Dotted lines show the expected $\Theta(c^{-1})$ and $\Theta(c^{-1/2})$ scalings in the $p(0) \neq p^*$ and $p(0) = p^*$ regimes respectively.

2.A.2 Predictability

In this section, we explore how the parameter β affects the adversary's prediction accuracy, as described in Section 2.6.2. For tied maximizers, the analysis there suggests that the excess accuracy above the naive IID baseline p_{max}^* scales like $\Theta(\sqrt{\beta})$. We compare it to our two naive baselines: IID sampling and deterministic sampling.

In IID sampling, the adversary will clearly always predict the index i which maximizes p_i^* , and hence its long-term accuracy will be p_{max}^* . In greedy sampling, we can make the sampler a bit more unpredictable by sampling the next index uniformly from $\arg \max_i p_i(n)$, rather than deterministically. In the uniform case $\beta_i \equiv \beta$, this gives an accuracy of approximately $\frac{\log(d)}{d}$, compared to the d^{-1} of IID sampling. In the non-uniform case, we estimate the accuracy under the greedy model through a simple Monte Carlo simulation. Namely, assume at step n the adversary has probability m_n^{-1} of correctly guessing the next sample, where

$$m_n = |\arg \max_i p_i(n)|,$$

is the number of outcomes that are maximally likely to be sampled. We generate samples until the average accuracy $N^{-1} \sum_{n=1}^N m_n^{-1}$ approximately converges. Similarly, we can likewise estimate the adversarial accuracy for the self-balancing sampler by averaging these estimates over many sampling trajectories.

The results are shown in Figure 2.3. In the uniform β case, prediction accuracy increases gradually as the bias scale (c) increases, and in the small- (c) regime the gap above the IID baseline roughly follows the predicted $\Theta(\sqrt{c})$ scaling. In the non-uniform case, prediction accuracy increases more sharply as (c) grows.

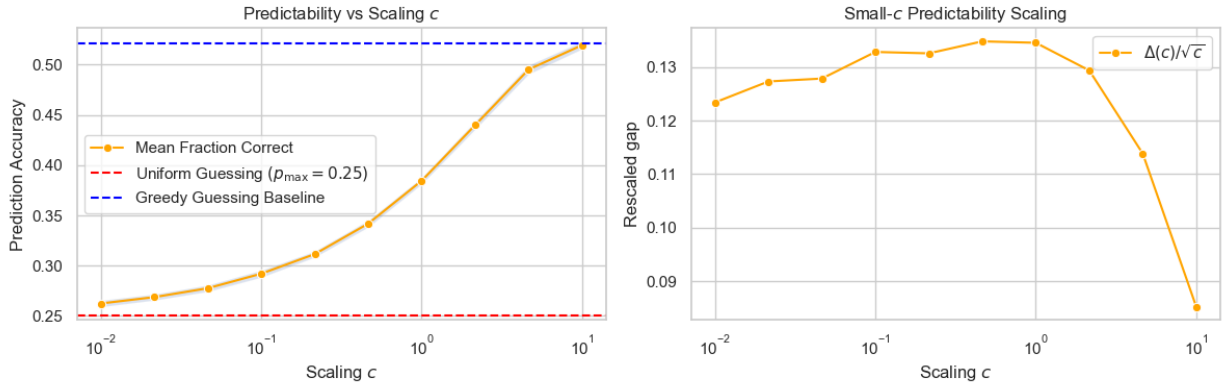
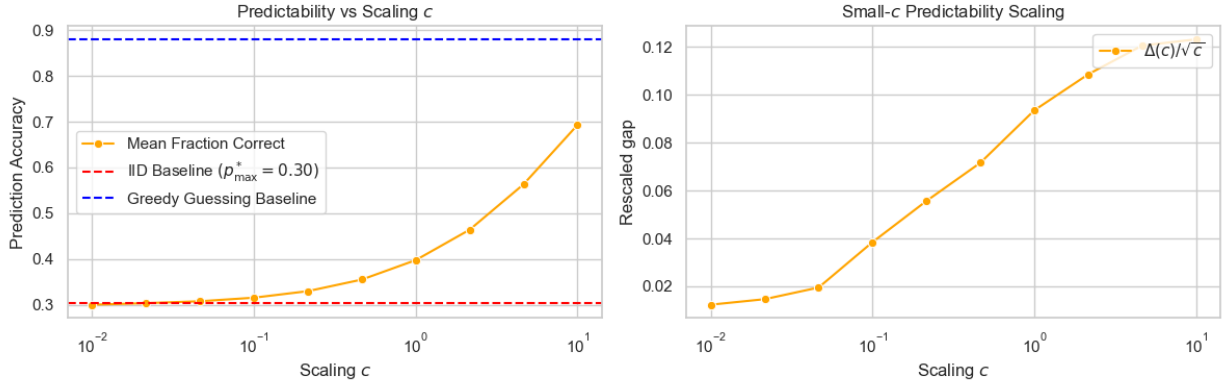
(a) Uniform $\beta = [0.5, 0.5, 0.5, 0.5]$.(b) Non-uniform, distinct maximizer $\beta \propto [1.1, 0.8, 0.9, 1.2]$.

Figure 2.3: Predictability plots for different choices of β . (left) Relationship between prediction accuracy and the scaling c of β . (right) The ratio of the gap $\Delta(c) := \text{accuracy} - p_{\max}^*$ from the uniform baseline and the first order $\sqrt{c}$ expected scaling.

2.A.3 Trade-Off

Lastly, we study the trade-off between predictability and the convergence rate. As discussed in the previous two sections, we quantify the former using the adversary's prediction accuracy, and the latter using $C(\beta) = \limsup_{n \rightarrow \infty} nd_n$. Figure 2.4 highlights the results.

Note the adversarial accuracy starts essentially at the baseline $p_{\max}^*$ and increases with the scaling c . It starts smaller in the uniform β case merely because $p_{\max}^*$ is smaller than in the non-uniform case. However, in the non-uniform case the accuracy increases much more slowly at first, before rising quickly as we scale β .

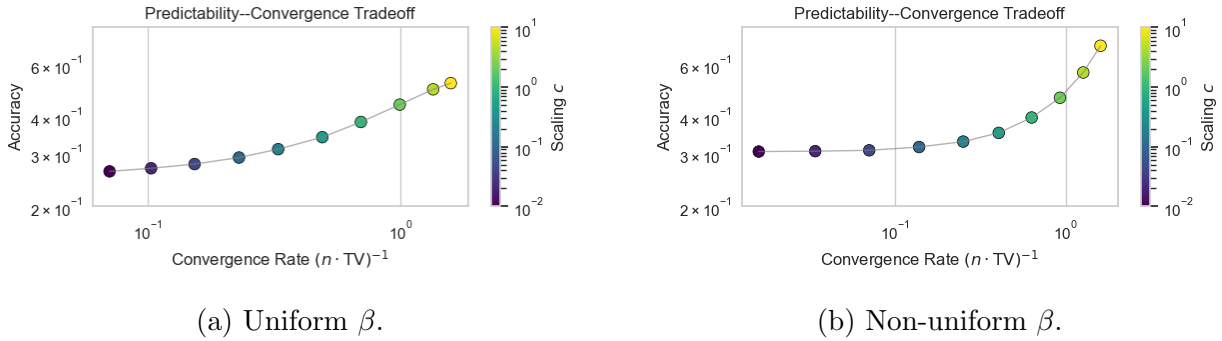


Figure 2.4: Log-log plots of the trade-off between adversarial accuracy and the convergence rate $(Nd_N)^{-1}$. (a) uniform $\beta = [0.5, 0.5, 0.5, 0.5]$ and (b) non-uniform $\beta \propto [1.1, 0.8, 0.9, 1.2]$.

2.A.4 Citizen's Assembly

In this section, we simulate the self-balancing sampler on the citizen's assembly problem discussed in Section 13 of [29]. The setting is as follows.

Suppose we want to select an assembly of $k = 200$ people that includes at least 99 members of each of the following categories: women, men, liberals, and conservatives. Let the pool consist of 1,000 conservative men, 999 liberal women, and 1 conservative woman. The goal is to generate an assembly satisfying these quotas while ensuring that each individual in the pool has a reasonable chance of being selected. More precisely, we want to maximize the minimum individual inclusion probability induced by the sampling scheme. We refer to this minimum inclusion probability as the *representativeness* of the scheme.

In this constructed population, the stratified sampling algorithm that selects 100 uniformly drawn women and 100 uniformly drawn men satisfies the quotas and selects each pool member with equal probability 10%. On the other hand, the *LEGACY* algorithm discussed in [29] has only a 0.2% chance of selecting the conservative woman. Thus, although LEGACY aims to balance covariates across the assembly, it provides poor ex ante representation for the rare class, since the conservative woman is almost never selected.

This stratified sampling strategy is tailored to the special structure of the example and does not provide a general solution. It works here because the population can be partitioned by gender into two groups of equal size, so selecting 100 women and 100 men simultaneously satisfies the gender quotas and gives every individual inclusion probability $100/1000 = 10\%$. In a more general population, the quota constraints may involve several overlapping attributes, and no single stratification need both satisfy the quotas and equalize individual inclusion probabilities. Our goal is therefore to test whether the self-balancing sampler can approximate the same effect in a more flexible way: biasing selection toward underrepresented attributes while preserving reasonable individual-level representativeness.

To apply the self-balancing sampler to this setting, we consider two sequential strategies:

1. **Stratified self-balancing over classes:** We first choose a class in

$$\{\text{man, woman}\} \times \{\text{conservative, liberal}\}$$

using the self-balancing sampler, and then choose an individual uniformly at random from the selected class. The relative weights $(\beta_i^{-1})_{i=1}^4$ are chosen proportional to the initial class sizes. This ensures that, at stationarity, individuals are selected uniformly from the total population.

2. **Attribute-balanced sampler:** Each individual is assigned weight proportional to

$$\exp(-\beta_{\text{gender}}N_{\text{gender}} - \beta_{\text{party}}N_{\text{party}}).$$

Equivalently, the sampler biases selection according to the current gender and party imbalances, and then samples among individuals with probabilities induced by these attribute-level weights.

By design, the attribute-based sampler aims to match the marginal distributions of the assembly to those of the population. This tends to make it better at matching desired marginal quotas, but at the cost of fairness: it tends to under-sample rare joint classes (e.g. the conservative woman) and biases the sampling distribution away from the population distribution. On the other hand, the stratified sampler aims to be as fair as possible. In settings where fairness is the main objective, we recommend the stratified sampler.

As a baseline, we additionally study the simple strategy where individuals are selected uniformly at random to be part of the assembly. We call this the *IID uniform baseline*. For each of these sampling strategies, we study the following quantities:

- The probability that the quota constraints are satisfied.
- The individual inclusion probabilities induced by the sampling scheme, especially the worst-case inclusion probability.

Moreover, we are interested in the influence the bias β has on these quantities. To investigate this tradeoff, we simulate many assemblies sampled *without replacement* according to these three sampling schemes. As before, we fix baseline bias parameters β and rerun the citizen's assembly experiment with rescaled parameters $\beta' = c\beta$ over a range of scalings c .

For each value of c and each sampler, we estimate both the probability that the quota constraints are met and the worst-case probability that a specific individual is selected. For the former, we use the fraction of simulated assemblies satisfying the quota constraints. For the latter, we estimate the inclusion probability $\hat{p}_i$ of each individual using the fraction of simulated assemblies in which that individual appears, and then take the minimum over individuals. To generate confidence bands, we bootstrap resample from the collection of simulated assemblies.

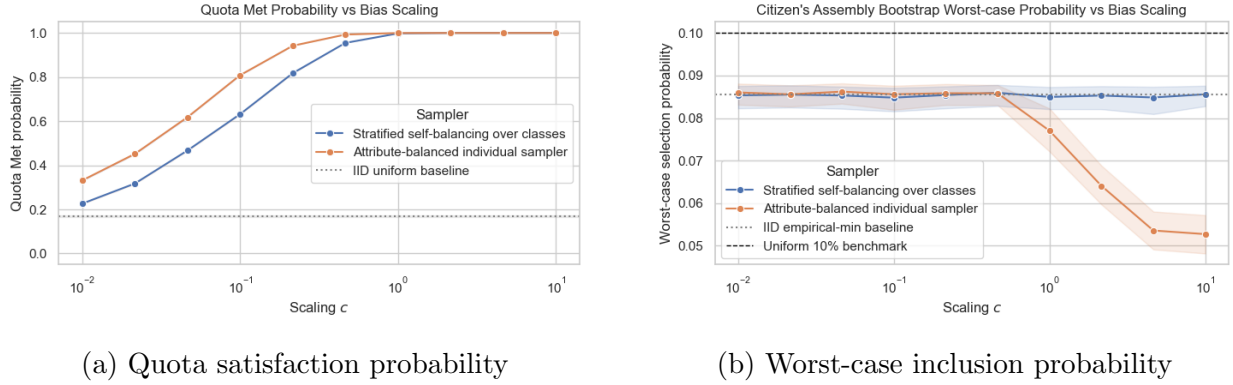


Figure 2.5: Citizen's assembly simulation results. The effect of the rescaled bias parameter $\beta' = c\beta$ on the probability that the quota constraints are met and on the worst-case individual inclusion probability.

The estimator of the minimum selection probability is downward biased, since it is the minimum of many noisy empirical estimates. Heuristically, uniformly sampling a fraction p of a population of size m across n assemblies, the empirical selection probability of individual i satisfies

$$\hat{p}_i \sim n^{-1} \text{Binom}(n, p) \approx N\left(p, \frac{p(1-p)}{n}\right),$$

for large n , and thus

$$\mathbb{E} \min_{i \leq m} \hat{p}_i \approx p + \mathbb{E} \min_{i \leq m} N\left(0, \frac{p(1-p)}{n}\right) \approx p - \sqrt{\frac{2p(1-p) \log m}{n}}.$$

Thus, the empirical minimum underestimates the population-level inclusion probability by an amount of order $\sqrt{\log(m)/n}$. This motivates the *IID* baseline.

The results are shown in Figure 2.5 for the class sizes described above, using 10^4 simulated assemblies and 1000 bootstrap samples. Figure 2.5a shows that quota-satisfaction probability increases with the bias scaling c and is already close to one for relatively small bias. Figure 2.5b shows a sharper distinction between the two self-balancing variants: the stratified sampler remains essentially optimally representative across the tested range, whereas the attribute-balanced sampler loses representativeness at moderate bias. This appears to be a parity effect. Because the population consists almost entirely of conservative men and liberal women, the attribute-based sampler tends to alternate between these two classes as $\beta \rightarrow \infty$, while the conservative woman receives little or no mass. The simulations confirm that the conservative woman is again underrepresented in this high-bias regime.

2.B Optimal Control

In this appendix section, we briefly review some basic optimal control theory, and discuss how Theorem 2.4.2 fits into the entropy-regularized optimal control framework. Suppose we have some discrete time dynamical system on some state space S

$$x_{t+1} = b(x_t, u_t), \quad x_t \in S, u_t \in U$$

where u_t is the input of some controller and U is the set of all allowable controls. In classical control theory, we have some loss function $\ell(x, u)$ and we wish to minimize the cumulative (discounted) loss over the entire trajectory $(x_t)_{t \in \mathbb{N}}$. Namely we aim to solve

$$V(x) := \inf_{(u_t)} \left[\sum_{s=1}^{\infty} \gamma^s \cdot \ell(x_s^u, u_s) \right],$$

for some discount $\gamma \in (0, 1)$, where $x_0 = x$ is the initial state of the system. Here (x_t^u) is the trajectory of the process when control sequence (u_t) is applied. The *value function* V and optimal control (u_t^*) can be determined using the standard Bellman equation

$$V(x) = \inf_{u \in U} \{ \ell(x, u) + \gamma V(b(x, u)) \}.$$

If we let $Q_\gamma(x, u) := \ell(x, u) + \gamma V(b(x, u))$ then we see the optimal policy u^* and its corresponding value function V satisfy

$$u^*(x) = \arg \min_{u \in U} Q_\gamma(x, u), \quad V(x) = Q_\gamma(x, u^*(x)). \quad (2.B.1)$$

We can generalize the above by allowing the control policy to select its control u_t at random, possibly depending on the current state. This paradigm is useful in the following settings:

- (i) *Adversarial or Strategic Environments*: In some settings, predictability can be exploited by an adversary or the environment. Entropy regularization encourages policies that remain sufficiently randomized, thereby mitigating worst-case or strategic responses to deterministic behavior [39].
- (ii) *Exploration Versus Exploitation*: In reinforcement learning, we often want to encourage the agent to explore the state space in order to learn about the environment and the corresponding reward process. Randomized policies explicitly encode this exploration into the objective function (as discussed in e.g [83]).
- (iii) *Robustness and Regularization*: Entropy regularization acts as a convex regularizer on the control problem, smoothing the optimization landscape and preventing degenerate (overly deterministic) solutions. This can significantly improve stability and sample efficiency in practice [34].

- (iv) *Information Constraints / Bounded Rationality*: Entropy regularization can also be interpreted as penalizing deviations from a reference policy via a Kullback–Leibler divergence. In this view, the controller faces a cost for using highly concentrated policies, reflecting limited information-processing capacity or a preference for staying close to a prior behavior [62].

Let U denote the set of available controls. Suppose there is some deterministic function $\pi(\cdot \mid x) : S \rightarrow \mathcal{P}(U)$ which maps the state-space S into the set $\mathcal{P}(U)$ of probability distributions on the set of controls. Define dynamics

$$X_{t+1} \mid X_t \sim b(X_t, U_t), \quad \mathbb{P}(U_t \in \cdot \mid \{X_i\}_{i \leq t}, \{U_i\}_{i < t}) = \pi(\cdot \mid X_t), \quad X_0 = x_0.$$

Namely, at time t our state X_t determines a distribution $\pi(\cdot \mid X_t)$ over the set of controls, and conditional on X_t we sample a new control independently from this distribution. This generalizes the Markovian sampler we discuss above. We then force a trade-off between the entropy of the control and the accrued loss

$$V(x_0) := \inf_{(\pi)} \mathbb{E} \left[\sum_{s=0}^{\infty} \gamma^s \cdot (\ell(X_s^u, U_s) - \tau H(\pi(\cdot \mid X_s))) \right],$$

In this case, Bellman's equation gives

$$V(x) = \inf_{\mu \in \mathcal{P}(U)} \left\{ \int_U [Q_\gamma(x, u) + \tau \log(\mu(u))] \mu(du) \right\}.$$

This can be optimized to find the optimal policy π pointwise, as well as the value function. In the case U is finite, these solutions can be expressed implicitly as

$$\pi^*(u \mid x) \propto \exp \left(-\frac{Q_\gamma(x, u)}{\tau} \right), \quad V(x) = -\tau \log \left(\sum_{u \in U} \exp \left[-\frac{Q_\gamma(x, u)}{\tau} \right] \right). \quad (2.B.2)$$

Lemma 2.B.1 (Entropy-Regularized Gibbs Optimizer). *Fix x , let U be finite, and let $\tau > 0$. Then*

$$\inf_{\mu \in \mathcal{P}(U)} \sum_{u \in U} \mu(u) [Q_\gamma(x, u) + \tau \log \mu(u)] = -\tau \log Z(x), \quad Z(x) := \sum_{u \in U} e^{-Q_\gamma(x, u)/\tau},$$

and the unique minimizer is

$$\pi^*(u \mid x) = Z(x)^{-1} e^{-Q_\gamma(x, u)/\tau}.$$

Proof. For any $\mu \in \mathcal{P}(U)$,

$$\begin{aligned} \sum_{u \in U} \mu(u) [Q_\gamma(x, u) + \tau \log \mu(u)] &= -\tau \log Z(x) + \tau \sum_{u \in U} \mu(u) \log \frac{\mu(u)}{\pi^*(u \mid x)} \\ &= -\tau \log Z(x) + \tau \text{KL}(\mu \parallel \pi^*(\cdot \mid x)). \end{aligned}$$

The result follows from nonnegativity of the KL divergence, with equality only when $\mu = \pi^*(\cdot \mid x)$. $\square$

As we can see, the optimal policy is naturally a soft-min / Gibbs distribution with Hamiltonian $-Q_\gamma(x, \cdot)/\tau$, and the value function is essentially the log-partition function of this distribution. Note how entropy-regularization essentially smooths out the strict minimization problem that occurs in equation (2.B.1). While equation (2.B.2) only implicitly defines the optimal policy, for certain choices of dynamics and loss functions we can be more explicit. Suppose we have some potential $\Phi : S \rightarrow \mathbb{R}$ and we define loss

$$\ell(x, u) := \Phi(x) + \tau \log \sum_{v \in U} \exp \left(-\frac{\gamma}{\tau} \cdot \Phi(b(x, v)) \right),$$

which is independent of u . Note that the second term is a simple soft-minimum. That is, we penalize spending time in states x that have high-potential $\Phi(x)$ and that do not have neighboring low-potential states. In this setting

$$\pi^*(u \mid x) \propto \exp \left(-\frac{\gamma(\Phi(b(x, u)) - \Phi(x))}{\tau} \right), \quad V(x) = \Phi(x).$$

The self-balancing sampler fits nicely into this framework. If we choose state-space $S = \mathbb{N}^d$, controls $u \in [d]$, dynamics $b(x, u) = x + \delta_u$, and potential $\Phi(x) = \frac{1}{2}\|x\|^2$ then we have $\pi^*(u|x) \propto e^{-\gamma x_u/\tau}$. Hence we recover the optimization problem in Theorem 2.4.1, the uniform $\beta_i \equiv \beta$ case. Analogously, if we instead choose $\Phi(x) = \frac{1}{2} \sum_i \beta_i (x_i - np_i^*)^2$, we recover the optimization problem in Theorem 2.4.2.

The former gives us another way of interpreting the role of β in the self-balancing sampler: $\beta = \gamma/\tau$ measures the trade-off between minimizing future imbalance through the value term Q_γ and maintaining randomness in the sampling policy via entropy-regularization.

Alternatively, if we choose a linear loss $\ell(x, u) = \beta_u x_u$, and myopic discounting $\gamma = 0$, we see the policy $\pi^*(u \mid x) \propto \exp(-\beta_u x_u)$ is the minimizer of

$$V(x) := \inf_{\pi} \left\{ \sum_{i=1}^d [\pi(i|x) \beta_i x_i] - H(\pi(\cdot \mid x)) \right\},$$

Hence the self-balancing sampler can be viewed as an entropy-regularized greedy mechanism that favors actions having smaller weighted current counts $\beta_i x_i$, analogous to Theorem 2.4.3.

2.C Technical Proofs

In this appendix, we collect the technical proofs deferred from the main text.

2.C.1 Positive Recurrence of the Projection-Centered Chain

Proof of Lemma 2.2.3

Lemma. For $\beta_i = rv_i^{-1}$ for $v \in \mathbb{N}_{>0}^d$ and $r > 0$, the projection-centered chain $\tilde{Y}^v(n) = (v_d N_i(n) - v_i N_d(n))_{i=1}^{d-1}$ associated to the self-balancing sampler (2.2.1) is irreducible and positive recurrent, and thus has a unique stationary distribution η on some subset of $\mathbb{Z}^{d-1}$.

Proof. Without loss of generality, assume $\gcd(v) = 1$, as we can always absorb shared factors into the constant r . Clearly the state-space of $\tilde{Y}^v$ is the subset of $\mathbb{Z}^{d-1}$ generated by nonnegative integer combinations of the vectors $\{v_d e_i\}_{i=1}^{d-1} \cup \{-v\}$. This does not always span $\mathbb{Z}^{d-1}$, even when $\gcd(v_i) = 1$. For example, if $v_1 = 2, v_2 = 3, v_3 = 4$ then the first coordinate of $\tilde{Y}^v$ is always a multiple of 2.

Nonetheless, we show the chain is still irreducible on this state-space. To see why, suppose

$$x = v_d \sum_{i=1}^{d-1} a_i e_i - a_d v, \quad a_i \geq 0.$$

Clearly we can go from 0 to x by sampling a_i times from index i for each $i \in [d-1]$ and sampling a_d times from index d . Conversely, we can go from x to 0 by sampling index $i \in [d-1]$ a total of $kv_i - a_i$ times, for some k large enough so $kv_i \geq a_i$ for all $i \in [d-1]$ and $kv_d \geq a_d$. Then sample coordinate d a total of $kv_d - a_d$ times. Every such finite path has positive probability since $p_i(0) > 0$ for every i .

Lastly, we show the chain is positive recurrent by developing a suitable Lyapunov function and proving a Foster-Lyapunov drift condition. See, for example, [56, Theorem 11.3.4], for more information about this technique. To this end, define

$$Q_i := \beta_i N_i - \log(p_i(0)\beta_i), \quad \bar{Q} = \frac{1}{d} \sum_{i=1}^d Q_i,$$

and let $Z_i := Q_i - \bar{Q}$. These new coordinates Z are merely the projection-centered version of the weighted counts $(\beta_i N_i)$ appearing in equation (2.2.2), and translated so that $Z = 0$ is the zero-drift location. After projection-centering, this translation agrees exactly with the correction c_β appearing in Proposition 2.6.1. Note

$$p_i = \frac{p_i(0)e^{-\beta_i N_i}}{\sum_j p_j(0)e^{-\beta_j N_j}} = \frac{\beta_i^{-1}e^{-Z_i}}{\sum_j \beta_j^{-1}e^{-Z_j}},$$

is a function of Z , which itself is a function of the projection-centered counts $(v_d N_i - v_i N_d)_{i=1}^{d-1}$. In fact, as (2.2.2) highlights, the map $x \mapsto Z(x)$ is a bijection from the reachable state space $\mathcal{X}_v \subseteq \mathbb{Z}^{d-1}$ to a discrete affine lattice contained in $\{z : \sum_i z_i = 0\} \subseteq \mathbb{R}^d$. For $x \in \mathcal{X}_v$, define the Lyapunov function

$$V(x) := \sum_{i=1}^d Z_i(x)^2,$$

which of course can be viewed as a function of Z directly. As

$$(\Delta Z)_i = \begin{cases} \frac{d-1}{d} \beta_i & \text{w.p. } p_i, \\ -\frac{\beta_j}{d} & \text{w.p. } p_j \text{ for } i \neq j, \end{cases}$$

Therefore, using the fact $\sum Z_i = 0$, the expected drift of V is

$$\begin{aligned} \mathbb{E}[\Delta V \mid Z = z] &= 2 \sum_{i=1}^d z_i \mathbb{E}[\Delta Z_i \mid Z = z] + \sum_{i=1}^d \mathbb{E}[(\Delta Z_i)^2 \mid Z = z], \\ &= 2 \sum_{i=1}^d z_i p_i(z) \beta_i + \frac{d-1}{d} \sum_{i=1}^d p_i(z) \beta_i^2. \end{aligned}$$

The latter term is at most $C = \max_i \beta_i^2$. The first term is sufficiently negative on

$$F := \left\{ x \in \mathcal{X}_v : \max_i Z_i(x) - \min_i Z_i(x) > R \right\}.$$

To see why, choose $M \in \arg \max_i Z_i$ and $m \in \arg \min_i Z_i$. The idea is that for the imbalance $z_M - z_m$ to be large, some coordinate must be significantly under-sampled, and hence be exponentially favored, reducing the imbalance. Specifically, for $x \in F$ and $z = Z(x)$, we have

$$\begin{aligned} \sum_{i=1}^d z_i p_i \beta_i &= \frac{1}{\sum_j \beta_j^{-1} e^{-z_j}} \sum_i z_i e^{-z_i} \\ &= \frac{1}{d \sum_j \beta_j^{-1} e^{-z_j}} \sum_{i < j} (z_i - z_j) (e^{-z_i} - e^{-z_j}) \\ &\leq \frac{(z_m - z_M) (e^{-z_m} - e^{-z_M})}{d \sum_j \beta_j^{-1} e^{-z_j}} \\ &\leq \frac{(z_m - z_M) (e^{-z_m} - e^{-z_M})}{d^2 e^{-z_m} \max_i \beta_i^{-1}} \\ &\leq -\frac{R(1 - e^{-R})}{d^2 \max_i \beta_i^{-1}}, \end{aligned}$$

where in the second line we use $\sum z_i = 0$ and in the third line we observed every term in the sum is nonpositive. Choose R so that

$$\frac{2R(1 - e^{-R})}{d^2 \max_i \beta_i^{-1}} > C + \epsilon$$

for some $\epsilon > 0$. Then the drift is at most $-\epsilon$ on F . On F^c , the zero-sum condition gives $|z_i| \leq R$ for every i , so discreteness of the reachable Z -states makes F^c finite. This gives the desired drift condition. $\square$

2.C.2 Proof of the Quantitative Drift Lemma

Proof of Lemma 2.3.1. The idea is to compare (a shifted version of) U_n to the “worst case” birth-death process with properties (i) and (ii), which will satisfy the desired bound. To this end, define the shifted version $X_n := (U_n - b)_+$ and the following birth-death process which makes jumps of size c

$$Q_0 := \max\{c, X_0\}, \quad Q_{n+1} := \max\{c, Q_n + Z_{n+1}\},$$

where $Z_1, Z_2, \dots$ are IID with

$$\mathbb{P}(Z_n = c) = \theta, \quad \mathbb{P}(Z_n = -c) = 1 - \theta, \quad \theta := \frac{c - a}{2c} \in [0, 1].$$

Note Z_i is chosen so $\mathbb{E}Z_i = -a$ and $|Z_i| \leq c$, so (i) and (ii) are satisfied by Q_n . Moreover, Z_i essentially maximizes the variance among all such random variables. Since the downside of Q_n is limited to c , this choice of increment in a sense maximizes the expectation of Q_n .

To compare X_n and Q_n , we claim for any increasing, convex F

$$\mathbb{E}[F(X_{n+1}) \mid \mathcal{F}_n] \leq \theta F(X_n + c) + (1 - \theta)F(\max\{c, X_n - c\}) = \mathbb{E}[F(Q_{n+1}) \mid \mathcal{F}_n]. \quad (2.C.1)$$

In words, this says $(Q_{n+1} \mid \mathcal{F}_n)$ stochastically dominates $(X_{n+1} \mid \mathcal{F}_n)$ in the order $\leq_{icx}$ induced by increasing, convex functions. Define an operator T by

$$(TF)(x) := \theta F(x + c) + (1 - \theta)F(\max\{c, x - c\}),$$

so that $TF(Q_n) = \mathbb{E}[F(Q_{n+1}) \mid \mathcal{F}_n]$.

To prove the claim, first note that if $X_n = 0$, then the inequality follows from (ii) as

$$F(X_{n+1}) = F((U_{n+1} - b)_+) \leq F(c).$$

When $X_n > 0$, let $G(z) := F(\max\{c, X_n + z\})$. Then G is increasing and convex, which gives

$$G(z) \leq \frac{c+z}{2c}G(c) + \frac{c-z}{2c}G(-c), \quad \forall |z| \leq c.$$

In particular, we know that $|U_{n+1} - U_n| \leq c$ almost surely, so applying this inequality and taking conditional expectations gives

$$\mathbb{E}[G(U_{n+1} - U_n) \mid \mathcal{F}_n] \leq \frac{c + \mathbb{E}[U_{n+1} - U_n \mid \mathcal{F}_n]}{2c}(G(c) - G(-c)) + G(-c).$$

Since G is increasing, $G(c) - G(-c) \geq 0$. We also know that $\mathbb{E}[U_{n+1} - U_n \mid \mathcal{F}_n] \leq -a$, so

$$\begin{aligned} &\leq \theta(G(c) - G(-c)) + G(-c) \\ &= \theta G(c) + (1 - \theta)G(-c) \\ &= \theta F(X_n + c) + (1 - \theta)F(\max\{c, X_n - c\}). \end{aligned}$$

Now, since $X_n > 0$, we have $X_n = U_n - b$, so that

$$G(U_{n+1} - U_n) = F(\max\{c, X_n + U_{n+1} - U_n\}) = F(\max\{c, U_{n+1} - b\}) \geq F(X_{n+1})$$

by the monotonicity of F ; this proves claim (2.C.1).

Taking expectations of both sides in the claim inequality gives $\mathbb{E}[F(X_{n+1})] \leq \mathbb{E}[TF(X_n)]$. Since T maps the set of increasing convex functions onto itself, we can apply claim (2.C.1) iteratively starting with $F(x) = x$ to get

$$\mathbb{E}[X_n] = \mathbb{E}[F(X_n)] \leq \mathbb{E}[TF(X_{n-1})] \leq \dots \leq \mathbb{E}[T^n F(X_0)].$$

Since $T^n F$ is increasing and $X_0 \leq Q_0$ we see

$$\leq \mathbb{E}T^n F(Q_0) = \mathbb{E}Q_n.$$

It now only remains to bound $\sup_n \mathbb{E}[Q_n]$. If we couple Q_n with Q'_n , a version started at $Q'_0 = c$ but with the same increments, then we get

$$\mathbb{E}[Q_n] \leq \mathbb{E}[Q'_n] + \mathbb{E}[Q_0 - c].$$

The stationary distribution of the Q'_n chain is

$$\pi(kc) := (1 - r)r^{k-1}, \quad r := \frac{\theta}{1 - \theta} = \frac{c - a}{c + a},$$

and the mean of the chain at stationarity is therefore

$$\sum_{k=1}^{\infty} ck(1 - r)r^{k-1} = \frac{c}{1 - r} = \frac{c}{2} + \frac{c^2}{2a}.$$

Since Q'_0 is less than a version of the chain started at stationarity and coupled to have the same increments, we get

$$\mathbb{E}[Q'_n] \leq \frac{c}{2} + \frac{c^2}{2a}.$$

So, in total, we get

$$\mathbb{E}[Q_n] \leq \mathbb{E}[Q'_n] + \mathbb{E}[Q_0 - c] \leq \mathbb{E}[Q_0] - \frac{c}{2} + \frac{c^2}{2a} \leq \mathbb{E}[Q_0] + \frac{c^2}{2a}.$$

Unwrapping the definitions gives the final bound

$$\mathbb{E}[U_n] \leq b + \mathbb{E}[(U_n - b)_+] \leq b + \mathbb{E}[\max\{c, U_0 - b\}] + \frac{c^2}{2a} \leq \mathbb{E}[U_0] + b + c + \frac{c^2}{2a},$$

uniformly in n . □

2.C.3 Proofs for Weak-Bias Diffusive Limits

Proof of Proposition 2.6.1 (Projection-Centered Diffusion Limit)

Proposition. Fix $i \in [d]$. If $Z_0^h \rightarrow z_0$, then $Z^h \Rightarrow Z$ in $D([0, T], \mathbb{R})$ for each $T < \infty$, where Z solves

$$dZ_t = -\frac{\beta_i \beta_d}{\beta_i + \beta_d} Z_t dt + \sqrt{\beta_i \beta_d} dW_t, \quad Z_0 = z_0.$$

Proof. For convenience, drop the subscript on c_β and let $q_i := p_i(0)$. Note

$$\frac{p_i^{\beta h}}{p_i^{\beta h} + p_d^{\beta h}} = \frac{q_i e^{-\beta_i h N_i}}{q_i e^{-\beta_i h N_i} + q_d e^{-\beta_d h N_d}} = \frac{e^{-Y_i^{\beta h}}}{q_d/q_i + e^{-Y_i^{\beta h}}}$$

Since $Z_{nh}^h = z$ implies $Y_i^{h\beta}(n) = z\sqrt{h} + c_\beta$, we have increments

$$\Delta Z_{nh}^h := Z_{(n+1)h}^h - Z_{nh}^h \mid \{Z_{nh}^h = z\} = \begin{cases} \beta_i \sqrt{h} & \text{w.p. } \frac{e^{-(z\sqrt{h}+c)}}{q_d/q_i + e^{-(z\sqrt{h}+c)}}, \\ -\beta_d \sqrt{h} & \text{w.p. } \frac{q_d/q_i}{q_d/q_i + e^{-(z\sqrt{h}+c)}}, \end{cases}$$

so since $e^{-c} = \frac{\beta_d q_d}{\beta_i q_i}$, our process has drift

$$\begin{aligned} \mu_h(z) &:= \mathbb{E}[\Delta Z_{nh}^h \mid Z_{nh}^h = z] \\ &= \frac{\sqrt{h}}{q_d/q_i + e^{-(z\sqrt{h}+c)}} \left[\beta_i e^{-(z\sqrt{h}+c)} - \beta_d q_d/q_i \right] \\ &= \frac{\beta_d \sqrt{h}}{1 + \frac{\beta_d}{\beta_i} e^{-z\sqrt{h}}} \left[e^{-z\sqrt{h}} - 1 \right]. \end{aligned}$$

Hence

$$\mu_h(z) = -\frac{\beta_d}{1 + \frac{\beta_d}{\beta_i}} zh + O(z^2 h^{3/2}) = -\frac{z}{\beta_d^{-1} + \beta_i^{-1}} h + O(z^2 h^{3/2}).$$

Hence $h^{-1}\mu_h(z)$ converges uniformly for z in a compact set. Similarly, for the variance

$$\begin{aligned} \sigma_h^2(z) &:= \mathbb{E}[(\Delta Z_{nh}^h)^2 \mid Z_{nh}^h = z] - \mu_h(z)^2 \\ &= \frac{h}{q_d/q_i + \frac{\beta_d q_d}{\beta_i q_i} e^{-z\sqrt{h}}} \left[\beta_i^2 \cdot \frac{\beta_d q_d}{\beta_i q_i} e^{-z\sqrt{h}} + \beta_d^2 \cdot \frac{q_d}{q_i} \right] + O(z^2 h^2) \\ &= \frac{\beta_i \beta_d + \beta_d^2}{1 + \frac{\beta_d}{\beta_i}} h + O(z^2 h^{3/2}) \\ &= \beta_i \beta_d h + O(z^2 h^{3/2}). \end{aligned}$$

Again $h^{-1}\sigma_h^2(z)$ converges uniformly for z in a compact set. Lastly, since

$$|\Delta Z_{nh}^h| \leq \max(\beta_i, \beta_d)\sqrt{h} \rightarrow 0,$$

we have no large jumps. Namely, for every $\epsilon > 0$

$$\mathbb{P}(|\Delta Z_{nh}^h| > \epsilon \mid Z_{nh}^h = z) = o(h),$$

independently of z , as this probability is zero for sufficiently small h . As the diffusion

$$dZ_t = -\frac{\beta_i\beta_d}{\beta_i + \beta_d}Z_t dt + \sqrt{\beta_i\beta_d} \cdot dW_t, \quad Z_0 = z_0,$$

has globally Lipschitz coefficients, there exists a unique strong solution, namely just the OU process. Standard theory shows the desired convergence. See for example Section 8.7 of [22]. $\square$

Proof of Proposition 2.6.2 (Mean-Centered Diffusion Limit)

Proposition. If $Z_0^h \rightarrow z_0$, then $Z^h \Rightarrow Z$ as $h \downarrow 0$, where

$$dZ_t = -\frac{1}{S_\beta}Z_t dt + \Sigma^{1/2}dW_t, \quad S_\beta = \sum_j \beta_j^{-1}, \quad D_\beta = \text{diag}(\beta), \quad \Sigma = S_\beta^{-1}D_\beta - S_\beta^{-2}\mathbf{1}\mathbf{1}^T,$$

and the limiting diffusion is supported on $\{z : \langle z, p^* \rangle = 0\}$.

Proof. For ease, drop the subscript of c_β and let $q_i := p_i(0)$. Since $\beta_i p_i^*$ is independent of i

$$p_i^{h\beta} = \frac{q_i e^{-h\beta_i N_i}}{\sum_j q_j e^{-h\beta_j N_j}} = \frac{q_i e^{-M_i^{h\beta}}}{\sum_j q_j e^{-M_j^{h\beta}}},$$

so the transitions are a function of our mean-centered counts alone. The value c_β is chosen so that $e^{-c_i} = \kappa(q_i\beta_i)^{-1}$ for $\kappa > 0$ independent of i , and so $\langle c_\beta, p^* \rangle = 0$ satisfies the same constraint as M^β . Since $Z_{nh}^h = z$ implies $M_n^{\beta h} = z\sqrt{h} + c$, note for standard basis vector e_i

$$\mathbb{P}\left(\Delta Z_{nh}^h = \sqrt{h}D_\beta(e_i - p^*)\right) = \frac{q_i e^{-(z_i\sqrt{h}+c_i)}}{\sum_{j=1}^d q_j e^{-(z_j\sqrt{h}+c_j)}} = \frac{\beta_i^{-1} e^{-z_i\sqrt{h}}}{\sum_{j=1}^d \beta_j^{-1} e^{-z_j\sqrt{h}}}.$$

Hence this process has drift

$$\begin{aligned} (\mu_h(z))_i &:= \mathbb{E}[(\Delta Z_{nh}^h)_i \mid Z_{nh}^h = z] \\ &= \sqrt{h} \left[\frac{e^{-z_i\sqrt{h}}}{\sum_{j=1}^d \frac{1}{\beta_j} e^{-z_j\sqrt{h}}} - \beta_i p_i^* \right] \end{aligned}$$

$$\begin{aligned}
 &= \sqrt{h} \left[\frac{1 - z_i \sqrt{h}}{\sum_{j=1}^d \frac{1}{\beta_j} (1 - z_j \sqrt{h})} - \beta_i p_i^* + O(h) \right] \\
 &= \sqrt{h} \left[\frac{\left(\sum_{j=1}^d p_j^* z_j - z_i \right) \sqrt{h}}{\sum_{j=1}^d \frac{1}{\beta_j} (1 - z_j \sqrt{h})} + O(h) \right] \\
 &= \frac{1}{\sum_{j=1}^d \frac{1}{\beta_j}} \left(\sum_{j=1}^d p_j^* z_j - z_i \right) h + o(h)
 \end{aligned}$$

As above, it is easy to see the convergence is uniform for z in compact sets. Similarly

$$\begin{aligned}
 \mathbb{E}[(\Delta Z_{nh}^h)_i^2 \mid Z_{nh}^h = z] &= h\beta_i^2 \left[(p_i^*)^2 + (1 - 2p_i^*) \cdot \frac{\frac{1}{\beta_i} e^{-z_i \sqrt{h}}}{\sum_{j=1}^d \frac{1}{\beta_j} e^{-z_j \sqrt{h}}} \right] \\
 &= h\beta_i^2 \left[(p_i^*)^2 + (1 - 2p_i^*) \cdot (p_i^* + O(\sqrt{h})) \right] \\
 &= h\beta_i^2 p_i^* (1 - p_i^*) + o(h)
 \end{aligned}$$

Moreover, for $i \neq j$

$$\begin{aligned}
 \mathbb{E}[(\Delta Z_{nh}^h)_i (\Delta Z_{nh}^h)_j \mid Z_{nh}^h = z] &= h\beta_i \beta_j \left[p_i^* p_j^* - \frac{\frac{p_i^*}{\beta_j} e^{-z_j \sqrt{h}} + \frac{p_j^*}{\beta_i} e^{-z_i \sqrt{h}}}{\sum_{\ell=1}^d \frac{1}{\beta_\ell} e^{-z_\ell \sqrt{h}}} \right] \\
 &= h\beta_i \beta_j \left[p_i^* p_j^* - \frac{\left(\frac{p_i^*}{\beta_j} + \frac{p_j^*}{\beta_i} - O(\sqrt{h}) \right)}{\left(\sum_{\ell=1}^d \frac{1}{\beta_\ell} - O(\sqrt{h}) \right)} \right] \\
 &= -\beta_i \beta_j p_i^* p_j^* h + o(h).
 \end{aligned}$$

Thus we have the desired covariance. As the coefficients of the limiting diffusion are globally Lipschitz, the associated SDE has a unique strong solution. Again standard theory, such as Section 8.7 of [22], gives the desired convergence result. This gives a limiting diffusion

$$dZ_t = -\frac{1}{S_\beta} (Z_t - \bar{Z}_t) dt + \Sigma^{1/2} dW_t, \quad \bar{Z}_t := \langle p^*, Z_t \rangle.$$

Lastly, we show that the limiting diffusion lives in the hyperplane

$$Z_t \in \{z : \langle z, p^* \rangle = 0\}.$$

Note that by construction

$$\langle M^\beta(n), p^* \rangle = \langle c_\beta, p^* \rangle = 0.$$

and hence $\langle Z_t^h, p^* \rangle = 0$ for all h and t , from which the result follows. In particular, $\bar{Z}_t = 0$, so our above diffusion simplifies to the desired form. $\square$

Chapter 3

Inference under the PCH Model

3.1 Introduction

In phylogenetics, we have a collection of objects whose ancestral history we hope to infer using only modern-day information. This shared history is summarized by a *phylogenetic tree*, a graphical tree whose leaves are labeled by these objects. Classically, these objects represented different biological species, the relevant data was their DNA, and the phylogenetic tree encoded the sequence of speciation events that occurred over time to produce the species we see today. Nonetheless, the tools of phylogenetics are far more broadly applicable: they have been used in medicine to better understand the progression of cancer [77], in astrophysics as a hierarchical clustering procedure for astral bodies [31], and in terrane analysis to study the geological history of specific regions [57].

In this chapter, we study a specific application to linguistics, where the objects of interest are different languages and the underlying data are *linguistic characters*. These characters encode features that vary across languages and are commonly divided into three broad categories: *phonological*, *morphological*, and *lexical*.

Phonological characters describe sound patterns, such as whether a language distinguishes tones or permits consonant clusters. Morphological characters describe grammatical structure, such as how past tense or plurality is expressed. Lexical characters describe the word forms used to express specific ideas, often by grouping languages according to whether their words for common concepts such as “fire,” or “mother” descend from a common historical source.

Lexical characters are particularly interesting as they can be multi-state—several different words can simultaneously share the same meaning—a property known as *polymorphism*. For example, in English the words “stone” and “rock” fill the same semantic slot. We are specifically interested in the model introduced in [14], which explicitly models this polymorphism. We refer to this model as the *Polymorphic CHaracter (PCH) model*, in a similar vein to the follow-up paper [28].

3.1.1 The PCH Model

The key feature of the PCH model is that it allows for polymorphism: a given character c may take on multiple different states simultaneously. To differentiate between the set-valued collection of states a character currently takes on and the individual states themselves, we call the former the *macro-state* of the character and the latter its individual *micro-states*. For example, if the macro-state of a character c at a given point of time is $\{\alpha, \beta\}$ then it currently exhibits two micro-states: α and β . Under the PCH model, it is assumed that micro-states may be born or die as the character evolves along the tree, so that the current *number* of micro-states present in the current macro-state evolves according to a standard birth-death process, which we refer to as the *cardinality process*.

More formally, suppose that $\mathcal{T}^*$ is the underlying rooted phylogenetic tree of languages. Given some collection of characters $\mathcal{C}$, associated to each character c is a set $\Sigma^{(c)}$ of possible states, birth/death rates $\{\lambda_i^{(c)}\}, \{\mu_i^{(c)}\}$ controlling the cardinality process, a probability measure $\nu^{(c)}$ on $\Sigma^{(c)}$ for selecting new states, and a probability measure $\rho^{(c)}$ on the set $\binom{\Sigma^{(c)}}{*}$ of nonempty finite subsets of $\Sigma^{(c)}$ for determining the macro-state of the root.

More specifically, let $[r, \ell]$ denote the unique path from the root r of $\mathcal{T}^*$ to some leaf $\ell \in \mathcal{T}^*$. For $x \in [r, \ell]$ we assume the macro-state $c(x)$ for the given character evolves down this path as follows

- (i) The macro-state $c(r)$ of the root is sampled from $\rho^{(c)}$.
- (ii) If $|c(x)| = i$, a new micro-state is born at rate $\lambda_i^{(c)}$ and an existing micro-state dies at rate $\mu_i^{(c)}$ respectively.
- (iii) If a death occurs, a micro-state $\alpha \in c(x)$ is chosen uniformly at random to delete. If a birth occurs and $\nu^{(c)}(\Sigma^{(c)} \setminus c(x)) > 0$, a new micro-state $\alpha \notin c(x)$ is chosen proportional to $\nu^{(c)}$, namely via

$$\nu_x^{(c)}(A) := \frac{\nu^{(c)}(A \setminus c(x))}{\nu^{(c)}(\Sigma^{(c)} \setminus c(x))}.$$

When a branch-point v of $\mathcal{T}^*$ is reached, conditional on $c(v)$ the macro-state evolves independently down each child branch starting from state $c(v)$.

A micro-state $\alpha \in \Sigma^{(c)}$ with $\nu^{(c)}(\{\alpha\}) > 0$ is called *homoplastic*. These states can appear independently in different lineages, rather than from shared ancestry. We denote the set of homoplastic states and non-homoplastic states by $\Sigma_H^{(c)}, \Sigma_N^{(c)}$ respectively. Homoplastic states in a sense complicate the inference process, as two languages can share the homoplastic trait without necessarily having any recent shared ancestry.

Throughout this chapter, we will assume characters follow the *immigration-death rates*

$$\lambda_i^{(c)} = \lambda^{(c)}, \quad \mu_i^{(c)} = i\mu^{(c)} \cdot 1\{i > 1\}, \quad (3.1.1)$$

for $i \geq 1$. Namely, the birth rate is constant, and the death rate is proportional to the number of states, except when only one state is present. These assumptions ensure that

each character has at least one micro-state associated to it at any point in $\mathcal{T}^*$ and that the number of states cannot blow-up in finite time.

Since the upcoming inference procedure is based solely on non-homoplastic states, we also make the following assumptions about $\nu^{(c)}$ and $\rho^{(c)}$.

Assumption (1). Non-homoplastic states occur sufficiently often, specifically

- (i) The root macro-state contains a non-homoplastic micro-state with positive probability

$$\rho^{(c)} \left(\left\{ A \in \binom{\Sigma^{(c)}}{*} : A \cap \Sigma_N^{(c)} \neq \emptyset \right\} \right) > 0.$$

- (ii) There is positive probability of adding a non-homoplastic micro-state to the current macro-state when a birth occurs

$$\nu^{(c)}(\Sigma_N^{(c)}) > 0.$$

The full PCH model additionally incorporates another mechanism of character transmission known as *borrowing* which allows characters to transfer laterally throughout the tree, analogous to horizontal gene transfer in biological contexts. In this chapter, we only consider the case where no borrowing occurs, but interested readers can find further details in [14].

3.1.2 ASTRAL

To infer the underlying tree from many independently evolving characters, we adapt ideas from quartet-based species-tree estimation. In phylogenetic inference, one often wants to combine information from many sources, such as genetic information across a collection of genes or, in our case, linguistic characters. In *consensus methods*, each source is used to generate an estimate g_i of the true phylogenetic tree $\mathcal{T}^*$, and then these estimates $\{g_i\}_{i=1}^n$ are aggregated. In the phylogenetics literature, the trees g_i are called the *gene trees*, while $\mathcal{T}^*$ is referred to as the *species tree*. We will occasionally utilize this vocabulary.

ASTRAL [59] is one commonly used aggregation procedure. It scores candidate phylogenetic trees $\mathcal{T}$ based on the number of *quartets*, four-leaf induced subtrees, they share with the estimated trees $\{g_i\}$. Specifically, it solves the optimization problem

$$\hat{\mathcal{T}} = \arg \max_{\mathcal{T}} \sum_{q \in \mathcal{Q}(\mathcal{T})} W(q), \quad (3.1.2)$$

where $W(q)$ is some measure of the support for the quartet q and $\mathcal{Q}(\mathcal{T})$ is the set of quartets present in the species-tree $\mathcal{T}$. Commonly,

$$W(q) = \sum_{i=1}^n 1\{q \in g_i\},$$

the number of gene trees in which the quartet appears. Later on, we will introduce other measures of the quartet support specific to the PCH model.

Since the set of phylogenetic trees is super-exponentially large, it is intractable to solve (3.1.2) exactly for trees with more than 20 or so leaves. For larger trees, a restricted search space is required. One possible strategy is to limit the bipartitions of the leaves the candidate trees $\mathcal{T}$ can induce.

A *bipartition* of a set S is a partition of $S = A \cup B$ into two nonempty, disjoint subsets. Every edge $e \in E(\mathcal{T})$ of our tree induces a bipartition of the leaves by considering the two connected components of the graph $(V(\mathcal{T}), E(\mathcal{T}) \setminus \{e\})$. Moreover, Buneman’s splits-equivalence theorem (e.g. [72, Theorem 3.1.4]) shows $\mathcal{T}$ is completely characterized by its set $\mathcal{X}(\mathcal{T})$ of bipartitions. Since each g_i is a noisy estimate of $\mathcal{T}^*$, we might expect that $\mathcal{X}(\mathcal{T}^*) \subseteq \mathcal{X}_g := \bigcup_i \mathcal{X}(g_i)$, suggesting the restricted search space

$$\left\{ \mathcal{T} : \mathcal{X}(\mathcal{T}) \subseteq \bigcup_{i=1}^n \mathcal{X}(g_i) \right\}.$$

More generally, given any set $\mathcal{X}$ of bipartitions, we can solve

$$\hat{\mathcal{T}} = \arg \max_{\mathcal{X}(\mathcal{T}) \subseteq \mathcal{X}} \sum_{q \in \mathcal{Q}(\mathcal{T})} W(q). \quad (3.1.3)$$

Using a clever dynamic-programming principle, ASTRAL efficiently solves this bipartition-constrained optimization problem. For polynomially-sized $\mathcal{X}$, the optimization problem (3.1.3) can be feasibly solved for trees with thousands of leaves. Moreover, when the trees g_i are generated under the multispecies coalescent (MSC) model [67], [36], its statistical properties are well-understood. In particular, $\mathcal{X}_g \subseteq \mathcal{X}$ is sufficient to guarantee asymptotic consistency.

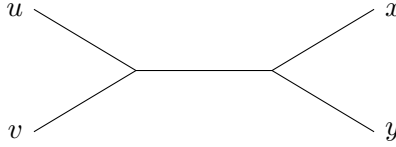
The paper [74] showed for the exact version (3.1.2) that $n = \Theta(f^{-2} \log(k))$ independent gene trees g_i are required to recover $\mathcal{T}^*$ with high probability under the MSC model, where f is the minimum branch length in the tree. It has been empirically observed that the relaxed version (3.1.3) requires significantly more data to obtain similar recovery rates [74]. Thus there is a trade-off between computational and sample efficiency.

As discussed in [80], [53], the main shortcoming of the relaxed problem is that it is quite unlikely that $\mathcal{X}(\mathcal{T}^*) \subseteq \mathcal{X}_g$, since the bipartitions induced by edges near the root are very difficult to recover. Several heuristics have been proposed to extend the set $\mathcal{X}_g$ in an effort to improve its coverage properties and the empirical performance of the relaxed problem [58], [87], [18]. Alternatively [86] proposes a weighted version of (3.1.2) which incorporates quartet-support information from the gene-tree estimation stage, which down-weights possibly misleading quartets.

ASTRAL recovers only the underlying unrooted tree. When an appropriate outgroup is available, it can be used to root the inferred tree.

3.1.3 PCH-ASTRAL: Phylogenetic Inference in the PCH Model

For data generated under the PCH model, quartet support can be defined directly from the shared non-homoplastic states of the observed languages. Let $q = uv||xy$ denote the unrooted quartet topology that groups u with v and x with y .



Given a character c , define the support $M_q^{(c)}$ of the quartet $q = uv||xy$ via

$$M_q^{(c)} := \left\{ (\alpha, \beta) \in \Sigma_N \times \Sigma_N : \alpha \neq \beta, \quad \alpha \in U \cap V \cap \bar{X} \cap \bar{Y}, \quad \beta \in \bar{U} \cap \bar{V} \cap X \cap Y \right\},$$

where $U := c(u)$, and so on. Namely, M_q consists of the pairs of non-homoplastic states that are seemingly “split” across the quartet q . For the true quartet topology, [28, Theorem 1] shows this support is larger in expectation than the support assigned to the two alternatives:

Theorem 3.1.1. *If $\mathcal{T}_q^* = uv||xy$ is the quartet topology of the leaves (u, v, x, y) in $\mathcal{T}^*$ then*

$$\mathbb{E}|M_{uv||xy}| > \mathbb{E}|M_{ux||vy}| = \mathbb{E}|M_{uy||vx}|,$$

so the correct topology has the highest expected support.

For each quartet q , they define its (empirical) support via

$$S_q := \frac{1}{n} \sum_{i=1}^n |M_q^{(c_i)}|. \quad (3.1.4)$$

Moreover, for each $s = \{u, v, x, y\} \in \binom{[k]}{4}$ they then let

$$q_s := \arg \max_{q \in A_s} S_q, \quad (3.1.5)$$

with ties broken uniformly, where A_s is the set of three quartets with leaf-set s . Namely, q_s is the quartet topology with maximal empirical support. By the law of large numbers and Theorem 3.1.1, the correct topology becomes overwhelmingly likely for large n . They then run ASTRAL on the quartet set $\{q_s\}$ to generate an estimate $\hat{\mathcal{T}}$ of $\mathcal{T}^*$. The authors of [28] call this algorithm *PCH-ASTRAL*, and additionally propose a weighted version, although we will only focus on the unweighted version described above.

Although not directly mentioned in [28], the proof of Theorem 3.1.1 directly implies the following. Recall, a phylogenetic tree (with branch lengths) is said to be *ultrametric* if every leaf is the same distance from the root. When this holds, the supports for the two incorrect topologies are identically distributed.

Corollary 3.1.2. *If $\mathcal{T}^*$ is ultrametric, then $M_{ux||vy} \stackrel{d}{=} M_{uy||vx}$.*

3.1.4 Summary of Contributions

In Section 3.2, we construct a coupling of the PCH model to a simple immigration-death process which allows for many direct computations and will be the basis of most of our rigorous analysis. Using this, we develop some basic properties of both the original PCH model and of this coupled process.

Afterward, in Section 3.3, we show how the search space for PCH-ASTRAL can be significantly restricted while maintaining statistical consistency. This vastly improves the run-time of the underlying ASTRAL procedure, enabling phylogenetic analysis of hundreds of different languages simultaneously. While the restriction increases the algorithm's *sample complexity*, the number of data samples required to recover $\mathcal{T}^*$ with a prescribed probability of success, we further introduce two heuristics intended to mitigate this loss in statistical efficiency.

Following this, in Section 3.4, we explore an alternative ASTRAL-based inference procedure, which we call *PCH-ASTRAL(sc)*. This procedure keeps character-level quartet information separate by converting each informative character-quartet pair into a single topology vote, rather than averaging support magnitudes across characters. We argue that PCH-ASTRAL(sc) may be more robust in heterogeneous or misspecified settings due to the heavy-tailed nature of the quartet-support distributions.

Lastly, in Section 3.5, we develop sample complexity bounds for the two ASTRAL-based procedures when characters are sampled under the simplified immigration-death process $ID(\theta, \lambda, \mu; \mathcal{T}^*)$. We show PCH-ASTRAL requires $O(L^{-2} \log(k) + L^{-1} \log^2(k))$ characters to recover $\mathcal{T}^*$ with high probability, where

$$L := \min_{s \in \binom{[k]}{4}} \min_{t \in A_s \setminus \{\mathcal{T}_s^*\}} \mathbb{E} [|M_{\mathcal{T}_s^*}| - |M_t|]$$

is the minimal margin the correct quartet topology has over incorrect ones. Similarly, PCH-ASTRAL(sc) requires $O(p_{\min}^{-1} \Delta^{-2} \log(k))$ total characters, equivalently $O(\Delta^{-2} \log(k))$ informative character votes per quartet, where

$$\Delta := \min_{s \in \binom{[k]}{4}} \min_{q \in A_s \setminus \{\mathcal{T}_s^*\}} [\mathbb{P}_s(\mathcal{T}_s^*) - \mathbb{P}_s(q)], \quad \mathbb{P}_s(q) := \mathbb{P} \left(q = \arg \max_{q' \in A_s} |M_{q'}| \mid \max_{q' \in A_s} |M_{q'}| > 0 \right).$$

again gives some notion of the margin.

3.1.5 Notation

Let $\mathcal{T}^*$ denote the true species tree underlying our PCH model, which has k leaves (species/languages), and let $\mathcal{L}(\mathcal{T}^*)$ denote its leaf set. We will often conflate $\mathcal{L}(\mathcal{T}^*)$ with $[k]$. Given any phylogenetic tree $\mathcal{T}$, let $\mathcal{Q}(\mathcal{T})$ denote its set of *quartets*, which are the subtrees induced by a collection of four leaves. Hence $|\mathcal{Q}(\mathcal{T})| = \binom{k}{4}$.

For random variables U and V , we write $U \succeq_{st} V$ if U first-order stochastically dominates V , meaning

$$\mathbb{E}f(U) \geq \mathbb{E}f(V)$$

for every bounded increasing function f .

We will always use $q = uv||xy$ to denote a quartet, which inherently comes with some (unrooted) topology, and $s = \{u, v, x, y\}$ to denote its underlying set of four leaves, which may not have a topology. Given a leaf set $s = \{u, v, x, y\}$, we will use

$$A_s := \{uv||xy, ux||vy, uy||vx\}$$

to denote the set of three different (unrooted) quartets on the same leaf set. We will use $\mathcal{T}_s^*$ to denote the quartet $q \in A_s$ which appears in the true phylogenetic tree $\mathcal{T}^*$.

Unless stated otherwise, assume the observed characters are independent and identically distributed, and assume any micro-states α, β referenced are non-homoplastic.

For ease of notation, when it is clear from context we will denote the macro-state $c(v)$ at a given point $v \in \mathcal{T}^*$ using upper-case letters. In particular, we will use $I := c(i)$, $J := c(j)$, and $K := c(k)$ down below.

Given two points $i, j \in \mathcal{T}^*$, let $[i, j]$ denote the unique path from i to j , and let ℓ_{ij} denote the length of this path. Unless otherwise specified, let r denote the root of $\mathcal{T}^*$.

Let $m_\theta(\ell) := \theta \exp(-\mu\ell) + \frac{\lambda}{\mu}(1 - \exp(-\mu\ell))$, for parameters μ, λ .

3.2 Basic Results

3.2.1 Coupling to Immigration-Death Processes

The constraint $\mu_1 = 0$ and the renormalization by $\nu^{(c)}(\Sigma^{(c)} \setminus c(x))$ make the PCH model hard to analyze as they add a certain dependence structure between the different micro-states of a specific character. If we drop these conditions, we get a classical immigration-death (ID) process, which is much more tractable. This approximation is reasonable in the parameter regime where the number of micro-states in a given branch is rarely one, which occurs when λ is sufficiently large relative to μ , and when homoplasy is fairly rare, so renormalization has little influence.

Our key tool will be a coupling of the PCH model to a standard ID process run along the branches of our species tree $\mathcal{T}^*$, and whose initial population size is Poissonian. It relies on two simple observations about how the set of nonhomoplastic states evolves over time:

- Births always occur at rate at least $\lambda_N := \lambda\nu(\Sigma_N)$ and at most λ .
- Deaths always occur at rate at most μ per individual.

In particular, we can couple an ID process $(Y_s)_{s \in \mathcal{T}^*}$ to our PCH model $(X_s)_{s \in \mathcal{T}^*}$ so

$$Y_s \subseteq X_s \cap \Sigma_N, \quad \forall s \in \mathcal{T}^*.$$

In this way, we only reduce the number of nonhomoplastic states we observe in the leaves, which heuristically makes inference more challenging.

To produce this coupling, we start by coupling the root states. Suppose $|X_0 \cap \Sigma_N| \succeq_{st} Z_\theta$ for some $Z_\theta \sim \text{Poisson}(\theta)$ and $\theta \geq 0$. Since we can always choose $\theta = 0$, such a θ always exists. Then there exists a coupling so $|X_0 \cap \Sigma_N| \geq Z_\theta$ almost surely. Let Y_0 be a uniformly at random chosen subset of size Z_θ from $X_0 \cap \Sigma_N$.

From here, we use rate- λ exponential clocks along the branches of $\mathcal{T}^*$ to determine possible births. If a clock rings at location $s \in \mathcal{T}^*$, sample a new state $\alpha \notin X_{s-}$ for the PCH model as usual, namely according to the conditional distribution

$$\frac{\nu(\cdot \cap (\Sigma \setminus X_{s-}))}{\nu(\Sigma \setminus X_{s-})}.$$

Let $X_s = X_{s-} \cup \{\alpha\}$, and update our ID process Y_s as follows: if $\alpha \notin \Sigma_N$, then let $Y_s = Y_{s-}$. Namely, ignore the homoplastic states. Otherwise, let

$$Y_s = \begin{cases} Y_{s-} \cup \{\alpha\} & \text{w.p. } \nu(\Sigma \setminus X_{s-}), \\ Y_{s-} & \text{w.p. } 1 - \nu(\Sigma \setminus X_{s-}). \end{cases}$$

This ensures that the birth rate in (Y_s) along a given branch is always λ_N . Similarly, assign each state α independent rate- μ exponential clocks along the branches of $\mathcal{T}^*$. If a clock for α rings at location $s \in \mathcal{T}^*$, let

$$X_s = \begin{cases} X_{s-} \setminus \{\alpha\} & X_{s-} \neq \{\alpha\}, \\ \{\alpha\} & \text{otherwise.} \end{cases}$$

Then let $Y_s = Y_{s-} \setminus \{\alpha\}$. Namely, (X_s) simply ignores the deaths causing the state set to become empty, while (Y_s) does not.

Heuristically, this coupling well-approximates the non-homoplastic states of the PCH model in the high-polymorphism regime where it is rare for there to be just a single state, and when there is little homoplasmy so the renormalization of ν does not dramatically affect the birth rate of non-homoplastic states over time.

To this end, let $ID(\theta, \lambda, \mu; \mathcal{T}^*)$ denote an immigration-death process on the species tree $\mathcal{T}^*$ with birth rate λ , death rate μ , and with an initial population of size $\text{Poisson}(\theta)$. The final assumption ensures that the number of states present at any point of $\mathcal{T}^*$ is Poisson distributed, making the process particularly simple to analyze.

In Appendix 3.B, we collect some properties of these immigration-death processes which will be useful for analyzing the PCH model. In particular, the following corollary of Lemma 3.B.2 shows bounding the quartet-supports reduces to bounding the means of some Poisson random variables.

Corollary 3.2.1. *Under the standalone $ID(\theta, \lambda, \mu; \mathcal{T}^*)$ process, the quartet-supports equal*

$$|M_{uv||xy}| = N_{uv} \cdot N_{xy}, \quad |M_{ux||vy}| = N_{ux} \cdot N_{vy}, \quad |M_{uy||vx}| = N_{uy} \cdot N_{vx},$$

for $\{N_{uv}, \dots, N_{vx}\}$ mutually independent Poisson random variables.

The coupling also gives a simple long-term estimate of its discrepancy from the non-homoplastic states of the PCH process. Along a branch, let

$$D_t := |(X_t \cap \Sigma_N) \setminus Y_t|, \quad \lambda_H := \lambda(1 - \nu(\Sigma_N)).$$

Proposition 3.2.2 (Coupling discrepancy). *If $\mathbb{E}D_0 < \infty$, then*

$$\mathbb{E}D_t \leq e^{-\mu t} \mathbb{E}D_0 + \frac{\lambda_H}{\mu} (1 - e^{-\mu t}) + \mu \int_0^t e^{-\mu(t-s)} \mathbb{P}(|X_s| = 1) ds.$$

Moreover,

$$\limsup_{t \rightarrow \infty} \mathbb{E}D_t \leq \frac{\lambda_H}{\mu} + \frac{\lambda/\mu}{e^{\lambda/\mu} - 1}.$$

Proof. Note $D_t \mapsto D_t + 1$ when either a non-homoplastic birth occurs in X_t and is ignored in Y_t or when $X_t = Y_t = \{\alpha\}$ for $\alpha \in \Sigma_N$, and a death clock rings. This occurs at rate

$$\frac{\lambda_N \nu(X_t)}{1 - \nu(X_t)} + \mu 1\{|X_t| = |Y_t| = 1\}.$$

Similarly, we see $D_t \mapsto D_t - 1$ when one of the states in $(X_t \cap \Sigma_N) \setminus Y_t$ dies, at rate

$$\mu D_t 1\{|X_t| > 1\}.$$

Thus we see

$$\frac{d}{dt} \mathbb{E}D_t = \lambda_N \mathbb{E} \left(\frac{\nu(X_t)}{1 - \nu(X_t)} \right) - \mu \mathbb{E}D_t + \mu \mathbb{P}(|X_t| = 1).$$

Since ν is diffuse on Σ_N , $\nu(X_t) \leq 1 - \nu(\Sigma_N)$. Thus

$$\frac{d}{dt} \mathbb{E}D_t \leq \lambda_H + \mu \mathbb{P}(|X_t| = 1) - \mu \mathbb{E}D_t.$$

Grönwall's inequality gives the finite-time bound. By Lemma 3.2.3,

$$\mathbb{P}(|X_t| = 1) \longrightarrow \frac{\lambda/\mu}{e^{\lambda/\mu} - 1},$$

so taking the limit superior gives the asymptotic bound. $\square$

Thus the coupled process is close in expected population discrepancy when homoplasmy is rare and λ/μ is large. In the low-polymorphism regime, the singleton term is instead close to one, reflecting the fact that the ID process may lose its final state while the PCH process cannot.

3.2.2 Properties of PCH Model

In this section, we derive some basic properties of the PCH model. The derivations often rely on the coupling we introduced in the previous section.

Given a path $[r, \ell]$ from the root r of $\mathcal{T}^*$ to some leaf ℓ , let $S_t = |c(t)|$ be the number of micro-states at location $t \in \mathcal{T}^*$. This is a simple birth-death process with birth rates $\lambda_i \equiv \lambda$ and death rates $\mu_i = i\mu 1\{i > 1\}$. Hence

Lemma 3.2.3. *The stationary distribution of S_t is $\pi = (Z|Z > 0)$ for $Z \sim \text{Poisson}(\lambda/\mu)$.*

$$\pi_i = \frac{(\lambda/\mu)^i}{i!(e^{\lambda/\mu} - 1)}, \quad \forall i \geq 1.$$

The ergodic theorem tells us as $T \rightarrow \infty$ the fraction of the time $\{S_t\}_{t \in [0, T]}$ spends in state 1 converges almost surely to $\pi_1 = \frac{\lambda/\mu}{e^{\lambda/\mu} - 1}$. Heuristically, over long-branches our effective death rate then becomes

$$\mu_{\text{eff}} = \mu(1 - \pi_1) = \mu \cdot \frac{e^{\lambda/\mu} - \lambda/\mu - 1}{e^{\lambda/\mu} - 1}.$$

If $\lambda \ll \mu$, then $\mu_{\text{eff}} \approx \frac{\lambda}{2}$ which is significantly less than μ in this low-polymorphism regime. If $\lambda \gg \mu$, then π_1 is exponentially small in λ/μ , so the process is rarely in state 1. Hence it becomes well-approximated by our ID process coupling.

To bound our quartet supports $|M_q|$, we will need two-sided bounds on the probability a given state α dies along a specific branch, which the next lemma provides.

Lemma 3.2.4 (Survival and Death Bounds). *Let $p_{n,T}$ be the probability that, starting with n micro-states, a given state α dies along a branch of length T in the PCH model. Then*

$$\frac{1}{2}(1 - e^{-\lambda T/2})(1 - e^{-\mu T}) =: \xi_T \leq p_{n,T} \leq 1 - \eta_T := 1 - e^{-\mu T}.$$

Thus ξ_T, η_T give lower bounds on the death and survival rates respectively. We will often be interested in applying these bounds to a specific path in the tree. Given two vertices i, j in the tree denote $\xi_{ij} := \xi_{\ell_{ij}}, \eta_{ij} := \eta_{\ell_{ij}}$ our corresponding lower bounds on death and survival along the path $[i, j]$.

Next, we aim to get some control over the tails of our supports $|M_q|$. For $q = uv||xy$, it is easy to see $M_q \subseteq (U \cap V) \times (X \cap Y)$. By Lemma 3.2.3, we roughly see then that $|M_q|$ is the product of two Poisson random variables, whose correlation depends on how closely related u, v are with x, y . This suggests tails $\mathbb{P}(|M_q| > t) \leq \exp(-\Omega(\sqrt{t} \log t))$, namely slightly lighter tails than Weibull(1/2) essentially. The paper [48] develops many useful concentration inequalities for such random variables, which we rely on in Lemma 3.B.3 to prove an analogous statement for the standalone ID model.

Later on, we will need to develop some control over the expected margin between the species tree topology and the other topologies

$$\Delta_q^{(c)} := \mathbb{E}D_t, \quad D_t := |M_q^{(c)}| - |M_t^{(c)}|, \quad q = \mathcal{T}_s^*, \quad t \in A_s \setminus \{\mathcal{T}_s^*\}.$$

This definition does not depend on the choice of incorrect topology t , by Theorem 3.1.1, and quantifies the expected-support gap. Let $\rho_N = \mathbb{E}|c(r) \cap \Sigma_N|$ be the expected number of non-homoplastic states present at the root. Note for a given (rooted) quartet, there are two possible topologies, as shown in Figure 3.1. Let i, j, k denote the most recent common ancestors of these leaves, as depicted in the figure.



Figure 3.1: The two rooted quartet topologies with split $uv||xy$.

Lemma 3.2.5 (Expected Margin Lower Bound). *If q induces a balanced tree in $\mathcal{T}^*$, then*

$$\Delta_q^{(c)} \geq \left(\frac{\lambda_N}{\mu} \right)^2 (1 - \eta_{ij})(1 - \eta_{ik})\eta_{uv}\eta_{xy}.$$

If q induces a caterpillar tree in $\mathcal{T}^$, then*

$$\Delta_q^{(c)} \geq \left[\rho_N \eta_{ri} + \frac{\lambda_N}{\mu} (1 - \eta_{ri}) \right] \frac{\lambda_N}{2\mu} (1 - \eta_{jk})^2 \eta_{uv}\eta_{xy}.$$

Remark 3.2.6. The same argument gives analogous expected-margin bounds under the standalone $ID(\theta, \lambda, \mu; \mathcal{T}^*)$ process; see Corollary 3.2.7.

More granular bounds are developed in the proof below. The above highlights exactly the quartets that are hard to recover: those where u, v and x, y are far from their respective most recent common ancestors j, k , and where these common ancestors j, k are close to their least common ancestor i .

In both cases, we see that the margin lower bound as a function of the minimal branch length f is of order f^2 , assuming all other tree parameters, such as the tree height, are held fixed. To see why this is natural, note that for a balanced quartet, the main outcome that produces support for the correct topology but not the incorrect topologies is if α is born on $[i, j]$ and β is born on $[i, k]$. When these branches are small, the probability each such birth occurs is $O(f)$.

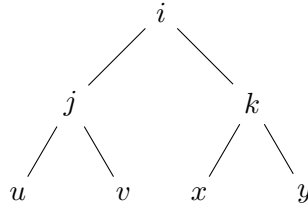
Likewise, for a caterpillar quartet, the main additional support for the correct topology comes from when α dies along $[j, k]$ and β is born along $[j, k]$. Again if branch $[j, k]$ is short, these events both have probability $O(f)$. Hence when $\mathcal{T}^*$ has short branches, the expected margin for the true topology can be quite small.

3.2.3 Proof of the Expected Margin Lower Bound

Proof of Lemma 3.2.5 (Expected support margin). Fix $t \in A_s \setminus \{q\}$ and recall that $D_t = |M_q^{(c)}| - |M_t^{(c)}|$ and $\Delta_q^{(c)} = \mathbb{E}D_t$. For an ordered pair of distinct non-homoplastic states, define

$$I_q(\alpha, \beta) := 1\{(\alpha, \beta) \in M_q^{(c)}\}.$$

Suppose first that the quartet $q = uv||xy$ induces a balanced tree in $\mathcal{T}^*$. Let i, j, k denote the internal vertices of this subtree like so:



Let $I := c(i)$ be the set of micro-states at vertex i , and likewise for J, K . The cancellation argument in the proof of Theorem 1 from [28] gives

$$\mathbb{E}[D_t \mid J, K] = \sum_{(\alpha, \beta) \in \mathcal{A}(J, K)} \mathbb{E}(I_{uv||xy}(\alpha, \beta) \mid J, K),$$

where

$$\mathcal{A}(J, K) = \{(\alpha, \beta) : \alpha \neq \beta \in \Sigma_N, \alpha \in J, \beta \in K, \{\alpha, \beta\} \not\subseteq J \cap K\}.$$

We see that

$$\mathcal{B}(J, K) := \{(\alpha, \beta) : \alpha \in J \setminus K, \beta \in K \setminus J\} \subseteq \mathcal{A}(J, K).$$

For $(\alpha, \beta) \in \mathcal{B}(J, K)$, since we just need α, β to survive until the leaves, by Lemma 3.2.4

$$\mathbb{E}(I_{uv||xy}(\alpha, \beta) \mid J, K) \geq \eta_{ju}\eta_{jv}\eta_{kx}\eta_{ky}.$$

Hence

$$\Delta_q^{(c)} \geq \mathbb{E}|\mathcal{B}(J, K)| \cdot \eta_{ju}\eta_{jv}\eta_{kx}\eta_{ky}.$$

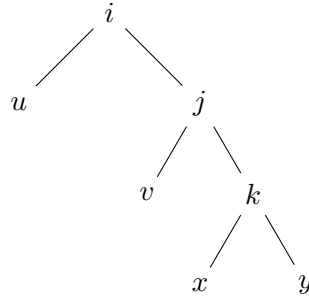
We lower-bound $|\mathcal{B}(J, K)|$ using only pairs for which state α is born along $(i, j]$ and survives until j , while state β is born along $(i, k]$ and survives until k . In this case the roles of J, K decouple. By Lemma 3.B.1, the numbers of such α and β are at least $\text{Poisson}(m_0(\ell_{ij}))$ and

Poisson($m_0(\ell_{ik})$) respectively, where $m_0(t) = \frac{\lambda_N}{\mu}(1 - \eta_t)$, and these are clearly independent. This produces on average $m_0(\ell_{ij}) \cdot m_0(\ell_{ik})$ pairs (α, β) . Thus

$$\mathbb{E}|\mathcal{B}(J, K)| \geq m_0(\ell_{ij})m_0(\ell_{ik}),$$

which gives the desired bound on $\Delta_q^{(c)}$ when the induced subtree is balanced.

If the subtree induced by q is a caterpillar tree like so



then the analogous cancellation argument from the proof of Theorem 1 in [28] gives

$$\mathbb{E}[D_t \mid I, J, K] = \sum_{(\alpha, \beta) \in \tilde{\mathcal{A}}(I, J, K)} \mathbb{E}(I_{uv||xy}(\alpha, \beta) \mid I, J, K),$$

where

$$\tilde{\mathcal{A}}(I, J, K) = \{(\alpha, \beta) : \alpha \neq \beta \in \Sigma_N, \alpha \in I \cap J, \beta \in K, \{\alpha, \beta\} \not\subseteq J \cap K\}.$$

Again, we see this set contains

$$\tilde{\mathcal{B}}(I, J, K) := \{(\alpha, \beta) : \alpha \in (I \cap J) \setminus K, \beta \in K \setminus J\}.$$

For $(\alpha, \beta) \in \tilde{\mathcal{B}}(I, J, K)$ we see $(\alpha, \beta) \in M_{uv||xy}$ if and only if:

- α survives on $(i, u]$ and $(j, v]$.
- β survives on $(k, x]$ and $(k, y]$.

so again by Lemma 3.2.4

$$\mathbb{E}(I_{uv||xy}(\alpha, \beta) \mid I, J, K) \geq \eta_{iu}\eta_{jv}\eta_{kx}\eta_{ky}.$$

In particular, $(\alpha, \beta) \in \tilde{\mathcal{B}}(I, J, K)$ can occur if

- α is born before i , survives along $(i, j]$, and dies on $(j, k]$.
- β is born on $(j, k]$ and survives until k .

Let $\rho_N := \mathbb{E}|c(r) \cap \Sigma_N|$ be the expected number of non-homoplastic states present at the root. Clearly we have

$$\mathbb{E}|I \cap \Sigma_N| \geq \rho_N \eta_{ri} + m_0(\ell_{ri})$$

as non-homoplastic states in I can come either from the root or be born along $[r, i]$. Of these, at least

$$\mathbb{E}|I \cap J \cap \Sigma_N| \geq (\rho_N \eta_{ri} + m_0(\ell_{ri})) \eta_{ij}.$$

survive until they reach j , giving our candidates for α . Condition on $I \cap J \cap \Sigma_N$ and fix $\alpha \in I \cap J \cap \Sigma_N$. Let

$$X_\alpha := \#\{\beta : (\alpha, \beta) \in \tilde{\mathcal{B}}(I, J, K)\}$$

be the number of pairs that α contributes to $\tilde{\mathcal{B}}$. If a non-homoplastic state β is born at location $s \in [j, k]$, it appears in X_α with probability at least

$$p(s) := \exp(-\mu \ell_{sk}) \cdot (1 - \exp(-\mu \ell_{sk})),$$

since this is the probability β 's death clock does not ring in $[s, k]$, but that α 's does. If N represents the coupled rate- λ_N Poisson process of non-homoplastic births in $[j, k]$, which is independent of the death clocks, then Campbell's formula gives

$$\mathbb{E}X_\alpha \geq \mathbb{E} \left(\sum_{s \in N} p(s) \right) = \lambda_N \int_0^{\ell_{jk}} \exp(-\mu s) (1 - \exp(-\mu s)) ds = \frac{\lambda_N}{2\mu} (1 - \exp(-\mu \ell_{jk}))^2.$$

Hence we see

$$\mathbb{E}|\tilde{\mathcal{B}}(I, J, K)| \geq \left[\rho_N \eta_{ri} + \frac{\lambda_N}{\mu} (1 - e^{-\mu \ell_{ri}}) \right] \eta_{ij} \cdot \left[\frac{\lambda_N}{2\mu} (1 - \eta_{jk})^2 \right],$$

giving the desired bound. $\square$

Corollary 3.2.7 (Expected Margin under the ID Process). *Suppose characters are generated under $ID(\theta, \lambda, \mu; \mathcal{T}^*)$. If q induces a balanced quartet, then*

$$\Delta_q \geq \left(\frac{\lambda}{\mu} \right)^2 (1 - \eta_{ij})(1 - \eta_{ik}) \eta_{uv} \eta_{xy}.$$

If q induces a caterpillar quartet, then

$$\Delta_q \geq \left[\theta \eta_{ri} + \frac{\lambda}{\mu} (1 - \eta_{ri}) \right] \frac{\lambda}{\mu} (1 - \eta_{jk})^2 \eta_{uv} \eta_{xy}.$$

Proof. The balanced bound follows from the proof of Lemma 3.2.5 with $\lambda_N = \lambda$ and $\rho_N = \theta$. For a caterpillar quartet, a candidate state α dies along $[j, k]$ with probability $1 - \eta_{jk}$, independently of the number of states β born along this branch and surviving to k , whose mean is $m_0(\ell_{jk}) = \lambda \mu^{-1} (1 - \eta_{jk})$. Multiplying these quantities and the relevant survival probabilities gives the result. $\square$

3.3 A Restricted Search-Space

The exact version of ASTRAL (3.1.2) becomes intractable to solve even for trees with a modest number of leaves. Instead, we must construct a set $\mathcal{X}$ of bipartitions which will limit the search space for the true tree $\mathcal{T}$, which is the content of this section.

To this effect, one useful class of sets when it comes to building $\mathcal{X}$ are the leaf sets

$$L_\alpha := \{\text{leaves } \ell : \alpha \in c(\ell)\}.$$

Say a subset $L \subseteq \mathcal{L}(\mathcal{T}^*)$ of leaves is *contained* within a given bipartition $\varphi := (A|B)$ if

$$L \subseteq A \quad \text{or} \quad L \subseteq B.$$

If $\alpha \in \Sigma_N$ is born on an edge e inducing bipartition $\varphi_e = (A_e|B_e)$, then L_α is contained in φ_e . Moreover, if no deaths of α occur after it is born then in fact $L_\alpha \in \{A_e, B_e\}$. Hence one choice for the candidate bipartitions is

$$\mathcal{X}_1 := \{(L_\alpha^{(c)} \mid \mathcal{L}(\mathcal{T}^*) \setminus L_\alpha^{(c)}) : c \in \mathcal{C}, \alpha \in \Sigma^{(c)}, \emptyset \subsetneq L_\alpha^{(c)} \subsetneq \mathcal{L}(\mathcal{T}^*)\}.$$

At stationarity, we typically have around $\lambda\mu^{-1}$ states in each leaf, so roughly we can see $|\mathcal{X}_1| \lesssim k|\mathcal{C}|\lambda\mu^{-1}$ grows only linearly in k . Under mild assumptions this alone is sufficient for ensuring statistical consistency, since with positive probability a given non-homoplastic α is born along an edge e and does not die at all. More precisely,

Theorem 3.3.1 (Bipartition Cover Bounds for PCH With No Borrowing). *Let $\mathcal{B}$ be the set of bipartitions of a phylogenetic tree $\mathcal{T}^*$ with k leaves and minimum/maximum branch lengths $f_{\min}, f_{\max}$ respectively. Suppose further we have independent PCH characters with parameters $\{\Sigma^{(c)}, \lambda^{(c)}, \mu^{(c)}, \nu^{(c)}, \rho^{(c)}\}_{c \in \mathcal{C}}$ on $\mathcal{T}^*$ with no borrowing and whose rates follow an immigration-death process. Let*

$$\mathcal{C}_N := \{c \in \mathcal{C} : \nu^{(c)}(\Sigma_N^{(c)}) > 0\},$$

be the set of characters with positive chance of non-homoplasy. Furthermore, assume we have the following assumptions on the parameters of the PCH model:

- (i) *Nonvanishing Birth Rates:* Suppose that $\lambda^{(c)}\nu^{(c)}(\Sigma_N^{(c)}) \geq b > 0$ for all $c \in \mathcal{C}_N$.
- (ii) *Bounded death rates:* Suppose $\mu^{(c)} \leq D < \infty$ for all $c \in \mathcal{C}_N$.

Then we have exponential convergence of the coverage probability

$$\mathbb{P}(\mathcal{B} \not\subseteq \mathcal{X}_1) \leq \sum_{\ell=2}^{k-2} q_\ell^{|\mathcal{C}_N|} \leq (k-3)q_{k-2}^{|\mathcal{C}_N|}, \quad q_\ell := \exp(-bf_{\min} \exp[-2\ell Df_{\max}]).$$

In particular, we have asymptotic consistency when $|\mathcal{C}_N| \rightarrow \infty$

$$\lim_{|\mathcal{C}_N| \rightarrow \infty} \mathbb{P}(\mathcal{B} \subseteq \mathcal{X}_1) = 1.$$

Proof. Fix an edge $e \in E(\mathcal{T})$ with k_e descendants and a given character c . For ease of notation, we drop all the (c) superscripts for now. If φ_e is the corresponding bipartition, we will lower bound the probability that $\varphi_e = (L_\alpha \mid \mathcal{L}(\mathcal{T}^*) \setminus L_\alpha)$ for some $\alpha \in \Sigma$.

First, note by our coupling to the ID process, the number N_e of non-homoplastic states born along the edge e satisfies

$$N_e \succeq_{st} \text{Poisson}(\lambda T_e \cdot \nu(\Sigma_N)) \succeq_{st} \text{Poisson}(\eta), \quad \eta := \lambda f_{\min} \nu(\Sigma_N),$$

where T_e is the length of the branch e and $f_{\min}$ is the minimum branch length in the tree.

For a given non-homoplastic α born along the edge e , we can lower bound the probability it has no deaths below e . By Lemma 3.2.4 the probability α survives along a branch of length T is at least $\exp(-\mu T)$. There are $2k_e - 1 \leq 2k_e$ edges (including e) that α can die on. Each is at most length $f_{\max}$, so we can lower bound the probability α has no deaths at all by

$$\mathbb{P}(\alpha \text{ has no deaths} \mid \alpha \in \Sigma_N, \alpha \text{ born on } e) \geq \exp(-\tau_{k_e}), \quad \tau_{k_e} := 2\mu k_e f_{\max}.$$

Using this result followed by $N_e \succeq_{st} \text{Poisson}(\eta)$ we see

$$\mathbb{P}(\exists \alpha \varphi_e = (L_\alpha \mid \mathcal{L}(\mathcal{T}^*) \setminus L_\alpha)) \geq \mathbb{E}_{n \sim N_e} (1 - (1 - \exp(-\tau_{k_e}))^n) \geq 1 - \exp(-\eta e^{-\tau_{k_e}}).$$

From here, the caterpillar bound from [53] applied to our $|\mathcal{C}|$ characters implies

$$\mathbb{P}(\mathcal{B} \subseteq \mathcal{X}_1) \geq 1 - \sum_{\ell=2}^{k-2} \prod_{c \in \mathcal{C}} \exp\left(-\eta^{(c)} e^{-\tau_\ell^{(c)}}\right) = 1 - \sum_{\ell=2}^{k-2} \exp\left(-\sum_{c \in \mathcal{C}} \eta^{(c)} e^{-\tau_\ell^{(c)}}\right).$$

Note none of the above uses our boundedness assumptions on the birth or death rates $\lambda^{(c)}, \mu^{(c)}$. Using these assumptions on the birth and death rates, we see

$$\mathbb{P}(\mathcal{B} \subseteq \mathcal{X}_1) \geq 1 - \sum_{\ell=2}^{k-2} q_\ell^{|\mathcal{C}_N|} \geq 1 - (k-3) q_{k-2}^{|\mathcal{C}_N|}, \quad q_\ell := \exp(-b f_{\min} \exp[-2\ell D f_{\max}]).$$

Hence we get the desired exponential convergence of the coverage probability. $\square$

In particular, this implies if we run the ASTRAL as proposed in [28], restricted to the bipartition set $\mathcal{X}_1$, this is a statistically consistent estimate of the true underlying phylogenetic tree T , and that we require $\exp(O(k))$ characters to achieve a bipartition cover with high probability. This is essentially the same rate bipartition-covers achieve under the standard multispecies-coalescent model to which ASTRAL is typically applied [80], [53].

This result should be viewed as a baseline consistency guarantee rather than a practical sample-complexity result. It is clear that $\mathcal{X}_1$ is bad at generating large bipartitions. Namely, if $\varphi_e = (A \mid B)$ the probability a character born in edge e has no deaths is exponentially small in $|A|$. To rectify this, we propose a few heuristics for expanding the size of $\mathcal{X}_1$ so it has better coverage probabilities. A natural solution is to allow multiple states α to contribute

$$\mathcal{X}_m := \left\{ (A \mid \mathcal{L}(\mathcal{T}^*) \setminus A) : A = \bigcup_{i=1}^r L_{\alpha_i}^{(c)}, c \in \mathcal{C}, \alpha_i \in \Sigma^{(c)}, 1 \leq r \leq m, \emptyset \subsetneq A \subsetneq \mathcal{L}(\mathcal{T}^*) \right\}.$$

It is easy to see that $\mathcal{X}_n \subseteq \mathcal{X}_m$ for $n \leq m$, so in particular $\mathcal{X}_1 \subseteq \mathcal{X}_m$. Hence asymptotic consistency using $\mathcal{X}_m$ follows from that of $\mathcal{X}_1$. To avoid combining leaf sets from completely different regions in the tree, we can also introduce some sort of compatibility condition like $L_{\alpha_i}^{(c)} \cap L_{\alpha_j}^{(c)} \neq \emptyset$.

Alternatively, for a nonnegative integer r , we can expand each observed leaf set by adding at most r leaves. Define

$$\mathcal{X}_{\leq r} := \{(A \mid \mathcal{L}(\mathcal{T}^*) \setminus A) : \exists c \in \mathcal{C}, \alpha \in \Sigma^{(c)} \text{ such that } L_\alpha^{(c)} \subseteq A \subsetneq \mathcal{L}(\mathcal{T}^*), |A \setminus L_\alpha^{(c)}| \leq r\}.$$

For a fixed leaf set $L_\alpha^{(c)}$, this generates at most $\sum_{j=0}^r \binom{k-|L_\alpha^{(c)}|}{j} = O(k^r)$ candidate bipartitions when r is fixed. Moreover, $\mathcal{X}_1 \subseteq \mathcal{X}_{\leq r}$, so this expansion preserves the asymptotic-consistency guarantee above.

3.4 Alternative ASTRAL Procedure

Another inference method, closer in spirit to the original ASTRAL framework, is to let each character independently vote on the topology of each quartet. These character-level quartet votes then play the role of the quartet information induced by individual gene trees in the traditional multispecies coalescent setting.

More formally, for each character c_i and each leaf set $s = \{u, v, x, y\}$, define

$$\tau_s^{(c_i)} := \arg \max_{q \in A_s} |M_q^{(c_i)}|, \quad I_s^{(c_i)} := \left\{ \max_{q \in A_s} |M_q^{(c_i)}| > 0 \right\},$$

with ties broken uniformly. We then solve (3.1.2) with quartet supports

$$W(q) = \sum_{i=1}^n 1\{\tau_s^{(c_i)} = q\} \cdot 1\{I_s^{(c_i)}\}.$$

The indicator $1\{I_s^{(c_i)}\}$ essentially just tells us to ignore characters which provide no information about the given quartet. We call this procedure *PCH-ASTRAL(sc)* for “separate characters”. This inference scheme is only really reasonable if for each quartet, the correct topology is the most likely one. Namely

$$\mathbb{P}(\tau_s^{(c)} = \mathcal{T}_s^*) > \mathbb{P}(\tau_s^{(c)} = q) \quad q \neq \mathcal{T}_s^*. \quad (3.4.1)$$

On I_s^c , all three supports vanish, so uniform tie-breaking assigns each topology probability $1/3$. Consequently, property (3.4.1) is equivalent to the corresponding strict inequalities conditional on I_s .

If property (3.4.1) holds for every quartet $\mathcal{T}_s^* \in \mathcal{Q}(\mathcal{T}^*)$, we say the PCH model is in the *anomaly-free zone*. When this occurs, the above ASTRAL-based procedure is asymptotically consistent and has sample-complexity essentially quadratic in the margin. Note the following makes no assumptions on the distribution of the characters beyond the anomaly-free property, so it applies equally to the PCH model and the standalone $ID(\theta, \lambda, \mu; \mathcal{T}^*)$ process.

Theorem 3.4.1 (Sample Complexity PCH-ASTRAL(sc)). *Suppose we have a collection of n iid characters $\{c_i\}$ in the anomaly-free zone, and let $0 < \varepsilon < 1$. If*

$$n > \frac{8}{p_{\min} \Delta^2} \log \left(\frac{3 \binom{k}{4}}{\varepsilon} \right),$$

where

$$\Delta := \min_{s \in \binom{[k]}{4}} \min_{t \in A_s \setminus \{\mathcal{T}_s^*\}} [\mathbb{P}(\tau_s^{(c)} = \mathcal{T}_s^* \mid I_s) - \mathbb{P}(\tau_s^{(c)} = t \mid I_s)], \quad p_{\min} := \min_{s \in \binom{[k]}{4}} \mathbb{P}(I_s),$$

then the above ASTRAL-based procedure recovers $\mathcal{T}^*$ with probability at least $1 - \varepsilon$.

Proof. Let $K := \binom{k}{4}$. For a fixed leaf set s , write $p_s := \mathbb{P}(I_s)$ and let n_s be the number of informative characters. Since $n_s \sim \text{Binomial}(n, p_s)$, Chernoff's bound gives

$$\mathbb{P} \left(n_s < \frac{np_s}{2} \right) \leq \exp \left(-\frac{np_{\min}}{8} \right).$$

Therefore, by union bound

$$\mathbb{P} \left(\min_{s \in \binom{[k]}{4}} n_s < \frac{np_{\min}}{2} \right) \leq K \exp \left(-\frac{np_{\min}}{8} \right).$$

Hence with high probability, every leaf set s has at least $n' := np_{\min}/2$ informative characters. Now fix s and label its two incorrect topologies t_1, t_2 . Conditional on I_s , define

$$p_s^* := \mathbb{P}(\tau_s^{(c)} = \mathcal{T}_s^* \mid I_s), \quad r_{s,j} := \mathbb{P}(\tau_s^{(c)} = t_j \mid I_s), \quad j = 1, 2.$$

For each informative character c_i , let

$$Z_i^{(j)} := 1\{\tau_s^{(c_i)} = \mathcal{T}_s^*\} - 1\{\tau_s^{(c_i)} = t_j\}, \quad j = 1, 2.$$

Conditional on n_s , the variables $Z_i^{(j)}$ are iid, take values in $[-1, 1]$, and satisfy

$$\mathbb{E}[Z_i^{(j)} \mid I_s] = p_s^* - r_{s,j} \geq \Delta.$$

Hence Hoeffding's inequality implies

$$\mathbb{P} \left(\sum_{i=1}^{n_s} Z_i^{(j)} \leq 0 \mid n_s \right) \leq \exp \left(-\frac{n_s \Delta^2}{2} \right).$$

Let F_s be the event that $\mathcal{T}_s^*$ is not the unique empirical maximizer for s . On F_s , the above event occurs for at least one $j \in \{1, 2\}$. Thus, on $n_s \geq np_{\min}/2$,

$$\mathbb{P}(F_s \mid n_s) \leq 2 \exp \left(-\frac{np_{\min} \Delta^2}{4} \right).$$

If the true topology is uniquely modal for every leaf set, exact ASTRAL returns $\mathcal{T}^*$. Combining the preceding estimate with the informative-count bound above and union bounding over the K leaf sets gives

$$\mathbb{P}(\hat{\mathcal{T}} \neq \mathcal{T}^*) \leq K \exp\left(-\frac{np_{\min}}{8}\right) + 2K \exp\left(-\frac{np_{\min}\Delta^2}{4}\right) \leq 3K \exp\left(-\frac{np_{\min}\Delta^2}{8}\right),$$

as $\Delta \leq 1$. Applying the assumed lower bound on n gives $\mathbb{P}(\hat{\mathcal{T}} = \mathcal{T}^*) \geq 1 - \varepsilon$ as desired. $\square$

The argument is analogous to the proof of Theorem 2.1 in [74]. In Appendix 3.A, we briefly discuss the anomaly-free zone and show that, at least for the $ID(\theta, \lambda, \mu; \mathcal{T}^*)$ process, it occurs in the low-polymorphism setting.

3.5 Sample-Complexity Bounds

3.5.1 PCH-ASTRAL Sample-Complexity Bounds

In this section, we aim to understand the number of independent characters required by PCH-ASTRAL to recover the true species tree $\mathcal{T}^*$ with high probability. To attempt to do so, we prove a sample complexity result for the PCH-ASTRAL procedure when samples are generated from the coupled immigration-death process $ID(\theta, \lambda, \mu; \mathcal{T}^*)$ on the true species tree $\mathcal{T}^*$ rather than the original PCH model. It is a simple consequence of our concentration bound Lemma 3.B.3.

Lemma 3.5.1 (Sample Complexity PCH-ASTRAL under ID Process). *Suppose we have a collection $\{c_i\}_{i=1}^n$ of iid characters whose macro-states are distributed according to the $ID(\theta, \lambda, \mu; \mathcal{T}^*)$ process. Let $\zeta = \theta \vee \frac{\lambda}{\mu}$ and*

$$\kappa := \max_{s \in \binom{[k]}{4}} \max_{uv || xy \in A_s} (1 + \zeta \eta_{uv}) (1 + \zeta \eta_{xy}).$$

Define our worst-case margin

$$L := \min_{s \in \binom{[k]}{4}} \min_{t \in A_s \setminus \{\mathcal{T}_s^*\}} \mathbb{E}[|M_{\mathcal{T}_s^*}| - |M_t|] > 0.$$

Then if we have at least

$$n \gtrsim \max \left\{ \frac{\kappa^2}{L^2} t_\delta, \frac{\kappa}{L} t_\delta^2 \right\}, \quad t_\delta := \log \left(\frac{4 \binom{k}{4}}{\delta} \right),$$

characters, we recover the true species tree with probability at least $1 - \delta$.

Proof of Lemma 3.5.1 (Sample Complexity PCH-ASTRAL). If for each leaf-set s the topology $q \in A_s$ maximizing the score S_q is the correct topology, then PCH-ASTRAL will trivially return the correct species tree $\mathcal{T}^*$, as the set of topologies $\{q_s\}_{s \in \binom{[k]}{4}}$ will exactly match those generated by the species tree $\mathcal{T}^*$. Hence if $\hat{\mathcal{T}}$ is the output of this inference procedure, by union bound

$$\mathbb{P}(\hat{\mathcal{T}} = \mathcal{T}^*) \geq 1 - \sum_s \mathbb{P}(q_s \neq \mathcal{T}_s^*).$$

Hence it suffices to upper bound $p_s := \mathbb{P}(q_s \neq \mathcal{T}_s^*)$. Note

$$p_s \leq \sum_{t \in A_s \setminus \{\mathcal{T}_s^*\}} \mathbb{P}(S_{\mathcal{T}_s^*} \leq S_t).$$

so it suffices to bound each term on the right hand side. For $t \in A_s \setminus \{\mathcal{T}_s^*\}$, define

$$D_i := |M_{\mathcal{T}_s^*}^{(c_i)}| - |M_t^{(c_i)}|, \quad S_{\mathcal{T}_s^*} - S_t = \frac{1}{n} \sum_{i=1}^n D_i.$$

Since $L \leq \mathbb{E}D_i$ we have

$$\mathbb{P}(S_{\mathcal{T}_s^*} \leq S_t) \leq \mathbb{P}\left(\left|\sum_{i=1}^n (D_i - \mathbb{E}D_i)\right| \geq nL\right).$$

Lemma 3.B.3 implies for universal constants

$$\|D_i - \mathbb{E}D_i\|_{\Psi_{1/2}} \lesssim \| |M_{\mathcal{T}_s^*}| \|_{\Psi_{1/2}} + \| |M_t| \|_{\Psi_{1/2}} \lesssim \kappa.$$

Applying the sub-Weibull concentration inequality of Theorem 3.1 in [48]

$$\mathbb{P}\left(\left|\sum_{i=1}^n (D_i - \mathbb{E}D_i)\right| \gtrsim \kappa(\sqrt{nt} + t^2)\right) \leq 2e^{-t}.$$

Choose $t_\delta = \log(4\binom{k}{4}/\delta)$. Note our assumptions on n imply

$$nL \gtrsim \kappa(\sqrt{nt_\delta} + t_\delta^2),$$

and we have

$$\mathbb{P}(S_{\mathcal{T}_s^*} \leq S_t) \leq 2\exp(-t_\delta).$$

Taking a union bound over quartets and the two incorrect topologies then gives

$$\mathbb{P}(\hat{\mathcal{T}} = \mathcal{T}^*) \geq 1 - 2\binom{k}{4} \cdot 2\exp(-t_\delta) = 1 - \delta,$$

as desired. $\square$

As discussed in the proof, the parameter κ in a sense measures how heavy-tailed the support sizes $|M_{uv||xy}|$ are. It has the trivial upper bound $\kappa \leq (1 + \zeta)^2$, which is a function merely of the expected number of states in a leaf of the $ID(\theta, \lambda, \mu; \mathcal{T}^*)$ process. Hence the worst-case margin L is the main parameter of interest. Corollary 3.2.7 lower bounds L under $ID(\theta, \lambda, \mu; \mathcal{T}^*)$ purely in terms of the underlying model parameters.

The worst-case margin L certainly depends on the topology of $\mathcal{T}^*$, although Corollary 3.2.7 gives a heuristic topology-free bound. If f is the minimal branch length in $\mathcal{T}^*$ and

$$d := \max_{uv||xy \in \mathcal{Q}(\mathcal{T}^*)} \ell_{uv} \vee \ell_{xy},$$

is the maximal distance between any two leaves on the same side of a quartet, then the margin lower bounds in Corollary 3.2.7 roughly become

$$\Delta_q \geq \left(\frac{\lambda}{\mu}\right)^2 (1 - e^{-\mu f})^2 (e^{-\mu d})^2 \gtrsim \lambda^2 f^2 e^{-2\mu d},$$

when the induced subtree of q in $\mathcal{T}^*$ is balanced and similarly

$$\Delta_q \gtrsim \lambda \mu f^2 e^{-2\mu d} \left(\theta \wedge \frac{\lambda}{\mu} \right),$$

when the induced subtree is a caterpillar. Thus, for a tree of fixed height and fixed model parameters, the displayed lower bounds scale quadratically in f , and the sample-complexity bound becomes

$$n \gtrsim \max \{ f^{-4} \log(k), f^{-2} \log^2(k) \}.$$

This remains significantly worse than the $\Theta(f^{-2} \log(k))$ of ASTRAL under the multi-species coalescent model [74]. Of course, our bound is naturally quite loose, and future improvements may significantly improve the rate.

3.6 Discussion and Concluding Remarks

In this chapter, we developed several potential candidates for bipartition covers which substantially reduce the search space of the original PCH-ASTRAL inference procedure. This modification is intended to make PCH-ASTRAL computationally feasible for much larger collections of languages. Although restricting the search space reduces sample efficiency, we proposed several heuristics designed to mitigate this loss in performance.

Beyond this, we derived sample-complexity bounds under the immigration-death process model $ID(\theta, \lambda, \mu; \mathcal{T}^*)$, both for the original PCH-ASTRAL procedure and for the character-voting procedure PCH-ASTRAL(sc). In both cases, the rates depend explicitly on a quartet-level margin measuring the advantage of the correct quartet topology over the incorrect ones. We further gave partial bounds relating these margins to the underlying model parameters.

There are several natural directions for future work. First, one would like to extend the sample-complexity results from the simplified $ID(\theta, \lambda, \mu; \mathcal{T}^*)$ process to the full PCH model. The main obstacle is concentration. The immigration-death coupling provides a convenient lower bound on non-homoplastic state counts, which is enough for expected-margin estimates, but the concentration argument also requires an upper bound on macro-state sizes at the leaves. Establishing such an upper bound for the full PCH model would therefore be an important first step toward extending Lemma 3.5.1 beyond the ID approximation.

Second, since the true tree $\mathcal{T}^*$ is unknown at inference time, it would be useful to develop topology-free sample-complexity bounds depending only on simple geometric parameters of $\mathcal{T}^*$, such as the minimum branch length, analogous to the results of [74]. Under the ID approximation, the Poisson means $m_{uv} = \mathbb{E}N_{uv}$ are explicit functions of the model parameters, so more careful bookkeeping in Lemma 3.5.1 may lead to relatively explicit bounds.

Third, the anomaly-free condition for PCH-ASTRAL(sc) remains somewhat opaque. Under the ID approximation, the condition reduces to a comparison of independent product-Poisson random variables, and is therefore numerically checkable from the quartet-level means. However, we do not currently have simple geometric sufficient conditions ensuring that the anomaly-free property holds.

Finally, several refinements of the present analysis should be possible. The margin bounds used here depend on coarse survival and death probability estimates from Lemma 3.2.4; sharper estimates may yield more granular constants.

Extensive empirical simulations would also be valuable, both to compare PCH-ASTRAL and PCH-ASTRAL(sc) under various model regimes and to assess how well the theoretical margins predict finite-sample behavior. It would also be interesting to study how well our proposed bipartition heuristics improve coverage properties.

3.A The Anomaly-Free Zone

In this section, we briefly discuss the anomaly-free property (3.4.1). Theorem 3.1.1 shows $\mathbb{E}|M_t|$ is maximized over $q \in A_s$ when $q = \mathcal{T}_s^*$, but this does not imply $\mathcal{T}_s^*$ maximizes

$$\mathbb{P} \left(q = \arg \max_{t' \in A_s} |M_{t'}| \right),$$

as it is possible $|M_{\mathcal{T}_s^*}|$ is zero most of the time, but takes on large values rarely. To avoid this, we must show the supports $|M_{\mathcal{T}_q^*}|$ are reasonably concentrated. We focus on the simpler $ID(\theta, \lambda, \mu; \mathcal{T}^*)$ setting.

Lemma 3.B.2 shows in this setting that the supports take the particular form

$$|M_{uv||xy}| = N_{uv}N_{xy}, \quad |M_{ux||vy}| = N_{ux}N_{vy}, \quad |M_{uy||vx}| = N_{uy}N_{vx},$$

where these six random variables $\{N_{uv}, \dots, N_{vx}\}$ are mutually independent Poissons whose means $m_{ab} := \mathbb{E}N_{ab}$ are reasonably simple to compute. The main difficulty is that these means depend on the underlying parameters $(\mathcal{T}^*, \lambda, \mu, \theta)$ in complicated, interwoven ways. For example, we know from Lemma 3.B.1 that $|I| \sim \text{Poisson}(m_\theta(\ell_{ri}))$. Hence if the true quartet topology is $q = uv||xy$ and it induces a balanced subtree in $\mathcal{T}^*$, then

$$m_{uv} = \eta_{uv} [m_0(\ell_{ij}) + m_\theta(\ell_{ri})\eta_{ij}(1 - \eta_{ik}(\eta_{ky} + \eta_{kx} - \eta_{kx}\eta_{ky}))],$$

$$m_{ux} = m_\theta(\ell_{ri})\eta_{ux} [(1 - \eta_{jv}) \cdot (1 - \eta_{ky})].$$

Even with these means known, these Poisson-products depend on these means in complicated ways. However, in the low-polymorphism regime $\lambda \ll \mu$, where these means are all close to zero, the latter issue simplifies drastically. In this regime, the supports are heavily concentrated on $\{0, 1\}$, and the maximum support is typically the unique nonzero one.

Lemma 3.A.1 (Low Polymorphism Implies Anomaly-Free). *Suppose $\theta > 0$. For an $ID(\epsilon\theta, \epsilon\lambda, \mu; \mathcal{T}^*)$ process, let $s = \{u, v, x, y\}$ be any set of four leaves. If*

$$\omega_*^{(\epsilon)} = \mathbb{E}|M_{\mathcal{T}_s^*}^{(\epsilon)}|, \quad \{\omega_1^{(\epsilon)}, \omega_2^{(\epsilon)}\} = \{\mathbb{E}|M_{q'}^{(\epsilon)}| : q' \in A_s \setminus \{\mathcal{T}_s^*\}\},$$

are the mean supports of the correct and two incorrect topologies respectively, then for every $q \in A_s$, as $\epsilon \downarrow 0$

$$\mathbb{P} \left(q = \arg \max_{q' \in A_s} |M_{q'}^{(\epsilon)}| \mid \max_{q' \in A_s} |M_{q'}^{(\epsilon)}| > 0 \right) = \frac{\omega_q^{(1)}}{\omega_*^{(1)} + \omega_1^{(1)} + \omega_2^{(1)}} + O(\epsilon).$$

In particular, for small enough $\epsilon > 0$ the anomaly-free property (3.4.1) holds.

Proof. Let $N_{uv}^{(\epsilon)}, \dots, N_{vx}^{(\epsilon)}$ be the corresponding Poisson random variables from Lemma 3.B.2, and $m_{uv}^{(\epsilon)}, \dots, m_{vx}^{(\epsilon)}$ their means. Observe $m_{uv}^{(\epsilon)} = \epsilon m_{uv}^{(1)}$, and similarly for the other means, by a simple Poisson-thinning argument. Letting $\omega_q := \omega_q^{(1)}$ and $m_{uv} := m_{uv}^{(1)}$, for $q = uv || xy$ let

$$p_q := \mathbb{P}(|M_q^{(\epsilon)}| > 0) = (1 - e^{-\epsilon m_{uv}})(1 - e^{-\epsilon m_{xy}}) = \epsilon^2 \omega_q + O(\epsilon^3),$$

and likewise for the other supports. More specifically, using $a \geq 1 - e^{-a} \geq a - a^2/2$ for $a \geq 0$ and the fact that $m_{uv}^{(1)} \leq \zeta := \theta \vee \lambda \mu^{-1}$, we see

$$-\epsilon^3 \zeta \omega_q \leq p_q - \epsilon^2 \omega_q \leq 0.$$

Since the supports are independent by Lemma 3.B.2, we see

$$-\sum_{q' < q} p_q p_{q'} \leq \mathbb{P}(\max |M_q^{(\epsilon)}| > 0) - \sum_{q \in A_s} p_q \leq 0.$$

For $W := \omega_* + \omega_1 + \omega_2$, the quadratic term is bound by

$$\sum_{q' < q} p_q p_{q'} \leq \frac{\epsilon^4 W^2}{2}.$$

Letting

$$\tau^{(\epsilon)} = \arg \max_{q \in A_s} |M_q^{(\epsilon)}|, \quad I_s^{(\epsilon)} = \left\{ \max_{q \in A_s} |M_q^{(\epsilon)}| > 0 \right\},$$

our above analysis shows

$$\begin{aligned} \mathbb{P}(\tau^{(\epsilon)} = q \mid I_s^{(\epsilon)}) &= \frac{\epsilon^2 \omega_q + O(\epsilon^3)}{1 - \prod_{q' \in A_s} (1 - \epsilon^2 \omega_{q'} + O(\epsilon^3))} \\ &= \frac{\omega_q}{\omega_* + \omega_1 + \omega_2} + O(\epsilon), \end{aligned}$$

as desired. Since the true topology has strictly highest expected support by Corollary 3.2.7, this implies the anomaly-free property when ϵ is sufficiently small. $\square$

Hence, the anomaly-free property holds in this low-polymorphism regime, at least under the simpler $ID(\theta, \lambda, \mu; \mathcal{T}^*)$ model. Of course, this is also the regime where the immigration-death process approximation is the least accurate, so it is not clear how this result translates to the full PCH model. Nevertheless, we see this as evidence the anomaly-free property is a reasonable constraint, and conjecture it holds in the low-polymorphism regime of the full PCH model.

3.B Immigration-Death Process Proofs

Let $ID(\theta, \lambda, \mu)$ denote an immigration-death process Y_t with $|Y_0| \sim \text{Poisson}(\theta)$, birth rate λ , and death rate μ per individual. We start by understanding the population size in such a process.

Lemma 3.B.1 (Population Sizes in ID Processes). *If $(Y_t)_{t \geq 0} \sim ID(\theta, \lambda, \mu)$ then*

$$|Y_t| \sim \text{Poisson}(m_\theta(t)), \quad m_\theta(t) = \theta e^{-\mu t} + \frac{\lambda}{\mu}(1 - e^{-\mu t}).$$

In particular, its stationary distribution is $\text{Poisson}(\lambda/\mu)$. Moreover for $s < t$

$$\mathbb{E}|Y_t \cap Y_s| = m_\theta(s) \cdot e^{-\mu(t-s)}.$$

Proof. Kolmogorov's forward equations tell us

$$\frac{d}{dt}p_k = \lambda p_{k-1} + \mu(k+1)p_{k+1} - (k\mu + \lambda)p_k, \quad p_k(0) = \frac{e^{-\theta}\theta^k}{k!}.$$

It is easy to see the Poisson probabilities $p_k(t) = \exp(-m_\theta(t))m_\theta(t)^k/k!$ satisfy these equations, and hence $|Y_t| \sim \text{Poisson}(m_\theta(t))$. For $s < t$, conditional on Y_s ,

$$|Y_t \cap Y_s| \sim \text{Binomial}(|Y_s|, e^{-\mu(t-s)}).$$

Taking expectations gives $\mathbb{E}|Y_t \cap Y_s| = m_\theta(s)e^{-\mu(t-s)}$. □

Lemma 3.2.3. Along a branch of the PCH model, let $S_t = |c(t)|$ have birth rate λ and death rate $i\mu$ in state $i > 1$, with no death from state 1. Its stationary distribution is a $\text{Poisson}(\lambda/\mu)$ random variable conditioned to be positive:

$$\pi_i = \frac{(\lambda/\mu)^i}{i!(e^{\lambda/\mu} - 1)}, \quad i \geq 1.$$

Proof of Lemma 3.2.3. This process is reversible, so by detailed balance

$$\pi_{i+1}(i+1)\mu = \lambda\pi_i, \quad \forall i \geq 1.$$

Recursively applying this shows $\pi_i \propto \frac{(\lambda/\mu)^i}{i!}$ as desired. □

Lemma 3.2.4. For a PCH branch of length T , let $p_{n,T}$ be the probability that a specified state α dies when the branch starts with $n \geq 1$ states and has birth and per-state death rates $\lambda, \mu > 0$. Then

$$\frac{1}{2}(1 - e^{-\lambda T/2})(1 - e^{-\mu T}) \leq p_{n,T} \leq 1 - e^{-\mu T}.$$

Proof of Lemma 3.2.4. Suppose along the given branch we spent a fraction $\tau \in [0, 1]$ of the time with at least two states, whose distribution implicitly depends on n, T . Then if α is a state present at time zero

$$\mathbb{P}(\alpha \text{ survives along branch} \mid \tau) = e^{-\mu T \cdot \tau}.$$

Hence the survival probability depends only on the Laplace transform φ of τ

$$\mathbb{P}(\alpha \text{ survives along branch}) = \varphi(\mu T).$$

Trivially $\tau \leq 1$, which gives the upper bound. For the lower bound, note that $p_{n,T}$ is increasing in n by a simple coupling argument. Hence WLOG we assume $n = 1$. One way for α to die in this setting is for another state β to be born at time $\gamma \sim \text{Exp}(\lambda)$. The first of these two states dies by time T with probability $1 - e^{-2\mu(T-\gamma)}$ and the death is equally likely to be α or β . Hence

$$p_{n,T} \geq \frac{1}{2} \cdot \int_0^T (1 - e^{-2\mu(T-\gamma)}) \lambda e^{-\lambda\gamma} d\gamma = \frac{1}{2} \left[(1 - e^{-\lambda T}) - \frac{\lambda(e^{-\lambda T} - e^{-2\mu T})}{2\mu - \lambda} \right].$$

An even simpler bound is

$$p_{n,T} \geq \frac{1}{2} (1 - e^{-\lambda T/2}) (1 - e^{-\mu T}),$$

which is the probability we have a birth of some state β by time $T/2$, a death of either α, β in the remaining time $T/2$, and that this death is α . $\square$

The following shows that our quartet supports $|M_{uv||xy}|$ take a particularly simple form under the standalone ID model.

Lemma 3.B.2 (Quartet Supports under ID Process). *Suppose $Y_t \sim ID(\theta, \lambda, \mu; \mathcal{T}^*)$. For any subset $\mathcal{U}$ of the leaves of $\mathcal{T}^*$, for each $A \subseteq \mathcal{U}$ define*

$$N_A^{\mathcal{U}} := |\{\alpha : \alpha \in Y_\ell \text{ for every } \ell \in A, \alpha \notin Y_\ell \text{ for every } \ell \in \mathcal{U} \setminus A\}|$$

to be the number of states α which appear exactly in the set of leaves A , among those in $\mathcal{U}$. Then $\{N_A^{\mathcal{U}}\}_{A \subseteq \mathcal{U}}$ are mutually independent Poisson random variables. In particular, for $s = \{u, v, x, y\}$ this implies

$$|M_{uv||xy}| = N_{uv}^{(s)} \cdot N_{xy}^{(s)}.$$

Thus, $|M_{uv||xy}|$ is the product of two independent Poisson random variables.

This Lemma makes analyzing the expected margin

$$\Delta_t := \mathbb{E}|M_{\mathcal{T}_s^*}| - \mathbb{E}|M_t|, \quad t \in A_s \setminus \{\mathcal{T}_s^*\}$$

particularly easy under the $ID(\theta, \lambda, \mu; \mathcal{T}^*)$ process: we just must try to understand the means $m_{uv}^{(s)}$ of these Poisson random variables $N_{uv}^{(s)}$. Calculating these exactly can be challenging, but Lemma 3.B.1 gives a simple upper bound

$$m_{uv}^{(s)} \leq \left(\theta \vee \frac{\lambda}{\mu} \right) \eta_{uv}.$$

The first term accounts for the number of states reaching the most recent common ancestor of u, v , and the second term thins these to only those reaching both u, v .

Proof of Lemma 3.B.2. Fix a point $x \in \mathcal{T}^*$. Each state α born at x appears on some subset $A \subseteq \mathcal{U}$ of the leaves in $\mathcal{U}$. For fixed tree topology $\mathcal{T}^*$ and model parameters θ, λ, μ , the probability $p_x(A)$ that it receives a particular mark A depends only on the location x , and is independent of the birth-process itself and any other states.

The number of states born along the tree form a Poisson point process with intensity

$$\nu(dz) := \theta \delta_r(dz) + \lambda dz,$$

where r is the root and dz denotes length measure on the branches of $\mathcal{T}^*$. Kingman's marking theorem then implies that the marked process forms a Poisson point process on $\mathcal{T}^* \times \{A : A \subseteq \mathcal{U}\}$ with intensity $\nu(dz)p_z(dA)$. See, for example, the chapter on marked Poisson processes in [47]. The desired result then follows immediately. $\square$

Lemma 3.B.3 (Tail Bounds for Quartet Supports). *Let $\zeta = \theta \vee \frac{\lambda}{\mu}$. Under $ID(\theta, \lambda, \mu; \mathcal{T}^*)$*

$$\mathbb{P}(|M_{uv||xy}| > t) \leq 2 \exp \left(-\zeta \min\{\eta_{uv}, \eta_{xy}\} \cdot h \left(\zeta^{-1} \sqrt{\frac{t}{\eta_{uv}\eta_{xy}}} \right) \right),$$

where $h(t) := (t \log(t) - t + 1)1\{t \geq 1\}$. Moreover, $|M_{uv||xy}|$ is sub-Weibull(1/2) with Orlicz pseudo-norm associated to $\Phi_\alpha(x) = \exp(x^\alpha) - 1$ given by

$$\| |M_{uv||xy}| \|_{\Psi_{1/2}} \lesssim (1 + \zeta \cdot \eta_{uv}) (1 + \zeta \cdot \eta_{xy}).$$

Proof of Lemma 3.B.3 (quartet-support tail bound). Note that

$$M_{uv||xy} \subseteq \{(\alpha, \beta) : \alpha \in U \cap V, \beta \in X \cap Y\} = (U \cap V) \times (X \cap Y).$$

Suppose that w, z are the most recent common ancestors of u, v and x, y , respectively, and let $W := Y_w$ and $Z := Y_z$. Lemma 3.B.1 implies that $\text{Poisson}(\zeta) \succeq_{st} |W|$ and $\text{Poisson}(\zeta) \succeq_{st} |Z|$. Then Poisson-thinning implies

$$\text{Poisson}(m_{uv}) \succeq_{st} |U \cap V|, \quad m_{uv} := \zeta \eta_{uv},$$

and analogously for $|X \cap Y|$. Chernoff's inequality implies

$$\mathbb{P}(\text{Poisson}(m) \geq mt) \leq \exp(-mh(t)).$$

By union bound, for any $\gamma > 0$

$$\mathbb{P}(|M_{uv||xy}| > t) \leq \mathbb{P}(|U \cap V| \geq \gamma\sqrt{t}) + \mathbb{P}(|X \cap Y| \geq \gamma^{-1}\sqrt{t}).$$

Setting $\gamma = \sqrt{m_{uv}/m_{xy}}$ weights the two terms according to their means, giving

$$\leq \exp\left(-m_{uv}h\left(\sqrt{\frac{t}{m_{uv}m_{xy}}}\right)\right) + \exp\left(-m_{xy}h\left(\sqrt{\frac{t}{m_{uv}m_{xy}}}\right)\right),$$

from which the desired bound directly follows.

To upper bound the Orlicz norm, we rely on the Hölder-type inequality appearing as equation (3.5) in [48]. This gives

$$\| |M_q| \|_{\Psi_{1/2}} \leq \|\text{Poisson}(m_{uv})\|_{\Psi_1} \cdot \|\text{Poisson}(m_{xy})\|_{\Psi_1}.$$

A simple calculation and the bound $\log(1+x) \geq \frac{x}{1+x}$ for $x \geq 0$ shows

$$\|\text{Poisson}(m)\|_{\Psi_1} = \frac{1}{\log(1 + \log(2)/m)} \leq 1 + \frac{m}{\log(2)},$$

from which the result follows. □

Chapter 4

Exchangeable Array Processes

4.1 Introduction

The goal of this chapter is to develop a general framework for constructing and studying consistent families of finite combinatorial structures, extending the program proposed in [2] toward a general theory of combinatorial limit objects. We aim to unify three ideas: Doob-Martin boundaries, graphons and hypergraphons, and the classical theory of exchangeable arrays. By encoding the finite states of a growing combinatorial process as restrictions of an exchangeable array, we simultaneously obtain a natural Markov chain and associated Martin boundary problem, a sampling representation of its limiting objects, and a natural notion of finite-substructure embedding. We briefly discuss these three fields.

Exchangeability arises naturally when modeling random systems composed of many interchangeable parts. The subject originates with de Finetti's representation theorem for exchangeable binary sequences and was subsequently extended to a wide variety of random combinatorial structures, including exchangeable partitions and higher-dimensional arrays [5, 41, 46]. A common feature of these theories is a representation theorem that generates an exchangeable object by sampling independent latent random variables and applying an appropriate measurable function. For the present chapter, the central result is the Aldous-Hoover-Kallenberg representation theorem for exchangeable arrays, stated below as Theorem 4.2.3. This theorem places many random combinatorial structures within a common sampling framework and reduces the construction of a corresponding limit theory to the problem of encoding their finite substructures as entries of an exchangeable array.

Martin boundary theory, whose foundations were set in [20], provides another perspective on this goal. Given a transient Markov chain $(X_n)_{n \in \mathbb{N}}$ on some countable state space E , Martin boundary theory aims to complete E into a compact metric space by adding "points at ∞ " to which the Markov chain is converging. These boundary points provide our first natural candidate for the desired limit objects, and this theory provides a mechanism for tilting the underlying chain in a way that forces it to converge to a given boundary point, analogous to how one can generate a sequence of graphs converging to a given graphon via

sampling. A short review of this topic and further references can be found in Appendix 4.B.

More recently, the theory of *graphons* provided a formalization for the limits of sequences of dense graphs. First developed in [10, 51], this theory topologizes the space of graphs by looking at a graph’s finite substructures. In particular, two graphs G, H are said to be close if for each “test” graph F , the weak embedding density $emb^W(F, G)$ is close to that of H , where

$$emb^W(F, G) := \frac{|\{\text{injective graph homomorphisms } \varphi : F \rightarrow G\}|}{(|V(G)|)^{|V(F)|}}$$

is the fraction of injections $\varphi : V(F) \rightarrow V(G)$ that preserve edges, and $(N)_k := N(N - 1) \cdots (N - k + 1)$. Since many other combinatorial objects have an analogous notion of embedding, this framework readily extends to other settings. A large body of work quickly followed, extending these results to hypergraphs [25, 88], sparse graphs [11], permutations [37], and beyond.

Partial progress has already been made in connecting these strands of work. The relationship between graphons and exchangeable arrays is well known: graphon sampling produces a dissociated binary exchangeable array, the underlying adjacency matrix of the graph sequence, while the corresponding exchangeable array representation recovers the graphon up to the usual measure-preserving equivalence [5, 78]. Moreover, Grübel and collaborators have shown that the usual graphon and permuton topologies agree with the Doob-Martin topologies of corresponding random growth chains [32, 33].

These identifications, however, have generally been established separately for each class of combinatorial structures. One first identifies the relevant finite substructures, constructs an appropriate growth chain which preserves these structures, and then proves that its Doob-Martin topology agrees with an independently defined limit topology. We aim to make this construction systematic. Once the relationships among finite collections of components are encoded by an exchangeable array, finite restrictions generate the labeled growth process, quotienting by relabeling generates the corresponding unlabeled chain, and finite array patterns provide a natural notion of substructure density.

Outline. In Section 4.2, we start by defining two simple Markov processes associated to a given exchangeable array: a *labeled* chain generated by successively revealing the “top left corner” of the given array, and an *unlabeled* chain by quotienting the former up to relabeling the indices of the array. Furthermore, we discuss how several existing examples of exchangeable combinatorial growth processes can be embedded into this framework.

In Section 4.3, we characterize the Doob-Martin boundary of the labeled exchangeable array process and the minimal boundary of the unlabeled process. In the former, we show the boundary is the set of deterministic arrays on which the exchangeable array is supported. In the latter, we show the minimal boundary is characterized by the set of ergodic arrays. Equivalently, these arrays admit Aldous–Hoover–Kallenberg representation functions that do not depend on the first coordinate, providing a natural analog of graphons and hypergraphons.

In Section 4.4, we define a few natural notions of embedding a smaller (unlabeled) array into a larger one. As in the case of graphons, these embeddings provide a way of topologizing the set of arrays: we say a sequence y_n of arrays converges if the embedding density $emb(x, y_n)$ converges for every finite array x . We show that these different notions of embedding generate the same topology, and that this topology agrees with the Doob-Martin topology. We define a way to extend these embedding densities to the limiting representation functions, and a way of embedding a finite array into this space of representation functions in a way that asymptotically preserves these densities.

Lastly, in Appendix 4.A, we discuss some partial progress on extending the classical cut metric from the theory of graphons to binary exchangeable array processes. We introduce a strong cut norm for which an analog of the classical counting lemma holds, and exhibit a random sequence that left-converges in probability while its canonical step-function representatives fail to be Cauchy in probability in this norm. Furthermore, we discuss how one might possibly extend a suitable metric from binary arrays to arrays with more general state spaces.

4.2 Exchangeable Array Processes

Given a set A , let $\binom{A}{d}$ denote the set of all subsets of size d of A , and let $[n] := \{1, 2, \dots, n\}$. A d -dimensional (jointly) exchangeable array is a collection of random variables $(X_e)_{e \in \binom{\mathbb{N}}{d}}$ indexed by d -subsets of $\mathbb{N}$ whose distribution is invariant under simultaneous permutation of the coordinates. Namely, letting $\text{Sym}(\mathbb{N})$ be the set of finite permutations of $\mathbb{N}$ then

$$(X_{\sigma(e)})_{e \in \binom{\mathbb{N}}{d}} \stackrel{d}{=} (X_e)_{e \in \binom{\mathbb{N}}{d}}, \quad \forall \sigma \in \text{Sym}(\mathbb{N}),$$

where $\sigma(e) = \{\sigma(i_1), \dots, \sigma(i_d)\}$ for $e = \{i_1, \dots, i_d\}$.

If $(X_e)_{e \in \binom{\mathbb{N}}{d}}$ is an exchangeable array there are two natural Markov processes associated to it: a “labeled” chain L_n and an “unlabeled” one M_n . Informally, the labeled chain L_n is produced by just revealing the entries with maximal coordinate n at step n . Namely, the finite labeled chain L_n is just the “top left corner” of the array: the restriction of the array to entries corresponding to $\binom{[n]}{d}$. In this framework, the full array (X_e) plays the role of the projective limit of the labeled chain which often appears in Doob–Martin theory.

The unlabeled chain M_n is produced from the labeled one by “forgetting the order” of the rows, columns, et cetera and just keeping track of the set of values appearing in each row, column, and so on. More formally:

Definition 4.2.1 (Exchangeable Array Processes). Given an exchangeable array $(X_e)_{e \in \binom{\mathbb{N}}{d}}$ with entries in some Borel space S , define a graded Markov chain L_n whose state space is $E := \bigsqcup_{n \in \mathbb{N}} E_n$ for $E_n \subseteq S^{\binom{[n]}{d}}$ by

$$(L_n)_e := X_e \quad \forall e \in \binom{[n]}{d}.$$

Define an equivalence relation $\sim$ on E_n by saying $A \sim B$ if there exists some $\sigma \in \text{Sym}([n])$ so that $A_e = B_{\sigma(e)}$ for all $e \in \binom{[n]}{d}$, where $\sigma(e) = \{\sigma(i_1), \sigma(i_2), \dots, \sigma(i_d)\}$ just permutes all elements of the subset e by the same permutation. With this, define the unlabeled chain

$$M_n := [L_n]_{\sim}.$$

In this chapter, we will almost exclusively think of S as countable, and often finite. In order to apply Doob-Martin theory, we require each state in E_n to be reachable from the empty array: $\mathbb{P}(L_n = x) > 0$ for all $x \in E_n$. In particular, it suffices to choose $E_n = \text{supp}(L_n)$.

Note that L_n is trivially a Markov process in the sense that any sequence of random variables becomes a Markov process if the state space is just the entire history of the process. In general, projections of Markov chains are not Markovian, but in this case, exchangeability ensures that the Markov property is preserved.

Lemma 4.2.2 (Markov Property of the Unlabeled Chain). *When the state space S is countable, the process M_n is a Markov chain.*

Proof. First, note that the definition of an exchangeable array implies

$$\mathbb{P}(L_n = x) = \mathbb{P}(L_n = x') \quad \forall x \sim x'.$$

Moreover if $x \sim x'$ are related by $\sigma \in \text{Sym}(\mathbb{N})$ then again exchangeability implies

$$\begin{aligned} \mathbb{P}(M_{n+1} = [y]_{\sim}, L_n = x) &= \sum_{y' \sim y} \mathbb{P}(L_{n+1} = y', L_n = x) \\ &= \sum_{y' \sim y} \mathbb{P}(L_{n+1} = \sigma(y'), L_n = \sigma(x)) \\ &= \sum_{z \sim y} \mathbb{P}(L_{n+1} = z, L_n = x') \\ &= \mathbb{P}(M_{n+1} = [y]_{\sim}, L_n = x'). \end{aligned}$$

Thus we see

$$\mathbb{P}(M_{n+1} = [y]_{\sim} \mid L_n = x) = \mathbb{P}(M_{n+1} = [y]_{\sim} \mid L_n = x') \quad \forall x \sim x', \quad [y]_{\sim} \in [E_{n+1}]_{\sim}.$$

Thus the transition probability from x into the class $[y]_{\sim}$ depends only on the equivalence class $[x]_{\sim}$. The projected transition kernel is therefore well-defined on equivalence classes, and hence M_n is a Markov chain. $\square$

The above lemma is true for any initial distribution, but we are almost exclusively interested in the case where M_0 is the empty array. The main benefit of this framework is that exchangeable arrays share a strong structural property: they have a representation through sampling from some hypercube.

Theorem 4.2.3 (Representation Theorem). *An exchangeable array on d -subsets of $\mathbb{N}$ has a representation of the form*

$$(X_e)_{e \in \binom{\mathbb{N}}{d}} \stackrel{d}{=} \left(f \left(U_\emptyset, (U_i)_{i \in e}, (U_a)_{a \in \binom{e}{2}}, \dots, (U_a)_{a \in \binom{e}{d-1}}, U_e \right) \right)_{e \in \binom{\mathbb{N}}{d}},$$

for some middle-symmetric $f : [0, 1]^{2^d} \rightarrow S$ and i.i.d. uniform random variables. Moreover, any array of this form is exchangeable.

A middle-symmetric function satisfies the following for any $\sigma \in \text{Sym}([d])$

$$f \left(x_\emptyset, (x_i)_{i \in [d]}, (x_a)_{a \in \binom{[d]}{2}}, \dots, x_{[d]} \right) = f \left(x_\emptyset, (x_{\sigma(i)})_{i \in [d]}, (x_{\sigma(a)})_{a \in \binom{[d]}{2}}, \dots, x_{[d]} \right).$$

In words, the same permutation is applied coherently to the indices at every level. More details can be found in [5] or Chapter 7 of [41].

Throughout this chapter, we will encounter many domains of the form $[0, 1]^{2^k}$ for some k . For ease of notation, we will assume the coordinates are always indexed by subsets of $[k]$ as in the definition of a middle-symmetric function above.

In the case of one-dimensional arrays, this representation theorem reduces to de Finetti's theorem, which says an exchangeable sequence is a mixture of i.i.d. sequences and thus takes the form $(X_n) \stackrel{d}{=} (f(U, U_n))$. Moreover, two-dimensional arrays take the form $f(U, U_i, U_j, U_{ij})$ for some function f with $f(w, x, y, z) = f(w, y, x, z)$.

As [2] points out, this quantifies the fundamental relationship between exchangeability and sampling that is already so well known: if we can get a hold of the representing function f , it gives us a simple procedure for generating the corresponding combinatorial objects. Next, we point out a simple way of generating new exchangeable arrays from old ones.

Lemma 4.2.4 (State-Space Projections of Exchangeable Arrays). *If (X_e) is an exchangeable array with entries in S and $g : S \rightarrow S'$ is measurable, then $(g(X_e))$ is an exchangeable array of the same dimension. Moreover, if f is the representing function of (X_e) , then $g \circ f$ is the representing function of $(g(X_e))$.*

Proof. The result follows immediately from applying g coordinatewise. If (X_e) is exchangeable, then for each finite permutation σ ,

$$(g(X_{\sigma(e)}))_{e \in \binom{\mathbb{N}}{d}} \stackrel{\text{law}}{=} (g(X_e))_{e \in \binom{\mathbb{N}}{d}}.$$

Moreover, if $X_e = f(U_\emptyset, (U_i)_{i \in e}, \dots, U_e)$ in distribution, then $g(X_e)$ is represented by $g \circ f$. $\square$

Here we are essentially defining an equivalence relation on the entries of the array where we let $s \sim s'$ if $g(s) = g(s')$. We call this the *state-space equivalence*.

Combined with Definition 4.2.1, this gives three layers of Markov processes associated with an exchangeable array: (i) the initial labeled process corresponding to (X_e) , (ii) the labeled process corresponding to $(g(X_e))$, obtained by applying an equivalence to the state space, and (iii) the unlabeled chain, obtained by quotienting the coordinates. The ballot-sequence example below uses all three layers.

Example 4.2.5 (Pólya’s Urn). In Pólya’s urn, an urn begins with $\alpha_i \in \mathbb{N}$ balls of color i for $i \in [m]$. At each step, we sample a ball from the urn with replacement and add another ball of the same color as the sampled ball. It is well known that the sequence of balls drawn from a Pólya urn is an exchangeable sequence with values in $[m]$, which is just a one-dimensional exchangeable array. In this case, the labeled chain L_n is the sequence of selected balls, and the unlabeled chain M_n encodes how many balls of each color have been sampled. Classical theory implies this exchangeable array has representation function

$$f(x, y) = \sum_{i=1}^m i \cdot \mathbf{1}\{y \in [t_{i-1}, t_i)\}, \quad t_0 := 0, \quad t_i := \sum_{k=1}^i p_k(x).$$

where $(p_1(U), \dots, p_m(U)) \sim \text{Dirichlet}(\alpha_1, \dots, \alpha_m)$ for α_i initial balls of color i .

Example 4.2.6 (Graphon Chain). Given any graphon w there is a sequence of finite graphs G_n converging to w in the cut metric (equivalently the Doob-Martin topology by [33]) constructed via sampling $U_i \sim \text{Unif}[0, 1]$ and then connecting vertex i to vertex j with probability $w(U_i, U_j)$. Thus a graphon can be represented as an “infinite adjacency matrix” which forms an exchangeable array with entries

$$X_e = \mathbf{1}\{U_e \leq w(U_i, U_j)\} \quad \text{for } e = \{i, j\} \in \binom{\mathbb{N}}{2}.$$

Here the representation function $f(x_\emptyset, x_i, x_j, x_{ij})$ does not depend on $x_\emptyset$.

Example 4.2.7 (d -Uniform Hypergraph Chain). We can extend the graphon example to d -uniform hypergraphs, which have hyperedges between subsets of $[n]$ of size d , initially explored in [24]. Namely, define a *hypergraphon* as a middle-symmetric function $W : [0, 1]^{2^d - 2} \rightarrow [0, 1]$ with coordinates indexed by proper, nonempty subsets of $[d]$. Then, analogous to graphons, this has representation function

$$f(x_\emptyset, (x_i)_{i \in [d]}, \dots, x_{[d]}) = \mathbf{1}\left\{x_{[d]} \leq W\left((x_i)_{i \in [d]}, \dots, (x_a)_{a \in \binom{[d]}{d-1}}\right)\right\}.$$

As we will see in Corollary 4.3.7 below, these are in a sense the only binary array processes.

Example 4.2.8 (Ballot Sequences). Given some total ordering of $I_0 := \bigcup_{i \in \mathbb{N}} \{a_i, b_i\}$ so $a_i < b_i$ for each i , there are six possible “interweavings” of the letters

$$a_i b_i a_j b_j \quad a_i a_j b_i b_j \quad a_i a_j b_j b_i \quad a_j a_i b_i b_j \quad a_j a_i b_j b_i \quad a_j b_j a_i b_i.$$

This ordering can be encoded by a 2-dimensional array X_{ij} with values in $\{1, \dots, 6\}$ describing which of these relationships the pairs $(a_i, b_i), (a_j, b_j)$ share. If we equate the interweavings above which have the a ’s and b ’s in the same locations, the unlabeled chain generates *ballot sequences*: strings of a ’s and b ’s where every prefix has at least as many a ’s as b ’s. In [17], the boundary is characterized through a pair of diffuse marginals (ζ, μ) encoding the relative density of a ’s and b ’s in the limit.

Exchangeable array processes also arise naturally in the theory of exchangeable partitions through Kingman's paintbox theorem [5], random permutations through permutons [32], planar binary trees through Rémy's chain [27], and growing random strings [16]. Many additional examples of such combinatorial processes and their associated sampling representations can be found in [65].

These and the preceding examples give a common strategy for embedding combinatorial processes as random arrays. If a random structure consists of certain "parts" (e.g. vertices of a graph) and is completely determined by the relationships within each subset of d parts (e.g. which vertices are connected by an edge), then it can be encoded as a d -dimensional array whose entries encode the relationship. Exchangeability then enforces the constraint that the distribution of the structure is invariant under relabeling of the parts.

4.3 Doob-Martin Boundary of Exchangeable Array Processes

The backwards dynamics of the labeled chain are deterministic: given L_n you simply restrict to $e \in \binom{[n-1]}{d}$ to obtain L_{n-1} . Thus, its Doob-Martin boundary is almost trivial.

Lemma 4.3.1 (Boundary of the Labeled Chain). *The boundary of L_n consists of the deterministic arrays $x \in \text{supp}(X)$.*

Proof. Fix $x \in E_m$ and $y \in E_n$ with $n \geq m$. Since the backwards dynamics of the labeled chain are deterministic,

$$\mathbb{P}(L_m = x \mid L_n = y) = \mathbf{1}\{y|_{[m]} = x\}.$$

Consequently, the Doob-Martin kernel of the labeled chain is

$$K_L(x, y) = \frac{\mathbf{1}\{y|_{[m]} = x\}}{\mathbb{P}(L_m = x)}.$$

Thus a sequence (y_n) with $y_n \in E_n$ converges in the Doob-Martin topology if and only if, for every m , the restriction $y_n|_{[m]}$ is eventually constant. These limiting restrictions are consistent and therefore define a unique deterministic infinite array in $\text{supp}(X_e)$. Conversely, the finite restrictions of any deterministic array in $\text{supp}(X_e)$ form a convergent sequence with that boundary point as its limit. $\square$

Given the exchangeable array (X_e) and a measurable function $g : S \rightarrow S'$, we can generate a new exchangeable array $G(X)$ by applying g coordinatewise: $G(X)_e := g(X_e)$. Although the boundary points of both labeled chains are deterministic arrays by Lemma 4.3.1, state-space projection may collapse a nontrivial mixture of boundary points to a single projected boundary point. Indeed, any random array X supported on $G^{-1}(y)$ trivially has $G(X) = y$ almost surely. Consequently, the bridge laws of the original chain that project to y are precisely the probability measures supported on the fiber $G^{-1}(y)$.

4.3.1 Boundary of the Unlabeled Chain

Now we seek to understand the boundary of the unlabeled chain M_n . Recalling the correspondence (4.B.6) between infinite bridges of a Markov chain and its nonnegative harmonic functions via the Doob h-transform (4.B.5) discussed in Appendix 4.B, we obtain the following lemma.

Lemma 4.3.2 (Exchangeable Bridges of the Labeled Chain). *Any infinite bridge of the labeled exchangeable array process is an array process, and it is exchangeable if and only if the corresponding harmonic function h is constant on the equivalence classes of $\sim$.*

Proof. Suppose the harmonic $h \geq 0$ with $h(x_0) = 1$ is the one corresponding to this particular bridge. If L_n is the original labeled exchangeable array process and L_n^h is the infinite bridge

$$\mathbb{P}(L_0^h = x_0, \dots, L_n^h = x_n) = h(x_n) \cdot \mathbb{P}(L_0 = x_0, \dots, L_n = x_n).$$

The right hand side is zero if the states $x_0, \dots, x_n$ are not nested in the sense of arrays, $x_m = (x_n)|_{\binom{[m]}{d}}$ for $m < n$, and thus the same is true of the left hand side. Thus, by taking a projective limit, the chain L_n^h a.s. defines an array (X_e^h) of the same dimension as the original exchangeable array (X_e) of which it is the corresponding array process.

Now if $\sigma \in \text{Sym}(\mathbb{N})$ let $\tilde{x} := \sigma(x)$ be the array generated by permuting the coordinates of x . Then the definition of the Doob h-transform and the exchangeability of the original array imply

$$\begin{aligned} \mathbb{P}(L_0^h = \tilde{x}_0, \dots, L_n^h = \tilde{x}_n) &= h(\tilde{x}_n) \cdot \mathbb{P}(L_0 = \tilde{x}_0, \dots, L_n = \tilde{x}_n) \\ &= h(\tilde{x}_n) \cdot \mathbb{P}(L_0 = x_0, \dots, L_n = x_n) \\ &= \frac{h(\tilde{x}_n)}{h(x_n)} \cdot \mathbb{P}(L_0^h = x_0, \dots, L_n^h = x_n). \end{aligned}$$

These two are equal if and only if $h(\tilde{x}_n) = h(x_n)$. Since x_n and σ were arbitrary, the result follows from the fact that the finite dimensional distributions of our array (X_e^h) are determined from the distribution of (L_n^h) . $\square$

This immediately implies the harmonic functions $\tilde{h}$ of the unlabeled chain are in direct correspondence with the bridges of labeled chain which come from exchangeable arrays.

Corollary 4.3.3 (Bridges of the Unlabeled Chain). *If B_n is an infinite bridge of the unlabeled chain M_n , then B_n equals in distribution the unlabeled process associated to some exchangeable array (X_e^h) . In particular, if $\tilde{h}$ is the harmonic corresponding to the bridge B_n , then $h(x) := \tilde{h}([x])$ is the harmonic corresponding to the array (X_e^h) .*

Proof. The bridges of (M_n) are in correspondence with harmonic functions $\tilde{h}$ of this chain. Define $h(x) := \tilde{h}([x])$ on the labeled state space. Since the transition kernel of M_n is obtained by summing the transition probabilities of L_n over equivalence classes, harmonicity of $\tilde{h}$ gives

$$h(x) = \tilde{h}([x]) = \sum_{[y] \sim} p_M([x]_{\sim}, [y]_{\sim}) \tilde{h}([y]_{\sim}) = \sum_y p_L(x, y) h(y).$$

Thus h is a harmonic function for the labeled chain and is constant on the equivalence classes of $\sim$. Applying Lemma 4.3.2 gives the corollary. $\square$

The relation $\sim$ extends to infinite d -dimensional arrays x in the obvious way: $x \sim y$ if there is $\sigma \in \text{Sym}(\mathbb{N})$ so $x = \sigma(y)$. We say a measurable collection I of arrays is ν -invariant if permutations do not change it up to ν -null sets

$$I \stackrel{\nu\text{-a.s.}}{=} \tau I, \quad \forall \tau \in \text{Sym}(\mathbb{N}).$$

Say an exchangeable array with distribution ν is *ergodic* if $\nu(I) \in \{0, 1\}$ for all ν -invariant sets I , and say it is *dissociated* if for any disjoint index sets $I_1, \dots, I_m \subset \mathbb{N}$ the arrays $\{X_e\}_{e \in \binom{I_i}{d}}$, $i \in [m]$, are independent. For example, the exchangeable array corresponding to a graphon is dissociated: since edges are added independently, the random subgraphs corresponding to disjoint collections of vertices are independent in the graphon model. The following shows these properties characterize the minimal Doob–Martin boundary of the unlabeled chain.

Theorem 4.3.4 (Extremal Bridges and Ergodic Arrays). *For any infinite bridge of the unlabeled exchangeable array process M_n , the following are equivalent:*

1. *The infinite bridge is extremal.*
2. *Its corresponding exchangeable array is ergodic.*
3. *Its corresponding exchangeable array is dissociated.*
4. *Its corresponding exchangeable array admits a representation function f that does not depend on $U_\emptyset$ almost surely.*

Remark 4.3.5. By Example 4.2.6, given a graphon w the representation function

$$f(q, x, y, u) := \mathbf{1}\{u \leq w(x, y)\},$$

of the associated binary exchangeable array process does not depend on q . Hence, Theorem 4.3.4 implies graphons correspond to the extremal infinite bridges, and hence the minimal Doob–Martin boundary, of the graphon chain, as noted in [33].

Proof. The last three statements are equivalent by Lemma 7.35 of [41]. For the implication (2) $\rightarrow$ (1), suppose for contraposition that we have a nonextremal bridge $M_n^{\tilde{h}}$ associated to the harmonic $\tilde{h}$. Then equation (4.B.6) applied to the Doob–Martin kernel $\tilde{K}$ of our unlabeled chain implies

$$\mathbb{P}((M_n^{\tilde{h}}) = [x]_\sim) = \int_{\partial_m \tilde{E}} \mathbb{P}^{\tilde{K}(\cdot, \alpha)}([x]_\sim) \mu_{\tilde{h}}(d\alpha),$$

for some nondegenerate measure $\mu_{\tilde{h}}$ on the minimal boundary $\partial_m \tilde{E}$ of the unlabeled chain. By construction of the Doob-Martin boundary, the functions $\tilde{K}(\cdot, \alpha)$ are harmonic for the unlabeled chain. Thus by Corollary 4.3.3 the distributions $\mathbb{P}^{\tilde{K}(\cdot, \alpha)}$ correspond to the unlabeled chains associated to exchangeable arrays (X_e^α) . Likewise, $M_n^{\tilde{h}}$ corresponds to some exchangeable array $(X_e^{\tilde{h}})$ as well. Thus we can rephrase the previous equation in terms of the associated exchangeable arrays

$$\mathbb{P}((X_e^{\tilde{h}}) \in [x]_{\sim}) = \int_{\partial_m \tilde{E}} \mathbb{P}(X_e^\alpha \in [x]_{\sim}) \mu_{\tilde{h}}(d\alpha).$$

These probabilities are constant on the equivalence classes by exchangeability. Hence

$$\mathbb{P}((X_e^{\tilde{h}}) = x) = \int_{\partial_m \tilde{E}} \mathbb{P}(X_e^\alpha = x) \mu_{\tilde{h}}(d\alpha).$$

Thus as $\mu_{\tilde{h}}$ is nondegenerate, $(X_e^{\tilde{h}})$ is a nondegenerate mixture of other exchangeable arrays, and hence not ergodic.

For the implication (1) $\rightarrow$ (2), suppose for contraposition that the exchangeable array associated with a bridge M_n^α has a nonergodic law ν . Then there is a ν -invariant event I with $0 < \nu(I) < 1$. Define the conditional laws

$$\nu_1(\cdot) := \nu(\cdot \mid I), \quad \nu_0(\cdot) := \nu(\cdot \mid I^c).$$

Because I is invariant and ν is exchangeable, both ν_0 and ν_1 are exchangeable. Let $M_n^{(0)}$ and $M_n^{(1)}$ be their associated unlabeled processes.

We claim that both processes are infinite bridges of M_n . Indeed, under any exchangeable array law, conditioning on $M_{n+1} = [y]_{\sim}$ makes the labeled array at level $n+1$ uniform over $[y]_{\sim}$. Consequently,

$$\mathbb{P}_{\nu_i}(M_n = [x]_{\sim} \mid M_{n+1} = [y]_{\sim}) = \frac{|\{z \in [y]_{\sim} : z|_{[n]} \in [x]_{\sim}\}|}{|[y]_{\sim}|}, \quad i \in \{0, 1\}.$$

This backwards transition probability depends only on the equivalence classes $[x]_{\sim}$ and $[y]_{\sim}$, not on the particular exchangeable law. Hence $M_n^{(0)}$ and $M_n^{(1)}$ have the same backwards transitions as M_n^α , and therefore as the original chain M_n .

Finally, projection from a labeled array to its unlabeled process is affine, so

$$\mathcal{L}((M_n^\alpha)_{n \geq 0}) = \nu(I) \mathcal{L}((M_n^{(1)})_{n \geq 0}) + (1 - \nu(I)) \mathcal{L}((M_n^{(0)})_{n \geq 0}).$$

This mixture is nontrivial. Indeed, if the two unlabeled process laws were equal, exchangeability and uniformity within each equivalence class would imply that all their finite labeled distributions were equal, and hence that $\nu_0 = \nu_1$. This is impossible because $\nu_1(I) = 1$ whereas $\nu_0(I) = 0$. Thus M_n^α is not extremal, proving the contraposition. $\square$

As a consequence, every unlabeled bridge admits a representation function obtained by mixing the extremal representation functions $f^{\tilde{K}(\cdot, \alpha)}$, which do not depend on the first coordinate by the previous theorem. More specifically, if we let $g : [0, 1] \rightarrow \overline{E}$ be a function so that $g(U)$ has distribution $\mu_{\tilde{h}}$, we may choose

$$f^{\tilde{h}}(U, (U_i), \dots, U_e) = f^{g(U)}((U_i), \dots, U_e).$$

Of course, the representation f of a given exchangeable array need not be unique. This is even apparent in the theory of graphons, which are defined as isomorphism classes of functions $w : [0, 1]^2 \rightarrow [0, 1]$ modulo Lebesgue isomorphisms $\lambda : [0, 1] \rightarrow [0, 1]$: we equate $w(\cdot, \cdot) \sim w(\lambda(\cdot), \lambda(\cdot))$. The idea is that applying a Lebesgue isomorphism to a uniform does not change its distribution, and thus will not influence the overall sampling distribution. More generally, the representation functions of exchangeable arrays have a similar uniqueness result “up to Lebesgue-measure-preserving maps”.

The slight technicality is that we have to allow such Lebesgue-measure-preserving maps to depend on the preceding coordinates. Namely, say a map $T : [0, 1]^{2^d} \rightarrow [0, 1]$ *preserves* λ *in the last coordinate* if for fixed $x_\emptyset, (x_i)_{i \in [d]}, \dots, (x_e)_{e \in \binom{[d]}{d-1}}$ the map

$$y \mapsto T(x_\emptyset, (x_i)_{i \in [d]}, \dots, (x_e)_{e \in \binom{[d]}{d-1}}, y),$$

preserves the Lebesgue measure. For a proof of the following, see Theorem 7.28 in [41], which proves a more general statement that does not restrict the array to a finite dimension d . For something easier to parse, the case $d = 2$ can also be found in Theorem 2.1' of [40].

Theorem 4.3.6 (Uniqueness of Representation Functions). *The following are equivalent:*

1. $f, g : [0, 1]^{2^d} \rightarrow S$ generate the same exchangeable array (in distribution).
2. There exists measurable functions $T_i, T'_i : [0, 1]^{2^i} \rightarrow [0, 1]$ for $i = 0, \dots, d$ which preserve λ in the last coordinate so that a.s.

$$\begin{aligned} f\left(T_0(x_\emptyset), (T_1(x_\emptyset, x_i))_{i \in [d]}, \dots, T_d(x_\emptyset, (x_i)_{i \in [d]}, \dots, x_{[d]})\right) \\ = g\left(T'_0(x_\emptyset), (T'_1(x_\emptyset, x_i))_{i \in [d]}, \dots, T'_d(x_\emptyset, (x_i)_{i \in [d]}, \dots, x_{[d]})\right). \end{aligned}$$

In the $d = 2$ case, this exactly recovers the uniqueness statement for graphons discussed above. An immediate corollary of Theorem 4.3.6 is that, in a sense, the d -uniform hypergraph chain mentioned in the previous examples is the *only* binary exchangeable array process.

Corollary 4.3.7 (Binary Representation Functions). *Any binary exchangeable array admits a (not necessarily unique) representation function of the form*

$$f = \mathbf{1}\left\{x_{[d]} \leq w\left(x_\emptyset, (x_i)_{i \in [d]}, \dots, (x_e)_{e \in \binom{[d]}{d-1}}\right)\right\},$$

for some $w : [0, 1]^{2^d-1} \rightarrow [0, 1]$.

Again, we are primarily interested in the extremal representation functions, which depend trivially on $x_\emptyset$ by Theorem 4.3.4. Hence our hypergraphons can be viewed as functions

$$w : [0, 1]^{2^d-2} \rightarrow [0, 1],$$

indexed by proper nonempty subsets of $[d]$. This recovers the original notion introduced previously in [25] and whose relationship to exchangeability was further explored in [6].

4.4 Embedding Density of the Representation Functions

The idea of embedding a smaller structure into a larger one plays a pivotal role in the theory of combinatorial processes. Two of our principal examples, graphons and permutons, already come equipped with natural notions of embedding: subgraphs and subpermutations respectively. Somehow these embeddings encode the “local” structure of the larger combinatorial object. We can then compare two such objects by comparing these local structures at various different scales.

The aim of this section is to recast these notions in the language of exchangeable arrays, with particular emphasis on binary arrays, and to extend the framework developed in [10]. For a finite binary d -dimensional array F of size k , let

$$\mathbf{1}(F) := \left\{ S \in \binom{[k]}{d} : F_S = 1 \right\}.$$

denote the collection of entries of our array which are one. Moreover, given a vector $x := (x_e)_{e \subseteq [n]} \in [0, 1]^{2^n}$ and a subset $A \in \binom{[n]}{d}$ denote

$$x_{(A)} := \left(x_\emptyset, (x_i)_{i \in A}, (x_{ij})_{ij \in \binom{A}{2}}, \dots, x_A \right) \in [0, 1]^{2^d}.$$

Namely, if x is indexed by all subsets of $[n]$ then $x_{(A)}$ restricts to only subsets of A . Lastly, for a finite d -dimensional array of size k let $|x| := k$ denote its size.

4.4.1 Embeddings of Labeled States

Suppose we have arrays $(x_e)_{e \in \binom{[n]}{d}}$ and $(y_e)_{e \in \binom{[n+m]}{d}}$. Then we say x is *embedded* in y if $y_e = x_e$ for all $e \in \binom{[n]}{d}$ and denote this by $x \leq y$. Namely, we obtain x_e by looking at the “top left corner” of the array y_e . From the viewpoint of our labeled Markov chain L_n , this just means that the array (y_e) is reachable starting in state (x_e) . This extends the standard notion of an induced subgraph beyond binary arrays.

We say a binary array x is *weakly embedded* in binary array y , denoting this by $x \lesssim y$, if $y_e = 1$ whenever $x_e = 1$. This is the standard notion of a graph embedding.

4.4.2 Embeddings of Unlabeled States

Again suppose we have arrays $(x_e)_{e \in \binom{[n]}{d}}$ and $(y_e)_{e \in \binom{[n+m]}{d}}$. Say $[x]_\sim$ is embedded in $[y]_\sim$ if there exists $\sigma \in \text{Sym}([n+m])$ so $(x_e) \leq (y_{\sigma(e)})$. Equivalently, there exists an injective function $\varphi : [n] \rightarrow [n+m]$ so $(x_e) = (y_{\varphi(e)})$.

In this case, there can be more than one embedding if there are several different choices of σ that produce an embedding. This suggests a notion of *embedding density* analogous to the theory of graphons and permutons.

$$\begin{aligned} \text{emb}(x, y) &:= \frac{|\{\sigma \in \text{Sym}([n+m]) : (x_e) \leq (y_{\sigma(e)})\}|}{(n+m)!} \\ &= \frac{|\{\varphi : [n] \xrightarrow{\text{injective}} [n+m] : (x_e) \leq (y_{\varphi(e)})\}|}{(n+m)(n+m-1) \cdots (m+1)} \end{aligned} \quad (4.4.1)$$

for $x \in E_n, y \in E_{n+m}$. Note that the embedding density is constant on each equivalence class of x so this is well-defined as a function on these equivalence classes. For binary arrays, we analogously define the *weak embedding density* emb^W by replacing $\leq$ with $\lesssim$. The exact and weak embedding densities satisfy the following relationships.

Lemma 4.4.1 (Exact and Weak Embedding Densities). *For binary arrays x, y we have*

- (i) $\text{emb}(x, y) \leq \text{emb}^W(x, y)$.
- (ii) $\text{emb}^W(x, y) = \sum_{x' \geq x} \text{emb}(x', y)$.
- (iii) $\text{emb}(x, y) = \sum_{x' \geq x} (-1)^{|\mathbf{1}(x')| - |\mathbf{1}(x)|} \cdot \text{emb}^W(x', y)$.

Proof. These proofs extend analogous arguments from graphon theory in [55]. First, (i) is immediate because every embedding is a weak embedding. For (ii), note that if $\sigma \in \text{Sym}(\mathbb{N})$ is such that $x \lesssim \sigma(y)$, then there is precisely one $x' \geq x$ so that $x' \leq y$, namely x' equals the array $\sigma(y)$ restricted to $\binom{\{1, \dots, |x|\}}{d}$. For (iii), if for each e where $x_e = 0$ we let A_e be the set of weak embeddings of $x' = x + \delta_e$ into y , then

$$\{\text{embeddings of } x \text{ into } y\} = \{\text{weak embeddings of } x \text{ into } y\} \setminus \bigcup_{e : x_e = 0} A_e.$$

The formula follows from inclusion–exclusion. □

Given any notion $t(x, y)$ of the embedding density of x into y , we can topologize our collection of combinatorial objects: we simply say a sequence y_n converges if

$$\lim_{n \rightarrow \infty} t(x, y_n) \text{ exists}$$

for every fixed x . Thus Lemma 4.4.1 shows that exact and weak embedding densities generate the same topology for binary arrays. Such convergence and its associated topology is often referred to as *left-convergence* in the graphon literature. In fact, left-convergence is closely related to the Doob-Martin topology.

Lemma 4.4.2 (Doob-Martin Convergence and Embedding Densities). *Let $y_N \in E_N$. The following are equivalent:*

- (i) *The sequence y_N converges in the Doob-Martin topology.*
- (ii) *The embedding density $\text{emb}(x, y_N)$ converges for all x .*

For binary arrays, convergence of the weak embedding density is also equivalent.

Proof. Exchangeability implies

$$\mathbb{P}(L_n = x \mid M_n = [x]_\sim) = \frac{\mathbb{P}(L_n = x)}{\mathbb{P}(M_n = [x]_\sim)} = \frac{1}{|[x]_\sim|}.$$

As L_n has deterministic backwards dynamics, this implies the unlabeled chain M_n has

$$\mathbb{P}(M_n = [x]_\sim \mid M_{n+m} = [y]_\sim) = \frac{|\{z \in [y]_\sim : z|_{[n]} \in [x]_\sim\}|}{|[y]_\sim|} \quad x \in E_n, y \in E_{n+m},$$

where $z|_{[n]}$ is the restriction of z to indices in $\binom{[n]}{d}$. Since any permutation of $[n]$ extends to a permutation of $[n+m]$, the number of elements $z \in [y]_\sim$ satisfying $z|_{[n]} = \tilde{x}$ is the same for every $\tilde{x} \in [x]_\sim$. Thus the above equals

$$= \frac{|\{z \in [y]_\sim : z|_{[n]} = x\}|}{|[y]_\sim|} \cdot |[x]_\sim| = \text{emb}(x, y) \cdot |[x]_\sim|.$$

As a result, we have Doob-Martin kernel for the unlabeled chain

$$K([x]_\sim, [y]_\sim) = \frac{\text{emb}(x, y) |[x]_\sim|}{\mathbb{P}(M_n = [x]_\sim)} = \frac{\text{emb}(x, y)}{\mathbb{P}(L_n = x)}.$$

Hence the embedding density and Doob-Martin kernel differ only up to a constant depending on x . Thus for fixed x , one converges if and only if the other does. For binary arrays, Lemma 4.4.1 shows that convergence of all exact embedding densities is equivalent to convergence of all weak embedding densities. $\square$

In the case of our graphon and permuton chains, the notion of embedding can be extended to the limiting object, graphons and permutons, via sampling. Namely, for a graphon w its embedding density $t(\cdot, w)$ is essentially defined so that for each n and finite graph F

$$t(F, w) = \mathbb{E} t(F, G_n(w)),$$

where $G_n(w)$ is the random graph generated by sampling n uniforms $U_1, \dots, U_n$ and connecting vertices i, j with probability $w(U_i, U_j)$.

Analogously, we would like to extend this notion of an embedding density to our limiting objects: the representation functions f of our exchangeable arrays. This suggests

$$\text{emb}(x, f) := \mathbb{E} \text{emb}(x, X_n(f)), \quad \text{emb}^W(x, f) := \mathbb{E} \text{emb}^W(x, X_n(f)), \quad (4.4.2)$$

where $X_n(f)$ is the array indexed by $\binom{[n]}{d}$ generated by

$$(X_n(f))_e := f(U_\emptyset, (U_i)_{i \in e}, \dots, U_e).$$

We will now show these expectations are independent of n and have an integral representation analogous to the embedding density of graphons.

Lemma 4.4.3 (Embedding Densities of Representation Functions). *Suppose $d \leq k \leq n$. If F is a size- k , d -dimensional array and f represents an exchangeable array X , then, independently of n ,*

$$\text{emb}(F, f) = \int_{[0,1]^{2^k}} \prod_{S \in \binom{[k]}{d}} \mathbf{1}\{f(x_{(S)}) = F_S\} d\lambda_{2^k}. \quad (4.4.3)$$

If F and f are binary-valued, then also

$$\text{emb}^W(F, f) = \int_{[0,1]^{2^k}} \prod_{S \in \mathbf{1}(F)} f(x_{(S)}) d\lambda_{2^k}. \quad (4.4.4)$$

Moreover, if F and f are binary-valued and f does not depend on $x_\emptyset$, we have

$$\begin{aligned} \mathbb{P}(|\text{emb}^W(F, X_n(f)) - \text{emb}^W(F, f)| > \epsilon) &\leq 2 \exp\left(-\frac{\epsilon^2}{2k^2}n\right), \\ \mathbb{P}(|\text{emb}(F, X_n(f)) - \text{emb}(F, f)| > \epsilon) &\leq 2 \exp\left(-\frac{\epsilon^2}{2k^2}n\right). \end{aligned}$$

Thus,

$$\lim_{n \rightarrow \infty} \text{emb}(F, X_n(f)) \stackrel{\text{a.s.}}{=} \text{emb}(F, f),$$

and

$$\lim_{n \rightarrow \infty} \text{emb}^W(F, X_n(f)) \stackrel{\text{a.s.}}{=} \text{emb}^W(F, f).$$

In particular, the results of Lemma 4.4.1 (i)–(iii) extend to $\text{emb}^W(F, f)$ and $\text{emb}(F, f)$.

Proof. First we prove the expectation formulas, starting with emb^W . Let $\varphi : [k] \rightarrow [n]$ be injective. Then

$$\mathbb{P}(\varphi \text{ is an array embedding}) = \int_{[0,1]^{2^n}} \prod_{S \in \mathbf{1}(F)} f(x_{(\varphi(S))}) d\lambda_{2^n}$$

By the injectivity of φ we can apply a change of coordinates,

$$= \int_{[0,1]^{2^n}} \prod_{S \in \mathbf{1}(F)} f(x_{(S)}) d\lambda_{2^n}$$

After integrating out all the superfluous coordinates,

$$= emb^W(F, f).$$

So, writing $emb^W(F, X_n(f))$ in terms of the indicator functions $\mathbf{1}_{\{\varphi \text{ is an array embedding}\}}$,

$$\mathbb{E}[emb^W(F, X_n(f))] = emb^W(F, f).$$

The proof for emb is identical.

For the concentration results, we follow exactly the proof of Theorem 2.5 in [51], except that we choose a different filtration meant to capture the randomness available at different levels of a multidimensional array. Namely, define filtration

$$\mathcal{F}_m := \sigma(\{U_A : A \subseteq [n], \max A \leq m\}), \quad \mathcal{F}_0 := \sigma(U_\emptyset).$$

This can be used to define a Doob martingale

$$B_m := \mathbb{E}(emb^W(F, X_n(f)) \mid \mathcal{F}_m).$$

Note that

$$B_0 = \mathbb{E}(emb^W(F, X_n(f)) \mid U_\emptyset) = emb^W(F, f_{U_\emptyset}), \quad B_n = emb^W(F, X_n(f)).$$

Here $f_{U_\emptyset}$ is the (random) representation function

$$f_{U_\emptyset}(u) := f(U_\emptyset, (u_i)_{i \in [d]}, \dots, u_{[d]}).$$

If f does not depend on $U_\emptyset$, then $f_{U_\emptyset} = f$ is deterministic. Bounding the increments using a bound on the number of injective functions $\varphi : [k] \rightarrow [n]$ and applying Azuma's inequality as in [51] gives the desired inequalities. Then apply an identical procedure for emb .

The convergence then follows immediately from Borel-Cantelli. $\square$

This is analogous to the graphon identity

$$\mathbb{E} emb^W(F, G_n(w)) = \int_{[0,1]^k} \prod_{\{i,j\} \in \mathbf{1}(F)} w(x_i, x_j) d\lambda_k(x).$$

To see why, note when $d = 2$

$$\int_{[0,1]^{2k}} \prod_{S \in \mathbf{1}(F)} f(x_S) d\lambda_{2k} = \int_{[0,1]^{2k}} \prod_{\{i,j\} \in \mathbf{1}(F)} f(x_\emptyset, x_i, x_j, x_{ij}) d\lambda_{2k}.$$

For graphons the terms inside the product become $\mathbf{1}\{x_{ij} \leq w(x_i, x_j)\}$ so we can integrate out the coordinate $x_\emptyset$ and the coordinates x_{ij} (which each only appear in one term of the product) to get the familiar formula.

In general, we can see that the integral in (4.4.4) only relies on the $\sum_{i=0}^d \binom{k}{i}$ coordinates corresponding to subsets $e \subseteq [k]$ of size at most d , so many of the coordinates in the integral are superfluous. We leave them in for simplicity of notation.

Lastly, we will show one strategy for embedding a *finite* binary exchangeable array z into the space of representation functions. A natural candidate is just a continuous version of the hypergraph adjacency matrix, analogous to how we associate a graphon to a finite graph

$$f_z(u_\emptyset, (u_i)_{i \in [d]}, \dots, u_{[d]}) := \mathbf{1} \{ \{ \lceil |z| u_i \rceil : i \in [d] \} \in \mathbf{1}(z) \}. \quad (4.4.5)$$

We call f_z the *sampling function* associated to a finite array z . Sampling from f_z chooses vertices of z independently and hence with replacement, whereas the embedding density samples without replacement. The resulting discrepancy is controlled by the probability of a collision, and vanishes asymptotically.

Lemma 4.4.4 (Sampling Functions Approximate Weak Embedding Densities). *Let F and z be finite binary d -dimensional arrays with $|F| \leq |z|$. Then*

$$|\text{emb}^W(F, f_z) - \text{emb}^W(F, z)| \leq \frac{1}{|z|} \binom{|F|}{2}.$$

Proof. Let $\Phi : [|F|] \rightarrow [|z|]$ be a uniformly random map. By the definition of f_z and the integral formula for weak embedding densities, $\text{emb}^W(F, f_z)$ is the probability that

$$\{\Phi(i) : i \in S\} \in \mathbf{1}(z) \quad \text{for every } S \in \mathbf{1}(F).$$

Conditional on Φ being injective, it is a uniformly random injection, so the conditional probability of this event is $\text{emb}^W(F, z)$. The difference between the unconditional and conditional probabilities is therefore at most

$$\mathbb{P}(\Phi \text{ is not injective}) \leq \frac{1}{|z|} \binom{|F|}{2},$$

where the last inequality is the union bound over pairs of indices in $[|F|]$. □

4.5 Conclusions and Future Directions

This chapter provides a framework for viewing a broad class of combinatorial growth processes through exchangeable arrays. Using the classical Aldous–Hoover–Kallenberg representation theorem for such arrays, we can identify such arrays with sampling functions on some hypercube, generalizing the familiar graphon picture. Applying the Doob–Martin boundary theory, we show that the points of the minimal boundary can equivalently be viewed as ergodic arrays or through representation functions that do not depend on the first coordinate $x_\emptyset$.

We define exact and weak notions of embedding one finite array into another and show that they generate the same topology for binary arrays. Using this, we show that this probabilistic viewpoint of the limit objects is equivalent to the more combinatorial notion of left-convergence built from these embedding densities, readily generalizing the graphon literature. In this way, we extend the relationship between the classical Doob-Martin boundary and these embedding densities previously explored in [32, 33]. This identifies a common compactified limit space, a feature that is far from obvious in the original construction of [51].

Our work so far omits one key success of graphon theory: the cut metric. For graphons, this metric metrizes the topology of left-convergence. It has a direct combinatorial interpretation, controlling discrepancies in edge densities across all pairs of vertex subsets and, through the counting lemma, the densities of every fixed finite graph. In higher dimensions and for nonbinary arrays, however, the analogous metric theory appears more delicate. In Appendix 4.A, we record some preliminary directions. In particular, we discuss a natural extension to hypergraphs and prove a corresponding hypergraph counting lemma. We then give a random sequence that left-converges in probability while its canonical representatives fail to be Cauchy in probability in the resulting strong cut norm. Furthermore, we discuss some ways one might be able to extend any such metric from these hypergraphs (e.g. binary exchangeable arrays) to exchangeable arrays with more general state spaces via a one-hot encoding.

4.A Candidate Cut Metrics for Exchangeable Array Processes

As Lemma 4.4.2 and [33] show, graphons and the Doob-Martin theory are topologically equivalent. While the latter already comes equipped with a metric, this metric is not naturally interpretable in terms of the underlying combinatorial process. The real success of the theory of graphons was the introduction of an alternative metric, the cut metric, which is reasonably interpretable, explicitly encodes graph-theoretic properties, and allows for exact computation of many quantities of interest.

This appendix records preliminary progress towards a cut-type metric for exchangeable array processes. First, we discuss the binary case, where d -dimensional arrays are closely related to d -uniform hypergraph chains. Then we describe a possible extension to nonbinary arrays by projecting the state space onto a collection of binary arrays.

We saw in the above example that the graphon w of a graphon chain is closely related to the representation function f of its associated exchangeable array. Namely,

$$f(q, x, y, u) = \mathbf{1}\{u \leq w(x, y)\}.$$

Hence we hope to reinterpret the cut metric in terms of a metric on these representation functions f , and from there find a way to extend this metric to arbitrary exchangeable array processes. For fixed graphon representatives, recall the cut norm distance

$$\|w - w'\|_{\square} := \sup_{0 \leq g, h \leq 1} \left| \int_{[0,1]^2} g(x)h(y)(w(x, y) - w'(x, y)) dx dy \right|.$$

This suggests a natural generalization to exchangeable array processes. Given a hypergraphon $W : [0, 1]^{2^{d-2}} \rightarrow [0, 1]$ associated with a d -dimensional binary exchangeable array, define

$$\|W\|_{\square^{d-1}} := \sup_{g_1, \dots, g_d} \left| \int_{[0,1]^{2^{d-2}}} W(x) \prod_{i=1}^d g_i((x_A)_{\emptyset \neq A \subseteq [d] \setminus \{i\}}) d\lambda_{2^{d-2}}(x) \right|,$$

where the supremum is over measurable functions $g_i : [0, 1]^{2^{d-1}-1} \rightarrow [0, 1]$.

While this form may seem strange at first, it arises naturally from attempting to extend the classical counting lemma to hypergraphons.

Lemma 4.A.1 (Counting Lemma). *Let F be a finite binary d -array of size $k \geq d$, and let W, W' be d -uniform hypergraphons. Then*

$$|\text{emb}^W(F, W) - \text{emb}^{W'}(F, W')| \leq |\mathbf{1}(F)| \|W - W'\|_{\square^{d-1}}.$$

Proof. Write $x_{(<S)} := (x_A)_{\emptyset \neq A \subseteq S}$. By the integral representation of the hypergraphon embedding density,

$$\text{LHS} = \left| \int_{[0,1]^{2^k}} \prod_{S \in \mathbf{1}(F)} W(x_{(<S)}) - \prod_{S \in \mathbf{1}(F)} W'(x_{(<S)}) d\lambda_{2^k} \right|$$

Enumerate $\mathbf{1}(F) = \{S_1, S_2, \dots, S_\ell\}$. Interpolating between the products one factor at a time and applying the triangle inequality gives

$$\leq \sum_{m=1}^{\ell} \left| \int_{[0,1]^{2k}} \left(\prod_{i=1}^{m-1} W(x_{(<S_i)}) \right) (W(x_{(<S_m)}) - W'(x_{(<S_m)})) \left(\prod_{i=m+1}^{\ell} W'(x_{(<S_i)}) \right) d\lambda_{2k} \right|.$$

For each summand, condition on all coordinates not indexed by nonempty proper subsets of S_m . Grouping the remaining factors according to their intersections with S_m puts the inner integral in the form defining $\|W - W'\|_{\square^{d-1}}$. Each summand is therefore at most this cut norm, and summing over the $\ell = |\mathbf{1}(F)|$ factors proves the result. $\square$

As in the case of graphons, a hypergraphon representation is not unique, so we minimize over possible representations

$$d_{\square}(W, W') := \inf_{V \sim W, V' \sim W'} \|V - V'\|_{\square^{d-1}}.$$

Here $V \sim W$ means that V and W generate the same exchangeable-array law. The counting lemma shows that convergence in the quotient distance $d_{\square}$ implies convergence in the topology generated by embedding densities, and hence also in the Doob-Martin topology by Lemma 4.4.2. The converse fails for the strong cut norm on a fixed choice of representatives: left-convergence need not make the canonical step representatives Cauchy in this norm. The following example, adapted from [88], illustrates this distinction.

Example 4.A.2. Let $(G_n)_{n \geq 1}$ be independent Erdős–Rényi graphs with $G_n \sim G(n, 1/2)$. Define a sequence of three-dimensional hypergraphs H_n by taking the triangles of G_n as hyperedges. Then, for any fixed three-dimensional hypergraph F ,

$$\text{emb}^W(F, H_n) \xrightarrow{p} 2^{-|\partial F|},$$

where ∂F is the set of unordered pairs of vertices contained in some hyperedge of F . Hence H_n left-converges in probability to the hypergraphon $W(x) = \mathbf{1}\{x_{12}, x_{13}, x_{23} \leq 1/2\}$. Let W_n be the step hypergraphon associated with H_n . Then

$$W_n(u_1, u_2, u_3, u_{12}, u_{13}, u_{23}) = A_n(u_1, u_2)A_n(u_1, u_3)A_n(u_2, u_3),$$

where $A_n : [0, 1]^2 \rightarrow \{0, 1\}$ is the graphon associated to G_n , namely

$$A_n(x, y) = \mathbf{1}\{(\lceil nx \rceil, \lceil ny \rceil) \in E(G_n)\}.$$

Hence, for $m > n$,

$$\|W_n - W_m\|_{\square^2} \geq \left| \int_{[0,1]^3} (W_n - W_m)A_n(x, y)A_n(x, z)A_n(y, z) dx dy dz \right|.$$

Note

$$\int_{[0,1]^3} W_n A_n(x, y) A_n(x, z) A_n(y, z) dx dy dz = emb^W(K_3, G_n) \xrightarrow{p} \frac{1}{8}.$$

Similarly, independence of G_n and G_m gives, as $n, m \rightarrow \infty$,

$$\int_{[0,1]^3} W_m A_n(x, y) A_n(x, z) A_n(y, z) dx dy dz = \frac{1}{64} + o_p(1),$$

since, outside a set of measure $O(n^{-1} + m^{-1})$, the integrand checks for three distinct edges in each of the two independent graphs. Hence $\|W_n - W_m\|_{\square^2} \geq 7/64 + o_p(1)$, so these canonical representatives are not Cauchy in probability in $\|\cdot\|_{\square^2}$.

This calculation concerns the displayed representatives W_n and does not by itself rule out convergence after minimizing over equivalent representations in the quotient distance $d_{\square}$.

The paper [88] attempts to circumvent this issue by replacing convergence in the strong cut norm with a weaker notion of *partitionable convergence*, based on compatible hierarchies of weak regularity partitions of lower-order faces. This notion admits subsequential limits and implies convergence of homomorphism densities. Since both the Doob-Martin and left-convergence constructions yield compact limit spaces, this is one step in the right direction. However, [88] still leaves the converse conjectural.

4.A.1 Cut metric for nonbinary arrays

In this section, we provide partial progress towards extending the cut norm beyond hypergraphons to nonbinary arrays. The idea is to characterize a nonbinary array in terms of a series of projections onto the space of binary arrays, and apply the existing cut metric to each projection separately.

First, we need to extend the notion of a weak embedding to the nonbinary case. For a subset of states $C \subseteq S$ define a C -weak embedding of x into y to be an injective map $\varphi : [|x|] \rightarrow [|y|]$ which preserves the states in C . Namely, if $s \in C$ then $x_e = s$ implies $y_{\varphi(e)} = s$. In the case $C = \{1\}$ and $S = \{0, 1\}$ this reduces to the standard notion of weak embeddings of a binary array. Denote the set of such embeddings by $\text{Emb}_C(x, y)$ and their density by $\text{emb}^C(x, y)$. Then we have the following extension of Lemmas 4.4.1 and 4.4.2.

Lemma 4.A.3 (Weak Embeddings for Nonbinary Arrays). *For finite S and any $C \subseteq S$:*

$$(i) \text{ emb}^C(x, y) = \sum_{x' \geq_C x} \text{emb}(x', y).$$

$$(ii) \text{ emb}(x, y) \leq \text{emb}^C(x, y).$$

Moreover, if $C = S \setminus \{s\}$ then:

$$(iii) \text{ emb}(x, y) = \sum_{x' \geq_C x} (-1)^{N_s(x) - N_s(x')} \cdot \text{emb}^C(x', y).$$

where $x' \geq_C x$ means x' can be obtained from x by changing some of the values x_e with $x_e \notin C$ and $N_s(x) := |\{e : x_e = s\}|$ is the number of entries of x which are s .

In particular, this shows that convergence of $\text{emb}(\cdot, y_N)$ implies convergence of $\text{emb}^C(\cdot, y_N)$ for any C , and the converse is true if $C = S \setminus \{s\}$.

Proof. Write $\text{Emb}(x, y)$ and $\text{Emb}_C(x, y)$ for the sets of embeddings and C -weak embeddings, respectively.

For (i), let φ be a C -weak embedding of x into y . There is exactly one array $x' \geq_C x$ for which φ is an embedding of x' into y : set $x'_e = x_e$ when $x_e \in C$, and set $x'_e = y_{\varphi(e)}$ when $x_e \notin C$. Thus the set of C -weak embeddings is the disjoint union of the sets of embeddings of $x' \geq_C x$ into y . Dividing by the number of injective maps $[|x|] \rightarrow [|y|]$ proves (i).

For (ii), every embedding of x into y is automatically a C -weak embedding. Consequently, the embedding set is contained in the set of C -weak embeddings. Dividing their cardinalities by the common number of injective maps gives

$$\text{emb}(x, y) \leq \text{emb}^C(x, y).$$

For (iii), suppose $C = S \setminus \{s\}$ and let $E_s(x) = \{e : x_e = s\}$. For $e \in E_s(x)$ and $t \in C$, let $A_{e,t}$ be the set of C -weak embeddings φ for which $y_{\varphi(e)} = t$. An exact embedding of x is a C -weak embedding that belongs to none of these sets. Hence inclusion-exclusion gives

$$|\text{Emb}(x, y)| = \sum_{x' \geq_C x} (-1)^{N_s(x) - N_s(x')} |\text{Emb}_C(x', y)|.$$

Indeed, choosing an intersection of the events $A_{e,t}$ amounts precisely to choosing the positions at which an entry s of x is replaced, along with its replacement state $t \in C$; the resulting array is the corresponding $x' \geq_C x$. Dividing by the common number of injective maps proves (iii). $\square$

There is a simple way of encoding a nonbinary array with finite state space S as a collection of binary arrays of the same dimension: simply apply a one-hot encoding to the state space. More specifically, for each $A \subseteq S$ define the function $g_A : S \rightarrow \{0, 1\}$ by $g_A(s) = \mathbf{1}\{s \in A\}$. Thus each d -dimensional array with values in S is encoded as $2^{|S|}$ binary arrays of the same size. First, we show there is a relationship between embeddings in the original state space and embeddings of the corresponding binary arrays.

Lemma 4.A.4 (Binary Projections of Nonbinary Arrays). *For d -dimensional arrays x, y with values in some finite set S ,*

$$\text{Emb}(x, y) = \bigcap_{s \in S} \text{Emb}(g_{\{s\}}(x), g_{\{s\}}(y)).$$

Moreover, if y_N left-converges to f , then $g_A(y_N)$ left converges to $g_A \circ f$ for all $A \subseteq S$.

Proof. Suppose $\varphi : [|x|] \rightarrow [|y|]$ is injective. Then $x_e = y_{\varphi(e)}$ if and only if $g_{\{s\}}(x_e) = g_{\{s\}}(y_{\varphi(e)})$ for every $s \in S$, proving the first statement.

For the second, fix a binary array b and let

$$\mathcal{Y}_A^b := \left\{ x \in S^{\left(\begin{smallmatrix} |b| \\ d \end{smallmatrix}\right)} : b_e = 1 \Rightarrow x_e \in A \right\}.$$

Every weak embedding of b into $g_A(y_N)$ induces a unique exact array $x \in \mathcal{Y}_A^b$, so

$$\text{emb}^W(b, g_A(y_N)) = \sum_{x \in \mathcal{Y}_A^b} \text{emb}(x, y_N).$$

Since S is finite and y_N left-converges to f , this converges to

$$\sum_{x \in \mathcal{Y}_A^b} \text{emb}(x, f) = \text{emb}^W(b, g_A \circ f).$$

$\square$

We conjecture that the converse is true as well.

Conjecture 4.A.5. *If $g_A(y_N)$ left-converges to $g_A \circ f$ for all $A \subseteq S$, then y_N left-converges to f .*

Given an exchangeable array process (X_e) represented by f , Lemma 4.2.4 implies the array process $(g_A(X_e))$ is a binary exchangeable array process represented by $g_A \circ f$. Thus if f, f' share the same state space S , if the given conjecture is true then it is natural to define

$$d(f, f') := \frac{1}{2^{|S|}} \sum_{A \subseteq S} d(g_A \circ f, g_A \circ f'). \quad (4.A.1)$$

4.B Background on Doob-Martin Boundary Theory

Here we aim to provide some of the essentials of Doob-Martin boundary theory. We restrict our attention to the case of discrete Markov chains relevant to this chapter. The primary reference on this topic is [20] but [71] and [27] also provide good overviews for the discrete case in more modern terminology.

Given a transient Markov chain $(X_n)_{n \in \mathbb{N}}$ with a countable state space E along with a base state $x_0 \in E$ from which the chain is irreducible, some of the goals of Martin boundary theory are:

- (i) Complete E into a compact metric space by adding “points at ∞ ” to which the Markov chain is converging.
- (ii) Characterize all the nonnegative/bounded harmonic functions $h : E \rightarrow [0, \infty)$.
- (iii) Develop a notion of conditioning the chain “at infinity” and understanding the behavior of these conditioned chains.

Suppose $(X_n)_{n \in \mathbb{N}}$ has transition kernel p . Define the *Green’s kernel* or *potential kernel* $g : E \times E \rightarrow \mathbb{R}_{\geq 0}$ by

$$g(x, y) := \sum_{n \geq 0} p^n(x, y) = \mathbb{E}_x(\text{number of times } X_n \text{ hits } y). \quad (4.B.1)$$

Transience implies $g(x, y) < \infty$. Assume further there exists a base state x_0 from which every state is accessible, namely $g(x_0, y) > 0$ for all $y \in E$. For this chapter, x_0 will typically be the empty array. Moreover, we will typically be interested in *graded* chains. These are Markov chains in which the state space $E = \bigsqcup E_n$ can be partitioned so transitions occur only from E_n to E_{n+1} . Here

$$g(x, y) = \mathbb{P}_x(X_n \text{ hits } y).$$

Define the *Doob-Martin kernel* $K : E \times E \rightarrow \mathbb{R}_{\geq 0}$ with respect to base state x_0 via

$$K(x, y) := \frac{g(x, y)}{g(x_0, y)} = \frac{\mathbb{P}_x(X_n \text{ hits } y)}{\mathbb{P}_{x_0}(X_n \text{ hits } y)}. \quad (4.B.2)$$

Observe for each fixed y the function $K(\cdot, y)$ is “almost harmonic” in the sense

$$\sum_{z \in E} p(x, z) K(z, y) = K(x, y) \quad x \neq y.$$

This suggests if we let y “go off to infinity” we ought to get a truly harmonic function in the limit. The Doob-Martin boundary will make this intuition precise.

Note for fixed x , the kernel $K(x, y)$ is a bounded function of y . To see why, note accessibility implies $p^m(x_0, x) > 0$ for some m , and the Markov property gives $g(x_0, y) \geq p^m(x_0, x)g(x, y)$. Thus if $g(x, y) = 0$, then $K(x, y) = 0$; otherwise,

$$0 \leq K(x, y) = \frac{g(x, y)}{g(x_0, y)} \leq \frac{1}{p^m(x_0, x)} =: C_x.$$

Hence we can view the collection of functions $\{K(\cdot, y)\}_{y \in E}$ as a subset of the product space

$$\{K(\cdot, y)\}_{y \in E} \subseteq \bigtimes_{x \in E} [0, C_x] \subseteq \mathbb{R}_{\geq 0}^E,$$

so $\{K(\cdot, y)\}_{y \in E}$ is precompact under the product topology. For the graded chains considered here, the Martin kernel separates states, so we can identify E with $\{K(\cdot, y)\}_{y \in E}$. The closure $\bar{E} \subseteq \mathbb{R}_{\geq 0}^E$ under this identification is compact. We call $\bar{E}$ the *Doob-Martin compactification* of E and its boundary $\partial E := \bar{E} \setminus E$ the *Doob-Martin boundary*.

In summary, the map $y \mapsto K(\cdot, y)$ embeds E into $\mathbb{R}_{\geq 0}^E$, and the closure of its image in the product topology defines the Doob-Martin compactification. Since E is countable, this topology is metrizable. For example, after enumerating $E = \{x_1, x_2, \dots\}$, a compatible metric is

$$d(y, z) := \sum_{i=1}^{\infty} 2^{-i} \min \{1, |K(x_i, y) - K(x_i, z)|\}.$$

Remark 4.B.1. By construction, a sequence $\{y_n\} \in E$ converges to a point $\alpha \in \bar{E}$ if and only if $\{K(\cdot, y_n)\}$ converges in $\mathbb{R}_{\geq 0}^E$, namely if and only if $K(x, y_n)$ converges in $\mathbb{R}_{\geq 0}$ for all $x \in E$. Thus for fixed x the function $K(x, \cdot) : E \rightarrow \mathbb{R}_{\geq 0}$ extends continuously to $\bar{E}$.

There is a close relationship between the Doob-Martin kernel K and the backwards transition probabilities of the chain. Namely, for $x \in E_n, y_N \in E_N$ Bayes rule implies

$$K(x, y_N) = \frac{\mathbb{P}(X_N = y_N \mid X_n = x)}{\mathbb{P}(X_N = y_N)} = \frac{\mathbb{P}(X_n = x \mid X_N = y_N)}{\mathbb{P}(X_n = x)}.$$

Thus $K(x, y_N)$ converges as $N \rightarrow \infty$ for all x if and only if the conditional probabilities $\mathbb{P}(X_n = x \mid X_N = y_N)$ converge for all x .

4.B.1 Minimal Boundary and Representation of Harmonics

Given a Markov chain X_n on state space E , we say a function $h : E \rightarrow \mathbb{R}$ is *harmonic* if

$$h(x) = \mathbb{E}[h(X_{n+1}) \mid X_n = x].$$

Note $K(\cdot, \alpha)$ is harmonic for every $\alpha \in \partial E$. The *minimal boundary* is the subset

$$\partial_m E := \{\alpha \in \partial E : K(\cdot, \alpha) \text{ is minimal harmonic}\}. \quad (4.B.3)$$

A *minimal harmonic* function is a harmonic function $u \geq 0$ so for any other harmonic w satisfying $0 \leq w \leq u$ we must have $w = cu$ for some constant $c \geq 0$.

The set $\mathbb{H}_{1,+}$ of nonnegative harmonics with $h(x_0) = 1$ is a compact, convex set whose extreme points are precisely the minimal harmonics. Clearly $K(\cdot, \alpha) \in \mathbb{H}_{1,+}$, so the Doob-Martin boundary can be thought of as a subset of $\mathbb{H}_{1,+}$, and the minimal boundary is precisely the set of its extreme points. Any nonnegative harmonic function u can be represented as an integral over these extreme points

$$u(x) = \int_{\partial_m E} K(x, \alpha) \mu_u(d\alpha). \quad (4.B.4)$$

Moreover, as $K(\cdot, \alpha)$ is a harmonic for $\alpha \in \partial_m E$ and $|K(x, \alpha)| \leq C_x$, equation (4.B.4) defines a nonnegative harmonic function u for any finite measure μ_u , and u satisfies $u(x_0) = 1$ if and only if μ_u is a probability measure.

In words, nonnegative harmonic functions are characterized by their distribution on the minimal boundary $\partial_m E$. For the special case of $u = K(\cdot, \alpha)$ for $\alpha \in \partial_m E$ clearly we have $\mu_u = \delta_\alpha$. In some cases, such as the R  my chain studied in [27], $\partial E = \partial_m E$, but in general this is not always the case. A full proof of these facts can be found in [71].

4.B.2 Using h -processes to Condition at Infinity

Given a harmonic function $h \in \mathbb{H}_{1,+}$, we can define a new transition mechanism p^h which “tilts” the dynamics using h . For any $y \in E$ and $x \in E_h := \{x : h(x) > 0\}$ let

$$p^h(x, y) := \frac{h(y)}{h(x)} p(x, y). \quad (4.B.5)$$

Thus the new chain reweights transitions of the original chain by a factor of $h(y)/h(x)$, favoring transitions towards states where h is large. When $h \equiv 1$ we recover the original chain.

We call the new Markov chain $X_n^{(h)}$ with transition probabilities p^h the *Doob h -process* or *Doob h -transform* of X_n corresponding to $h(x)$; it has state space E_h . The h -processes $X_n^{(K(\cdot, \alpha))}$ are of natural interest because they characterize the distributions of all other h -processes:

$$\mathbb{P}_{x_0}^h(B) = \int_{\partial_m E} \mathbb{P}_{x_0}^{K(\cdot, \alpha)}(B) \mu_h(d\alpha). \quad (4.B.6)$$

for the unique measure μ_h supported on the minimal boundary as given in (4.B.4). In particular, these h -processes converge almost surely in the Doob-Martin topology, and

$$\mathbb{P}_x(\lim X_n^{(h)} \in A) = \frac{1}{h(x)} \int_{A \cap \partial_m E} K(x, \alpha) \mu_h(d\alpha), \quad (4.B.7)$$

Since $K(x_0, \alpha) = h(x_0) = 1$, an h -process started at our basepoint x_0 is merely a chain conditioned to have distribution μ_h on the minimal boundary. Again, if $\alpha \in \partial_m E$ then

$\mu_{K(\cdot, \alpha)} = \delta_\alpha$ so

$$X_n^{K(\cdot, \alpha)} \xrightarrow{a.s.} \alpha.$$

Thus we can think of the chain $X_n^{(K(\cdot, \alpha))}$ to be our original Markov chain conditioned to equal $\alpha \in \partial_m E$ “at infinity”.

4.B.3 Infinite Bridges

A finite-step Markov process $(Y_n)_{n=1}^N$ with $Y_0 = x_0$ which has the same backwards transition probabilities as $(X_n)_{n \in \mathbb{N}}$ is called a (finite) *bridge* for X_n .

$$\mathbb{P}(X_n = i \mid X_{n+1} = j) = \mathbb{P}(Y_n = i \mid Y_{n+1} = j), \quad \forall i, j \in E, n < N.$$

Similarly, if the index set is $\mathbb{N}$ we call $(Y_n)_{n \in \mathbb{N}}$ an *infinite bridge*. Let $(X_0^{(y)}, X_1^{(y)}, \dots, X_{N(y)}^{(y)})$ be the finite-step chain obtained from X_n by conditioning on $X_{N(y)} = y$. Then

$$\begin{aligned} \mathbb{P}(X_n^{(y)} = i \mid X_{n+1}^{(y)} = j) &= \frac{\mathbb{P}_{x_0}(X_n = i, X_{n+1} = j, X_{N(y)} = y)}{\mathbb{P}_{x_0}(X_{n+1} = j, X_{N(y)} = y)} \\ &= \frac{\mathbb{P}_{x_0}(X_n = i, X_{n+1} = j) \cdot \mathbb{P}_j(X_{N(y)} = y)}{\mathbb{P}_{x_0}(X_{n+1} = j) \cdot \mathbb{P}_j(X_{N(y)} = y)} \\ &= \mathbb{P}_{x_0}(X_n = i \mid X_{n+1} = j). \end{aligned}$$

Namely, $X_n^{(y)}$ is a finite bridge of X_n . Moreover,

$$\begin{aligned} \mathbb{P}(X_{n+1}^{(y)} = i \mid X_n^{(y)} = j) &= \frac{\mathbb{P}_{x_0}(X_n = j, X_{n+1} = i, X_{N(y)} = y)}{\mathbb{P}_{x_0}(X_n = j, X_{N(y)} = y)} \\ &= \frac{\mathbb{P}_{x_0}(X \text{ hits } j) \cdot p(j, i) \cdot \mathbb{P}_i(X \text{ hits } y)}{\mathbb{P}_{x_0}(X \text{ hits } j) \cdot \mathbb{P}_j(X \text{ hits } y)} \\ &= K(j, y)^{-1} \cdot p(j, i) \cdot K(i, y). \end{aligned}$$

This is essentially the definition of an h -process with $h = K(\cdot, y)$, except for the fact that $K(\cdot, y)$ is not harmonic for $y \in E$. However, as before, intuition suggests we ought to take the limit as y “goes to infinity” to get a truly harmonic function $K(\cdot, \alpha)$. Namely, by Remark 4.B.1 we know a sequence (y_k) in E converges to a point $\alpha \in \partial E$ if and only if $K(x, y_k)$ converges in $\mathbb{R}_{\geq 0}$ to $K(x, \alpha)$ for all $x \in E$ which of course implies

$$\mathbb{P}(X_{n+1}^{(y_k)} = i \mid X_n^{(y_k)} = j) \xrightarrow{k \rightarrow \infty} K(j, \alpha)^{-1} \cdot p(j, i) \cdot K(i, \alpha) =: p^{K(\cdot, \alpha)}(j, i).$$

As observed in [30], such a sequence converges in the Doob Martin topology if and only if finite initial segments of the corresponding bridges converge in distribution, namely

$$(X_0^{(y_k)}, X_1^{(y_k)}, \dots, X_M^{(y_k)}) \Rightarrow (X_0^{K(\cdot, \alpha)}, X_1^{K(\cdot, \alpha)}, \dots, X_M^{K(\cdot, \alpha)}) \quad (4.B.8)$$

as $k \rightarrow \infty$, for every finite M . More generally, given any nonnegative harmonic function with $h(x_0) = 1$, its Doob h -transform is an infinite bridge

$$\begin{aligned} \mathbb{P}(X_n^h = x \mid X_{n+1}^h = y) &= \frac{\mathbb{P}(X_{n+1}^h = y \mid X_n^h = x) \cdot \mathbb{P}(X_n^h = x)}{\mathbb{P}(X_{n+1}^h = y)} \\ &= \frac{\frac{h(y)}{h(x)} \mathbb{P}(X_{n+1} = y \mid X_n = x) \cdot \frac{h(x)}{h(x_0)} \mathbb{P}(X_n = x)}{\frac{h(y)}{h(x_0)} \mathbb{P}(X_{n+1} = y)} \\ &= \mathbb{P}(X_n = x \mid X_{n+1} = y). \end{aligned}$$

Conversely, given any infinite bridge X_n^∞ , then (4.B.6) tells us there exists a measure μ_h so that X_n^∞ is a mixture of the extremal bridges $\mathbb{P}^{K(\cdot, \alpha)}$ and the corresponding harmonic h is given by (4.B.4).

Call an infinite bridge *extremal* if it has a trivial tail σ -field, or equivalently if it cannot be written as a nondegenerate mixture of other infinite bridges. By (4.B.6) and the correspondence between bridges and harmonic functions, all infinite bridges are just mixtures of the extremal bridges. Hence characterizing the extremal bridges allows us to characterize all infinite bridges.

Our observations from Section 4.B.2 tell us that for $\alpha \in \partial_m E$ the conditioned chain $X_n^{K(\cdot, \alpha)}$ is extremal as it converges almost surely to the constant α . Moreover, the converse is true: $X_n^{K(\cdot, \alpha)}$ is extremal only if $\alpha \in \partial_m E$. Hence characterizing the extremal bridges will allow us to characterize the minimal boundary, and thus by (4.B.4) characterize all nonnegative harmonic functions for this chain.

In summary, points of the Doob–Martin boundary can be described as limits of sequences in E , or equivalently through limits of the corresponding finite bridges as in (4.B.8). Within this boundary, the points of the minimal boundary correspond to extremal infinite bridges and extremal nonnegative harmonic functions.

Bibliography

- [1] G.M. Adelson-Velsky and E.M. Landis. “An algorithm for the organization of information”. In: *Soviet Mathematics Doklady* 3 (1962), pp. 1259–1262.
- [2] David J. Aldous. “More Uses of Exchangeability: Representations of Complex Random Structures”. In: *Probability and Mathematical Genetics: Papers in Honour of Sir John Kingman*. Ed. by N. H. Bingham and C. M. Goldie. Vol. 378. London Mathematical Society Lecture Note Series. Cambridge University Press, 2010, pp. 1–36. DOI: 10.1017/CB09780511777414.001. URL: <https://doi.org/10.1017/CB09780511777414.001>.
- [3] Alessandro Baldi Antognini and Alessandra Giovagnoli. “A new biased coin design for the sequential allocation of two treatments”. In: *Journal of the Royal Statistical Society: Series C (Applied Statistics)* 53.4 (2004), pp. 651–664. DOI: <https://doi.org/10.1111/j.1467-9876.2004.00436.x>. eprint: <https://rss.onlinelibrary.wiley.com/doi/pdf/10.1111/j.1467-9876.2004.00436.x>. URL: <https://rss.onlinelibrary.wiley.com/doi/abs/10.1111/j.1467-9876.2004.00436.x>.
- [4] Anthony C. Atkinson. “The comparison of designs for sequential clinical trials with covariate information”. In: *Journal of the Royal Statistical Society: Series A (Statistics in Society)* 165.2 (2002), pp. 349–373. DOI: <https://doi.org/10.1111/1467-985X.00564>. eprint: <https://rss.onlinelibrary.wiley.com/doi/pdf/10.1111/1467-985X.00564>. URL: <https://rss.onlinelibrary.wiley.com/doi/abs/10.1111/1467-985X.00564>.
- [5] Tim Austin. *Exchangeable Random Arrays*. Online. Available at: https://www.stat.berkeley.edu/~aldous/206-Exch/Papers/austin_arrays.pdf. 2013.
- [6] Tim Austin. “On Exchangeable Random Variables and the Statistics of Large Graphs and Hypergraphs”. In: *Probability Surveys* 5 (2008), pp. 80–145.
- [7] Antar Bandyopadhyay and Gursharn Kaur. “Linear de-preferential urn models”. In: *Advances in Applied Probability* 50.4 (2018), pp. 1176–1192. DOI: 10.1017/apr.2018.55.
- [8] Nikhil Bansal et al. “The Gram–Schmidt Walk: A Cure for the Banaszczyk Blues”. In: *Theory of Computing* 15.21 (2019), pp. 1–27. DOI: 10.4086/toc.2019.v015a021.

- [9] Dennis D. Boos. “A Converse to Scheffé’s Theorem”. In: *Annals of Statistics* 13.1 (Mar. 1985), pp. 423–427. DOI: 10.1214/aos/1176346604. URL: <https://doi.org/10.1214/aos/1176346604>.
- [10] C. Borgs et al. “Convergent Sequences of Dense Graphs I: Subgraph Frequencies, Metric Properties and Testing”. In: *Advances in Mathematics* 219.6 (2008), pp. 1801–1851. DOI: 10.1016/j.aim.2008.07.008. URL: <https://doi.org/10.1016/j.aim.2008.07.008>.
- [11] Christian Borgs et al. “Sampling perspectives on sparse exchangeable graphs”. In: *The Annals of Probability* 47.5 (Sept. 2019). ISSN: 0091-1798. DOI: 10.1214/18-aop1320. URL: <http://dx.doi.org/10.1214/18-AOP1320>.
- [12] U. Buchholz et al. “A risk-based restaurant inspection system in Los Angeles County”. In: *Journal of Food Protection* 65.2 (2002), pp. 367–372. DOI: 10.4315/0362-028X-65.2.367.
- [13] Meitner Cadena. *A note on Tauberian Theorems of Exponential Type*. 2015. arXiv: 1503.08793 [math.CA]. URL: <https://arxiv.org/abs/1503.08793>.
- [14] Marc E. Canby et al. “Addressing Polymorphism in Linguistic Phylogenetics”. In: *Transactions of the Philological Society* 122.2 (2024), pp. 191–222. DOI: 10.1111/1467-968X.12289.
- [15] Hua Chen and Kun Chen. “Asymptotic Distributions of Coalescence Times and Ancestral Lineage Numbers for Populations with Temporally Varying Size”. In: *Genetics* 194.3 (2013), pp. 721–736. DOI: 10.1534/genetics.113.151522.
- [16] Hye Soo Choi and Steven N. Evans. “Doob-Martin compactification of a Markov chain for growing random words sequentially”. In: *Stochastic Processes and their Applications* 127.7 (2017), pp. 2428–2445. DOI: 10.1016/j.spa.2016.11.006. URL: <https://doi.org/10.1016/j.spa.2016.11.006>.
- [17] Hye Soo Choi et al. “Doob–Martin Boundaries and Growing Ballot Sequences”. Unpublished manuscript. 2023.
- [18] Payam Dibaieinia, Shayan Tabe-Bordbar, and Tandy Warnow. “FASTRAL: improving scalability of phylogenomic analysis”. In: *Bioinformatics* 37.16 (2021), pp. 2317–2324. DOI: 10.1093/bioinformatics/btab093. URL: <https://doi.org/10.1093/bioinformatics/btab093>.
- [19] M. D. Donsker and S. R. S. Varadhan. “Asymptotic evaluation of certain markov process expectations for large time, I”. In: *Communications on Pure and Applied Mathematics* 28.1 (1975), pp. 1–47. DOI: <https://doi.org/10.1002/cpa.3160280102>. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/cpa.3160280102>. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/cpa.3160280102>.
- [20] J.L. Doob. “Discrete potential theory and boundaries”. In: *J. Math. Mech.* (1959), pp. 433–458.

- [21] Vishwaraj Doshi, Jie Hu, and Do Young Eun. “Self-Repellent Random Walks on General Graphs – Achieving Minimal Sampling Variance via Nonlinear Markov Chains”. In: *Proceedings of the 40th International Conference on Machine Learning (ICML)*. Vol. 202. Proceedings of Machine Learning Research. PMLR, 2023, pp. 8403–8423. URL: <https://proceedings.mlr.press/v202/doshi23a.html>.
- [22] Richard Durrett. *Stochastic Calculus: A Practical Introduction*. Probability and Stochastics Series. Boca Raton, FL: CRC Press, 1996. ISBN: 0-8493-8071-5.
- [23] Bradley Efron. “Forcing a Sequential Experiment to be Balanced”. In: *Biometrika* 58.3 (1971), pp. 403–417. ISSN: 00063444, 14643510. URL: <http://www.jstor.org/stable/2334377> (visited on 07/21/2026).
- [24] Gabor Elek and Balazs Szegedy. “A Measure-Theoretic Approach to the Theory of Dense Hypergraphs”. In: *Advances in Mathematics* 231.3–4 (2012), pp. 1731–1772. DOI: 10.1016/j.aim.2012.06.022. URL: <https://doi.org/10.1016/j.aim.2012.06.022>.
- [25] Gábor Elek and Balázs Szegedy. “A Measure-Theoretic Approach to the Theory of Dense Hypergraphs”. In: *Advances in Mathematics* 231.3–4 (2012), pp. 1731–1772.
- [26] Gary D. Eppen. “Convexity-Preserving Transformations with Stochastic Matrices”. In: *SIAM Journal on Applied Mathematics* 14.2 (1966). Accessed: 2025-11-04, pp. 234–236. URL: <http://www.jstor.org/stable/2946261>.
- [27] Steven N. Evans, Rudolf Grübel, and Anton Wakolbinger. “Doob–Martin boundary of Rémy’s tree growth chain”. In: *The Annals of Probability* 45.1 (2017), pp. 225–277. DOI: 10.1214/15-AOP1078. URL: <https://doi.org/10.1214/15-AOP1078>.
- [28] Steven N. Evans et al. “Estimating Language Phylogenies from Polymorphic Characters”. Unpublished manuscript. 2026.
- [29] Bailey Flanigan et al. “Fair algorithms for selecting citizens’ assemblies”. In: *Nature* 596 (2021), pp. 548–552. DOI: 10.1038/s41586-021-03788-6.
- [30] Hans Föllmer. “Phase transition and Martin boundary”. In: *Séminaire de Probabilités IX Université de Strasbourg*. Ed. by P. A. Meyer. Berlin, Heidelberg: Springer Berlin Heidelberg, 1975, pp. 305–317. ISBN: 978-3-540-37518-0.
- [31] Didier Fraix-Burnet, Marc Thuillard, and Asis K. Chattopadhyay. “Multivariate approaches to classification in extragalactic astronomy”. In: *Frontiers in Astronomy and Space Sciences* 2 (2015). ISSN: 2296-987X. DOI: 10.3389/fspas.2015.00003.
- [32] R. Grübel. “Ranks, copulas, and permutons”. In: *Metrika* 87.2 (May 2023), pp. 155–182. ISSN: 1435-926X. DOI: 10.1007/s00184-023-00908-2. URL: <http://dx.doi.org/10.1007/s00184-023-00908-2>.

- [33] Rudolf Grübel. “Persisting randomness in randomly growing discrete structures: graphs and search trees”. In: *Discrete Mathematics; Theoretical Computer Science* Vol. 18 no. 1. Analysis of Algorithms (Oct. 2015). ISSN: 1365-8050. DOI: 10.46298/dmtcs.644. URL: <http://dx.doi.org/10.46298/dmtcs.644>.
- [34] Tuomas Haarnoja et al. “Soft Actor-Critic: Off-Policy Maximum Entropy Deep Reinforcement Learning with a Stochastic Actor”. In: *Proceedings of the 35th International Conference on Machine Learning (ICML)*. Vol. 80. Proceedings of Machine Learning Research. PMLR, 2018, pp. 1861–1870. URL: <https://proceedings.mlr.press/v80/haarnoja18b.html>.
- [35] Christopher Harshaw et al. “Balancing Covariates in Randomized Experiments with the Gram–Schmidt Walk Design”. In: *Journal of the American Statistical Association* 119.548 (2024), pp. 2934–2946. DOI: 10.1080/01621459.2023.2285474. eprint: <https://doi.org/10.1080/01621459.2023.2285474>. URL: <https://doi.org/10.1080/01621459.2023.2285474>.
- [36] Max Hill, Brandon Legried, and Sebastien Roch. *Species tree estimation under joint modeling of coalescence and duplication: sample complexity of quartet methods*. 2020. arXiv: 2007.06697 [math.PR]. URL: <https://arxiv.org/abs/2007.06697>.
- [37] Carlos Hoppen et al. “Limits of Permutation Sequences”. In: *Journal of Combinatorial Theory, Series B* 103.1 (2013), pp. 93–113. DOI: 10.1016/j.jctb.2012.09.003.
- [38] Yanqing Hu. “Generalized Efron’s biased coin design and its theoretical properties”. In: *Journal of Applied Probability* 53.2 (2016), pp. 327–340. DOI: 10.1017/jpr.2016.3.
- [39] Hisham Husain, Kamil Ciosek, and Ryota Tomioka. “Regularized Policies Are Reward Robust”. In: *Proceedings of the 24th International Conference on Artificial Intelligence and Statistics*. Vol. 130. Proceedings of Machine Learning Research. 2021, pp. 64–72.
- [40] Olav Kallenberg. “On the representation theorem for exchangeable arrays”. In: *Journal of Multivariate Analysis* 30.1 (1989), pp. 137–154. ISSN: 0047-259X. DOI: [https://doi.org/10.1016/0047-259X\(89\)90092-4](https://doi.org/10.1016/0047-259X(89)90092-4). URL: <https://www.sciencedirect.com/science/article/pii/0047259X89900924>.
- [41] Olav Kallenberg. *Probabilistic Symmetries and Invariance Principles*. 1st ed. New York: Springer, 2005. ISBN: 978-0387257174.
- [42] Samuel Karlin and James McGregor. “Coincidence Probabilities”. In: *Pacific Journal of Mathematics* 9.4 (1959), pp. 1141–1164.
- [43] Gursharn Kaur. “Negatively reinforced balanced urn schemes”. In: *Advances in Applied Mathematics* 105 (2019), pp. 48–82.
- [44] Götz Kersting and Anton Wakolbinger. “The time to absorption in Λ -coalescents”. In: *Advances in Applied Probability* 50.A (2018), pp. 177–190. DOI: 10.1017/apr.2018.78. URL: <https://doi.org/10.1017/apr.2018.78>.

- [45] Sophie J Kersting, Kristina Wicke, and Mareike Fischer. “Tree balance in phylogenetic models”. In: *Philosophical Transactions of the Royal Society B: Biological Sciences* 380.1919 (2025), p. 20230303. DOI: 10.1098/rstb.2023.0303. URL: <https://doi.org/10.1098/rstb.2023.0303>.
- [46] J. F. C. Kingman. “The Representation of Partition Structures”. In: *Journal of the London Mathematical Society. Second Series* 18.2 (1978), pp. 374–380. DOI: 10.1112/jlms/s2-18.2.374.
- [47] J.F.C Kingman. *Poisson Processes. Oxford Studies in Probability*. eng. 1st ed. Vol. 3. Oxford: Oxford University Press, 1992. ISBN: 0198536933.
- [48] Arun Kumar Kuchibhotla and Abhishek Chakraborty. “Moving Beyond Sub-Gaussianity in High-Dimensional Statistics: Applications in Covariance Estimation and Linear Regression”. In: *Information and Inference: A Journal of the IMA* 11.4 (2022), pp. 1383–1456.
- [49] Fredrik Lauritzen and Geir Holden. “Intelligence-based doping control planning improves testing effectiveness: Perspectives from a national anti-doping organisation”. In: *Drug Testing and Analysis* 15.5 (2023), pp. 506–515. DOI: <https://doi.org/10.1002/dta.3435>. eprint: <https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/pdf/10.1002/dta.3435>. URL: <https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/abs/10.1002/dta.3435>.
- [50] Nicholas J. Loman and Mark J. Pallen. “Sequencing of bacterial genomes: principles and insights”. In: *Microbiology and Molecular Biology Reviews* 79.3 (2015), pp. 363–385. DOI: 10.1128/MMBR.00009-15. URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC3927574/>.
- [51] Laszlo Lovasz and Balazs Szegedy. “Limits of dense graph sequences”. In: *Journal of Combinatorial Theory, Series B* 96.6 (2006), pp. 933–957. DOI: 10.1016/j.jctb.2006.05.002. URL: <https://doi.org/10.1016/j.jctb.2006.05.002>.
- [52] Tigran Markaryan and William F. Rosenberger. “Exact properties of Efron’s biased coin randomization procedure”. In: *The Annals of Statistics* 38.3 (2010), pp. 1546–1567. DOI: 10.1214/09-AOS758. URL: <https://doi.org/10.1214/09-AOS758>.
- [53] Zachary McNulty. *An Improved Bipartition Cover Bound for the Multispecies Coalescent Model*. 2026. arXiv: 2604.09509 [math.PR]. URL: <https://arxiv.org/abs/2604.09509>.
- [54] Zachary McNulty. *msc_bipartition_cover: Code repository*. https://github.com/zackmcnulty/msc_bipartition_cover. GitHub repository. Accessed: 2025-12-21.
- [55] Pierre-Loïc Méliot. “A Survey of the Theory of Graphons and Permutons”. In: (). URL: https://www.imo.universite-paris-saclay.fr/~pierre-loic.meliot/surveys/graphon_survey.pdf.
- [56] Sean P. Meyn and Richard L. Tweedie. *Markov Chains and Stochastic Stability*. 2nd ed. Cambridge University Press, 2009.

- [57] Bernard Michaux, Malte C. Ebach, and Elizabeth M. Dowding. “Cladistic methods as a tool for terrane analysis: a New Zealand example”. In: *New Zealand Journal of Geology and Geophysics* 61.2 (2018), pp. 127–135. DOI: 10.1080/00288306.2018.1441159.
- [58] Siavash Mirarab and Tandy Warnow. “ASTRAL-II: coalescent-based species tree estimation with many hundreds of taxa and thousands of genes”. In: *Bioinformatics* 31.12 (2015), pp. i44–i52. DOI: 10.1093/bioinformatics/btv234. URL: <https://doi.org/10.1093/bioinformatics/btv234>.
- [59] Siavash Mirarab et al. “ASTRAL: genome-scale coalescent-based species tree estimation”. In: *Bioinformatics* 30.17 (2014), pp. i541–i548. DOI: 10.1093/bioinformatics/btu462. URL: <https://doi.org/10.1093/bioinformatics/btu462>.
- [60] Martin Möhle and Helmut Pitters. “Absorption Time and Tree Length of the Kingman Coalescent and the Gumbel Distribution”. In: *Markov Processes and Related Fields* 21.2 (2015), pp. 317–338. arXiv: 1502.03983 [math.PR].
- [61] J. Nievergelt and E.M. Reingold. “Binary search trees of bounded balance”. In: *SIAM Journal on Computing* 2.1 (1973), pp. 33–43.
- [62] Pedro A. Ortega et al. *Information-Theoretic Bounded Rationality*. 2015. arXiv: 1512.06789 [stat.ML]. URL: <https://arxiv.org/abs/1512.06789>.
- [63] Robin Pemantle and Jeffrey S. Rosenthal. “Moment conditions for a sequence with negative drift to be uniformly bounded in L^r ”. In: *Stochastic Processes and their Applications* 82.1 (1999), pp. 143–155. DOI: 10.1016/S0304-4149(99)00003-6.
- [64] Mihaela Pertea and Steven L. Salzberg. “The status of the human gene catalogue”. In: *Nature* 617.7960 (2023), pp. 687–694. DOI: 10.1038/s41586-023-06490-x. URL: <https://www.nature.com/articles/s41586-023-06490-x>.
- [65] Jim Pitman. *Combinatorial Stochastic Processes*. Ed. by Jean Picard. Vol. 1875. Lecture Notes in Mathematics. Berlin, Heidelberg: Springer, 2006. ISBN: 978-3-540-30990-1. DOI: 10.1007/b11601500.
- [66] S. J. Pocock and R. Simon. “Sequential treatment assignment with balancing for prognostic factors in the controlled clinical trial”. In: *Biometrics* 31.1 (1975), pp. 103–115.
- [67] Bruce Rannala and Ziheng Yang. “The Multispecies Coalescent Model and Species Tree Inference”. In: *Phylogenetics in the Genomic Era*. Ed. by Cédric Scornavacca, Frédéric Delsuc, and Nicolas Galtier. Available at <https://hal.inria.fr/PGE>. No commercial publisher — Authors open access book, 2020. Chap. 3.3. URL: <https://hal.inria.fr/PGE>.
- [68] Hannes Risken. *The Fokker–Planck Equation: Methods of Solution and Applications*. 2nd ed. Vol. 18. Springer Series in Synergetics. Berlin, Heidelberg: Springer, 1996. ISBN: 978-3-540-61530-9. DOI: 10.1007/978-3-642-61544-3.

- [69] Rafael A. Rosales, Fernando P.A. Prado, and Benito Pires. “Vertex reinforced random walks with exponential interaction on complete graphs”. In: *Stochastic Processes and their Applications* 148 (2022), pp. 353–379. ISSN: 0304-4149. DOI: <https://doi.org/10.1016/j.spa.2022.03.007>. URL: <https://www.sciencedirect.com/science/article/pii/S0304414922000631>.
- [70] Adrien Saumard and Jon A. Wellner. “Log-concavity and strong log-concavity: a review”. In: *Statistics Surveys* 8 (2014), pp. 45–114. DOI: 10.1214/14-SS107. URL: <https://doi.org/10.1214/14-SS107>.
- [71] Stanley A. Sawyer. “Martin Boundaries and Random Walks”. In: (). URL: <https://www.math.wustl.edu/~sawyer/hmhandouts/martbrwf.pdf>.
- [72] Charles Semple and Mike Steel. *Phylogenetics*. Oxford Lecture Series in Mathematics and Its Applications 24. Oxford: Oxford University Press, 2003. ISBN: 9780198509424. DOI: 10.1093/oso/9780198509424.001.0001. URL: <https://doi.org/10.1093/oso/9780198509424.001.0001>.
- [73] Moshe Shaked and J. George Shanthikumar. *Stochastic Orders*. 1st. Springer Series in Statistics. Springer, 2007. ISBN: 978-0-387-32915-4. DOI: 10.1007/978-0-387-34675-5.
- [74] Shameek Shekhar, Sebastien Roch, and Siavash Mirarab. “Species Tree Estimation Using ASTRAL: How Many Genes Are Enough?”. In: *IEEE/ACM Transactions on Computational Biology and Bioinformatics* 15.5 (Sept. 2018). Epub 2017 Sep 29, pp. 1738–1747. DOI: 10.1109/TCBB.2017.2757930.
- [75] Richard L. Smith. “Sequential Treatment Allocation Using Biased Coin Designs”. In: *Journal of the Royal Statistical Society. Series B (Methodological)* 46.3 (1984), pp. 519–543. ISSN: 00359246. URL: <http://www.jstor.org/stable/2345691> (visited on 07/21/2026).
- [76] Oswin So, Ziyi Wang, and Evangelos A. Theodorou. “Maximum Entropy Differential Dynamic Programming”. In: *2022 International Conference on Robotics and Automation (ICRA)*. IEEE, 2022, pp. 3422–3428. DOI: 10.1109/ICRA46639.2022.9812228. URL: <https://doi.org/10.1109/ICRA46639.2022.9812228>.
- [77] Jason A. Somarelli et al. “PhyloOncology: Understanding cancer through phylogenetic analysis”. In: *Biochimica et Biophysica Acta (BBA) - Reviews on Cancer* 1867.2 (2017), pp. 101–108. DOI: 10.1016/j.bbcan.2016.10.006.
- [78] Terence Tao. “A Correspondence Principle Between (Hyper)graph Theory and Probability Theory, and the (Hyper)graph Removal Lemma”. In: *Journal d’Analyse Mathématique* 103.1 (2007), pp. 1–45. DOI: 10.1007/s11854-008-0001-0.
- [79] Emanuel Todorov. “Linearly-solvable Markov decision problems”. In: *Advances in Neural Information Processing Systems*. Ed. by B. Schölkopf, J. Platt, and T. Hoffman. Vol. 19. MIT Press, 2006. URL: https://proceedings.neurips.cc/paper_files/paper/2006/file/d806ca13ca3449af72a1ea5aedbed26a-Paper.pdf.

- [80] Lawrence H Uricchio, Tandy Warnow, and Noah A Rosenberg. “An analytical upper bound on the number of loci required for all splits of a species tree to appear in a set of gene trees”. In: *BMC Bioinformatics* 17.Suppl 14 (Nov. 2016), p. 417. DOI: 10.1186/s12859-016-1266-4.
- [81] Pranjal Vachaspati and Tandy Warnow. “FastRFS: Fast and Accurate Robinson–Foulds Supertrees Using Constrained Exact Optimization”. In: *Bioinformatics* 33.5 (2017), pp. 631–639. DOI: 10.1093/bioinformatics/btw600.
- [82] Roman Vershynin. *High-Dimensional Probability: An Introduction with Applications in Data Science*. 2nd ed. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge: Cambridge University Press, 2026. ISBN: 9781009490672. DOI: 10.1017/9781009490672. URL: <https://doi.org/10.1017/9781009490672>.
- [83] Haoran Wang, Thaleia Zariphopoulou, and Xun Yu Zhou. “Reinforcement Learning in Continuous Time and Space: A Stochastic Control Approach”. In: *Journal of Machine Learning Research* 21.198 (2020), pp. 1–34.
- [84] George P. Yanev. “Exponential and Hypoexponential Distributions: Some Characterizations”. In: *Mathematics* 8.12 (Dec. 2020), p. 2207. ISSN: 2227-7390. DOI: 10.3390/math8122207. URL: <http://dx.doi.org/10.3390/math8122207>.
- [85] Ziheng Yang and Bruce Rannala. “Bayes Estimation of Species Divergence Times and Ancestral Population Sizes Using DNA Sequences from Multiple Loci”. In: *Genetics* 164.4 (2003), pp. 1645–1656. URL: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1462670/pdf/12930768.pdf>.
- [86] Chao Zhang and Siavash Mirarab. “Weighting by Gene Tree Uncertainty Improves Accuracy of Quartet-based Species Trees”. In: *Molecular Biology and Evolution* 39.12 (2022), msac215. ISSN: 1537-1719. DOI: 10.1093/molbev/msac215. URL: <https://doi.org/10.1093/molbev/msac215>.
- [87] Chao Zhang et al. “ASTRAL-III: polynomial time species tree reconstruction from partially resolved gene trees”. In: *BMC Bioinformatics* 19.Suppl 6 (2018), p. 153. DOI: 10.1186/s12859-018-2129-y.
- [88] Yufei Zhao. “Hypergraph Limits: A Regularity Approach”. In: *Random Structures & Algorithms* 47.2 (2015), pp. 205–226. DOI: 10.1002/rsa.20537.